

GDM V1.0.1

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2026-02-23

Eligibility Criteria

Variable: redcap_record_id

Type: Free Text Entry (text)

Description: REDCap Record ID

Reference: n/a

Data Source: REDCap system-generated record identifier

Curation Directives:

- **Definition:** The unique, system-assigned identifier for a patient record within the REDCap database. This ID is used to distinguish and track individual records and is generated automatically by REDCap at the time of record creation.
- **Source Documents:**
 - REDCap project interface (record status dashboard or data entry form header)
- **Curation Instructions:**
 1. This field is typically auto-populated by REDCap and should not require manual entry.
 2. If manual entry is needed (e.g., during migration, template generation, or manual imports), ensure the value entered matches the REDCap-assigned ID exactly.
 3. Do not alter, abbreviate, or reuse record IDs across different patients or forms.
 4. If duplicate IDs are encountered, escalate to project administrator immediately.
- **Notes:**
 - This is a system-level identifier used for internal tracking; it is not a patient identifier and is safe for use in de-identified exports.
 - Always check the REDCap branching logic or record status to ensure the record is active and properly classified before proceeding with further curation.

Variable: curator_name

Type: Free Text Entry (text)

Description: Curator Name

Reference: n/a

Data Source: Curator (person entering or verifying the data)

Curation Directives:

- **Definition:** Records the full name of the individual responsible for curating or verifying the data in this form. Used for internal tracking, accountability, and quality assurance. This field is not shared outside of the data coordinating center or project team.

- **Source Documents:**
 - Not applicable; entered directly by the curator
- **Curation Instructions:**
 - Enter your name in the format: **First Name, Last Name** (e.g., Jane Smith).
- **Notes:**
 - This field supports internal traceability and quality control. Do not leave it blank.

Variable: `curation_cohort`

Type: Categorical (Dropdown)

Description: GENIE Cohort / Curation Project

Reference: n/a

Data Source: Project assignment list; institutional or study-level documentation indicating assigned GENIE or related curation cohort

Curation Directives:

- **Definition:** Identifies the specific GENIE cohort or curation project under which this patient's data is being curated. This classification aligns the record with a defined study population or institutional submission effort.
- **Permitted Values:**
 - 1 = Study 1
 - 2 = Study 2
 - 3 = Study 3
 - *(Additional values may be added to the dropdown in the future as new cohorts are launched. Consult the current project configuration or curation guidance for full list.)*
- **Source Documents:**
 - Study Protocol
- **Curation Instructions:**
 - Select the correct cohort or project that this patient was assigned to for curation.

Variable: `eligibility_yn`

Type: Binary (Yes/No)

Description: Indicates whether the patient is eligible for the study

Reference: n/a

Data Source: clinical eligibility checklist, relevant EHR documentation

Curation Directives:

- **Definition:** Indicates whether the patient meets all criteria for eligibility based on the protocol-defined inclusion and exclusion criteria for the study.
- **Permitted Values:**
 - 1 = Yes
 - 0 = No
- **Source Documents:**

- EHR documentation for inclusion / exclusion criteria specified by study protocol.
- IRB-approved protocol documents (for criteria reference)

- **Curation Instructions:**

1. Review documentation of eligibility determination for the study in the source documents.
2. If the patient meets all inclusion and no exclusion criteria, select **Yes (1)**.
3. If the patient fails to meet inclusion or meets any exclusion criteria, select **No (0)**.
4. Confirm that eligibility status is final (i.e., not under review or pending labs).
5. If eligibility is unclear or conflicting across documents, escalate to study coordinator or supervisor for adjudication.

- **Notes:**

- Eligibility must be explicitly documented aligning with the inclusion/exclusion criteria as specified in the study protocol.

Variable: `ineligible_reason`

Type: Free Text Entry (text)

Description: Reason for ineligibility

Reference: n/a

Data Source: clinical eligibility checklist, relevant EHR documentation

Curation Directives:

- **Definition:** Free-text documentation of the specific reason a patient was determined to be ineligible for the study, based on protocol-defined criteria. This field is only completed if the patient is marked as ineligible (`eligibility_yn` = No).

- **Source Documents:**

- EHR documentation for inclusion / exclusion criteria specified by study protocol.
- IRB-approved protocol documents (for criteria reference)

- **Curation Instructions:**

1. Review documentation to confirm the patient was marked as ineligible (`eligibility_yn` = No).
2. Identify the documented reason(s) the patient did not meet eligibility criteria (e.g., abnormal lab result, stage at diagnosis, age outside protocol range, comorbid condition, prior treatment).
3. Enter the reason exactly, using complete sentences or key phrases for clarity.
4. If multiple reasons are documented, include all relevant reasons.

- **Curator Quality Review:**

- Confirm alignment between the ineligibility reason and protocol-defined inclusion/exclusion criteria.
- Ensure the reason is not vague (e.g., “not eligible” alone is insufficient. Include specifics).

- **Notes:**

- This field is conditionally displayed and should only be completed when `eligibility_yn` = No.
- Avoid interpretations. Only transcribe or summarize directly documented reasons.

Variable: `curator_role`

Type: Categorical (Dropdown)

Description: Role of the individual performing or verifying curation

Reference: n/a

Data Source: Curator declaration (entered manually)

Curation Directives:

- **Definition:** Identifies the organizational role of the person responsible for data curation or quality review. Used for tracking and auditing purposes.
- **Permitted Values:**
 - 1 = In-House Curator
 - 2 = External Curator
 - 3 = QA Manager
 - 4 = PI
- **Curation Instructions:**
 1. Select the option that best represents your role when performing the curation or review.
 2. Use “In House Curator” or “QA Manager” if you are part of the submitting institution, depending on your role.
 3. Use “External Curator” if you are curating through the third-party vendor.
 4. Use “PI” only if you are the principal investigator performing or verifying the curation directly.
- **Curator Quality Review:**
 - Ensure that the selected role matches internal rosters or project tracking logs if required for audit.
 - Do not leave blank unless your system populates this automatically.
- **Notes:**
 - This field supports audit trails and workload tracking. Use your assigned role, even if performing a different task temporarily.

Variable: curation_type

Type: Categorical (Dropdown)

Description: Type of curation activity performed

Reference: n/a

Data Source: Curator declaration (entered manually)

Curation Directives:

- **Definition:** Indicates the classification of the curation activity, whether for production, training, validation, or other purposes.
- **Permitted Values:**
 - 1 = Curation Case
 - 2 = Training Curation Case
 - 3 = SDV Case
 - 4 = Other
- **Source Documents:**
 - Case selection list, internal documentation
- **Curation Instructions:**
 1. Select the category that best describes the type of case you are curating.
 2. Choose **Curation Case** for standard production cases intended for data delivery or submission.
 3. Choose **Training Curation Case** if the case is part of onboarding or skills development.
 4. Choose **SDV Case** for source data verification or QA audit checks.
 5. Use **Other** only if none of the above categories apply

- **Curator Quality Review:**
 - Verify consistency with any assigned logs or training identifiers, if applicable.
 - Ensure the curation type aligns with the expected workflow or batch assignment.
- **Notes:**
 - This field is important for downstream tracking of data inclusion in final datasets.

Cancer Patient Information

For Site Internal Use Only:

The following variables are for site internal use only.

Variable: `patient_mrn`

Type: Free Text Entry (text)

Description: Internal Patient Medical Record Number (MRN)

Reference: n/a

Data Source: Medical record system (institution-specific EHR or clinical data warehouses)

Curation Directives:

- **Definition:** Record the internal Medical Record Number (MRN) assigned to the patient by your site or center. This identifier is site-specific and used to internally link the patient to their clinical, biospecimen, or sequencing records. It is not shared outside of the institution or data coordinating center. This identifier is site-specific and used to internally link the patient to their clinical, biospecimen, or sequencing records. It is not shared outside of the institution.
- **Source Documents:**
 - EHR patient demographics or header section
 - Admission/registration summary
- **Curation Instructions:**
 1. Locate the internal MRN used by your institution for the patient associated with the sample.
 2. Record the MRN exactly as shown in the medical record, including any leading zeros or formatting characters (e.g., dashes, if applicable).
 3. Confirm that the MRN corresponds to the correct patient intended for curation.
 4. Do not enter research IDs, study codes, or identifiers from external institutions.
- **Notes:**
 - This field is marked as an identifier and should not be used for de-identified exports.
 - Used for internal linkage and audit trails only.

Variable: `patient_study_id`

Type: Free Text Entry (text)

Description: Site-Specific Patient Study ID

Reference: n/a

Data Source: Institutional enrollment or study tracking systems

Curation Directives:

- **Definition:** Records the patient-specific study identifier assigned by the submitting institution for internal tracking and linkage. Unlike the MRN, this identifier is typically used for research or project-level documentation and does not correspond to system-wide clinical identifiers. It is site-generated and not shared outside the institution.
 - **Source Documents:**
 - Site-specific case tracking logs
 - **Curation Instructions:**
 - Enter the patient’s study-specific ID exactly as assigned by your site or data management team.
 - **Curator Quality Review:**
 - Confirm consistency with other internal logs or documentation.
 - **Notes:**
 - This field is considered an internal-use-only identifier and will not be shared outside of your institution’s REDCap.
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Piped In - Tier1A Data Elements: Patient

The following variables are *piped in* from the existing Tier1A dataset and are not manually curated. These fields are auto-populated via data integration from the GENIE Tier1A repository via the provided REDCap export file, sourced and maintained by Sage Bionetworks. They are included for reference, audit trails, or linkage purposes, and are typically marked as @READONLY in the data collection system. Curators should not modify these fields, but may use them to validate or contextualize other data points.

Piped-in Tier1A Variables:

- `patient_id` – GENIE Patient ID
 - `center` – Submitting Center
 - `sex` – Sex assigned at birth [NAACCR #220]
 - `primary_race` – Primary Race [NAACCR #160]
 - `secondary_race` – Secondary Race [NAACCR #161]
 - `tertiary_race` – Tertiary Race [NAACCR #162]
 - `ethnicity` – Hispanic/Latino Ethnicity [NAACCR #190]
 - `birth_year` – Year of Birth
 - `year_contact` – Year of Last Contact
 - `int_contact` – Interval (days) from DOB to Last Contact
 - `dead` – Vital Status (Deceased: Yes/No)
 - `year_death` – Year of Death
 - `int_dod` – Interval (days) from DOB to Date of Death
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Curated Patient-Level Information

For all fields: complete below fields utilizing clinical documentation closest to date of diagnosis

Note: Although certain patient demographic variables (such as sex, race, ethnicity, birth year, and vital status) can be piped in from the Tier1A dataset, curators are required to complete all corresponding curated fields below in the data entry form, even if the values match those that were piped in. These curated fields

must be independently abstracted from the patient’s medical record and should reflect the most current and accurate documentation available.

Do not alter any piped-in values as these are maintained for audit and consistency with the original Tier1A submission. However, if curated values differ from the piped-in fields, record the curated values exactly as documented into the curated variable fields. Discrepancies should be captured transparently. The curated variable fields will represent the most accurate and current abstraction based on all available clinical documentation, and may reflect documentation updates, corrections, or greater specificity than was available at the time of Tier1A data submission.

Variable: patient_id_curated

Type: Free Text Entry (text)

Description: Curated GENIE Patient ID

Reference: n/a

Data Source: Main GENIE registry

Curation Directives:

- **Definition:** Records the patient’s GENIE ID
- **Source Documents:**
 - GENIE submission tracker
 - Institutional REDCap or internal case list
- **Curation Instructions:**
 - Ensure no leading or trailing spaces are included.
 - Do not reuse GENIE IDs across patients
- **Curator Quality Review:**
 - If unclear or inconsistent, cross-check with site coordinator, or data manager.
- **Notes:**
 - This field is used in downstream data integration with genomic data and external partners (e.g., Sage Synapse, AACR GENIE).
 - While this field is typically considered de-identified, it remains a key linking field and must be accurate.

Variable: birth_date

Type: Date (MM/DD/YYYY)

Description: Date of Birth

Reference: NAACCR #240

Data Source: Clinical/ EHR documentation

Curation Directives:

- **Definition:** Records the patient’s date of birth as documented in the medical record. This field captures the actual birth date and is used for age-related calculations and eligibility verification.
- **Source Documents:**
 - Demographics or patient header section of the electronic health record (EHR)
 - Intake or admission forms

- Registration or official identity documentation
- **Curation Instructions:**
 1. Record the date of birth exactly as documented in official demographic or registration sources.
 2. Enter the date in **MM-DD-YYYY** format.
 3. If multiple dates appear, prioritize the one in official registration/demographics documentation.
 4. If not available, follow estimation guidance (see Appendix: Date Estimation Rules). Do not leave blank unless explicitly directed.
 5. If the date of birth is not documented, leave the field blank and note the absence in your curation log.
- **Curator Quality Review:**
 - Ensure the date format is correct (MM-DD-YYYY).
 - Verify there are no typographical errors (e.g., transposed digits or wrong century).
 - If conflicting dates are present, escalate to a QA manager for adjudication.
- **Notes:**
 - Do not alter the Tier1A-piped birth year, but you must still complete this curated field even if the values match.
 - Estimation is allowed as per NAACCR registry standards (see [here](#))

Variable: `birth_year_curated`

Type: Integer (Calculated)

Description: Four-digit birth year, automatically derived from full date of birth

Reference: n/a

Data Source: Derived from `birth_date` (curated field)

Curation Directives:

- **Definition:** Automatically calculates the patient’s year of birth from the manually curated `birth_date` field. Used for consistency checks and age-based validation.
- **Source Documents:**
 - Not directly abstracted; sourced from the `birth_date` field already curated by the user
- **Curation Instructions:**
 1. Curators do not need to enter or modify this field. It is automatically calculated using the REDCap function `year([birth_date])`.
 2. Ensure that the `birth_date` field is completed accurately, as this field’s accuracy depends entirely on that input.
 3. If the calculated birth year appears incorrect, double-check the `birth_date` entry for formatting or typographical errors.
- **Curator Quality Review:**
 - Confirm that the calculated year matches the four-digit year portion of the entered `birth_date`.

Variable: `sex_curated`

Type: Categorical (Dropdown)

Description: Sex

Reference: NAACCR #220

Data Source: Medical record, patient-reported documentation, or administrative records

Curation Directives:

- **Definition:** Code for the sex of the patient as recorded in the medical record or official registration documentation. Refers to the biological assignment at birth, not gender identity.
- **Permitted Values:**
 - 1 = Male
 - 2 = Female
 - 3 = Other (intersex, disorders of sexual development/DSD)
 - 4 = Transsexual, NOS
 - 5 = Transsexual, natal male
 - 6 = Transsexual, natal female
 - 9 = Not stated/Unknown
- **Source Documents:**
 - Demographic or face sheet from the EHR
 - Admission or intake forms
 - Patient-reported history if clearly stated and dated
- **Curation Instructions:**
 1. Abstract the value that reflects the patient’s sex assigned at birth, as documented closest to the date of diagnosis.
 2. If multiple values are found, prioritize the one from the official registration or demographics section.
 3. If the field is blank, not stated, or unclear in the record, select code 9.
- **Curator Quality Review:**
 - Ensure the selected code aligns with clinical documentation.
- **Notes:**
 - Do not infer sex based on names, pronouns, or phenotype—only use explicitly documented sources.
 - If a discrepancy exists between curated and Tier1A values, record the curated value exactly as documented and do not alter the piped-in data.

Variable: gender_curated

Type: Categorical (dropdown)

Description: Gender Identity

Reference: USCDI v2

Data Source: Medical record (intake notes, social history, patient-reported forms)

Curation Directives:

- **Definition:** A person’s self-identified internal sense of being a man, woman, both, neither, or another gender. This variable captures gender identity as recorded in the medical record, which may differ from sex assigned at birth.
- **Permitted Values:**
 - 1 = Male
 - 2 = Female
 - 3 = Female-to-Male (FTM)/Transgender Male/Trans Man
 - 4 = Male-to-Female (MTF)/Transgender Female/Trans Woman
 - 5 = Genderqueer, neither exclusively male nor female
 - 88 = Not documented
 - 99 = Unknown
- **Source Documents:**
 - Social history or intake sections of the EHR

- Patient-reported forms or questionnaires
 - Psychosocial assessments or mental health notes
 - Provider notes that explicitly document gender identity
- **Curation Instructions:**
 1. Review the medical record for explicit documentation of the patient’s gender identity.
 2. Select the code that most closely matches the recorded identity.
 3. If “transgender,” choose the option based on the patient’s identified gender (not natal sex).
 4. If the patient identifies as genderqueer or similar, use code 5.
 5. If gender identity is not documented at all, use code 88.
 6. If the record contains ambiguous or conflicting information, use code 99 and document the issue in your curation log.
 - **Curator Quality Review:**
 - Confirm that the value reflects gender identity and not sex assigned at birth.
 - Do not assume gender identity based on pronouns or appearance—only use explicit documentation.
 - **Notes:**
 - This variable is distinct from `sex_curated` and must be curated independently.

Variable: `primary_race_curated`

Type: Categorical (dropdown)

Description: Primary Race

Reference: NAACCR #160

Data Source: Medical record (demographics, intake forms, patient-reported documentation)

Curation Directives:

- **Definition:** Code the patient’s primary race as documented in the medical record. This should reflect the first-listed or most prominently documented race and is distinct from ethnicity or Hispanic origin.
- **Permitted Values:**
 - 1 = White
 - 2 = Black
 - 3 = American Indian, Aleutian, or Eskimo (includes all indigenous populations of the Western hemisphere)
 - 4 = Chinese
 - 5 = Japanese
 - 6 = Filipino
 - 7 = Hawaiian
 - 8 = Korean
 - 10 = Vietnamese
 - 11 = Laotian
 - 12 = Hmong
 - 13 = Kampuchean (Cambodian)
 - 14 = Thai
 - 15 = Asian Indian or Pakistani, NOS (code 09 prior to Version 12)
 - 16 = Asian Indian
 - 17 = Pakistani
 - 20 = Micronesian, NOS
 - 21 = Chamorro/Chamoru
 - 22 = Guamanian, NOS

- 25 = Polynesian, NOS
- 26 = Tahitian
- 27 = Samoan
- 28 = Tongan
- 30 = Melanesian, NOS
- 31 = Fiji Islander
- 32 = New Guinean
- 96 = Other Asian, including Asian, NOS and Oriental, NOS
- 97 = Pacific Islander, NOS
- 98 = Other
- 99 = Unknown

- **Source Documents:**

- Demographics section of the EHR
- Patient intake or registration forms
- Clinical notes (if race is stated)
- Social history or patient-reported outcomes documentation

- **Curation Instructions:**

1. Abstract the race value explicitly documented in the medical record, prioritizing official registration or intake documentation.
2. If multiple races are listed, select the one identified as primary or listed first.
3. Match the documented value to the closest permitted code above.
4. If race is not documented, select code 99 (Unknown).
5. If race is explicitly stated as “declined,” “not provided,” or equivalent, use code 98 (Other).

- **Curator Quality Review:**

- Confirm the selected race aligns with documented terminology.
- Ensure the field is completed even if the value matches the Tier1A-piped version.

- **Notes:**

- Race and ethnicity must be curated and coded separately.

Variable: ethnicity_curated

Type: Categorical (dropdown)

Description: Spanish/Hispanic Origin (Ethnicity)

Reference: NAACCR #190

Data Source: Medical record (demographics, intake forms, patient-reported documentation)

Curation Directives:

- **Definition:** Code identifying whether the patient is of Spanish or Hispanic origin. This includes self-reported ethnicity or documentation by a provider or registrar. This variable is used for classification and population-level cancer surveillance.
- **Permitted Values:**
 - 0 = Non-Spanish; non-Hispanic
 - 1 = Mexican (includes Chicano)
 - 2 = Puerto Rican
 - 3 = Cuban
 - 4 = South or Central American (except Brazil)
 - 5 = Other specified Spanish/Hispanic origin
 - 6 = Spanish, NOS / Hispanic, NOS / Latino, NOS

- 7 = Spanish surname only
- 8 = Dominican Republic
- 9 = Unknown whether Spanish or not

- **Source Documents:**

- Demographics section of the EHR
 - Intake or registration forms
 - Patient-reported outcome forms
 - Clinical notes that reference Hispanic/Latino identity

- **Curation Instructions:**

1. Identify whether the patient is documented as being of Spanish or Hispanic origin.
2. Select the code that most closely aligns with the specific origin stated (e.g., Mexican, Puerto Rican, Cuban).
3. Use code 5 if a specific Spanish origin is noted but not listed among the other codes.
4. If only a general term such as “Hispanic,” “Latino,” or “Spanish” is used, select code 6.
5. If the record only mentions a Spanish surname without explicit ethnicity documentation, use code 7.
6. If ethnicity is not stated or is unclear, use code 9.

- **Curator Quality Review:**

- Ensure that race and ethnicity are curated independently.
 - Verify that the selection matches the terminology used in the source document.

- **Notes:**

- This field must be completed even if the value matches the Tier1A-piped version.
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Variable: bmi_yn

Type: Categorical (dropdown)

Description: Were height and weight recorded in the patient’s record?

Reference: n/a

Data Source: Medical record (vitals, intake notes, nursing assessments)

Curation Directives:

- **Definition:** Indicates whether the patient’s height and weight were recorded in the medical record and, if so, whether they were documented in imperial or metric units. This field determines whether body mass index (BMI) can be calculated based on available data.

- **Permitted Values:**

- 1 = Yes (Imperial)
 - 2 = Yes (Metric)
 - 3 = No

- **Source Documents:**

- Vitals section of the EHR
 - Intake or nursing assessment notes
 - Clinic visit summaries
 - Hospital admission forms

- **Curation Instructions:**

1. Search for documentation of **both height and weight** recorded either on or after the date of cancer diagnosis.

2. If both values are found and are recorded in **feet/inches and pounds**, select code 1 (Yes, Imperial).
 3. If both values are found and are recorded in **centimeters and kilograms**, select code 2 (Yes, Metric).
 4. If one or both values are missing, or no documentation exists for either, select code 3 (No).
 5. If multiple values are found, use the earliest complete set recorded on or after diagnosis to determine the coding.
- **Curator Quality Review:**
 - Confirm that both height and weight are present and clearly documented.
 - **Notes:**
 - This field only indicates availability of data. It does not capture the actual height or weight values.
 - Do not use self-reported values.

Variable: `bmi_date`

Type: Date (MM/DD/YYYY)

Description: Date of Height and Weight

Reference: n/a

Data Source: Medical record (vitals, intake notes, nursing assessments)

Curation Directives:

- **Definition:** The date on which the patient's height and weight were both recorded, used for BMI calculation. This should be the earliest available date on or after the cancer diagnosis when both values were documented.
- **Source Documents:**
 - Vitals section of the EHR
 - Intake or nursing assessment notes
 - Clinic visit summaries
 - Hospital admission forms
- **Curation Instructions:**
 1. Locate the earliest documented date on or after the date of cancer diagnosis when **both height and weight** were recorded.
 2. Enter the date in **MM-DD-YYYY** format.
 3. Do not use dates when only one of the two measurements is available.
 4. If multiple qualifying dates exist, select the earliest complete entry following cancer diagnosis.
 5. This field should only be completed if `bmi_yn` is coded as 1 (Imperial) or 2 (Metric).
- **Curator Quality Review:**
 - Ensure that both height and weight values are documented on the same date.
 - Verify that the date falls on or after the date of diagnosis.
- **Notes:**
 - Leave blank if no date includes both height and weight together, and ensure `bmi_yn` is coded as 3 (No).

Variable: bmi_date_int

Type: Integer (Calculated)

Description: Date of Height and Weight in Days from DOB (Interval)

Reference: n/a

Data Source: Automatically calculated from curated fields birth_date and bmi_date

Curation Directives:

- **Definition:** System-generated calculation representing the number of days between the patient's date of birth (birth_date) and the date on which both height and weight were recorded (bmi_date).
 - **Source Documents:**
 - Not directly curated; derived from birth_date and bmi_date entered by the curator
 - **Curation Instructions:**
 1. Ensure both birth_date and bmi_date are correctly entered in MM-DD-YYYY format.
 2. This field will auto-populate using the formula: datediff([birth_date], [bmi_date], "d", "mdy", true).
 3. Do not manually enter or override this field.
 4. If either birth_date or bmi_date is missing or incorrectly formatted, this field will be blank .
 - **Notes:**
 - This is a read-only calculated field and should never be manually modified.
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Variable: height_inch

Type: Numeric

Description: Height in Inches

Reference: n/a

Data Source: Medical record (vitals, intake assessments)

Curation Directives:

- **Definition:** The patient's height recorded in inches, abstracted from clinical documentation and used for BMI calculation. This value should only be completed if height and weight are documented in imperial units.
- **Source Documents:**
 - Vitals section of the EHR
 - Intake or nursing assessment notes
 - Clinic visit summaries
 - Hospital admission forms
- **Curation Instructions:**
 1. Locate the patient's height documented in inches, on or after the date of cancer diagnosis.
 2. Enter the height as a whole number in inches (e.g., 65 for 5'5").
 3. Do not convert metric measurements to inches. This field is for source data recorded in imperial units only.
 4. Complete this field only if bmi_yn is coded as 1 (Yes, Imperial).
 5. If height is not recorded in inches, leave this field blank.
- **Curator Quality Review:**
 - Ensure date of measurement corresponds with bmi_date.
- **Notes:**

- Do not estimate height.
 - If multiple values are available, choose the earliest complete set corresponding with **bmi_date**.
 - Ensure that weight is also recorded in pounds for consistency.
-

Variable: weight_lb

Type: Numeric

Description: Weight in Pounds

Reference: n/a

Data Source: Medical record (vitals, intake assessments)

Curation Directives:

- **Definition:** The patient's weight recorded in pounds, abstracted from clinical documentation and used for BMI calculation. This value should only be completed if weight and height are documented in imperial units.
 - **Source Documents:**
 - Vitals section of the EHR
 - Intake or nursing assessment notes
 - Clinic visit summaries
 - Hospital admission forms
 - **Curation Instructions:**
 1. Locate the patient's weight documented in pounds, on or after the date of cancer diagnosis.
 2. Enter the weight as a whole number in pounds (e.g., 150).
 3. Do not convert kilograms to pounds. This field is for source data recorded in imperial units only.
 4. Complete this field only if **bmi_yn** is coded as 1 (Yes, Imperial).
 5. If weight is not recorded in pounds, leave this field blank.
 - **Curator Quality Review:**
 - Ensure date of measurement corresponds with **bmi_date**.
 - **Notes:**
 - Do not estimate weight.
 - If multiple values are available, choose the earliest complete set corresponding with **bmi_date**.
 - Ensure that height is also recorded in inches for consistency.
-

Variable: height_cm

Type: Numeric

Description: Height in Centimeters

Reference: n/a

Data Source: Medical record (vitals, intake assessments)

Curation Directives:

- **Definition:** The patient's height recorded in centimeters, abstracted from clinical documentation and used for BMI calculation. This value should only be completed if height and weight are documented in metric units.
- **Source Documents:**
 - Vitals section of the EHR

- Intake or nursing assessment notes
- Clinic visit summaries
- Hospital admission forms

- **Curation Instructions:**

1. Locate the patient’s height documented in **centimeters**, on or after the date of cancer diagnosis.
2. Enter the height as a whole number in centimeters (for example, 170).
3. Do not convert imperial measurements to centimeters. Only enter values that were originally documented in metric units.
4. Complete this field only if **bmi_yn** is coded as 2 (Yes, Metric).
5. If height is not recorded in centimeters, leave this field blank.

- **Curator Quality Review:**

- Ensure date of measurement corresponds with **bmi_date**.

- **Notes:**

- Do not estimate height.
- If multiple values are available, choose the earliest complete set corresponding with **bmi_date**.
- Ensure that weight is also recorded in kilograms for consistency.

Variable: weight_kg

Type: Numeric

Description: Weight in Kilograms

Reference: n/a

Data Source: Medical record (vitals, intake assessments)

Curation Directives:

- **Definition:** The patient’s weight recorded in kilograms, abstracted from clinical documentation and used for BMI calculation. This value should only be completed if weight and height are documented in metric units.

- **Source Documents:**

- Vitals section of the EHR
- Intake or nursing assessment notes
- Clinic visit summaries
- Hospital admission forms

- **Curation Instructions:**

1. Locate the patient’s weight documented in **kilograms**, on or after the date of cancer diagnosis.
2. Enter the weight as a whole number in kilograms.
3. Do not convert pounds to kilograms. Only enter values that were originally documented in metric units.
4. Complete this field only if **bmi_yn** is coded as 2 (Yes, Metric).
5. If weight is not recorded in kilograms, leave this field blank.

- **Curator Quality Review:**

- Ensure date of measurement corresponds with **bmi_date**.

- **Notes:**

- Do not estimate weight.
- If multiple values are available, choose the earliest complete set corresponding with **bmi_date**.
- Ensure that height is also recorded in centimeters for consistency.

Variable: smoking_status

Type: Categorical (Dropdown)

Description: What is the smoking status of the patient?

Reference: NAACCR #344

Data Source: Medical record (social history, intake forms, clinic notes)

Curation Directives:

- **Definition:** Indicates the patient's history of tobacco smoking (cigarettes, cigars, pipes) as documented in the medical record. This variable excludes vaping, smokeless tobacco, and marijuana use.
- **Permitted Values:**
 - 0 = Never smoker
 - 1 = Current smoker
 - 2 = Former smoker
 - 3 = Smoker: current status unknown
 - 88 = Not documented
 - 99 = Unknown
- **Source Documents:**
 - Social history or lifestyle section of the EHR
 - Intake or admission forms
 - Physician or nursing notes
 - Oncology or primary care progress notes
 - Nursing Interview Guide, Flow Chart, Vital Stats, or Nursing Assessment
- **Curation Instructions:**
 1. Review the medical record for documentation of tobacco smoking history.
 2. Select code 0 if the record states the patient has never smoked.
 3. Select code 1 if the patient is currently smoking at the time of documentation.
 4. Select code 2 if the patient has a history of smoking and the most recent documentation in the record states that the patient quit >1 year prior. If patient quit <1 year prior (e.g. "3 months ago") code, select code 1.
 5. Select code 3 if the patient is documented as a smoker but current status (active or former) is unclear.
 6. Use code 88 if smoking is not addressed in the record.
 7. Use code 99 only if documentation is ambiguous or contradictory.
- **Curator Quality Review:**
 - Ensure the selected code is supported by explicit documentation.
 - Capture the most recent smoking status as documented at time of curation.
- **Notes:**
 - This field only captures tobacco smoking. Do not include vaping, smokeless tobacco, or marijuana use.
 - If multiple entries exist, prioritize the most complete and specific documentation available.

Variable: patient_marital_status

Type: Categorical (Dropdown)

Description: Marital Status at Time of Diagnosis

Reference: NAACCR #150

Data Source: Medical record (demographics, admission forms, intake notes)

Curation Directives:

- **Definition:** Patient’s marital status at or closest to the date of cancer diagnosis, as documented in the medical record.
- **Permitted Values:**
 - 1 = Single (never married)
 - 2 = Married (including common law)
 - 3 = Separated
 - 4 = Divorced
 - 5 = Widowed
 - 6 = Unmarried or Domestic Partner (same or opposite sex, registered or unregistered, other than common law marriage)
 - 88 = Not documented
 - 99 = Unknown
- **Source Documents:**
 - Demographic section of the EHR
 - Intake or admission forms
 - Social history documentation
 - Provider notes (if marital status is explicitly mentioned)
- **Curation Instructions:**
 1. Identify documentation of marital status closest to the date of cancer diagnosis.
 2. Select the appropriate code that most accurately represents the patient’s status at that time.
 3. Use code 6 for any type of domestic partnership not qualifying as common law or legal marriage.
 4. Use code 88 if the field is not addressed in available documentation.
 5. Use code 99 only if documentation is ambiguous or conflicting.
- **Curator Quality Review:**
 - Ensure consistency with related documentation such as emergency contact forms or household structure.
 - Do not assume current status based on notes written significantly after diagnosis unless retrospective context is clear.
 - Escalate discrepancies with other records (e.g., spouse named but marital status marked as single) to supervisor if needed.
- **Notes:**
 - n/a

Variable: patient_education

Type: Categorical (Dropdown)

Description: Highest level of education attained

Reference: n/a

Data Source: Medical record (social history, intake forms, or demographic section)

Curation Directives:

- **Definition:** Captures the highest level of formal education completed by the patient, if documented. Education level may be used in population-level analyses of health disparities, socioeconomic status, and social determinants of health.
- **Permitted Values:**
 - 1 = Elementary School
 - 2 = High School or GED
 - 3 = College, no degree

- 4 = Associate's degree (vocational)
- 5 = Associate's degree (academic)
- 6 = Bachelor's degree
- 7 = Master's degree
- 8 = Professional degree
- 9 = Doctorate degree
- 88 = Not documented
- 99 = Unknown

- **Source Documents:**

- Intake or demographic forms
- Social history or psychosocial assessment
- Case management or nursing notes
- Oncology or primary care notes with SDOH screening

- **Curation Instructions:**

1. Review the medical record for documentation of the patient's highest level of completed education.
2. Select the most specific value supported by documentation.
3. If the record mentions a degree type (e.g., "completed nursing degree"), match to the appropriate coded value.
4. If multiple levels are documented, use the highest completed level.
5. If documentation states only "college" with no degree, select code 3.
6. Use code 88 if the field is not addressed in the record.
7. Use code 99 only if the documentation is unclear or ambiguous.

- **Curator Quality Review:**

- Confirm the selected code reflects the most complete and specific level documented.
- If multiple education levels are mentioned, verify which was completed.
- Do not infer education level based on occupation or income alone.

- **Notes:**

- This field should reflect completed education only, not current enrollment.

Variable: patient_employment

Type: Categorical (Dropdown)

Description: Patient's employment status

Reference: USCDI 3

Data Source: Medical record (demographics, intake forms, or social history)

Curation Directives:

- **Definition:** Captures the patient's employment status as documented in the medical record at or near the time of diagnosis. This variable supports analysis of socioeconomic context and care accessibility.
- **Permitted Values:**
 - 1 = Employed
 - 2 = Unemployed
 - 3 = Retired
 - 4 = Student
 - 88 = Not documented
 - 99 = Unknown
- **Source Documents:**
 - Intake or demographic forms

- Social history or psychosocial assessments
- Nursing or case management notes
- Oncology or primary care notes with SDOH documentation
- **Curation Instructions:**
 1. Review the record for explicit documentation of the patient’s employment status.
 2. Select code 1 if the patient is currently working (full- or part-time).
 3. Select code 2 if the patient is not currently working and not retired or a student.
 4. Select code 3 if the record indicates the patient is retired.
 5. Select code 4 if the patient is actively enrolled as a student.
 6. Use code 88 if employment status is not addressed.
 7. Use code 99 only if documentation is contradictory or ambiguous.
- **Curator Quality Review:**
 - Ensure employment status reflects the time of diagnosis, not past history.
 - Do not infer status based on occupation fields or insurance type unless explicitly stated.
 - Confirm that employment-related notes refer to the patient, not family members.
- **Notes:**
 - Use the status most relevant to the date of diagnosis if multiple timepoints are documented.
 - Part-time or per diem work should be coded as “Employed” if currently active.

Variable: `patient_payer`

Type: Categorical (Check-Box)

Description: Patient’s insurance coverage at time of diagnosis or treatment

Reference: NAACCR #630

Data Source: Medical record (demographics, insurance section, admission documentation)

Curation Directives:

- **Definition:** Captures all known types of insurance coverage or payer sources associated with the patient at the time of initial cancer diagnosis and/or first course of treatment at the reporting facility. More than one selection may apply.
- **Permitted Values:**
 - 01 = Not insured
 - 02 = Not insured, self-pay
 - 10 = Insurance, NOS
 - 20 = Private Insurance: Managed care, HMO, or PPO
 - 21 = Private Insurance: Fee-for-service
 - 31 = Medicaid
 - 35 = Medicaid - administered through a Managed Care plan
 - 60 = Medicare/Medicare, NOS
 - 61 = Medicare with supplement, NOS
 - 62 = Medicare - Administered through a Managed Care Plan
 - 63 = Medicare with private supplement
 - 64 = Medicare with Medicaid eligibility
 - 65 = TRICARE
 - 66 = Military
 - 67 = Veterans Affairs
 - 68 = Indian/Public Health Service
 - 88 = Not documented
 - 99 = Unknown

- **Source Documents:**
 - Insurance or billing section of the EHR
 - Admission or registration summaries
 - Financial counseling notes
 - Treatment authorizations
- **Curation Instructions:**
 1. Review all insurance documentation near or at the date of cancer diagnosis or first treatment.
 2. Select all applicable insurance types based on the available documentation.
 3. If the patient is uninsured, select either 01 or 02 depending on whether self-pay is noted.
 4. Use 88 if no insurance information is documented in the chart.
 5. Use 99 only if payer information is ambiguous, contradictory, or illegible.
- **Curator Quality Review:**
 - Confirm that all selected options are supported by documentation.
 - Avoid assumptions based on age (e.g., Medicare for patients over 65) without confirmation.
 - If payer coverage changes frequently, use the coverage closest to the cancer diagnosis date.
- **Notes:**
 - This field supports SDOH analyses and access-to-care metrics.
 - For multi-coverage scenarios (e.g., Medicare plus Medicaid), ensure both codes are checked.
 - Do not code third-party payers used for one-time services unless they represent active coverage.

Variable: zipcode_full

Type: Numeric

Description: Full 5-digit ZIP code of the patient’s residence at diagnosis

Reference: NAACCR #100

Data Source: Medical record (demographics, admission forms, address documentation)

Curation Directives:

- **Definition:** Captures the 5-digit ZIP code of the patient’s home address at the time of cancer diagnosis, if patient resides within the United States. For consolidated records, this may reflect a corrected or updated address based on available documentation.
- **Permitted Values:**
Free-text field restricted to 5 numeric characters. Do not include ZIP+4.
- **Source Documents:**
 - Demographic section of the EHR
 - Admission or intake forms
 - Billing or registration records
 - Clinical correspondence with listed address
- **Curation Instructions:**
 1. Locate the patient’s residential address closest to the time of cancer diagnosis.
 2. Extract the 5-digit ZIP code from the address.
 3. Enter the ZIP code as a five-digit number with leading zeros if applicable (e.g., 00501).
 4. If multiple addresses are present, use the one most clearly identified as primary or residential.
 5. If no ZIP code is documented, leave blank. Do not enter “unknown” or placeholder values.
- **Curator Quality Review:**
 - Verify that the ZIP code reflects a residential rather than facility or mailing address.

- Confirm accuracy and presence of leading zeroes.
- Ensure consistency across other related address fields, if available.

- **Notes:**

- This field supports geographic stratification for population health and SDOH analyses.
- Do not infer ZIP code from city or state if not explicitly documented.

Variable: `zipcode_first_three`

Type: Integer (Calculated)

Description: First three digits of the patient's ZIP code

Reference: Derived from NAACCR #100

Data Source: Derived variable

Curation Directives:

- **Definition:** Automatically extracts the first three digits of the 5-digit ZIP code recorded in the `zipcode_full` field. Used for regional or geographic analysis while reducing identifiability.
- **Permitted Values:**
3-digit numeric string (e.g., 100, 941, 005)
- **Source Documents:**
 - Not applicable — value is auto-calculated from `zipcode_full`
- **Curation Instructions:**
 1. Do not enter data manually; this field is populated via REDCap's calculated field function: `left([zipcode_full], 3)`.
 2. Confirm that `zipcode_full` is complete and valid.
 3. If `zipcode_full` is missing or invalid, this field will remain blank or return an error.
- **Curator Quality Review:**
 - Ensure that the calculated output matches the first three digits of `zipcode_full`.
 - Review for formatting or truncation issues, especially with leading zeroes.
- **Notes:**
 - Used for less granular geospatial mapping and to support linkage with regional datasets (e.g., Census tract approximations).
 - This is a system-generated field and should not be altered or overridden manually.

Curated Patient-Level Follow Up Dates

Variable: `hospice_yn`

Type: Binary (Yes/No)

Description: Hospice Admission or Referral Mentioned

Reference: n/a

Data Source: Clinical provider note (Consultation, H&P, Progress Note, Discharge Summary)

Curation Directives:

- **Definition:** Indicates whether the provider documented a referral to or admission into hospice services during the clinical visit.
- **Permitted Values / Codes:**
 - 1, Yes, Hospice Admission
 - 2, Yes, Hospice Referral
 - 3, No, not mentioned
- **Source Documents:**
 - Clinical provider note (Assessment, Plan, or Social Work section)
 - Discharge summary, if part of clinical documentation
 - Case management notes or palliative care consultations, if included
- **Curation Instructions:**
 - Review the clinical record for explicit documentation of hospice services.
 - Assign “Yes, Hospice Admission” (1) if the record documents a hospice admission.
 - Assign “Yes, Hospice Referral” (2) if the record documents a hospice referral.
 - If both a hospice admission and hospice referral exist in the clinical record, document only the hospice admission event.
 - Assign “No, not mentioned” (3) if no mention of hospice admission or referral is made, or if palliative care is discussed without specific mention of hospice.
 - Do not infer from general decline, poor prognosis, or do-not-resuscitate orders alone.
 - If uncertain whether language refers to hospice (e.g., vague discussions of end-of-life planning), flag for review.
- **Curator Quality Review:**
 - Confirm that the language in the note explicitly refers to hospice services and not general palliative care.
 - Cross-check with other visit fields for consistency (e.g., advanced care planning, code status).
 - Document rationale in notes if decision is ambiguous.
- **Notes:**
 - This is a required field.

Variable: hospice_admit_date

Type: Date (MM/DD/YYYY)

Description: Date of Hospice Admission

Reference: n/a

Data Source: Medical record (progress notes, discharge summaries, follow-up documentation)

Curation Directives:

- **Definition:** The date the patient’s hospice admission as documented in the clinical record.
- **Permitted Values:**

Date in MM-DD-YYYY format
- **Source Documents:**
 - Any clinical visit note documented in the patient record (for example: oncology progress notes, palliative care notes, care coordination notes).
 - Inpatient discharge summaries.
 - Referral/order documentation when present in the chart (for example: hospice referral order, consult note, discharge planning documentation).
- **Curation Instructions:**

1. Review the clinical record for explicit documentation of hospice admission.
2. If multiple hospice admissions are recorded, document the earliest admission event in the clinical record.
3. Enter the date in MM-DD-YYYY format.
4. For all dates, use the Data Estimation Rules defined within the Curation Directives.

- **Curator Quality Review:**

- Confirm the documented event is an admission and not a hospice referral.
- Cross-check with `hospice_yn` to ensure consistency (this field should only be completed when `hospice_yn` = 1, Yes, Hospice Admission).
- If both referral and admission are present, ensure the referral date is not mistakenly entered as the admission date (and vice versa).
- Check that approximated dates follow the standardized rules.

- **Notes:**

- This field supports survival analysis and outcome tracking.

Variable: `hospice_admit_int`

Type: Integer (Calculated)

Description: Date of Hospice Admission in Days from Date of Birth

Reference: n/a

Data Source: Derived from curated hospice referral date (`hospice_admit_date`) and date of birth (`birth_date`)

Curation Directives:

- **Definition:** Automatically calculates the number of days between the patient's date of birth and their hospice admission date.
 - **Permitted Values:**
Integer (number of days)
 - **Source Documents:**
 - No direct source — derived from curated fields:
 - * `birth_date`
 - * `hospice_admit_date`
 - **Curation Instructions:**
 1. No manual entry required; value is system-generated.
 2. Calculated using the REDCap function:


```
datediff([birth_date],[hospice_admit_date],"d","mdy",true)
```
 3. If `hospice_admit_date` is missing, this field will remain blank.
 - **Curator Quality Review:**
 - Ensure `birth_date` and `hospice_admit_date` are correctly curated prior to reviewing this field.
 - **Notes:**
 - This interval is used for time-to-event analyses and survival modeling.
 - Field is read-only and should not be overwritten manually.
-

Variable: hospice_referral_date

Type: Date (MM/DD/YYYY)

Description: Date of Hospice Referral

Reference: n/a

Data Source: Medical record (progress notes, discharge summaries, follow-up documentation)

Curation Directives:

- **Definition:** The date the patient was referred to hospice, as explicitly documented in the clinical record (referral/order/consult/request), regardless of whether hospice admission occurred.
- **Permitted Values:**
Date in MM-DD-YYYY format
- **Source Documents:**
 - Any clinical visit note documented in the patient record (for example: oncology progress notes, palliative care notes, care coordination notes).
 - Inpatient discharge summaries.
 - Referral/order documentation when present in the chart (for example: hospice referral order, consult note, discharge planning documentation).
- **Curation Instructions:**
 1. Review the clinical record for explicit documentation of hospice referral (for example: “referred to hospice,” “hospice referral placed,” “hospice consult requested,” “hospice informational visit ordered”).
 2. If multiple hospice referrals are recorded (without subsequent admission), document the earliest referral event in the clinical record.
 3. Enter the date in MM-DD-YYYY format.
 4. For all dates, use the Data Estimation Rules defined within the Curation Directives.
- **Curator Quality Review:**
 - Confirm the documented event is a referral and not a hospice admission.
 - Cross-check with hospice_yn to ensure consistency (this field should only be completed when hospice_yn = 2, Yes, Hospice Referral).
 - Check that approximated dates follow the standardized rules.
- **Notes:**
 - Hospice referral can occur without hospice admission; this field supports tracking end-of-life planning and care transitions.

Variable: hospice_referral_int

Type: Integer (Calculated)

Description: Date of Hospice Referral in Days from Date of Birth

Reference: n/a

Data Source: Derived from curated hospice referral date (hospice_referral_date) and date of birth (birth_date)

Curation Directives:

- **Definition:** Automatically calculates the number of days between the patient’s date of birth and their hospice referral date.
- **Permitted Values:**
Integer (number of days)
- **Source Documents:**

- No direct source — derived from curated fields:
 - * `birth_date`
 - * `hospice_referral_date`
- **Curation Instructions:**
 1. No manual entry required; value is system-generated.
 2. Calculated using the REDCap function:


```
datediff([birth_date],[hospice_referral_date],"d","mdy",true)
```
 3. If `hospice_referral_date` is missing, this field will remain blank.
- **Curator Quality Review:**
 - Ensure `birth_date` and `hospice_referral_date` are correctly curated prior to reviewing this field.
- **Notes:**
 - This interval is used for time-to-event analyses and survival modeling.
 - Field is read-only and should not be overwritten manually.

Variable: `follow_up_date_clinical`

Type: Date (MM/DD/YYYY)

Description: Date of Last Clinical Follow-Up

Reference: n/a

Data Source: Medical record (progress notes, discharge summaries, follow-up documentation)

Curation Directives:

- **Definition:** The most recent documented date when the patient was evaluated clinically by a health-care provider. This may include oncology clinic visits, general medicine visits, or inpatient discharges, provided there is documentation of patient status.
- **Permitted Values:**

Date in MM-DD-YYYY format
- **Source Documents:**
 - Any clinical visit note documented in the patient record.
 - Inpatient discharge summaries.
- **Curation Instructions:**
 1. Identify the most recent clinical documentation that includes clinical findings or treatment plans.
 2. Use the date of this documentation as the follow-up date.
 3. Enter the date in **MM-DD-YYYY** format.
- **Curator Quality Review:**
 - Confirm that the selected date is the latest available clinical encounter.
 - Ensure documentation clearly supports the patient being alive at that time.
 - Check that approximated dates follow the standardized rules.
- **Notes:**
 - This field supports survival analysis and outcome tracking.
 - Do not use future-dated appointments or administrative updates (e.g., insurance renewals).

Variable: follow_up_date_clin_int

Type: Integer (Calculated)

Description: Date of Last Clinical Follow-Up in Days from Date of Birth

Reference: n/a

Data Source: Derived from curated clinical follow-up date (follow_up_date_clinical) and date of birth (birth_date)

Curation Directives:

- **Definition:** Automatically calculates the patient's age in days at the time of their last documented clinical follow-up.
- **Permitted Values:**
Integer (number of days)
- **Source Documents:**
 - No direct source — derived from curated fields:
 - * birth_date
 - * follow_up_date_clinical
- **Curation Instructions:**
 1. No manual entry required; value is system-generated.
 2. Calculated using the REDCap function:
`datediff([birth_date],[follow_up_date_clinical],"d","mdy",true)`
 3. If follow_up_date_clinical is missing, this field will remain blank.
- **Curator Quality Review:**
 - Ensure birth_date and follow_up_date_clinical are correctly curated prior to reviewing this field.
- **Notes:**
 - This interval is used for time-to-event analyses and survival modeling.
 - Field is read-only and should not be overwritten manually.

Variable: lastalive_date

Type: Date (MM/DD/YYYY)

Description: Date of Last Known Alive

Reference: n/a

Data Source: Medical record (clinic notes, discharge summaries, follow-up communications)

Curation Directives:

- **Definition:** Most recent date on which the patient was known to be alive, based on documentation in the medical record.
- **Source Documents:**
 - Oncology, primary care, or survivorship clinic notes
 - Discharge or transfer summaries
 - Follow-up calls, telehealth visits, or communications documented in the chart
 - Death certificate or obituary (if confirms patient was alive until a known date prior to death)
- **Curation Instructions:**

1. Locate the most recent documented evidence that the patient was alive.
 2. Use the exact date from that documentation as the `lastalive_date`.
 3. Enter the date in **MM-DD-YYYY** format.
 4. Do not use the date of death or inference from lack of follow-up as a proxy for this field.
- **Curator Quality Review:**
 - Confirm that the date is supported by a specific and dated note or document.
 - Ensure that the selected date precedes the date of death, if known.
 - **Notes:**
 - This date is used for survival analyses and censoring in time-to-event studies.
-

Variable: `lastalive_date_int`

Type: Integer (Calculated)

Description: Date of Last Known Alive in Days from DOB

Reference: n/a

Data Source: Derived from `birth_date` and `lastalive_date` fields

Curation Directives:

- **Definition:** The number of days from the patient's date of birth to the date they were last known to be alive.
 - **Source Documents:**
 - This is a calculated field and does not rely on direct abstraction.
 - Values are derived using: `datediff([birth_date], [lastalive_date], "d", "mdy", true)`
 - **Curation Instructions:**
 1. Ensure both `birth_date` and `lastalive_date` fields are populated and properly formatted.
 2. Confirm that `lastalive_date` occurs after `birth_date`.
 3. No manual entry is required.
 - **Curator Quality Review:**
 - Verify that both input fields (`birth_date` and `lastalive_date`) are accurately entered.
 - **Notes:**
 - This interval is used for time-to-event analyses and survival modeling.
 - Field is read-only and should not be overwritten manually.
-

Variable: `dead_curated`

Type: Binary (Yes/No)

Description: Vital Status: Is patient known to be deceased at time of curation?

Reference: n/a

Data Source: Medical record (death summary, EHR flags, clinical notes)

Curation Directives:

- **Definition:** Indicates whether the patient is confirmed to be deceased at the time of clinical data curation, based on documentation in the medical record.

- **Permitted Values:**

0 = No

1 = Yes

- **Source Documents:**

- Death summary or final hospitalization notes
- Correspondence in clinical notes (e.g., oncologist documentation, palliative care notes)
- EHR flags indicating deceased status
- Documentation of death in vital status sections of the chart
- Tumor Registry: Information sourced from institutional tumor registry systems
- NDI: Information from the National Death Index
- Commercial Database: Information from commercial mortality databases
- Other: Any other validated source not listed above

- **Curation Instructions:**

1. Review all available documentation for explicit mention of the patient’s death.
2. Select “Yes” (1) if there is unambiguous evidence that the patient is deceased.
3. Select “No” (0) if there is no documentation of death or if the patient is explicitly noted as alive.
4. Do not infer death based on absence of follow-up or treatment discontinuation alone.

- **Curator Quality Review:**

- Ensure death status is clearly documented and not assumed.
- Cross-check with `lastalive_date`, `follow_up_date_clinical`, or any death-related notes for consistency.
- Confirm that “Yes” is not used based solely on lack of recent records.

- **Notes:**

- This field supports interpretation of follow-up completeness and survival outcomes.

Variable: `death_date`

Type: Date (MM/DD/YYYY)

Description: Date of Death

Reference: n/a

Data Source: Medical record (death summary, discharge records, EHR indicators)

Curation Directives:

- **Definition:** The date on which the patient is documented to have died. Only applicable for patients confirmed as deceased (`dead_curated` = 1).

- **Source Documents:**

- Death summary or final hospitalization notes
- Correspondence in clinical notes (e.g., oncologist documentation, palliative care notes)
- EHR flags indicating deceased status
- Documentation of death in vital status sections of the chart
- Tumor Registry: Information sourced from institutional tumor registry systems
- NDI: Information from the National Death Index
- Commercial Database: Information from commercial mortality databases
- Other: Any other validated source not listed above

- **Curation Instructions:**

1. Record the date of death as documented in the medical record.
2. Enter the date in **MM-DD-YYYY** format.

3. Do not use inference from lack of follow-up as a proxy for this field.
- **Curator Quality Review:**
 - Verify date consistency with other variables such as `lastalive_date` and `follow_up_date_clinical`.
 - **Notes:**
 - This date is used in survival analyses and endpoint censoring.
 - Do not populate if vital status is unknown or patient is alive.
-

Variable: `death_year_curated`

Type: Integer (Calculated)

Description: Year of Death

Reference: n/a

Data Source: Calculated from death date in medical record

Curation Directives:

- **Definition:** The calendar year extracted from the full date of death. This field is auto-calculated and should not be manually entered.
 - **Source Documents:**
 - Derived directly from `death_date`
 - **Curation Instructions:**
 1. Do not manually enter this field, it is automatically calculated from `death_date`.
 2. Ensure that `death_date` is accurately curated and formatted in MM-DD-YYYY.
 3. If `death_date` is blank, this field should remain blank.
 - **Curator Quality Review:**
 - Ensure the calculated year is logical and does not precede `birth_date` or follow the abstraction date.
 - Review for consistency with `lastalive_date` and `follow_up_date_clinical`.
 - **Notes:**
 - This field supports grouped survival and mortality analyses.
-

Variable: `death_date_int`

Type: Integer (Calculated)

Description: Date of Death in Days from DOB (Interval)

Reference: n/a

Data Source: Calculated from medical record documentation of date of birth and death

Curation Directives:

- **Definition:** Integer value representing the number of days between the patient's date of birth and date of death. This interval reflects total lifespan in days and is auto-calculated.
- **Source Documents:**
 - Derived directly from:
 - * `birth_date` (date of birth)
 - * `death_date` (documented date of death)

- Interval calculated using REDCap’s `datediff()` function with format:
`datediff([birth_date], [death_date], "d", "mdy", true)`
- **Curation Instructions:**
 1. Do not manually enter this field. It is automatically calculated from `birth_date` and `death_date`.
 2. Ensure `death_date` is accurately curated in MM-DD-YYYY format.
 3. If either `birth_date` or `death_date` is missing, this field should remain blank.
- **Curator Quality Review:**
 - Confirm that the interval is logically consistent with known life events and dates.
 - Ensure this field is blank if the patient is not deceased or death date is not documented.
- **Notes:**
 - This field supports time-to-event analyses and age-at-death calculations.

Variable: `death_source_curated`

Type: Categorical (Dropdown)

Description: What is the source of the patient’s vital status?

Reference: n/a

Data Source: Medical record, registry, and supplemental mortality data sources

Curation Directives:

- **Definition:** Indicates the source from which the patient’s death status was confirmed. This field captures whether death information was obtained through clinical documentation, structured EHR fields, registry linkage, or external mortality databases.
- **Permitted Values / Codes:**
 - 1 = Curated – documented in clinical note
 - 2 = EHR – documented in structured EHR field
 - 3 = Tumor Registry
 - 4 = NDI (National Death Index)
 - 5 = Commercial Database
 - 6 = Other
- **Source Documents:**
 - Clinical documentation (progress notes, discharge summaries)
 - Structured EHR fields (e.g., vital status flag, problem list)
 - Tumor registry records
 - National Death Index (NDI) linkage reports
 - Commercial data sources
 - Other institutional mortality feeds or communications
- **Curation Instructions:**
 1. Review all available documentation to determine the source of confirmed death.
 2. If the death is confirmed in narrative notes (e.g., “patient expired”), select code 1.
 3. If death is recorded in a structured EHR field, select code 2.
 4. If sourced from a tumor registry match, select code 3.
 5. If confirmed via NDI linkage, select code 4.
 6. If sourced from a commercial mortality data provider, select code 5.
 7. If source does not fit the above categories, select code 6 and document source if needed.
 8. This field should only be completed if `dead_curated` = ‘Yes’.
- **Curator Quality Review:**

- Ensure consistency with the date of death and source documentation.
 - Prioritize the most definitive source if multiple are available.
 - **Notes:**
 - Supports downstream mortality analyses.
-

Variable: death_source_other

Type: Free Text Entry (text)

Description: If vital status source is ‘other’, please provide description

Reference: n/a

Data Source: Medical record or external documentation

Curation Directives:

- **Definition:** Free-text field used to specify the source of death confirmation when the predefined options in death_source_curated do not apply.
 - **Source Documents:**
 - Unstructured clinical communication (e.g., faxed death notices)
 - Institution-specific feeds or proprietary data systems
 - Emails or internal messages documenting vital status
 - Manual uploads or reports not falling into standard categories
 - **Curation Instructions:**
 1. Complete only if death_source_curated is coded as 6 (Other).
 2. Enter a brief but specific description of the source used to determine patient death.
 - Examples: “Social worker phone call,” “Faxed obituary from hospice,” “Internal cancer registry note”
 3. Do not use vague descriptions such as “unknown” or “unspecified.”
 4. If more than one nonstandard source exists, list the primary and indicate others as appropriate.
 - **Curator Quality Review:**
 - Confirm the description is present if death_source_curated = 6.
 - Ensure the entry is not duplicative of standard categories.
 - **Notes:**
 - Supports documentation transparency for nonstandard death ascertainment.
 - Facilitates audit trails and validation of mortality data when outside sources are used.
-

Variable: cause_death

Type: Free Text Entry (text)

Description: If noted, what is the described cause of death?

Reference: n/a

Data Source: Medical record (death summary, discharge note, clinical notes)

Curation Directives:

- **Definition:** Free-text field capturing the primary or contributing cause of death as documented in the patient’s record. This may include a specific disease, condition, or event noted by the provider.
- **Source Documents:**

- Death summary or discharge/death note
- Oncology or palliative care provider notes
- Hospice communication or summary
- Any provider documentation indicating cause of death

- **Curation Instructions:**

1. Complete only if `dead_curated = 1` (Yes, patient is deceased).
2. Abstract the most specific documented cause of death.
 - Example: “Progressive metastatic breast cancer” rather than just “cancer”
3. If multiple causes are listed, enter the primary cause if specified.
 - If not, include all causes as listed.
4. Do not interpret or infer a cause of death if it is not explicitly stated in the record.
5. If the only documentation is “unknown” or “not documented,” leave the field blank.

- **Curator Quality Review:**

- Confirm alignment with documentation in source notes.
- Verify that the cause is relevant to the clinical picture and not copied from unrelated sections.
- Ensure free-text entries are spelled correctly and clearly describe the cause.

- **Notes:**

- Do not use ICD codes in this field.
- This field supports downstream mortality analyses and review of cancer-related versus non-cancer deaths.

Variable: `cause_death_ndi`

Type: Free Text Entry (text)

Description: Cause of Death (ICD-10 code)

Reference: NAACCR #1910

Data Source: National Death Index (NDI) or other linked death certificate source

Curation Directives:

- **Definition:** Captures the official cause of death using a standardized ICD code, as reported on the death certificate and made available through NDI linkage or similar verified mortality source.
- **Permitted Values / Codes:**
Valid ICD-7, ICD-8, ICD-9, or ICD-10 codes.
- **Source Documents:**
 - National Death Index (NDI) matched dataset
 - Death certificate data from state cancer registry
 - Other linked mortality databases that provide structured cause of death coding
- **Curation Instructions:**
 1. Complete only if `dead_curated = 1` (Yes, patient is deceased).
 2. Enter the primary cause of death code exactly as reported in the NDI or death certificate source.
 3. Do not interpret or modify the code.
 4. If multiple codes are listed, enter the first (underlying) cause of death.
 5. If no code is available, leave the field blank.
- **Curator Quality Review:**
 - Ensure the code is valid per ICD standard and transcribed accurately.
 - Confirm that the source is verified (e.g., not self-reported or from EHR fields).

- Do not populate based on free-text death notes or inferred causes.
- **Notes:**
 - This variable is used for population-level mortality and cause-specific survival analyses.
 - Free-text causes of death should be captured separately in `cause_death`.
 - If both `cause_death` and `cause_death_ndi` are available, retain both.

Family History

Is there clinical documentation of any of the following for the patient?

Variable: `fhx_other_cancer_yn`

Type: Categorical (Radio)

Description: Family history of other cancers?

Reference: n/a

Data Source: Medical record (family history section, genetics consults, intake questionnaires)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented family history of cancers in biologic relatives.
 - **Permitted Values / Codes:**
 - 1 = Yes
 - 0 = No
 - 88 = Not documented
 - 99 = Unknown
 - **Source Documents:**
 - Family history section of oncology/primary care notes.
 - Genetics consult notes / genetic counseling documentation.
 - Patient intake forms listing family history of cancer.
 - **Curation Instructions:**
 1. Review the record for explicit documentation of cancer diagnoses in family members (for example: “mother with breast cancer,” “uncle with colon cancer”).
 2. Do not include the patient’s own cancers.
 3. Select Yes if any cancer is reported in a biologic relative.
 4. Select No only if the record explicitly documents a negative family history for cancer (and there is no mention of cancer in relatives).
 5. Select Not documented if family history is not addressed in the record.
 6. Select Unknown if family history is mentioned but cancer type/history is unclear or conflicting (for example: “family history of cancer, type unknown”).
 - **Curator Quality Review:**
 - Confirm that “Yes” is supported by documentation of at least one relative with cancer (even if type is unknown).
 - If `fhx_other_cancer_yn` = Yes, check that `fhx_other_cancer_type` is completed (or set to Unknown when appropriate).
-

Variable: fhx_other_cancer_type

Type: Categorical (Checkbox; multi-select)

Description: Cancer type(s) in family

Reference: n/a

Data Source: Medical record (family history section, genetics consults)

Curation Directives:

- **Definition:** Captures the cancer type(s) reported among biologic relatives when the patient has a family history of cancer
- **Permitted Values / Codes:**
 - 1 = Breast
 - 2 = Colorectal
 - 3 = Prostate
 - 4 = Lung
 - 5 = Brain/CNS
 - 6 = Bone/Sarcoma
 - 7 = Other solid tumor
 - 8 = Hematologic malignancy
 - 99 = Unknown
- **Source Documents:**
 - Family history section of clinical notes (oncology, primary care).
 - Genetics consult / genetic counseling notes.
 - Patient questionnaires documenting cancer types in relatives.
- **Curation Instructions:**
 1. Complete this field only when fhx_other_cancer_yn = 1 (Yes).
 2. Select all cancer types that are explicitly documented in relatives.
 3. Use category guidance:
 1. Brain/CNS: glioma, meningioma, medulloblastoma, CNS tumor.
 2. Bone/Sarcoma: osteosarcoma, Ewing sarcoma, soft tissue sarcoma, “sarcoma” when specified.
 3. Other solid tumor: melanoma, pancreatic, ovarian, endometrial, gastric, renal, bladder, head and neck, etc., when not otherwise listed.
 4. Hematologic malignancy: leukemia, lymphoma, myeloma, MDS/MPN.
 4. Select Unknown (99) when cancer is reported in the family but the type is not specified.
 5. Do not infer cancer type from vague terms (for example: “female cancer,” “stomach issues,” “tumor”) unless the cancer type is explicitly stated.
- **Curator Quality Review:**
 - Ensure selections are consistent with documented cancer types and not over-specified.
 - If only nonspecific “cancer” is documented, Unknown should be selected rather than choosing a type.
- **Notes:**
 - Multi-select allows capturing multiple cancer types across relatives; this field does not capture number of relatives, ages at diagnosis, or lineage.

Comorbidities

Collection of Patient’s Comorbidities (NCI Comorbidity Index)

Is there clinical documentation of any of the following conditions for the patient?

{% embed url="https://healthcaredelivery.cancer.gov/seermedicare/considerations/comorbidity.html" %}

Variable: comorb_ami

Type: Categorical (Dropdown)

Description: Acute Myocardial Infarction (AMI)

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, discharge summaries, admission notes, cardiology consults)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented history of acute myocardial infarction (heart attack), as defined by the NCI Comorbidity Index.
 - **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
 - **Source Documents:**
 - Hospital discharge summaries
 - Cardiologist notes or echocardiogram/imaging results
 - Problem list or past medical history in clinic notes
 - ICD-10 coded fields in structured EHR (e.g., I21.x, I22.x)
 - **Curation Instructions:**
 1. Review the full medical record for any documentation of AMI.
 2. Accept any indication of AMI based on clinical diagnosis, EHR problem list, or coded diagnosis using ICD-10: I21.x (acute MI) or I22.x (subsequent MI).
 3. Do not infer AMI from symptom descriptions (e.g., chest pain) without a clinical diagnosis.
 4. If the record includes ambiguous or conflicting information, select “Unknown.”
 5. If no documentation is found, select “No.”
 - **Curator Quality Review:**
 - Confirm presence of valid diagnosis and alignment with ICD-10 codes.
 - Cross-check date of diagnosis if available to ensure it predates cancer diagnosis or treatment, if relevant.
 - Do not confuse with angina or other cardiovascular conditions unless explicitly stated as AMI.
-

Variable: comorb_mi

Type: Categorical (Dropdown)

Description: History of Myocardial Infarction (MI)

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, cardiology notes, admission/discharge summaries)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented history of prior (non-acute) myocardial infarction, specifically chronic or old MI, not representing the acute event.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown

- **Source Documents:**

- Problem list or past medical history in clinic notes
- Hospital discharge summaries
- Cardiology consult notes or imaging reports
- ICD-10 coded fields in structured EHR (e.g., I25.2)

- **Curation Instructions:**

1. Identify documentation of previous myocardial infarction, including references to “old MI” or “prior infarct.”
2. Accept coded diagnosis of chronic MI using ICD-10: I25.2.
3. Do not conflate with acute MI (see `comorb_ami`), which should be captured separately.
4. If language is ambiguous or there is conflicting information, select “Unknown.”
5. If no documentation is found, select “No.”

- **Curator Quality Review:**

- Ensure diagnosis is not an acute event.
-

Variable: `comorb_chf`

Type: Categorical (Dropdown)

Description: Congestive Heart Failure (CHF)

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, cardiology notes, admission/discharge summaries)

Curation Directives:

- **Definition:** Indicates whether the patient has clinical documentation of congestive heart failure, including both systolic and diastolic dysfunction.

- **Permitted Values / Codes:**

- 1 = Yes
- 2 = No
- 3 = Unknown

- **Source Documents:**

- Problem list or past medical history
- Hospital discharge summaries
- Cardiology or internal medicine consult notes
- Echocardiogram or imaging reports
- ICD-10 coded fields (e.g., I09.9, I11.0, I13.0, I13.2, I25.5, I42.0, I42.5–I42.9, I43.x, I50.x, P29.0)

- **Curation Instructions:**

1. Accept any documentation of CHF, including terms like “congestive heart failure,” “heart failure with reduced ejection fraction,” or “diastolic heart failure.”
2. Confirm diagnosis through either narrative or structured data (e.g., relevant ICD-10 codes).
3. Do not infer from symptoms alone (e.g., edema, dyspnea) unless explicitly linked to a CHF diagnosis.
4. If documentation is vague or inconsistent, select “Unknown.”
5. If no evidence of CHF is found, select “No.”

- **Curator Quality Review:**

- Cross-check problem list with hospital admissions and cardiology notes.
-

Variable: comorb_pvd

Type: Categorical (Dropdown)

Description: Peripheral Vascular Disease (PVD)

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, vascular surgery or cardiology notes, discharge summaries)

Curation Directives:

- **Definition:** Indicates whether there is clinical documentation of peripheral vascular disease, including arterial disorders affecting the extremities, abdomen, or other non-cardiac vasculature.
 - **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
 - **Source Documents:**
 - Problem list or past medical history
 - Vascular surgery or cardiology notes
 - Imaging reports (e.g., angiograms, Doppler studies)
 - Hospital discharge summaries
 - ICD-10 coded fields (e.g., I70.x, I71.x, I73.1, I73.8, I73.9, I77.1, I79.0, I79.2, K55.1, K55.8, K55.9, Z95.8, Z95.9)
 - **Curation Instructions:**
 1. Include diagnoses such as peripheral artery disease (PAD), intermittent claudication, aortic aneurysm, and chronic limb ischemia.
 2. Accept documented history of revascularization or vascular stent placement (e.g., Z95.8, Z95.9).
 3. Exclude isolated venous insufficiency or unrelated vascular findings.
 4. If documentation only suggests suspicion or workup without confirmed diagnosis, select “Unknown.”
 5. If no relevant history is identified, select “No.”
 - **Curator Quality Review:**
 - Prioritize clear documentation of diagnosis or relevant ICD-10 codes.
 - Watch for mentions of vascular bypass, angioplasty, or amputation due to vascular disease.
-

Variable: comorb_cvd

Type: Categorical (Dropdown)

Description: Cerebrovascular Disease (CVD)

Reference: NCI Comorbidity Index

Data Source: Medical record (neurology notes, discharge summaries, imaging reports, problem list)

Curation Directives:

- **Definition:** Indicates whether the patient has documented cerebrovascular disease, including ischemic or hemorrhagic stroke, transient ischemic attack (TIA), or other related cerebrovascular conditions.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Neurology consult or inpatient notes

- Hospital discharge summaries
- Head imaging reports (CT, MRI)
- Problem list or past medical history
- ICD-10 coded fields (e.g., G45.x, G46.x, H34.0, I60.x–I69.x)

- **Curation Instructions:**

1. Include documented history of ischemic stroke, hemorrhagic stroke, TIA, or cerebrovascular accidents.
2. Include mentions of long-term sequelae (e.g., hemiplegia following stroke, I69.x).
3. Exclude non-cerebral vascular disease (e.g., peripheral vascular disease, unless also CVD).
4. If the patient has an ambiguous mention (e.g., “rule out stroke”) without confirmatory evidence, select “Unknown.”
5. If no relevant documentation exists, select “No.”

- **Curator Quality Review:**

- Confirm diagnosis through narrative notes or imaging reports.
- Watch for over-reliance on patient-reported history without corroboration.
- Cross-reference with ICD-10 codes where possible.

- **Notes:**

- “CVD” in this context refers specifically to cerebrovascular pathology, not cardiovascular disease more broadly.
- TIA and resolved strokes still count as “Yes.”

Variable: comorb_copd

Type: Categorical (Dropdown)

Description: Chronic Obstructive Pulmonary Disease (COPD)

Reference: NCI Comorbidity Index

Data Source: Medical record (pulmonology notes, problem list, admission history, imaging, or pulmonary function testing)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented diagnosis of chronic obstructive pulmonary disease, including chronic bronchitis, emphysema, or related chronic pulmonary disorders.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Pulmonology consult or clinic notes
 - Discharge summaries or admission history
 - Chest imaging reports
 - Pulmonary function test (PFT) results
 - Problem list or past medical history
 - ICD-10 coded fields (e.g., I27.8, I27.9, J40.x–J47.x, J60.x–J67.x, J68.4, J70.1, J70.3)
- **Curation Instructions:**
 1. Select “Yes” if the medical record explicitly documents a diagnosis of COPD, chronic bronchitis, or emphysema.
 2. Include chronic pulmonary diseases such as interstitial lung disease (ILD) or pneumoconiosis if listed among the referenced codes.

3. Exclude isolated asthma unless the record specifies a co-diagnosis with COPD.
4. Use “Unknown” if the record mentions pulmonary symptoms (e.g., chronic cough) without a formal diagnosis.
5. Select “No” if there is no evidence or explicit mention of COPD or related conditions.

- **Curator Quality Review:**

- Confirm the diagnosis from narrative notes or diagnostic test results when ICD codes are unavailable.
- Ensure “Yes” selections are supported by chronic, not acute, pulmonary diagnoses.
- Avoid coding “Yes” for one-time respiratory infections or transient bronchitis.

- **Notes:**

- Patients with co-existing COPD and lung cancer should still have COPD coded if separately documented.
- Chronic pulmonary diseases from occupational exposures (e.g., asbestosis) are included if listed under specified ICD codes.

Variable: comorb_dementia

Type: Categorical (Dropdown)

Description: Dementia

Reference: NCI Comorbidity Index

Data Source: Medical record (neurology consults, cognitive assessment notes, problem list, discharge summaries)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented diagnosis of dementia or related neurodegenerative disorders, including Alzheimer’s disease.

- **Permitted Values / Codes:**

- 1 = Yes
- 2 = No
- 3 = Unknown

- **Source Documents:**

- Neurology consults or outpatient clinic notes
- Cognitive assessment or memory evaluation reports
- Discharge summaries or admission history
- Problem list or past medical history
- ICD-10 coded fields: F00.x–F03.x (dementia subtypes), F05.1 (delirium superimposed on dementia), G30.x (Alzheimer’s), G31.1 (senile degeneration)

- **Curation Instructions:**

1. Select “Yes” if the medical record documents a formal diagnosis of dementia, Alzheimer’s disease, or related cognitive disorders.
2. Include vascular or mixed-type dementias if specified.
3. Select “No” only when records explicitly rule out dementia or contain no evidence or suspicion of cognitive impairment.
4. Use “Unknown” if cognitive impairment is not mentioned or documented.
5. Do not include temporary or reversible conditions (e.g., medication-induced confusion).

- **Curator Quality Review:**

- Validate that diagnosis is supported by physician documentation or relevant ICD-10 codes.

- Exclude age-related memory complaints not formally assessed or diagnosed.
 - Ensure Alzheimer’s is only coded “Yes” if explicitly stated as a diagnosis.
-

Variable: comorb_paralysis

Type: Categorical (Dropdown)

Description: Paralysis (Hemiplegia or Paraplegia)

Reference: NCI Comorbidity Index

Data Source: Medical record (neurology consults, physical therapy/rehab notes, discharge summaries, problem list)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented diagnosis of paralysis, including hemiplegia, paraplegia, quadriplegia, or other motor deficits resulting from neurological injury or disease.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Neurology or rehabilitation consults
 - Hospital discharge summaries
 - Physical or occupational therapy evaluations
 - Problem list or diagnosis list in EHR
 - ICD-10 codes:
 - * G04.1 (Tropical spastic paraparesis)
 - * G11.4 (Hereditary spastic paraplegia)
 - * G80.1–G80.2 (Cerebral palsy with motor impairment)
 - * G81.x (Hemiplegia and hemiparesis)
 - * G82.x (Paraplegia and tetraplegia)
 - * G83.0–G83.4, G83.9 (Other paralytic syndromes)
- **Curation Instructions:**
 1. Select “Yes” if the medical record contains a diagnosis or clinical evidence of motor paralysis affecting one or more limbs.
 2. Include conditions such as hemiplegia, paraplegia, quadriplegia, and other permanent or chronic paralysis.
 3. Select “No” if there is clear documentation of motor function being intact and no diagnosis of paralysis.
 4. Use “Unknown” if clinical notes are ambiguous or if motor status is not assessed.
 5. Do not include temporary limb weakness due to fatigue, transient neurologic symptoms, or non-paralytic conditions.
- **Curator Quality Review:**
 - Confirm that the term “paralysis,” “hemiplegia,” “paraplegia,” or equivalent is documented in association with a diagnosis or neurological condition.
 - If based solely on ICD-10 codes, ensure they correspond to valid documentation in the medical record.
 - Exclude reports of mild weakness without a diagnosis of a paralytic syndrome.
- **Notes:**
 - May be caused by stroke, spinal cord injury, cerebral palsy, demyelinating disease, or neurodegenerative disorders.

- Consider functional assessments (e.g., mobility scores) as supportive evidence when diagnosis is implied.

Variable: `comorb_diabetes`

Type: Categorical (Dropdown)

Description: Diabetes

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, endocrinology notes, primary care documentation, medication list, lab results)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented diagnosis of diabetes mellitus, without mention of complications.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Problem list or structured diagnosis fields
 - Clinic notes (primary care, endocrinology)
 - ICD-10 codes:
 - * E10.0, E10.1, E10.6, E10.8, E10.9 (Type 1)
 - * E11.0, E11.1, E11.6, E11.8, E11.9 (Type 2)
 - * E13.0, E13.1, E13.6, E13.8, E13.9 (Other specified diabetes)
- **Curation Instructions:**
 1. Select “Yes” if diabetes is clearly documented with no mention of complications.
 2. Exclude gestational diabetes and prediabetes.
 3. If both uncomplicated and complicated codes are present, prioritize `comorb_diabetes_comp`.
 4. Use “Unknown” if there is insufficient documentation to determine diabetic status.
- **Curator Quality Review:**
 - Confirm diagnosis is supported by provider note.
 - Check for mention of insulin or oral hypoglycemics in the absence of specific diagnosis codes.
 - Ensure no complications are described before selecting this variable.
- **Notes:**
 - Do not double-code as both uncomplicated and complicated.
 - If complications are documented, use `comorb_diabetes_comp` instead.

Variable: `comorb_diabetes_comp`

Type: Categorical (Dropdown)

Description: Diabetes with Complications

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, endocrinology notes, specialist consultations, diagnostic test results)

Curation Directives:

- **Definition:** Indicates whether the patient has diabetes mellitus with documented complications such as retinopathy, nephropathy, neuropathy, or peripheral vascular disease.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Specialty consults (endocrinology, nephrology, ophthalmology)
 - Problem list or diagnosis list
 - Diagnostic imaging, fundus exams, EMG reports, urine protein tests
 - ICD-10 codes:
 - * E10.2–E10.5, E10.7 (Type 1 with complications)
 - * E11.2–E11.5, E11.7 (Type 2 with complications)
 - * E13.2–E13.5, E13.7 (Other diabetes with complications)
- **Curation Instructions:**
 1. Select “Yes” if diabetes is documented alongside complications such as retinopathy, nephropathy, neuropathy, ketoacidosis, or other organ involvement.
 2. Prioritize this variable over `comorb_diabetes` if both codes are present.
 3. Use “No” only if diabetes is explicitly noted to be without complications.
 4. Use “Unknown” if complications are suspected but not confirmed in documentation.
- **Curator Quality Review:**
 - Ensure complications are clearly linked to diabetes, not from unrelated causes.
 - Validate against ICD-10 code range provided.
- **Notes:**
 - “With complications” takes precedence over “uncomplicated” diabetes in coding.
 - Include microvascular and macrovascular complications when attributed to diabetes.

Variable: `comorb_renal`

Type: Categorical (Dropdown)

Description: Renal Disease

Reference: NCI Comorbidity Index

Data Source: Medical record (nephrology consults, problem list, lab reports, dialysis records, procedure history)

Curation Directives:

- **Definition:** Indicates whether the patient has documented evidence of chronic renal disease, including end-stage renal disease, dialysis, kidney transplant, or other forms of significant kidney dysfunction.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Nephrology consult notes
 - Problem list or structured diagnosis fields
 - Dialysis center documentation or procedure notes
 - Lab reports indicating chronic kidney disease (CKD) staging or kidney function abnormalities
 - ICD-10 codes:

- * I12.0, I13.1 (Hypertensive renal disease)
- * N03.2–N03.7, N05.2–N05.7 (Nephritic/nephrotic syndromes)
- * N18.x (CKD stages 1–5, ESRD)
- * N19.x (unspecified renal failure)
- * N25.0 (disorders from renal tubular dysfunction)
- * Z49.0–Z49.2 (dialysis procedures)
- * Z94.0 (renal transplant)
- * Z99.2 (dependence on dialysis)

- **Curation Instructions:**

1. Select “Yes” if there is documentation of CKD stage 3, ESRD, dialysis, or renal transplant.
2. Do not include transient or acute renal conditions unless clearly part of chronic disease.
3. Select “No” if documentation explicitly rules out renal disease.
4. Use “Unknown” if renal function is not discussed or documentation is insufficient to classify.

- **Curator Quality Review:**

- Confirm presence of qualifying ICD-10 codes or clinical narrative describing chronic kidney dysfunction.
- Confirm duration and chronicity to rule out acute injury.

Variable: comorb_liver_mild

Type: Categorical (Dropdown)

Description: Mild Liver Disease

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, hepatology notes, labs, imaging, procedures)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented diagnosis of mild chronic liver disease, including hepatitis, steatosis, or early-stage fibrosis without evidence of cirrhosis or decompensation.

- **Permitted Values / Codes:**

- 1 = Yes
- 2 = No
- 3 = Unknown

- **Source Documents:**

- Problem list or structured diagnosis fields
- Hepatology consultation notes
- Lab results (elevated LFTs, hepatitis serologies)
- Imaging (evidence of steatosis or fibrosis without cirrhosis)
- Procedure reports (liver biopsy)
- ICD-10 codes:
 - * B18.x
 - * K70.0–K70.3, K70.9
 - * K71.3–K71.5, K71.7
 - * K73.x, K74.x
 - * K76.0, K76.2–K76.4, K76.8–K76.9
 - * Z94.4

- **Curation Instructions:**

1. Select “Yes” if there is documented evidence of chronic liver disease that is **not cirrhosis** and without clinical or imaging signs of liver failure or portal hypertension.

2. Include diagnoses such as nonalcoholic fatty liver disease (NAFLD), chronic viral hepatitis, and hepatic fibrosis.
 3. Use “No” if the patient has no history of liver disease or if documentation clearly rules it out.
 4. Use “Unknown” if the documentation is insufficient or ambiguous.
 5. If the patient has **moderate/severe liver disease**, select “No” for this variable and refer to the `comorb_liver_mod_severe` field.
- **Curator Quality Review:**
 - Confirm diagnosis timing and chronicity.
 - Exclude if only transient or reversible liver abnormalities are present (e.g., drug-induced LFT elevation).
 - Validate against ICD codes and provider documentation.
 - **Notes:**
 - Do not double-code with `comorb_liver_mod_severe`.
-

Variable: `comorb_liver_mod_severe`

Type: Categorical (Dropdown)

Description: Moderate/Severe Liver Disease

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, hepatology notes, labs, imaging, procedures)

Curation Directives:

- **Definition:** Indicates whether the patient has documented evidence of moderate to severe liver disease, such as cirrhosis, portal hypertension, liver failure, or hepatic encephalopathy.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Problem list or structured diagnosis fields
 - Hepatology consultation notes
 - Imaging (e.g., nodular liver contour, ascites, splenomegaly, varices)
 - Procedure reports (e.g., liver biopsy showing cirrhosis)
 - ICD-10 codes:
 - * I85.0, I85.9 (esophageal varices)
 - * I86.4, I98.2 (portal hypertension)
 - * K70.4 (alcoholic hepatic failure)
 - * K71.1, K72.1, K72.9 (hepatic necrosis and failure)
 - * K76.5 (hepatic veno-occlusive disease)
 - * K76.6 (portal vein thrombosis)
 - * K76.7 (hepatorenal syndrome)
- **Curation Instructions:**
 1. Select “Yes” if there is documentation of cirrhosis, decompensated liver disease, or portal hypertension.
 2. Prioritize imaging and biopsy findings over vague clinical impressions.
 3. Use “No” if liver disease is mild or absent.
 4. Use “Unknown” if there is insufficient or conflicting evidence.
 5. If this variable is coded as “Yes,” `comorb_liver_mild` must be coded as “No.”

- **Curator Quality Review:**
 - Look for explicit mentions of cirrhosis, ascites, varices, or liver decompensation.
 - Confirm that ICD codes align with documentation.
 - Check for contradictory evidence between clinical and imaging findings.
- **Notes:**
 - This variable overrides `comorb_liver_mild` in cases where both may apply.

Variable: `comorb_pud`

Type: Categorical (Dropdown)

Description: Peptic Ulcer Disease

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, gastroenterology notes, EGD reports, hospital discharge summaries)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented history of peptic ulcer disease (gastric, duodenal, or unspecified), either active or chronic.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Structured problem list
 - Gastroenterology consultation notes
 - Endoscopy (EGD) reports
 - Hospital discharge summaries
 - Pathology reports (gastric/duodenal biopsies)
 - ICD-10 codes:
 - * K25.x (Gastric ulcer)
 - * K26.x (Duodenal ulcer)
 - * K27.x (Peptic ulcer, site unspecified)
 - * K28.x (Gastrojejunal ulcer)
- **Curation Instructions:**
 1. Select “Yes” if there is documentation of any gastric, duodenal, or peptic ulcer diagnosed via imaging, endoscopy, or clinical impression.
 2. Select “No” if there is clear documentation stating no history of ulcers or if other causes of dyspepsia (e.g., GERD) are explicitly noted instead.
 3. Select “Unknown” if ulcer history is not mentioned or is too ambiguous to determine.
 4. Do not include patients with only NSAID prophylaxis or empiric acid suppression (e.g., PPI use) without confirmed ulcer diagnosis.
- **Curator Quality Review:**
 - Verify date of ulcer diagnosis and ensure it reflects a chronic or recurrent condition (not just acute gastritis or dyspepsia).
 - Confirm ICD-10 codes align with provider documentation.
- **Notes:**
 - *Helicobacter pylori* infection alone is not sufficient unless associated with ulcer findings.

Variable: comorb_rheu

Type: Categorical (Dropdown)

Description: Rheumatologic Disease

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, rheumatology notes, discharge summaries, specialty consults)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented history of autoimmune or systemic connective tissue disease, including rheumatoid arthritis, systemic lupus erythematosus, or other rheumatologic disorders.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Structured problem list
 - Rheumatology consultation notes
 - Hospital discharge summaries
 - Specialty clinic notes
 - ICD-10 codes:
 - * M05.x (Seropositive rheumatoid arthritis)
 - * M06.x (Other rheumatoid arthritis)
 - * M31.5 (Giant cell arteritis with polymyalgia rheumatica)
 - * M32.x (Systemic lupus erythematosus)
 - * M33.x–M34.x (Dermatomyositis, systemic sclerosis)
 - * M35.1 (Other overlap syndromes)
 - * M35.3 (Polymyalgia rheumatica)
 - * M36.0 (Dermato/poly/autoimmune syndromes in neoplastic disease)
- **Curation Instructions:**
 1. Select “Yes” if there is documentation of any systemic autoimmune connective tissue disorder as defined by the ICD-10 codes above.
 2. Select “No” if the record explicitly states absence of autoimmune disease or if the patient is being evaluated for symptoms without diagnosis.
 3. Select “Unknown” if there is insufficient or ambiguous documentation.
- **Curator Quality Review:**
 - Confirm that ICD codes or narrative documentation support a chronic systemic condition.
 - Be cautious of EHR carry-forward errors or rule-out language without confirmatory diagnosis.
- **Notes:**
 - Do not include isolated osteoarthritis, fibromyalgia, or localized joint pain without confirmed systemic rheumatologic diagnosis.

Variable: comorb_aids

Type: Categorical (Dropdown)

Description: AIDS

Reference: NCI Comorbidity Index

Data Source: Medical record (problem list, infectious disease notes, discharge summaries, HIV/AIDS clinic notes)

Curation Directives:

- **Definition:** Indicates whether the patient has a documented diagnosis of Acquired Immune Deficiency Syndrome (AIDS), as defined by ICD-10 codes B20.x–B22.x, B24.x.
- **Permitted Values / Codes:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
- **Source Documents:**
 - Problem list
 - Infectious disease consultation notes
 - HIV/AIDS clinic notes
 - Hospital discharge summaries
 - ICD-10 codes:
 - * B20.x (HIV disease resulting in infectious/parasitic disease)
 - * B21.x (HIV disease resulting in malignant neoplasms)
 - * B22.x (HIV disease resulting in other specified diseases)
 - * B24.x (Unspecified HIV disease)
- **Curation Instructions:**
 1. Select “Yes” if there is any documentation of AIDS or HIV-related illness with ICD-10 codes listed above.
 2. Select “No” if the record explicitly documents that the patient is HIV-negative.
 3. Select “Unknown” if HIV status is not mentioned or only suspected without confirmation.
 4. Do not select “Yes” solely based on HIV positivity — only code as “Yes” if AIDS-defining illness is documented (i.e., diagnosis falls under ICD-10 B20–B22 or B24).
 5. For patients with HIV but no documentation of AIDS-related conditions, consider whether the diagnosis is coded as B20 or later — if only Z21 (asymptomatic HIV), exclude.
- **Curator Quality Review:**
 - Verify that HIV-related documentation includes an AIDS-defining diagnosis (e.g., Kaposi’s sarcoma, pneumocystis pneumonia).
- **Notes:**
 - Do not include if only a lab value or positive HIV test is noted without confirmatory AIDS diagnosis.
 - Consider treatment history (e.g., antiretroviral therapy) only as supporting evidence if diagnosis is clearly stated.

Cancer Diagnosis

Piped In - Tumor Registry Sections

Piped In – Tumor Registry Data Elements: Cancer Diagnosis The following variables are piped in from institutional tumor registry feeds and are not manually curated. These fields are auto-populated via data integration from registry-derived sources, which follow standardized NAACCR coding guidelines. They are included to support consistency, auditability, and downstream analytic use, but curators should

not alter these values. These fields are typically marked as **@READONLY** in the data collection system. However, curators may reference these variables to validate diagnosis-level information or assess consistency across sources.

Piped-in Tumor Registry Variables:

- `naaccr_version_tr` – NAACCR Record Version [NAACCR #50]
 - `ca_seq_tr` – Cancer Sequence Number [NAACCR #380]
 - `dx_date_tr` – Date of Diagnosis [NAACCR #390]
 - `dx_date_tr_int` – Diagnosis Date Interval from DOB (Calculated)
 - `age_dx_tr` – Age at Diagnosis [NAACCR #230]
 - `diag_confirm_tr` – Diagnostic Confirmation Method [NAACCR #490]
 - `primary_site_tr` – Primary Site Code (ICD-O-3) [NAACCR #400]
 - `histology_tr` – Histologic Type (ICD-O-3) [NAACCR #522]
 - `behavior_tr` – Behavior Code (ICD-O-3) [NAACCR #523]
 - `laterality_tr` – Tumor Laterality [NAACCR #410]
 - `grade_clinical_tr` – Grade Clinical [NAACCR #3843]
 - `grade_clinical_post_tr` – Grade Post Therapy Clinical (yc) [NAACCR #1068]
 - `grade_path_tr` – Grade Pathological [NAACCR #3844]
 - `grade_path_post_tr` – Grade Post Therapy Pathological (yp) [NAACCR #3845]
 - `grade_summary_tr` – Derived Summary Grade 2018 [NAACCR #1975]
 - `tumor_size_tr` – Tumor Size Summary [NAACCR #756]
-

Piped In – Tumor Registry Data Elements: Historical Staging

The following variables are piped in from institutional tumor registry systems and represent historically abstracted staging and grading information. These fields reflect clinical and pathologic staging based on older AJCC editions and SEER Summary Stage criteria. They are included to support longitudinal comparisons, facilitate data harmonization, and allow analysts to assess historical staging practices or trends across AJCC versions. These variables are not manually curated and are typically marked as **@READONLY** in the data collection system. Curators should reference but not modify these values.

Piped-in Tumor Registry Historical Variables:

- `ajcc_edition_tr` – AJCC Edition Number Used for Staging [NAACCR #1060]
 - `grade_trhis` – Historical Grade (pre-2018) [NAACCR #440]
 - `stage_sum_trhis` – SEER Summary Stage [NAACCR #759]
 - `clin_t_trhis` – Clinical T (Tumor) [NAACCR #940]
 - `clin_n_trhis` – Clinical N (Nodes) [NAACCR #950]
 - `clin_m_trhis` – Clinical M (Metastases) [NAACCR #960]
 - `stage_clin_trhis` – Clinical Stage Group [NAACCR #970]
 - `path_t_trhis` – Pathologic T (Tumor) [NAACCR #880]
 - `path_n_trhis` – Pathologic N (Nodes) [NAACCR #890]
 - `path_m_trhis` – Pathologic M (Metastases) [NAACCR #900]
 - `stage_path_trhis` – Pathologic Stage Group [NAACCR #910]
-

Piped In – Tumor Registry Data Elements: Stage of Disease at Diagnosis

The following variables are piped in from institutional tumor registries and represent stage of disease at the time of diagnosis, captured using the **NAACCR Extent of Disease (EOD) 2018** and **Summary Stage 2018** frameworks. These fields are not manually curated and are derived or abstracted using standardized EOD algorithms and coding guidelines. They are included to enable uniform staging comparisons across tumor types and reporting centers. These fields are typically marked as **@READONLY** in the data collection system and should not be modified by curators.

Piped-in Stage of Disease Variables:

- `eod_primary_tr` – EOD Primary Tumor: Contiguous tumor growth/extension [NAACCR #772]
 - `eod_regional_tr` – EOD Regional Nodes: Regional lymph node involvement [NAACCR #774]
 - `eod_mets_tr` – EOD Mets: Distant metastases present at diagnosis [NAACCR #776]
 - `stage_sum_tr` – Summary Stage 2018 (directly assigned) [NAACCR #764]
 - `stage_sum_der_tr` – Derived Summary Stage 2018 (algorithmically calculated) [NAACCR #762]
-

Piped In – Tumor Registry Data Elements: Stage-Related Data Items

The following variables are piped in from institutional tumor registries and capture stage-related pathology findings and site-specific metastatic involvement at the time of diagnosis. These fields are not manually curated and are intended to support staging accuracy, risk stratification, and cohort identification in downstream analyses. They are typically marked as **@READONLY** in the data collection system and should not be modified by curators.

Piped-in Stage-Related Variables:

- `lymph_inv_tr` – Lymphovascular Invasion (LVI) [NAACCR #1182]
 - `mets_bone_tr` – Metastatic Site at Diagnosis: Bone [NAACCR #1112]
 - `mets_brain_tr` – Metastatic Site at Diagnosis: Brain [NAACCR #1113]
 - `mets_liver_tr` – Metastatic Site at Diagnosis: Liver [NAACCR #1115]
 - `mets_lung_tr` – Metastatic Site at Diagnosis: Lung [NAACCR #1116]
 - `mets_distant_lymph_tr` – Metastatic Site at Diagnosis: Distant Lymph Nodes [NAACCR #1114]
 - `mets_other_tr` – Metastatic Site at Diagnosis: Other/Unspecified Sites [NAACCR #1117]
-

Curated Cancer Diagnosis Section

For all fields: complete the curated diagnosis variables using clinical documentation closest to the date of cancer diagnosis.

Note: Although many diagnosis-related variables (such as date of diagnosis, cancer sequence number, histology, primary site, and stage) are piped in from institutional tumor registries, curators are required to independently abstract and complete all corresponding **curated diagnosis fields** in the data entry form. These curated values must be sourced directly from the patient’s medical record, pathology reports, imaging

summaries, or multidisciplinary notes, and should reflect the most current, comprehensive, and clinically verified documentation available.

Do not modify any piped-in values, as they are maintained for traceability, consistency with the original tumor registry feed, and audit purposes. However, if the curated diagnosis-level values differ from the piped-in tumor registry fields, record the curated values exactly as documented in the medical record. All discrepancies should be preserved as-is and not reconciled. The curated fields are considered the authoritative abstraction for research and analysis, and may reflect updated clinical understanding, correction of earlier documentation, or greater granularity than the registry's original submission.

Variable: `dx_date`

Type: Date (MM/DD/YYYY)

Description: Curated Date of Diagnosis. Date of initial diagnosis by a recognized medical practitioner for the tumor being reported, whether clinically or microscopically confirmed.

Reference: NAACCR Item #390 (Date of Diagnosis)

Data Source: EHR documentation of cancer diagnosis (clinical or pathological)

Curation Directives:

- **Definition:** The date when the patient was first diagnosed with the malignancy being reported, based on clinical or pathological confirmation by a qualified healthcare provider. For more discussion on determining date of diagnosis, consult the *SEER Program Coding and Staging Manual* or *CoC STORE* manual. Aligned with NAACCR #390.
- **Source Documents:**
 - Pathology report (date of diagnostic biopsy or resection)
 - Oncology consultation note
 - Clinic visit note documenting initial cancer diagnosis
- **Curation Instructions:**
 1. Abstract the earliest documented date of diagnosis for the tumor being reported, not recurrence or progression.
 2. Prefer the date of pathologic confirmation (e.g., biopsy) if available. If no pathology is available, use date of clinical diagnosis.
 3. If multiple documents list different dates, choose the earliest date when the cancer diagnosis was clinically or pathologically established.
 4. Record the full date in MM/DD/YYYY format.
 5. Use date estimation rules (if applicable) when a full date is not documented.
 6. Do not use dates of recurrence, progression, or follow-up diagnoses.
- **Curator Quality Review:**
 - Confirm consistency with pathology and clinical documentation.
 - If multiple tumors are present, ensure the diagnosis date corresponds to the tumor under review.
- **Notes:**
 - This field drives the `dx_year` calculated field and is critical for time-based eligibility and cohort inclusion.
 - If exact date is unknown, follow project-specific guidelines for partial or estimated dates.
 - For patients with multiple primary tumors, record the diagnosis date **specific to the tumor type** being curated (not the first-ever cancer diagnosis unless it is the index tumor).

Variable: dx_year

Type: Calculated Field (read-only)

Description: Year of Diagnosis

Reference: n/a

Data Source: System-calculated from date of initial cancer diagnosis (dx_date)

Curation Directives:

- **Definition:** Represents the calendar year in which the patient received their initial cancer diagnosis. This field is automatically calculated based on the dx_date field using the formula year([dx_date]). It is used for cohort stratification, timeline alignment, and eligibility assessments.
 - **Source Documents:**
 - Derived from the curated dx_date value
 - **Curation Instructions:**
 - Do not enter or modify this field directly; it is a system-calculated value and marked as @READ-ONLY.
 - **Curator Quality Review:**
 - Confirm that the dx_year value correctly reflects the four-digit year portion of the dx_date.
 - **Notes:**
 - Do not override calculated values
-

Variable: dx_date_int

Type: Integer (Calculated)

Description: Interval in Days from Date of Birth to Date of Diagnosis

Reference: n/a

Data Source: Derived from birth_date and dx_date fields

Curation Directives:

- **Definition:** Automatically calculated field representing the number of days between the patient's date of birth and the date of cancer diagnosis.
 - **Source Documents:**
 - Not directly curated — derived using values from:
 - * birth_date (Date of Birth)
 - * dx_date (Curated Date of Diagnosis)
 - **Curation Instructions:**
 1. Ensure that both birth_date and dx_date are valid and properly formatted as MM/DD/YYYY.
 2. Confirm that dx_date occurs after birth_date; if not, flag for data quality review.
 3. The field is read-only and auto-populated using the REDCap function:
datediff([birth_date],[dx_date],"d","mdy",true)
 4. No manual data entry is required.
 - **Curator Quality Review:**
 - Ensure underlying birth_date and dx_date fields are not missing or estimated without justification.
-

Variable: dx_age_yr

Type: Integer (Calculated)

Description: Age at Diagnosis (in Years)

Reference: n/a

Data Source: Derived from dx_date_int

Curation Directives:

- **Definition:** Age in years at time of cancer diagnosis, calculated by dividing the interval in days from date of birth to diagnosis by 365.25 to account for leap years.
- **Source Documents:**
 - Not directly curated. Derived from:
 - * dx_date_int (DOB to Diagnosis Date, in days)
- **Curation Instructions:**
 1. Confirm that the underlying dx_date_int value is correctly calculated and reflects a valid, positive interval.
 2. This field is calculated using the REDCap expression:
round([dx_date_int] / 365.25, 4)
 3. No manual data entry required.
 4. If age appears biologically implausible (e.g., < 0 or > 120), verify input values (birth_date, dx_date) for errors.
- **Curator Quality Review:**
 - Review for accuracy and consistency with dx_date_int.
- **Notes:**
 - Used in eligibility determination, cohort stratification, and age-based epidemiologic analyses.

Variable: dx_institution

Type: Categorical (Dropdown)

Description: Facility Where Cancer Was Diagnosed

Reference: n/a

Data Source: Medical record (clinical notes, pathology reports, referral documentation)

Curation Directives:

- **Definition:** Indicates whether the patient's cancer diagnosis was made at the GENIE participating institution or at an external facility.
- **Source Documents:**
 - Admission notes
 - Initial oncology consults
 - Pathology reports
 - Outside records scanned into EMR
- **Curation Instructions:**
 1. Review documentation from the time of diagnosis to determine location of the diagnosing facility.
 2. Code as:
 - 1 = Internal Institution: Diagnosis confirmed at the participating cancer center.
 - 2 = External Institution: Diagnosis made elsewhere and referred to the center.
 3. Do not infer based solely on treatment location.

- **Curator Quality Review:**

- Ensure consistency with diagnosis timeline and institutional referral notes.
- Flag ambiguous cases for clinician or data manager adjudication.

- **Notes:**

- This field is used to stratify patients by origin of diagnosis and may influence data availability (e.g., staging, pathology).

Variable: dx_zip_code

Type: Free Text Entry (text)

Description: Zip Code of Facility That Diagnosed the Cancer

Reference: n/a

Data Source: The five-digit ZIP code of the facility where the patient received their initial cancer diagnosis. This refers to the physical location of the diagnosing institution (e.g., hospital, clinic, or pathology lab) and is used for geographic and site-level analyses.

Curation Directives:

- **Definition:** Coded value indicating the best method used to confirm the cancer diagnosis. May be based on microscopic, laboratory, imaging, or clinical evidence.

- **Source Documents:**

- Pathology report or diagnostic biopsy summary (check facility letterhead or header)
- Initial oncology consult note or referral documentation
- Admission or encounter record associated with cancer diagnosis

- **Curation Instructions:**

1. Abstract the five-digit ZIP code of the facility where the cancer was first diagnosed.
2. Prioritize the ZIP code of the facility listed on the pathology report if diagnosis was made histologically.
3. If only a clinical diagnosis was made (e.g., via imaging or exam), use the ZIP code of the facility where the diagnosis was first recorded by a qualified provider.

- **Notes:**

- This field may be used for mapping regional diagnosis patterns, facility clustering, and demographic comparisons.
- If the diagnosis occurred outside the U.S., leave blank or enter “00000” if permitted by project rules (confirm with coordinator).
- If multiple facilities were involved, prioritize the first diagnosing entity.

Variable: dx_confirmation

Type: Categorical (Dropdown)

Description: Diagnostic Confirmation Method

Reference: NAACCR #490

Data Source: Medical record (pathology reports, radiology reports, clinical documentation, death certificate)

Curation Directives:

- **Definition:** Coded value indicating the best method used to confirm the cancer diagnosis. May be based on microscopic, laboratory, imaging, or clinical evidence.

- **Permitted Values / Codes:**
 - 1 = Positive histology
 - 2 = Positive cytology
 - 3 = Positive histology plus immunophenotyping and/or genetic studies (Hematopoietic and Lymphoid Neoplasms only)
 - 4 = Positive microscopic confirmation, method not specified
 - 5 = Positive laboratory test/marker study
 - 6 = Direct visualization without microscopic confirmation
 - 7 = Radiology and other imaging techniques without microscopic confirmation
 - 8 = Clinical diagnosis only
 - 9 = Unknown whether or not microscopically confirmed; death certificate only
- **Source Documents:**
 - Pathology reports
 - Hematopathology or flow cytometry reports
 - Radiology reports
 - Clinic notes with final diagnosis
 - Molecular/genetic testing results
 - Death certificate (if diagnosis only appears post-mortem)
- **Curation Instructions:**
 1. Review all available documentation to identify the highest level of diagnostic certainty.
 2. Prioritize histology or cytology when available.
 3. For hematologic malignancies, select code 3 only if both histology and immunophenotyping/genetic studies are positive.
 4. Use imaging or clinical diagnosis codes (6–8) only when microscopic or laboratory confirmation is unavailable.
 5. Use code 9 only when diagnosis is based solely on death certificate or confirmation status is unknown.
 6. Select the **best** confirmation method ever documented, even if it occurred after initial diagnosis.
- **Curator Quality Review:**
 - Verify consistency with primary site and histology fields.
 - Ensure correct application of code 3 for hematologic cases.
 - Flag any case coded 9 for possible follow-up or review.
 - Confirm that method of confirmation aligns with documented timeline.
- **Notes:**
 - This field is used for quality metrics and incidence calculations.
 - Accurate classification supports population-level evaluations of diagnostic patterns.

Variable: dx_primary_site

Type: Categorical (Dropdown)

Description: Primary Site of Cancer Diagnosis (ICD-O-3 Topography Code)

Reference: NAACCR #400

Data Source: Pathology report, radiology report, operative notes, clinic documentation

Curation Directives:

- **Definition:** Code corresponding to the primary anatomical site of the cancer using the International Classification of Diseases for Oncology, Third Edition (ICD-O-3) topography codes.

- **Permitted Values / Codes:**

Extensive list mapped to ICD-O-3 topography codes (e.g., C50.4 = Upper-outer quadrant of breast, C34.9 = Lung NOS, C61.9 = Prostate, etc.). Autocomplete dropdown is used for ease of search and selection.

- **Source Documents:**

- Final pathology reports
- Operative and procedure notes
- Imaging reports confirming primary tumor location
- Oncology consult or multidisciplinary team notes
- Tumor registry abstract if available

- **Curation Instructions:**

1. Extract the anatomical location of the primary tumor from pathology or clinical diagnosis documentation.
2. Use imaging and surgical reports to confirm or refine location, particularly if pathology is ambiguous.
3. Assign the most specific ICD-O-3 topography code available (e.g., C50.4 vs. C50.9 for breast cancer).
4. Use code C80.9 only when all attempts to identify the primary site are exhausted and inconclusive.
5. If primary site is fluid-based (e.g., peritoneal, CSF), use F-codes accordingly.
6. Do not enter metastatic sites; this field is reserved for the origin of the cancer.

- **Curator Quality Review:**

- Ensure only one topography code is recorded per primary tumor.
- Review clinical notes if pathology lacks specificity.

- **Notes:**

- This variable is foundational for site-specific incidence, survival, and outcome analyses.

Variable: dx_morphology

Type: Categorical (Dropdown)

Description: Histology and Behavior (ICD-O-3 Morphology Code)

Reference: NAACCR #522 (Histology) and #523 (Behavior)

Data Source: Pathology report, cytology report, oncology consult, tumor registry abstract

Curation Directives:

- **Definition:** ICD-O-3 morphology code representing the microscopic histologic type (four digits) and behavior (one digit) of the tumor being reported. The morphology describes the tumor cell type, while the behavior denotes whether the tumor is benign, in situ, or malignant.

- **Source Documents:**

- Final pathology report
- Cytology report
- Surgical pathology or bone marrow biopsy report
- Oncology consultation note
- Tumor registry abstract (if pathology not available)

- **Curation Instructions:**

1. Review the final pathology or cytology report for a confirmed diagnosis.
2. Locate the ICD-O-3 morphology code listed in the report, or assign based on the histologic diagnosis using the WHO ICD-O-3 manual.

3. Record the full five-digit code in the format **####/x**:
 - First four digits = histology (e.g., 8140 = adenocarcinoma)
 - Fifth digit (after slash) = behavior:
 - * 0 = benign
 - * 1 = uncertain/borderline
 - * 2 = in situ
 - * 3 = malignant, primary site
 4. Do not enter obsolete morphology codes; refer to the most recent ICD-O-3 updates.
 5. If multiple morphologies are reported, code the most specific, aggressive, or clinically relevant type per SEER/ICD-O-3 guidance.
- **Curator Quality Review:**
 - Confirm the morphology code is compatible with the primary site (topography).
 - Ensure the format matches the required **####/x** structure.
 - Validate behavior code matches the documentation of invasiveness.
 - Flag unspecified or NOS morphology codes for potential review if a more specific code is supported by text.
 - **Notes:**
 - This is a core variable for cancer classification, used in surveillance and research.
 - Always code the most definitive diagnosis; prioritize histology confirmed by biopsy.
 - Refer to SEER Site/Histology Validation List for site-morphology consistency checks.

Variable: dx_laterality

Type: Categorical (Dropdown)

Description: Laterality

Reference: NAACCR #410

Data Source: Pathology report, operative note, imaging report, clinical documentation, tumor registry abstract

Curation Directives:

- **Definition:** Indicates the side of a paired organ or the side of the body on which the reportable tumor originated. This variable applies only to paired sites and primary tumors.
- **Source Documents:**
 - Final pathology report
 - Operative report
 - Imaging report (e.g., CT, MRI, mammogram)
 - Clinical encounter or H&P note
 - Tumor registry abstract (if present)
- **Curation Instructions:**
 1. Confirm whether the primary site is a paired organ (e.g., breast, kidney, lung).
 2. Code based on the side of origin as described in source documents:
 - 0 = Not a paired site (use for unpaired organs)
 - 1 = Right: origin of primary
 - 2 = Left: origin of primary
 - 3 = Only one side involved, right or left origin unspecified
 - 4 = Bilateral involvement at time of diagnosis, lateral origin unknown for a single primary; or both ovaries involved simultaneously, single histology; bilateral retinoblastomas; bilateral Wilms tumors

- 5 = Paired site: midline tumor (effective with 01/01/2010 dx)
 - 9 = Paired site, but no information concerning laterality
- 3. If documentation states “only one side involved” but does not specify which, use code 3.
- 4. Use code 9 only if laterality cannot be determined and no documentation is available.
- 5. Do not infer laterality from context; explicit documentation is required.
- **Curator Quality Review:**
 - Verify that bilateral tumors are coded appropriately (not duplicated across entries).
- **Notes:**
 - Required for all paired-site cancers.

Variable: dx_grade

Type: Categorical (Dropdown)

Description: Grade or Differentiation of the Cancer

Reference: NAACCR #440 and #3843 (for diagnoses after 2018)

Data Source: Pathology report, tumor board documentation, oncology consult, tumor registry abstract

Curation Directives:

- **Definition:** Pre-treatment histologic grade or level of differentiation of the tumor, reflecting how much the tumor resembles normal tissue.
 - **Source Documents:**
 - Final pathology report
 - Tumor board notes or staging documentation
 - Oncology consult notes
 - Tumor registry abstract (if present)
 - **Curation Instructions:**
 1. Record the grade **prior to** any treatment (surgery, chemo, radiation).
 2. Use the grade provided in the pathology report or staging documents:
 - 1 = Well differentiated (Grade I)
 - 2 = Moderately differentiated (Grade II)
 - 3 = Poorly differentiated (Grade III)
 - 4 = Undifferentiated/anaplastic (Grade IV)
 - 5 = Unknown, not stated, or not applicable
 - 6 = Low grade (if explicitly documented)
 - 7 = High grade (if explicitly documented)
 3. Do not abstract grade from post-treatment specimens.
 - **Curator Quality Review:**
 - Confirm grade is from a pre-treatment specimen.
 - **Notes:**
 - Grade 5 should only be used if no documentation is available or if grade is explicitly stated as unknown.
 - Avoid overuse of “high” and “low” grade codes unless clearly documented.
-

Variable: dx_clin_t

Type: Categorical (Dropdown)

Description: Clinical T Stage

Reference: NAACCR #940

Data Source: Oncology staging forms, clinical documentation, imaging reports, pathology reports (if used to inform clinical staging)

Curation Directives:

- **Definition:** Clinical assessment of the extent of the primary tumor (T) before any definitive treatment, assigned by the managing physician using AJCC staging guidelines. Includes tumor size, invasion into adjacent structures, or special T categories (e.g., Tis, T1mic).
- **Source Documents:**
 - Initial oncology consult notes
 - Imaging reports (CT, MRI, PET, ultrasound)
 - Endoscopy or surgical exploration findings
 - Pre-treatment pathology (if influencing clinical staging)
- **Curation Instructions:**
 1. Record the pre-treatment clinical T stage as documented by the managing physician.
 2. If both a physician-assigned stage and supporting documentation exist, prioritize the documented physician-assigned value.
 3. Choose the most specific T category available (e.g., T1a1 over T1a).
 4. Use:
 - TX if primary tumor cannot be assessed
 - T0 for no evidence of primary tumor
 - Ta, Tis, T1mic for in situ or microinvasive tumors
 - 88 if staging is not applicable (e.g., leukemia)
 - 99 if staging is applicable but not documented
 5. Do **not** derive T stage if not explicitly documented.
- **Curator Quality Review:**
 - Confirm clinical timing. Must be pre-treatment.
 - Verify that Tis/Ta/T1mic are only used for relevant cancer types (e.g., bladder, cervical, breast).
- **Notes:**
 - Do not abstract post-surgical or pathologic T stage into this field.
 - Avoid using 99 unless staging clearly applies but is absent

Variable: dx_clin_n

Type: Categorical (Dropdown)

Description: Clinical N Stage

Reference: NAACCR #950

Data Source: Oncology staging forms, clinical documentation, imaging reports, pathology reports (if used to inform clinical staging)

Curation Directives:

- **Definition:** Clinical assessment of regional lymph node involvement prior to treatment. Includes size, number, laterality, and confirmation method (imaging or biopsy), as per AJCC guidelines.
- **Source Documents:**

- Initial oncology consult notes
- Imaging reports (CT, MRI, PET, ultrasound)
- Endoscopy or surgical exploration findings
- Pre-treatment pathology (if influencing clinical staging)

- **Curation Instructions:**

1. Abstract the physician-assigned clinical N stage prior to definitive treatment.
2. Record specific N subclassifications when documented (e.g., N1a, N2b, N3c).
3. Use:
 - NX if nodes cannot be assessed
 - N0 for no regional node involvement
 - N1–N4 for increasing levels of nodal disease
 - 88 if staging is not applicable
 - 99 if staging is applicable but not documented
4. Use N0(i+), N0(i-), etc., only when clearly specified.
5. Do **not** infer from pathology unless it was used in assigning clinical stage.

- **Curator Quality Review:**

- Confirm N stage is clinical, not pathologic.
- Verify site-specific appropriateness of detailed codes (e.g., N3c not valid for all cancers).

- **Notes:**

- Clinical N stage should reflect information available at diagnosis, not after surgery.

Variable: dx_clin_m

Type: Categorical (Dropdown)

Description: Clinical M Stage

Reference: NAACCR #960

Data Source: Oncology staging forms, clinical documentation, imaging reports, pathology reports (if used to inform clinical staging)

Curation Directives:

- **Definition:** Clinical assessment of distant metastasis (M) prior to any treatment, based on physical examination, imaging, and/or biopsy of distant sites as defined by AJCC guidelines.

- **Source Documents:**

- Initial oncology consult notes
- Imaging reports (CT, MRI, PET, ultrasound)
- Endoscopy or surgical exploration findings
- Pre-treatment pathology (if influencing clinical staging)

- **Curation Instructions:**

1. Abstract the physician-assigned clinical M stage.
2. Use:
 - M0 for no distant metastasis
 - M1 (and subcategoriesI M1a–M1c) for documented distant spread
 - M0(i+) when isolated tumor cells in blood/marrow are noted
 - MX if metastasis cannot be assessed
 - 88 if staging is not applicable (e.g., leukemia)
 - 99 if staging is applicable but undocumented
3. Do not infer metastasis without physician documentation or biopsy confirmation.

4. Use the most specific M subcategory if stated.

- **Curator Quality Review:**

- Confirm staging is clinical and not based on post-treatment findings.
- Ensure M1 is supported by radiologic or pathologic evidence.
- Validate correct subcategories based on site-specific AJCC staging.

- **Notes:**

- Clinical M stage is critical for determining overall stage group,
- Documentation must be clear; avoid assumptions based on imaging alone.

Variable: `dx_clin_group_stage`

Type: Categorical (Dropdown)

Description: AJCC Clinical Stage Group

Reference: NAACCR #970

Data Source: Oncology staging summary, AJCC staging form, clinical documentation

Curation Directives:

- **Definition:** Composite clinical stage group (Stage 0–IVC) determined by combining clinical T, N, and M categories as defined by AJCC. Assigned by the physician based on clinical assessment prior to treatment.

- **Source Documents:**

- Physician staging summary (AJCC Staging Form or equivalent)
- Oncology consultation not
- Electronic medical record clinical staging section

- **Curation Instructions:**

1. Record the physician-documented clinical stage group (e.g., Stage IIIB) determined before treatment.
2. Use the most specific stage provided (e.g., Stage IA2 over Stage I).
3. Do not derive the stage group based on individual TNM values unless the physician has explicitly documented it.
4. Use:
 - Code 88 if AJCC staging is not applicable (e.g., hematologic malignancies).
 - Code 99 if staging is applicable but no documentation is present in the record.
5. If multiple stage groups are documented, prioritize:
 - The staging form
 - Most recent pre-treatment clinical documentation
6. Do not abstract pathologic or post-treatment stage here.

- **Curator Quality Review:**

- Confirm that stage group is clinical, not pathologic or restaged.
- Ensure consistency with recorded `dx_clin_t`, `dx_clin_n`, and `dx_clin_m` values.

- **Notes:**

- The clinical stage group is essential for prognosis, treatment planning, and staging-based analyses.
-

Variable: dx_path_t

Type: Categorical (Dropdown)

Description: Pathologic T Category (Tumor)

Reference: NAACCR #880

Data Source: Pathology report, operative report, physician staging summary

Curation Directives:

- **Definition:** The most definitive pathologic assessment of the primary tumor’s extent of invasion, based on surgical resection and histologic evaluation. This variable corresponds to the “T” component of the AJCC TNM pathologic staging.
- **Source Documents:**
 - Final pathology report from surgical resection
 - Operative note (if referenced in staging)
 - Oncology staging summary or clinic note
- **Curation Instructions:**
 1. Abstract the T category based on the most complete pathologic evaluation (post-resection), not imaging or clinical assessment alone.
 2. Use only the final diagnosis section of the pathology report for T category unless the clinician explicitly modifies it.
 3. Do not infer a higher T category than what is explicitly stated by the pathologist or oncologist.
 4. Use detailed subcategories (e.g., T1a1, T2a2) when available in the documentation.
 5. Use code 88 for cancers without an applicable AJCC staging schema or when no surgery was performed to assess tumor size/invasion.
 6. Use code 99 if no T category is documented in the medical record despite surgery.
 7. If both clinical and pathologic T categories are documented, only abstract the pathologic version here.
- **Curator Quality Review:**
 - Confirm that the selected T value aligns with AJCC guidelines for the diagnosis year and edition used.
 - Flag any cases where code 99 is selected despite presence of resection/pathology report.
 - Ensure subcategories are used when available; do not collapse to broader T value unless necessary.
- **Notes:**
 - The pathologic T category may affect treatment planning and survival analysis.

Variable: dx_path_n

Type: Categorical (Dropdown)

Description: Pathologic N Category (Regional Lymph Nodes)

Reference: NAACCR #890

Data Source: Pathology report, operative report, physician staging summary

Curation Directives:

- **Definition:** The extent of regional lymph node involvement based on pathologic examination, including resection and microscopic evaluation of lymph nodes. This corresponds to the “N” component of the AJCC TNM pathologic staging.
- **Source Documents:**
 - Final pathology report from surgical resection

- Operative note detailing lymph node dissection
- Oncology staging summary (must specify pathologic basis)
- **Curation Instructions:**
 1. Abstract the N category based on microscopic confirmation of lymph node involvement after surgical resection.
 2. Code NX if no regional lymph nodes were removed or evaluated pathologically.
 3. Use detailed subcategories when specified (e.g., N1mi, N2b).
 4. Code 88 if the AJCC staging system does not apply to the case.
 5. Code 99 if pathologic staging is applicable but not available in the medical record.
 6. Do not infer N stage from imaging, physical exam, or clinical notes alone.
 7. Do not assign pathologic N category based on biopsy of a single node unless it is documented as representative.
 8. Select the most definitive pathologic N category even if clinical N is also documented elsewhere.
- **Curator Quality Review:**
 - Confirm N value is consistent with pathologist’s node count and involvement.
 - Validate that N0 is not used if positive nodes are noted.
 - Flag code 99 if lymph node dissection was performed but no N category is abstracted.
- **Notes:**
 - Pathologic N category is required for AJCC staging when surgical node assessment is done.

Variable: dx_path_m

Type: Categorical (Dropdown)

Description: Pathologic M Category (Distant Metastasis)

Reference: NAACCR #900

Data Source: Pathology report, operative report, physician staging summary

Curation Directives:

- **Definition:** The pathologic assessment of the presence or absence of distant metastasis at the time of diagnosis, as defined by AJCC. This reflects direct microscopic confirmation of distant metastases during the course of surgical resection or tissue sampling.
- **Source Documents:**
 - Final pathology report from biopsy or resection of a metastatic site
 - Operative report (if includes distant site biopsy)
 - Physician’s pathologic staging summary
 - Tumor board documentation indicating pathologic M status
- **Curation Instructions:**
 1. Code MX if the presence of distant metastasis cannot be assessed pathologically.
 2. Code M0 if there is definitive microscopic evidence that no distant metastasis is present.
 3. Code M0(i+) if isolated tumor cells are found in distant sites (e.g., circulating blood, bone marrow) but without clinical/radiographic evidence of metastasis.
 4. Code M1 if pathologic confirmation of distant metastasis is present but no subcategory is provided.
 5. Use M1a, M1b, M1c, or M1d if the physician or pathology report assigns a specific AJCC-defined M1 substage.
 6. Use 88 for cancers where AJCC staging does not apply (e.g., hematologic malignancies).
 7. Use 99 if AJCC staging applies but there is no documentation of pathologic M status and distant metastasis was not evaluated microscopically.

8. Do not infer pathologic M category based on imaging alone or clinical documentation without tissue confirmation.
9. Pathologic M category should only be coded if metastasis is proven by microscopic examination.

- **Curator Quality Review:**

- Verify M1 coding is supported by pathology documentation of distant metastatic site.
- Flag cases with M0 if distant metastasis is mentioned elsewhere in clinical notes or imaging.
- Confirm code 88 is used only for cases truly exempt from AJCC staging.

- **Notes:**

- The pathologic M component is infrequently assigned and requires tissue confirmation from a distant site.

Variable: dx_path_group_stage

Type: Categorical (Dropdown)

Description: AJCC Pathologic Stage Group

Reference: NAACCR #910

Data Source: Pathology report, operative report, physician's staging note, tumor board summary

Curation Directives:

- **Definition:** The pathologic stage group assigned based on the tumor's extent, regional lymph node involvement, and presence of distant metastases, as assessed through surgical resection and microscopic examination. This staging reflects the most definitive AJCC classification derived after surgery.

- **Source Documents:**

- Final pathology report
- Operative report
- Physician's post-surgical staging summary
- AJCC-compliant staging worksheets (if included)

- **Curation Instructions:**

1. Assign the pathologic stage group using the AJCC edition documented in the medical record.
2. Use this field only if T, N, and M components have been assessed pathologically.
3. Use Stage 0/0A/0is for carcinoma in situ or preinvasive disease when confirmed pathologically.
4. Use Stage I–IV and subgroups (e.g., IA, IIIB, IVB) according to the AJCC manual version used by the clinician.
5. Code 88 if the cancer is not staged by AJCC (e.g., hematologic malignancies, certain CNS tumors).
6. Code 99 if staging is applicable but no documentation of pathologic stage group is available.
7. If both clinical and pathologic stage groups are present, ensure you abstract the *pathologic* value here.
8. Do not infer a pathologic stage group if documentation is absent or incomplete.

- **Curator Quality Review:**

- Cross-check pathologic T, N, and M components to confirm consistency with assigned stage group.
- Ensure the edition of the AJCC manual used is documented and consistent across staging fields.
- Confirm that pathologic stage group is not assigned in cases treated with neoadjuvant therapy unless explicitly documented.

- **Notes:**

- The AJCC pathologic stage group should be abstracted exactly as documented by the physician.
-

Variable: ajcc_edition

Type: Categorical (Dropdown)

Description: AJCC Edition Number

Reference: NAACCR #1060

Data Source: Physician staging documentation, pathology report, operative report, tumor board summary, AJCC-compliant staging templates

Curation Directives:

- **Definition:** Specifies the edition of the AJCC Cancer Staging Manual used by the provider to assign TNM components and stage group for the case. Applies only to manually assigned AJCC fields, not to derived staging.
- **Source Documents:**
 - Provider-stated AJCC edition in staging note or assessment
 - Pathology report or operative report indicating AJCC version used
 - Tumor board or multidisciplinary team summary
 - Abstractor-added metadata if edition confirmed from context (e.g., staging format matches a specific edition year)
- **Curation Instructions:**
 1. Abstract the edition number explicitly stated in the documentation used to assign clinical or pathologic stage.
 2. Code 06 for Sixth Edition, 07 for Seventh Edition, 08 for Eighth Edition.
 3. Code 00 if AJCC staging is applicable but was not performed.
 4. Code 88 if the cancer site or histology is not staged using AJCC (e.g., hematologic malignancies, CNS).
 5. Code 99 if AJCC staging was performed but the edition is not specified and cannot be inferred from context.
 6. Do not guess or backfill the edition based solely on diagnosis year unless clearly aligned with institutional policy and documented practice at the time.
 7. Use caution in cases staged at outside facilities or referred with partial documentation—only assign edition if source is reliable.
- **Curator Quality Review:**
 - Confirm consistency between edition and format of TNM/stage group codes documented.
 - Flag cases for clinical review if AJCC staging appears performed but edition is not stated or mismatched.
- **Notes:**
 - AJCC editions are periodically updated and can significantly change staging definitions—accurate edition coding is essential for interpretability.
 - This variable enables downstream analysis of stage-specific outcomes across staging eras.

Curated Sites of Metastasis at Diagnosis

For all fields: abstract based on clinical documentation available at or closest to the time of initial cancer diagnosis.

Curators are required to complete each of the following curated fields indicating whether specific metastatic sites were involved at diagnosis. These curated fields must be abstracted from source documentation such as imaging reports, pathology reports, oncology consults, or multidisciplinary tumor board notes. While

corresponding variables may be piped in above from tumor registry feeds, the curated values should reflect the most current, complete, and clinically verified evidence available in the patient’s medical record.

If there is no documentation of metastatic involvement at a given site, select “0 = None.” If the record explicitly confirms involvement, select “1 = Yes.” Use “99 = Unknown” only when documentation is absent or ambiguous. If a site is not applicable for a particular cancer type (e.g., “brain” for hematologic malignancies), use “88 = Not Applicable.”

Curated values should not be modified to match piped-in registry data; instead, curators should independently document findings based on source evidence. These variables support consistent staging, stratification, and disease burden classification for research and analysis.

Variable: dx_mets_bone

Type: Categorical (Dropdown)

Description: Mets at Diagnosis – Bone

Reference: NAACCR #1112

Data Source: Medical record (imaging reports, pathology reports, oncology consults, multidisciplinary team notes)

Curation Directives:

- **Definition:** Indicates whether distant bone metastases were present at the time of initial cancer diagnosis. This field is part of the curated set of “Mets at Diagnosis” indicators, which document involvement of specific metastatic sites.
- **Permitted Values / Codes:**
 - 0 = None; no bone metastases
 - 1 = Yes, distant bone metastases
 - 88 = Not applicable
 - 99 = Not available in patient record, or unknown whether bone is involved metastatic site
- **Source Documents:**
 - Radiology reports (e.g., bone scan, PET/CT, MRI, X-ray)
 - Pathology reports (e.g., bone biopsy)
 - Oncologist’s or radiologist’s diagnostic assessment
 - Staging summaries
- **Curation Instructions:**
 1. Code **1** if bone metastases are confirmed through imaging or biopsy at time of diagnosis.
 2. Code **0** if bone metastases are explicitly ruled out or not identified in imaging and clinical evaluation at diagnosis.
 3. Code **88** if the cancer is not one where bone metastasis is applicable (e.g., hematologic malignancies or early-stage localized cancers where staging or evaluation does not consider bone metastasis).
 4. Code **99** if documentation does not mention bone status, or if available information is inconclusive.
 5. Only include distant metastasis to bone; do not count direct extension or contiguous invasion from the primary tumor site.
 6. Use the status at initial diagnosis, not subsequent disease progression or recurrence data.
 7. If the record mentions “suspicious for bone metastases” without definitive confirmation and no further clarification is available, code as **99**.
- **Curator Quality Review:**
 - Validate consistency with other staging data (e.g., M1 vs M0) and overall AJCC stage group.

Variable: dx_mets_brain

Type: Categorical (Dropdown)

Description: Mets at Diagnosis – Brain

Reference: NAACCR #1113

Data Source: Medical record (imaging reports, pathology reports, oncology notes, multidisciplinary team summaries)

Curation Directives:

- **Definition:** Indicates whether distant brain metastases were present at the time of initial cancer diagnosis. This is part of the structured “Mets at Diagnosis” variable set, capturing disease spread to specific distant sites.
- **Permitted Values / Codes:**
 - 0 = None; no brain metastases
 - 1 = Yes, distant brain metastases
 - 88 = Not applicable
 - 99 = Not available in patient record, or unknown whether brain is involved metastatic site
- **Source Documents:**
 - Brain imaging reports (MRI, CT, PET/CT)
 - Neurologic consult notes
 - Pathology reports confirming brain lesion (if biopsied)
 - Oncologist or radiologist assessment
 - Staging summaries
- **Curation Instructions:**
 1. Code **1** if brain metastases are confirmed at diagnosis via imaging or histopathologic evaluation.
 2. Code **0** if brain metastases are ruled out or not identified during initial staging workup.
 3. Code **88** for primary cancers where brain metastasis assessment is not applicable or not clinically indicated based on disease type and stage.
 4. Code **99** if brain status is not documented or cannot be determined from available records.
 5. Only capture distant metastases to the brain; exclude contiguous invasion from adjacent structures.
 6. Record based on initial diagnosis timeframe; do not update based on later disease progression or follow-up findings.
 7. If radiology notes indicate “suspicious” or “indeterminate” brain findings without resolution in the diagnostic window, default to **99** unless clarified elsewhere.
- **Curator Quality Review:**
 - Confirm that imaging studies were performed near the time of diagnosis.
 - Ensure proper differentiation between primary CNS tumors and brain metastases from non-CNS primaries.
 - Cross-check with AJCC M stage and clinical group stage fields for consistency

Variable: dx_mets_liver

Type: Categorical (Dropdown)

Description: Mets at Diagnosis – Liver

Reference: NAACCR #1115

Data Source: Medical record (imaging reports, pathology reports, oncology notes, multidisciplinary team summaries)

Curation Directives:

- **Definition:** Indicates whether distant liver metastases were present at the time of initial cancer diagnosis. This is one of the site-specific variables used to document metastatic disease at diagnosis.
- **Permitted Values / Codes:**
 - 0 = None; no liver metastases
 - 1 = Yes, distant liver metastases
 - 88 = Not applicable
 - 99 = Not available in patient record, or unknown whether liver is involved metastatic site
- **Source Documents:**
 - Liver imaging reports (CT, MRI, PET/CT)
 - Abdominal ultrasound
 - Pathology reports (e.g., biopsy of liver lesion)
 - Operative reports or liver resection documentation
 - Oncologist or radiologist staging assessment
 - Multidisciplinary tumor board notes
- **Curation Instructions:**
 1. Code **1** if liver metastases are explicitly documented or confirmed by imaging or pathology at the time of initial diagnosis.
 2. Code **0** if liver metastases are ruled out or not identified in the initial staging workup.
 3. Code **88** when liver metastasis assessment is not applicable for the cancer type (e.g., in situ cancers, hematologic malignancies).
 4. Code **99** if no liver assessment is documented or if metastatic status cannot be confirmed from available records.
 5. Do not capture local liver extension from contiguous structures (e.g., gallbladder or colon invasion) unless explicitly described as distant metastatic disease.
 6. Record findings from the initial diagnostic and staging period only, prior to initiation of definitive treatment.
 7. If imaging describes “suspicious,” “possible,” or “indeterminate” liver lesions without conclusive resolution, default to **99** unless clarified elsewhere.
- **Curator Quality Review:**
 - Confirm that liver imaging occurred during initial staging and not solely in follow-up.
 - Check for consistency between this field and AJCC M stage or clinical/pathologic group stage.
 - Ensure correct differentiation between hepatic metastases and benign lesions (e.g., cysts, hemangiomas).

Variable: dx_mets_lung

Type: Categorical (Dropdown)

Description: Mets at Diagnosis – Lung

Reference: NAACCR #1116

Data Source: Medical record (imaging reports, pathology reports, clinical staging notes)

Curation Directives:

- **Definition:** Indicates whether distant lung metastases were present at the time of initial cancer diagnosis. This variable captures site-specific metastatic involvement to support cancer staging and population-level analysis.
- **Permitted Values / Codes:**
 - 0 = None; no lung metastases
 - 1 = Yes, distant lung metastases

- 88 = Not applicable
 - 99 = Not available in patient record, or unknown whether lung is involved metastatic site
 - **Source Documents:**
 - Chest imaging reports (CT, PET/CT, chest X-ray)
 - Pathology reports from lung biopsies
 - Operative reports or thoracic surgical consults
 - Oncologist’s staging summary
 - **Curation Instructions:**
 1. Code **1** if lung metastases are confirmed by imaging, biopsy, or documented by the treating physician at diagnosis.
 2. Code **0** if the lung is evaluated during staging and no metastases are identified.
 3. Code **88** if the primary cancer type is not staged for metastases (e.g., in situ cancers, hematologic malignancies).
 4. Code **99** if the presence or absence of lung metastases cannot be determined from the medical record.
 5. Do not include direct extension from a nearby tumor (e.g., esophageal tumor invading bronchus) as metastatic unless it is explicitly described as distant spread.
 6. Record only findings available at the time of diagnosis and prior to the initiation of treatment.
 7. If imaging reports describe indeterminate or suspicious nodules without resolution or biopsy, default to **99** unless clarified elsewhere.
 - **Curator Quality Review:**
 - Confirm consistency with AJCC M stage and overall staging fields.
-

Variable: dx_mets_lymph

Type: Categorical (Dropdown)

Description: Mets at Diagnosis – Distant Lymph Nodes

Reference: NAACCR #1114

Data Source: Medical record (imaging reports, operative notes, pathology reports, clinical staging documentation)

Curation Directives:

- **Definition:** Indicates whether distant lymph node(s) were involved by metastasis at the time of initial cancer diagnosis. This refers to nodal metastases that are not regional, based on AJCC guidelines for the primary tumor site.
- **Permitted Values / Codes:**
 - 0 = None; no distant lymph node metastases
 - 1 = Yes; distant lymph node metastases
 - 88 = Not applicable
 - 99 = Unknown whether distant lymph node(s) are an involved metastatic site. Not documented in patient record.
- **Source Documents:**
 - Imaging reports (e.g., PET/CT, CT chest/abdomen/pelvis)
 - Pathology reports (e.g., fine-needle aspiration, lymph node dissection)
 - Operative reports or surgical oncology consults
 - Staging workup in oncology notes
- **Curation Instructions:**

1. Code **1** if distant nodal involvement is confirmed by pathology, or clearly described in imaging or clinical documentation.
 2. Code **0** if distant nodal sites are evaluated during staging and found negative.
 3. Use AJCC site-specific guidelines to differentiate between regional and distant nodes. For example:
 - Supraclavicular nodes are considered distant for some abdominal or pelvic primaries.
 - Para-aortic nodes may be regional or distant depending on the tumor site.
 4. Code **88** if staging is not applicable to the cancer type (e.g., hematologic malignancies or in situ cancers).
 5. Code **99** if documentation does not specify whether distant lymph nodes were assessed or involved.
 6. If findings are ambiguous (e.g., “suspicious adenopathy” without further clarification), and there is no pathologic confirmation or staging summary, default to **99**.
- **Curator Quality Review:**
 - Confirm correct application of regional vs. distant nodal classification using AJCC guidelines.
 - Cross-check for consistency with M stage and other Mets at Dx fields.
 - **Notes:**
 - Distant lymph node metastasis often indicates **M1 disease** and influences both prognosis and treatment.
 - This variable supports tumor staging and metastatic pattern analysis in population-level datasets.

Variable: dx_mets_other

Type: Categorical (Dropdown)

Description: Mets at Diagnosis – Other

Reference: NAACCR #1117

Data Source: Medical record (imaging reports, pathology reports, clinical staging notes, operative reports)

Curation Directives:

- **Definition:** Indicates whether the patient had distant (discontinuous) metastases to sites other than bone, brain, liver, lung, or distant lymph nodes at the time of initial cancer diagnosis. This field captures all other metastatic sites not recorded in the other Mets at Dx fields and includes uncommon or generalized metastatic presentations.
- **Permitted Values / Codes:**
 - 0 = Patient has no distant metastasis at all; or no additional metastases beyond bone, brain, liver, lung, or distant lymph nodes
 - 1 = Patient has distant metastases in any site(s) other than bone, brain, liver, lung, or distant lymph nodes
 - 88 = Not applicable
 - 99 = Cannot be determined from the medical record
- **Source Documents:**
 - Imaging reports (CT, PET/CT, MRI, bone scan, ultrasound)
 - Pathology reports (biopsies of non-standard metastatic sites)
 - Clinical documentation
 - Operative or procedural reports (e.g., omental biopsy, pleural tap)
 - Staging summaries
- **Curation Instructions:**
 1. Code **1** if distant metastatic disease is confirmed in any site not already captured by Mets at Dx
 - Bone, Brain, Liver, Lung, or Distant Lymph Nodes.

2. Code **0** if there are no distant metastases at all, or if metastases are only present in bone, brain, liver, lung, or distant lymph nodes (already captured by other fields).
3. Code **88** if metastatic staging is not applicable (e.g., hematologic malignancies, in situ tumors).
4. Code **99** if the presence or absence of other distant metastases cannot be determined due to insufficient documentation.
5. Do not code based on suspicion alone (e.g., “rule out adrenal metastasis”) unless confirmed by imaging, pathology, or clinical staging documentation.

- **Curator Quality Review:**

- Review imaging reports thoroughly for less common sites of spread (e.g., adrenal, skin, peritoneal seeding).
- Confirm whether the metastatic site is distinct from primary or regional spread per AJCC site-specific definitions.
- Cross-check with the M stage and ensure no duplication across other Mets at Dx fields.
- Flag cases with vague documentation (e.g., “extensive metastases”) for team review if specific sites are not mentioned.

- **Notes:**

- This variable helps capture non-traditional and rarer sites of metastasis, which may impact treatment plans and prognostication.
- Always prioritize confirmed findings at diagnosis and avoid post-treatment progression data.

Curated Site Specific Data Items

For all fields: complete based on clinical documentation and testing available at or closest to the time of diagnosis and/or definitive treatment.

This section captures tumor-specific variables that provide additional clinical, molecular, and prognostic information beyond core staging elements. These site-specific data items (SSDIs) are curated in alignment with the NAACCR SSDI Manual v3.2 and are tailored to reflect nuances in disease classification, biomarker status, genomic testing, and site-specific staging systems (e.g., FIGO for ovarian cancer).

Abstract each field based on source documentation such as pathology reports, molecular testing results, imaging interpretations, and operative or oncology consult notes. Do not infer or assume values. If the information is not documented, select the appropriate code (e.g., “99 = Not documented in medical record”). These curated SSDIs enable more granular clinical characterization, support cohort stratification, and enhance data quality for downstream analysis.

Variable: dx_ca_type

Type: Categorical (Dropdown)

Description: Primary Cancer Type

Reference: NAACCR; Site-Specific Data Item Manual v3.2

Data Source: Medical record (pathology reports, diagnostic imaging, oncology consults, surgical reports)

Curation Directives:

- **Definition:** The type of cancer most relevant to the diagnosis. This field captures the *primary cancer type* and should align with the anatomical site, histology, and clinical context of the primary diagnosis.
- **Permitted Values / Codes:**

- 1 = Lung
- 2 = Prostate
- 3 = Stomach
- 4 = Colorectal
- 5 = Liver
- 6 = Melanoma
- 7 = Bladder
- 8 = Kidney
- 9 = Breast
- 10 = Ovarian
- 11 = Pancreas
- 12 = Brain / CNS
- 13 = Chordoma
- 88 = Other

- **Source Documents:**

- Pathology reports (final diagnosis, synoptic templates)
- Imaging reports (radiologist’s impression, cancer staging)
- Oncology or surgical consult notes
- Operative reports and procedure notes

- **Curation Instructions:**

1. Review all diagnostic documentation and identify the primary site of cancer.
2. Select the cancer type that corresponds with the confirmed primary diagnosis.
 - Example: A diagnosis of “invasive ductal carcinoma of the left breast” should be coded as 9 = **Breast**.
3. If the primary diagnosis corresponds to a cancer type not listed (e.g., cervical, esophageal, hematologic), select **88 = Other**.
4. Do not use metastatic site or secondary involvement to determine the cancer type—this variable should reflect the *primary tumor*.
5. If multiple primary cancers are present, select the one most relevant to the form being abstracted or the surgical context, if applicable.

- **Curator Quality Review:**

- Confirm that the selected cancer type is consistent with ICD-O-3 site and histology codes documented elsewhere in the chart.
- Cross-reference with pathology and staging data to ensure accuracy.

- **Notes:**

- This variable may be used to trigger site-specific validation logic or downstream SSDI fields.
- When in doubt, prioritize pathology-confirmed diagnosis over clinical impressions.

Variable: breast_ot_inv_dcis

Type: Categorical (Dropdown)

Description: Indicates whether the breast cancer diagnosis represents invasive carcinoma or ductal carcinoma in situ (DCIS)

Reference: n/a

Data Source: Pathology report and/or diagnostic documentation

Curation Directives:

- **Definition:** Captures whether the breast cancer diagnosis refers to invasive carcinoma or ductal carcinoma in situ (DCIS). This classification is required to distinguish in situ vs. invasive histologies in breast cancer cases for clinical, analytical, and eligibility purposes.
- **Permitted Values / Codes:**
 - 1 = Invasive
 - 2 = DCIS
- **Source Documents:**
 - Final pathology report
- **Curation Instructions:**
 1. Review the pathology report to determine whether the diagnosis is invasive breast carcinoma or ductal carcinoma in situ (DCIS).
 2. If both invasive and in situ components are present, select Invasive (1), as this represents the dominant clinical behavior.
 3. If only DCIS is documented with no invasive component, select DCIS (2).
 4. Do not select based solely on imaging or clinical suspicion, use confirmed pathology documentation whenever possible.
 5. This field is only displayed if [dx_ca_type] = 9, indicating a breast cancer case. Do not complete for other primary cancer types.
- **Curator Quality Review:**
 - Ensure that the selection aligns with the terminology in the pathology report (e.g., “invasive ductal carcinoma,” “ductal carcinoma in situ”).
 - Double-check that DCIS is not selected if any invasive component is mentioned, even if minimal.
 - Cross-validate with TNM staging or AJCC stage, if available (e.g., Stage 0 typically indicates DCIS).

Variable: breast_ot_inv_score

Type: Categorical (Dropdown)

Description: Breast - Oncotype Dx Recurrence Score-Invasive

Reference: NAACCR #3904

Data Source: Pathology report, genomic testing report, laboratory information system

Curation Directives:

- **Definition:** The Oncotype Dx Recurrence Score-Invasive is a genomic assay result derived from testing 21 genes in breast tumor tissue. It predicts the likelihood of distant recurrence in early-stage, estrogen receptor-positive invasive breast cancer and helps guide treatment decisions regarding adjuvant chemotherapy.
- **Permitted Values / Codes:**
 - 1 = 100: Enter actual recurrence score between 0 and 100
 - 2 = XX4: Stated as less than 11
 - 3 = XX5: Stated as equal to or greater than 11
 - 4 = XX6: Not applicable: in situ case
 - 5 = XX7: Test ordered, results not in chart
 - 6 = XX9: Not documented in medical record
- **Source Documents:**
 - Genomic test report (e.g., Genomic Health Oncotype Dx)
 - Final pathology report

- Oncology consult or treatment planning notes
- Laboratory information system reports

- **Curation Instructions:**

1. This field should only be completed for breast cancer cases: ensure `dx_ca_type` = 9.
2. Review the genomic test report for the Oncotype Dx recurrence score.
3. If an exact numeric score (0–100) is provided, select code 1 and record the score in the paired detail field (`breast_ot_inv_score_details`).
4. If the score is only stated as a risk category (e.g., “low risk, <11”), select code 2 = `XX4` or 3 = `XX5` as applicable.
5. Use code 4 = `XX6` if the case is *in situ* (e.g., DCIS) and not eligible for the test.
6. Use code 5 = `XX7` if the test was ordered but no results are present in the chart.
7. Use code 6 = `XX9` if there is no documentation of the test or its results anywhere in the record.
8. Do not infer or estimate the score from treatment decisions alone; score must be documented.

- **Curator Quality Review:**

- Verify that Oncotype Dx is specific to the invasive component of breast cancer.
- Confirm that the numeric value (if applicable) is transcribed exactly as shown in the lab report.
- Ensure *in situ* cases (e.g., pure DCIS) are assigned code 4.
- If multiple recurrence scores are documented, abstract the most recent or final score used for treatment planning.

- **Notes:**

- This variable informs decisions about adjuvant chemotherapy and is often used in treatment algorithms.

Variable: `breast_ot_inv_score_details`

Type: Numeric

Description: Breast – Oncotype Dx Recurrence Score-Invasive – details

Reference: NAACCR #3904

Data Source: Genomic test report, laboratory results, pathology report

Curation Directives:

- **Definition:** This variable captures the exact Oncotype Dx Recurrence Score (range: 0–100) for invasive breast cancer, as reported by the genomic assay. It is only applicable when the test result is documented as a specific numeric value.
- **Permitted Values / Codes:**
Numeric input only (1–100)
- **Source Documents:**
 - Oncotype Dx genomic test report (e.g., from Genomic Health or electronic lab system)
 - Final pathology report
 - Oncology consult note
 - Treatment planning documents
 - Tumor board summaries
- **Curation Instructions:**
 1. Complete this field only if the coded field `breast_ot_inv_score` = 1 (“Enter actual recurrence score between 0 and 100”) was selected.
 2. Locate the recurrence score in the Oncotype Dx report; it is typically reported as a numeric value under “Recurrence Score.”

3. Enter the exact numeric value (e.g., 23) into this field. Do not round or convert categories (e.g., “low risk,” “high risk”).
4. Do not complete this field if the score is only described in qualitative terms (e.g., “<11” or “high risk”) without a precise number.
5. If multiple scores are listed (e.g., from repeat testing), enter the most recent or final result used for treatment decision-making.

- **Curator Quality Review:**

- Confirm that the score falls within the valid range (1–100).
- Cross-check against the coded variable (`breast_ot_inv_score = 1`).
- Ensure source is an official lab or pathology report, not inferred from clinical notes alone.
- Confirm that score is for invasive breast cancer, not DCIS or in situ disease.

- **Notes:**

- This score may influence adjuvant treatment plans and should be entered with precision.

Variable: `breast_ot_inv_risk`

Type: Categorical (Dropdown)

Description: Breast – Oncotype Dx Risk Level-Invasive

Reference: NAACCR #3906

Data Source: Genomic test report, pathology report, oncology consult note

Curation Directives:

- **Definition:** Categorical risk classification derived from the Oncotype Dx Recurrence Score-Invasive, indicating the likelihood of distant recurrence in invasive breast cancer patients. Based on pre-defined score thresholds, cases are assigned a low, intermediate, or high risk level.

- **Permitted Values / Codes:**

- 0 = Low risk (recurrence score 0–17)
- 1 = Intermediate risk (recurrence score 18–30)
- 2 = High risk (recurrence score 31)
- 6 = Not applicable: DCIS case
- 7 = Test ordered, results not in chart
- 8 = Not applicable: Information not collected for this case
- 9 = Not documented in medical record; Oncotype Dx Risk Level-Invasive not assessed or unknown if assessed

- **Source Documents:**

- Oncotype Dx genomic test report (e.g., from Genomic Health)
- Pathology report
- Oncology consult notes
- EMR lab results or genomics tab

- **Curation Instructions:**

1. Complete this field only for breast cancer cases (`dx_ca_type = 9`).
2. Determine whether an Oncotype Dx test was performed and locate the recurrence score.
3. If the exact recurrence score is available, assign the risk level using the following cut points:
 - 0–17 = Low (code 0)
 - 18–30 = Intermediate (code 1)
 - 31 = High (code 2)
4. If the case is for DCIS only, and not invasive breast cancer, select code 6.

5. If the test was ordered but results are not available in the chart, select code 7.
6. If the site does not collect this information, select code 8.
7. If the test is not documented, not assessed, or unknown whether it was assessed, select code 9.

- **Curator Quality Review:**

- Cross-validate this field with the numeric recurrence score (`breast_ot_inv_score_details`) if present.
- Ensure that DCIS-only cases are not assigned a risk level unless specifically reported.
- Confirm that the correct range-based category is used and not inferred.
- Use only documented score categories from official genomic test reports.

- **Notes:**

- Do not abstract based on risk language alone (e.g., “low risk”) unless the corresponding numeric score or category is explicitly linked to the Oncotype Dx test.

Variable: `breast_ot_dcis_score`

Type: Categorical (Dropdown)

Description: Breast – Oncotype Dx Recurrence Score-DCIS

Reference: NAACCR #3903

Data Source: Genomic test report, pathology report, oncology consult note

Curation Directives:

- **Definition:** Oncotype Dx Recurrence Score-DCIS is a numeric result from a genomic assay designed to estimate the risk of local recurrence in patients with ductal carcinoma in situ (DCIS). It is based on gene expression profiling of tumor tissue.

- **Permitted Values / Codes:**

- 1 = 000–100: Enter actual recurrence score between 0 and 100
- 2 = XX6: Not applicable – invasive case
- 3 = XX7: Test ordered, results not in chart
- 4 = XX8: Not applicable – information not collected for this case
- 5 = XX9: Not documented in medical record

- **Source Documents:**

- Oncotype Dx DCIS Score report (e.g., Genomic Health)
- Pathology report
- Oncology consult notes
- Tumor board summaries
- EMR lab or genomics section

- **Curation Instructions:**

1. Complete this field only for breast cancer cases (`dx_ca_type` = 9).
2. Confirm that the case is DCIS only, not invasive cancer.
3. If the test was performed and the numeric score is reported, select code 1 and enter the actual score (use companion field if available).
4. If the case is invasive only and DCIS scoring does not apply, use code 2.
5. If the test was ordered but no results are found in the chart, use code 3.
6. If the institution does not collect this information, use code 4.
7. If there is no documentation of test performance or results, use code 5.

- **Curator Quality Review:**

- Confirm whether the test performed was specifically the DCIS version of the Oncotype Dx assay.

- Do not confuse DCIS scores with invasive Oncotype Dx scores.
- Ensure recurrence scores are not inferred from risk language alone; abstract only when documented.
- **Notes:**
 - This field supports clinical and research efforts related to treatment decision-making for non-invasive breast cancer.
 - DCIS-specific Oncotype Dx scoring is used less frequently than the invasive version and may not be ordered routinely.

Variable: `breast_ot_dcis_score_details`

Type: Numeric

Description: Breast – Oncotype Dx Recurrence Score-DCIS – details

Reference: NAACCR #3903

Data Source: Genomic test report

Curation Directives:

- **Definition:** This field captures the exact numeric value of the Oncotype Dx Recurrence Score-DCIS, ranging from 0 to 100, when reported. It provides precision beyond the categorical coding in the primary `breast_ot_dcis_score` field.
- **Permitted Values / Codes:**
Integer values from 1 to 100 only
&#xNAN;(*Only required if `breast_ot_dcis_score` = 1*)
- **Source Documents:**
 - Oncotype Dx DCIS Score report (e.g., Genomic Health)
 - Pathology report addenda or molecular diagnostics
 - Oncology progress notes
 - EMR lab/genomics module
- **Curation Instructions:**
 1. This field should only be completed if `breast_ot_dcis_score` is coded as 1 (“000–100: Enter actual recurrence score”).
 2. Enter the exact integer recurrence score documented in the report.
 3. Do not round or infer from qualitative descriptors (e.g., “low risk”)—abstract only if the numeric value is clearly documented.
 4. If score is given as a range or includes decimals, abstract the nearest whole number **only if** explicitly stated or a single discrete score is provided.
 5. Leave blank if score is not present in source documentation, or if any code other than 1 is selected in `breast_ot_dcis_score`.
- **Curator Quality Review:**
 - Cross-check that `breast_ot_dcis_score` is correctly coded as 1 before abstracting details.
 - Ensure the score pertains specifically to **DCIS** and not to the invasive Oncotype Dx assay.
 - Confirm accuracy by reviewing all pages of genomic reports, as scores may be buried in graphs or appendices

Variable: `breast_ot_dcis_risk`

Type: Categorical (Dropdown)

Description: Breast – Oncotype Dx Risk Level-DCIS

Reference: NAACCR #3905

Data Source: Genomic test report

Curation Directives:

- **Definition:** This field captures the risk category (low, intermediate, high) assigned to a patient based on the Oncotype Dx Recurrence Score-DCIS, a genomic assay predicting risk of **local** recurrence in patients with ductal carcinoma in situ (DCIS). Risk categories are determined by defined score thresholds.
- **Permitted Values / Codes:**
 - 0 = Low risk (recurrence score 0–38)
 - 1 = Intermediate risk (recurrence score 39–54)
 - 2 = High risk (recurrence score 55)
 - 6 = Not applicable: invasive case
 - 7 = Test ordered, results not in chart
 - 8 = Not applicable: Information not collected for this case
 - 9 = Not documented in medical record; Oncotype Dx Risk Level-DCIS not assessed or unknown if assessed
- **Source Documents:**
 - Oncotype Dx DCIS Score report (e.g., Genomic Health)
 - Pathology report addenda
 - Genomics section of the EMR
 - Oncology progress or treatment planning notes
- **Curation Instructions:**
 1. Abstract the coded risk level based on the stated interpretation from the Oncotype Dx DCIS Score report.
 2. If only a numeric recurrence score is provided without explicit risk category, use the defined thresholds to assign the appropriate code.
 - Score 0–38 = code 0
 - Score 39–54 = code 1
 - Score 55 = code 2
 3. If the report refers to the risk level using language such as “low risk of local recurrence” or similar, map to the appropriate code.
 4. Do not use this field if the case is invasive breast cancer only (code as 6) or if the score pertains to the invasive Oncotype Dx assay (use invasive risk field instead).
 5. Use codes 7–9 only when documentation is explicitly incomplete or missing, per definitions above.
- **Curator Quality Review:**
 - Ensure consistency with `breast_ot_dcis_score` value and whether case is truly DCIS (in situ).
 - Confirm the risk level pertains specifically to the DCIS assay, not the invasive version.
 - Validate source document includes either the risk category or numeric score to derive the level.
- **Notes:**
 - Use caution when score and risk category are discordant; defer to explicit report labeling if provided.

Variable: `breast_multigene_method`

Type: Categorical (Dropdown)

Description: Breast – Multigene Signature Method

Reference: NAACCR #3894

Data Source: Genomic test report, pathology report, EMR (molecular diagnostics section)

Curation Directives:

- **Definition:** This field captures the specific multigene assay (excluding Oncotype Dx) used to assess recurrence risk or guide treatment decisions in breast cancer. This includes commercially available tests like Mammaprint, PAM50, Breast Cancer Index, and EndoPredict.
- **Permitted Values / Codes:**
 - 1 = Mammaprint
 - 2 = PAM50 (Prosigna)
 - 3 = Breast Cancer Index
 - 4 = EndoPredict
 - 5 = Test performed, type of test unknown
 - 6 = Multiple tests, any tests in codes 1–4
 - 7 = Test ordered, results not in chart
 - 8 = Not applicable: Information not collected for this case
 - 9 = Not documented in medical record
- **Source Documents:**
 - Multigene assay reports (e.g., Agendia, Biotheranostics, NanoString)
 - Pathology report or addendum
 - Oncology progress notes
 - Molecular diagnostics or genomics section of the EMR
- **Curation Instructions:**
 1. Review the medical record for documentation of multigene assays used in breast cancer staging or treatment planning, excluding Oncotype Dx (coded separately).
 2. Select the code that corresponds to the named test if explicitly stated in the record.
 - For Mammaprint, code as 1.
 - For PAM50 (Prosigna), code as 2.
 - For Breast Cancer Index, code as 3.
 - For EndoPredict, code as 4.
 3. If multiple assays from codes 1–4 were used, code as 6.
 4. If a test is known to have been performed but the type is not specified, use code 5.
 5. Use code 7 if the test was ordered but the results are not available in the chart.
 6. Use codes 8 or 9 only if test use is not applicable or not documented, respectively.
- **Curator Quality Review:**
 - Ensure that multigene test type is not confused with Oncotype Dx (abstracted in a separate variable).
 - Verify consistency with other breast genomic testing fields.
 - Confirm that the test was performed on a primary breast tumor, not a metastatic site.
- **Notes:**
 - These tests are typically ordered for hormone receptor-positive, HER2-negative, early-stage breast cancers to inform decisions on adjuvant therapy.
 - Reports often include recurrence scores, molecular subtype classification, and treatment recommendations; these may assist in verifying the test type.

Variable: breast_multigene_results

Type: Categorical (Dropdown)

Description: Breast – Multigene Signature Results

Reference: NAACCR #3895

Data Source: Genomic test report, pathology report, EMR (molecular diagnostics or lab results section)

Curation Directives:

- **Definition:** This field captures the **result** of a multigene signature assay (excluding Oncotype Dx) performed on breast cancer tissue. These results may be reported numerically or as risk categories, depending on the assay used. This does not include Oncotype Dx results, which are abstracted separately.
- **Permitted Values / Codes:**
 - 1 = 00–99: Enter numeric recurrence score (if available)
 - 2 = X1: Score is 100
 - 3 = X2: Low risk
 - 4 = X3: Moderate (intermediate) risk
 - 5 = X4: High risk
 - 6 = X7: Test ordered, results not in chart
 - 7 = X8: Not applicable: Information not collected for this case
 - 8 = X9: Not documented in medical record; Multigene Signature Results not assessed or unknown if assessed
- **Source Documents:**
 - Multigene assay reports (e.g., MammaPrint, PAM50/Prosigna, EndoPredict, Breast Cancer Index)
 - Pathology report or addendum
 - Oncology progress notes
 - Molecular diagnostics section of the EMR
- **Curation Instructions:**
 1. Confirm the cancer type is breast (`dx_ca_type = 9`).
 2. Identify the result of the multigene test (excluding Oncotype Dx) used. Do not confuse this with the test name (`breast_multigene_method`).
 3. If a numeric score is provided (0–99), code as 1 and enter value in `breast_multigene_details`.
 4. If a score of 100 is documented, code as 2.
 5. If the result is described using risk categories (e.g., “low risk,” “intermediate,” or “high risk”) without a numeric score, code as:
 - 3 = Low risk
 - 4 = Moderate/intermediate risk
 - 5 = High risk
 6. If test was ordered but no results are in the chart, use code 6.
 7. Use codes 7 or 8 if the test is not applicable or not documented in the medical record, respectively.
 8. Do not enter results from Oncotype Dx—these are recorded in the Oncotype Dx fields only.
 9. Cross-reference this field with `breast_multigene_method` to ensure the result corresponds to a valid test type.
- **Curator Quality Review:**
 - Confirm that the result reflects the correct non-Oncotype test.
 - Ensure numeric vs categorical results are appropriately coded.
 - If both score and risk category are reported, prioritize numeric score and enter category in notes if needed.
- **Notes:**
 - For MammaPrint, results are typically “High Risk” or “Low Risk”.
 - For PAM50 (Prosigna), results may include subtype classifications (e.g., Luminal A) but risk levels should be abstracted here.

- For Breast Cancer Index and EndoPredict, both scores and risk groupings may be reported—follow instructions above to assign proper code.
-

Variable: `breast_multigene_details`

Type: Numeric

Description: Breast – Multigene Signature Results – details

Reference: NAACCR #3895 (linked to result details)

Data Source: Genomic test report, pathology report, EMR (molecular diagnostics or lab results section)

Curation Directives:

- **Definition:** This field captures the **exact numeric recurrence score** reported from a multigene signature test (excluding Oncotype Dx) if available. It provides granularity beyond the dropdown classification captured in `breast_multigene_results`.
 - **Permitted Values / Codes:**
Acceptable values: 00–99
(Must be used **only** if `breast_multigene_results` = 1, indicating a numeric recurrence score was abstracted.)
 - **Source Documents:**
 - Multigene test reports (e.g., MammaPrint, PAM50/Prosigna, Breast Cancer Index, EndoPredict)
 - Pathology reports with attached genomic result appendices
 - Molecular diagnostics reports or EMR lab results with detailed genomic assay readouts
 - **Curation Instructions:**
 1. Confirm that the cancer type is breast (`dx_ca_type` = 9).
 2. Only abstract this field if `breast_multigene_results` was coded as 1 (00–99: Recurrence Score).
 3. Enter the exact numeric value as reported in the test documentation.
 - If a range is reported, do **not** enter a value; instead consult with clinical lead.
 - If the score is rounded, enter as documented.
 4. Do not extrapolate or infer a value based on risk category descriptions (e.g., “low risk” without a score).
 5. Do not use this field for Oncotype Dx scores; use `breast_ot_inv_score_details` or `breast_ot_dcis_score_details` as appropriate.
 - **Curator Quality Review:**
 - Ensure score is entered only if `breast_multigene_results` = 1.
 - Confirm that score falls within valid range (00–99).
 - Verify that score corresponds to the test listed in `breast_multigene_method`.
 - **Notes:**
 - Always prioritize direct transcription from original source—do not derive from interpretation or free-text impressions.
-

Variable: `ovarian_figo`

Type: Categorical (Dropdown)

Description: FIGO Stage for Ovarian Cancer

Reference: NAACCR #3836

Data Source: Pathology report, operative report, staging summary, clinical notes

Curation Directives:

- **Definition:** The FIGO (International Federation of Gynecology and Obstetrics) Stage classifies the extent of ovarian cancer based on clinical and pathological evaluation. FIGO staging is used worldwide for female reproductive cancers and reflects tumor size, spread to lymph nodes, and distant metastases. Do not confuse with FIGO grade.
- **Permitted Values / Codes:**
 - 1 = FIGO Stage I
 - 2 = FIGO Stage IA
 - 3 = FIGO Stage IB
 - 4 = FIGO Stage IC
 - 5 = FIGO Stage IC1
 - 6 = FIGO Stage IC2
 - 7 = FIGO Stage IC3
 - 8 = FIGO Stage II
 - 9 = FIGO Stage IIA
 - 10 = FIGO Stage IIB
 - 11 = FIGO Stage III
 - 12 = FIGO Stage IIIA
 - 13 = FIGO Stage IIIA1
 - 14 = FIGO Stage IIIA1i
 - 15 = FIGO Stage IIIA1ii
 - 16 = FIGO Stage IIIA2
 - 17 = FIGO Stage IIIB
 - 18 = FIGO Stage IIIC
 - 19 = FIGO Stage IV
 - 20 = FIGO Stage IVA
 - 21 = FIGO Stage IVB
 - 97 = Not applicable: Carcinoma in situ (intraepithelial, noninvasive, preinvasive)
 - 98 = Not applicable: Information not collected for this case
 - 99 = Not documented in medical record; FIGO stage unknown, not assessed, or unknown if assessed
- **Source Documents:**
 - Pathology report (final diagnosis or staging section)
 - Operative report (staging laparotomy details)
 - Oncology staging summary (manual or EHR-integrated)
 - Clinician notes if explicitly state FIGO stage
- **Curation Instructions:**
 1. Confirm the cancer type is ovarian (`dx_ca_type = 10`).
 2. Abstract the most definitive FIGO stage available (prefer pathology-confirmed if present).
 3. Use Roman numeral-to-code mapping provided in the dropdown.
 4. Substage (e.g., IC1, IIIA1ii) should be captured precisely if documented.
 5. If only the main stage (e.g., “Stage III”) is documented without substage, use the broader category code (e.g., 11 = Stage III).
 6. Do not infer stage based on TNM or spread alone. Code only if FIGO stage is explicitly stated.
 7. If staging refers to pre-invasive disease only (e.g., intraepithelial carcinoma), use code 97.
 8. If staging was not performed or information is missing, use codes 98 or 99 as appropriate.
- **Curator Quality Review:**
 - Ensure the code selected matches the Roman numeral stage and substage terminology.
 - Confirm documentation is specific to FIGO (not TNM or AJCC stage unless it explicitly aligns with FIGO).
 - Validate that code 97 is only used for noninvasive/preinvasive cases.
- **Notes:**

- FIGO staging is distinct from grade and TNM staging. Always ensure the correct system is applied.
- If both clinical and pathological FIGO stages are available, abstract the pathological stage.

Variable: `ovarian_residual_vol`

Type: Categorical (Dropdown)

Description: Residual Tumor Volume Post Cytoreduction for Ovarian Cancer

Reference: NAACCR #3921

Data Source: Operative report, pathology report, oncology consult notes, postoperative summary

Curation Directives:

- **Definition:** Captures the size of residual tumor left behind following primary cytoreductive surgery for ovarian cancer. Residual disease after debulking surgery is a key prognostic indicator in advanced ovarian cancer. This field is specific to gross macroscopic residual disease and should not be inferred from imaging alone.
- **Permitted Values / Codes:**
 - 00 = No gross residual tumor nodules
 - 50 = Residual tumor nodule(s) 1 centimeter (cm) or less
 - 60 = Residual tumor nodule(s) greater than 1 cm
 - 70 = Macroscopic residual tumor, size not stated
 - 80 = Procedure described as optimal debulking, size of residual tumor nodule(s) not given
 - 97 = No cytoreductive surgery performed, Non-invasive neoplasm (behavior /2)
 - 98 = Not applicable: Information not collected for this case
 - 99 = Not documented in medical record; residual tumor status not assessed or unknown
- **Source Documents:**
 - Operative report (post-surgical findings, surgeon’s impression)
 - Pathology report (residual disease noted in intra-abdominal/peritoneal samples)
 - Postoperative notes or oncology consultation summarizing residual tumor volume
 - Tumor board notes (if explicitly referencing residual disease post-surgery)
- **Curation Instructions:**
 1. Confirm the diagnosis is ovarian cancer (`dx_ca_type` = 10).
 2. Only abstract macroscopic residual tumor data documented after primary cytoreduction (not interval debulking unless explicitly stated).
 3. Code 00 only if explicitly stated “no gross residual disease” or “R0 resection.”
 4. Code 50 if residual tumor nodules are 1 cm in maximum dimension.
 5. Code 60 if >1 cm residual tumor nodules are documented.
 6. Use code 70 if “macroscopic residual disease” is mentioned but no measurement is provided.
 7. Code 80 if operative notes say “optimal debulking” but no size is documented.
 8. Use 97 for non-invasive tumors or when no cytoreduction was performed.
 9. Code 98 if the case falls outside the ovarian cancer scope or residual tumor info is intentionally not collected.
 10. Use 99 if no documentation is available regarding residual tumor post-surgery.
- **Curator Quality Review:**
 - Ensure the surgery type is cytoreductive and not a biopsy or diagnostic laparoscopy.
 - Do not infer from radiology reports or CA-125 levels.
- **Notes:**
 - “Optimal debulking” is sometimes used inconsistently—prioritize size-based documentation if available.

- If both size and descriptive terminology (e.g., “optimal debulking”) are available, code based on size.

Variable: brain_tumor_loc

Type: Categorical (Dropdown)

Description: Brain/CNS – Primary Tumor Location within Brain Stem (C717)

Reference: NAACCR #3964

Data Source: Imaging reports (MRI, CT), neurology/neurosurgery consult notes, pathology report (if available)

Curation Directives:

- **Definition:** Captures the anatomical subsite of the brain stem for tumors coded to ICD-O topography code C717 (Brain Stem). Differentiating whether the tumor arises from the Pons or other brain stem subsites is clinically relevant, especially in pediatric neuro-oncology. This variable supports finer stratification of brain stem tumors for research and treatment planning.
- **Permitted Values / Codes:**
 - 1 = Pons
 - 2 = Subsite other than Pons
 - 8 = Not applicable: Information not collected for this case
 - 9 = Brain Stem, NOS; Unknown subsite of Brain Stem
 - 10 = Primary site is not the Brain Stem (C717)
- **Source Documents:**
 - Radiology reports (preferably MRI brain with contrast)
 - Neurologist or neurosurgeon consultation notes
 - Clinical documentation of tumor location
 - Pathology reports (only if a biopsy or autopsy was performed)
- **Curation Instructions:**
 1. Code 1 if the tumor originates in or is explicitly described as arising from the Pons.
 2. Code 2 if the tumor is in the Brain Stem, but explicitly not in the Pons (e.g., medulla, midbrain).
 3. Code 9 if the tumor is located in the Brain Stem but the subsite is not specified or labeled “NOS.”
 4. Code 10 if the primary site is not C717 – i.e., brain tumor in other regions such as cerebrum (C711) or cerebellum (C716).
 5. Code 8 if this data element was not collected intentionally for the case.
 6. Prioritize radiology reports over clinical notes unless the latter provide more specific subsite identification.
 7. Do not infer subsite from clinical presentation (e.g., cranial nerve symptoms) unless explicitly stated.
- **Curator Quality Review:**
 - Double-check that the primary site is C717 before abstracting this variable.
 - If site is unclear in the imaging, verify whether “pons” is mentioned anywhere in the impression or description.
 - Do not code based on laterality (e.g., left/right brain stem) — this field concerns subsite only.
 - If both “pons” and another subsite are mentioned, code based on the stated origin of the tumor.
 - Avoid assuming tumor spread or extension equals origin.
- **Notes:**
 - Tumors like diffuse intrinsic pontine gliomas (DIPGs) will be coded as 1 (Pons) if identified as such in radiology.

- Many brain stem tumors are inoperable and lack histologic confirmation — radiology will be the primary source for abstraction.
- This variable improves alignment with AJCC, COG, and pediatric neuro-oncology classification systems.

Variable: `brain_tumor_loc_details`

Type: Categorical (Dropdown)

Description: Brain/CNS – Brain Primary Tumor Location: Subsite Details (for tumors coded to Brain Stem subsites other than Pons)

Reference: Internal (custom site sub-classification based on NAACCR #3964)

Data Source: Radiology reports (preferably MRI), neurology/neurosurgery notes, operative or clinical documentation

Curation Directives:

- **Definition:** This variable specifies the detailed anatomical subsite within the brainstem when the primary tumor location is coded as a subsite other than Pons (i.e., `brain_tumor_loc` = 2). These anatomical distinctions are important in neuro-oncology, particularly for pediatric brain tumors where precise location informs prognosis and treatment.
- **Permitted Values / Codes:**
 - 1 = Basis peduncle
 - 2 = Cerebral peduncle
 - 3 = Choroid plexus of fourth ventricle
 - 4 = Fourth ventricle, NOS
 - 5 = Infratentorial brain, NOS
 - 6 = Medulla oblongata
 - 7 = Midbrain
 - 8 = Olive
 - 9 = Pyramid
- **Source Documents:**
 - Brain MRI reports (with and without contrast)
 - CT imaging reports (less preferred)
 - Clinical documentation from neurology or neurosurgery
 - Tumor board notes when available
- **Curation Instructions:**
 1. This field should only be completed if `brain_tumor_loc` is coded as 2 (Subsite other than Pons).
 2. Carefully review imaging reports for specific reference to anatomical structures listed above.
 3. Choose the most specific subsite mentioned as the point of tumor origin.
 4. If multiple subsites are listed, code the site identified as the origin; do not code based on tumor extension alone.
 5. If none of the listed subsites are explicitly mentioned, do not guess; leave blank.
 6. Do not use this field if the tumor is coded as `brain_tumor_loc` = 1 (Pons), 9 (NOS), 10 (Not brain stem), or 8 (Not applicable).
- **Curator Quality Review:**
 - Confirm that `brain_tumor_loc` = 2 before abstracting this variable.
 - If the imaging lists “brainstem tumor” without subsite clarification, do not complete this field.
 - Avoid inferring subsite from symptoms or clinical syndrome unless explicitly tied to anatomical description.
 - If “midbrain” and “cerebral peduncle” are both mentioned, and the tumor is said to arise from the peduncle, code as 2.

- Use the imaging impression section as the primary source for subsite designation.
- Notes:
 - This field supports research efforts in pediatric CNS tumors, such as DIPG vs. non-DIPG brain-stem gliomas.
 - Subsite detail may inform prognosis and treatment classification in cooperative group trials.
 - Tumors in these regions often do not undergo surgical resection; imaging is the gold standard source.

Variable: `cns_brain_mgmt`

Type: Categorical (Dropdown)

Description: Brain/CNS – Methylation of O6-Methylguanine-Methyltransferase (MGMT)

Reference: NAACCR #3889

Data Source: Pathology report (molecular diagnostics section), molecular profiling results, neuro-oncology documentation

Curation Directives:

- **Definition:** This variable captures the methylation status of the MGMT gene promoter in central nervous system tumors. MGMT methylation is a predictive biomarker for response to alkylating agents (e.g., temozolomide) and a prognostic factor in high-grade gliomas, especially glioblastoma and anaplastic astrocytoma.
- **Permitted Values / Codes:**
 - 0 = MGMT methylation absent/not present, unmethylated MGMT
 - 1 = MGMT methylation present, low level; hypomethylated; partial methylated
 - 2 = MGMT methylation present, high level; hypermethylated
 - 3 = MGMT methylation present, level unspecified
 - 6 = Benign or borderline tumor
 - 7 = Test ordered, results not in chart
 - 8 = Not applicable: Information not collected for this case
 - 9 = Not documented in medical record; cannot be determined by pathologist; MGMT not assessed or unknown if assessed
- **Source Documents:**
 - Pathology report (molecular diagnostics or special studies section)
 - Molecular profiling reports (e.g., FoundationOne, Tempus)
 - Oncology progress notes referencing molecular results
 - Tumor board or precision medicine notes
- **Curation Instructions:**
 1. Abstract this variable only for primary CNS tumors when `dx_ca_type` = 12.
 2. Review the pathology report for MGMT methylation testing; search terms may include “MGMT,” “methylation,” “hypermethylated,” or “unmethylated.”
 3. Assign:
 - **Code 0** if explicitly stated as “unmethylated,” “not methylated,” or “methylation absent.”
 - **Code 1** if described as “low-level methylation,” “hypomethylated,” or “partially methylated.”
 - **Code 2** if stated as “hypermethylated” or “high-level methylation.”
 - **Code 3** if methylation is confirmed but the level is not specified.
 - **Code 6** if the tumor is benign or borderline and MGMT testing is not clinically indicated.
 - **Code 7** if the test was ordered but the result is not yet available.
 - **Code 8** if this information is not collected for the tumor type.
 - **Code 9** if there is no documentation of testing or if the report states that the methylation status could not be determined.

4. Do not infer methylation status based on treatment decisions alone.
5. If multiple reports exist, abstract from the most definitive or recent molecular result available.

- **Curator Quality Review:**

- Confirm that MGMT testing is applicable based on histology and grade (typically high-grade gliomas).
- Cross-check molecular testing dates to ensure results correspond to the current diagnosis.
- Flag cases where MGMT is documented but the result is discordant across reports.
- If both qualitative (e.g., “methylated”) and quantitative (% methylation) results are available, rely on the qualitative interpretation.

- **Notes:**

- MGMT methylation may impact prognosis and treatment decisions, particularly eligibility for temozolomide in glioblastoma.
- Standard methods include methylation-specific PCR (MSP) and pyrosequencing; abstract only the interpretive result, not method details.
- Do not confuse MGMT mutation with methylation—this variable captures epigenetic silencing, not sequence alterations.

Imaging

Radiologic Imaging Procedure Information

Curators are required to abstract radiologic imaging data from all imaging reports available in the patient’s medical record, including those originating from external institutions. Each imaging event should be documented independently, regardless of whether it was performed internally or externally. The abstraction should include procedural details (date, modality, body location), the radiologist’s interpretation of disease status (e.g., progression, response, or stability), and the diagnostic impressions (evidence of disease, presence of primary or metastatic cancer sites). If multiple cancer sites are described in a single report, each must be recorded in accordance with the form structure. Use the information explicitly stated in the finalized radiology report; do not infer findings or rely on clinical notes referencing prior imaging. Accurate capture of imaging-based findings provides essential temporal and anatomical context for disease progression and treatment response evaluation.

Variable: imaging_yn

Type: Binary (Yes/No)

Description: Has the patient undergone any radiographic imaging at any point during their treatment?

Reference: N/A

Data Source: Medical record (radiology reports, imaging orders, oncology progress notes)

Curation Directives:

- **Definition:** This variable captures whether any radiographic imaging was performed at any point during the patient’s cancer care. Imaging includes diagnostic, staging, treatment planning, response assessment, or surveillance scans across all modalities (e.g., CT, MRI, PET, X-ray, ultrasound).

- **Permitted Values / Codes:**
 - 1 = Yes
 - 0 = No
- **Source Documents:**
 - Radiology reports (initial or follow-up)
 - Imaging orders or requisitions
 - Oncology or multidisciplinary team (MDT) progress notes
 - Treatment summaries or operative reports referencing imaging
- **Curation Instructions:**
 1. Abstract this field for all cancer types.
 2. Select **Yes (1)** if any radiographic imaging is documented at any point during the cancer continuum (diagnosis through surveillance).
 3. Include any imaging used to guide biopsy, assess tumor burden, monitor treatment response, or evaluate recurrence.
 4. Select **No (0)** only if the medical record explicitly states that no imaging was performed or if there is no documentation of any imaging.
 5. Do not include non-radiographic procedures (e.g., endoscopy) or imaging unrelated to cancer evaluation (e.g., unrelated injury).
 6. If the presence of imaging is unclear or documentation is insufficient, leave the field blank and flag for quality review.
- **Curator Quality Review:**
 - If no imaging is identified in a case of advanced-stage disease or radiotherapy, recheck for imaging documentation or omissions.
- **Notes:**
 - If historical imaging was performed in the month prior to diagnosis, it should be included.

Variable: imaging_institution

Type: Categorical (Dropdown)

Description: Radiologic Imaging Procedure Information – Imaging Facility

Reference: N/A

Data Source: Radiology reports, imaging orders, EHR metadata, provider documentation

Curation Directives:

- **Definition:** This variable captures the facility where radiographic imaging was performed. It distinguishes between imaging conducted at the patient's primary oncology institution (internal) versus an outside or referring institution (external).
- **Permitted Values / Codes:**
 - 1 = Internal Institution
 - 2 = External Institution
- **Source Documents:**
 - Radiology reports (facility header or footer, institutional logo)
 - PACS or image viewer metadata (study origin site)
 - Scanned documents or CDs of external imaging uploaded to the EHR
- **Curation Instructions:**
 1. Abstract this variable for each imaging event included in the dataset.

2. Select **Internal Institution (1)** if the imaging was performed at the GENIE-participating cancer center or its satellite clinics.
 3. Select **External Institution (2)** if the imaging was performed elsewhere (e.g., community hospitals, referring centers, or prior treating facilities).
 4. Use the radiology report header/footer or metadata to determine location. Institutional logos, headers, or dictating physician affiliations may also assist.
 5. If a report was scanned into the record but originated from another institution, code as **External** even if the image was uploaded into the internal system.
- **Curator Quality Review:**
 - Confirm consistency between imaging reports and source documentation.
 - Cross-reference facility names or scan source metadata when available.
 - For patients with both internal and external scans, verify each is correctly categorized.
 - If reports were imported but dictation occurred internally (e.g., internal radiologist re-read an external image), code based on the location where the imaging was performed, not where it was interpreted.
 - **Notes:**
 - This field is important for downstream analyses of imaging availability, scan quality, and institutional interpretation consistency.
 - Coding should reflect scan acquisition location, not interpretation site.
 - Use discretion when multiple institutions are involved (e.g., transferred care)—flag ambiguous cases.
-

Variable: `imaging_date`

Type: Free Text Entry (text)

Description: Date of Radiology Imaging

Reference: N/A

Data Source: Radiology report (header/footer), imaging orders, PACS metadata

Curation Directives:

- **Definition:**
Captures the actual date the imaging procedure was performed, not the date of report completion or interpretation.
- **Permitted Values / Codes:**
Free text date field (validated in MM/DD/YYYY format)
- **Source Documents:**
 - Radiology report (date of procedure usually at the top)
 - PACS metadata or imaging viewer
 - Imaging order requisition
 - Clinical notes referencing scan date
- **Curation Instructions:**
 1. Record the date the scan was performed, not when it was ordered or interpreted.
 2. Use the exact date found in the radiology report, typically listed near the top.
 3. If multiple imaging dates are present (e.g., multi-sequence MRI), record the earliest date.
 4. Do not record the date of final report or dictation.
 5. If the date is missing or ambiguous, leave the field blank and flag for review.
- **Curator Quality Review:**
 - Validate against clinical timeline to ensure logical sequence (e.g., date precedes report date).

- Flag inconsistent or retroactively dictated scans.
- **Notes:**
 - Critical for temporal alignment of imaging with treatment and disease progression.

Variable: `imaging_date_int`

Type: Integer (Calculated)

Description: Date of Radiology Imaging in Days from DOB

Reference: N/A

Data Source: Derived from `imaging_date` and `birth_date`

Curation Directives:

- **Definition:** Automatically calculates the number of days between the patient's date of birth and the imaging date to support deidentified, age-relative analyses.
- **Permitted Values / Codes:**
Integer (positive number of days)
- **Source Documents:**
 - Automatically derived by REDCap
 - Based on previously curated `birth_date` and `imaging_date` fields
- **Curation Instructions:**
 1. No manual entry required. Ensure that `birth_date` and `imaging_date` are accurately abstracted.
 2. System-calculated using: `datediff([birth_date],[imaging_date],"d","mdy",true)`
 3. This field is **read-only** and should not be modified manually.
- **Curator Quality Review:**
 - Spot-check intervals to ensure calculations are logical.

Variable: `imaging_report_date`

Type: Free Text Entry (text)

Description: Date of Radiology Imaging Final Report

Reference: N/A

Data Source: Radiology report (finalized timestamp)

Curation Directives:

- **Definition:** Captures the date the radiology report was finalized and signed by the interpreting radiologist.
- **Permitted Values / Codes:**
Free text date field (validated in MM/DD/YYYY format)
- **Source Documents:**
 - Radiology report footer or electronic signature
 - Imaging information system metadata
 - Date of dictation or electronic finalization
- **Curation Instructions:**
 1. Record the date the report was finalized, not the date the scan was performed.
 2. If both preliminary and final reports exist, use the final report date.

3. If date is not explicit, use metadata timestamp or radiologist signature date.
4. If no date can be confirmed, leave blank and flag for quality review.

- **Curator Quality Review:**

- Confirm final report status vs. preliminary.
 - Ensure date logically follows the scan date.
-

Variable: `imaging_report_date_int`

Type: Integer (Calculated)

Description: Date of Radiology Imaging Final Report in Days from DOB

Reference: N/A

Data Source: Derived from `imaging_report_date` and `birth_date`

Curation Directives:

- **Definition:** Automatically calculates the number of days between the patient's birth date and the date the final imaging report was completed.
 - **Permitted Values / Codes:**
Integer (positive number of days)
 - **Source Documents:**
 - Automatically derived from REDCap
 - Dependent on `birth_date` and `imaging_report_date`
 - **Curation Instructions:**
 1. No manual entry required.
 2. System calculates interval using: `datediff([birth_date],[imaging_report_date],"d","mdy",true)`
 3. Do not override this value manually.
 - **Curator Quality Review:**
 - Ensure inputs to calculation (`birth_date` and `imaging_report_date`) are valid.
 - Flag anomalies where report date predates imaging date.
-

Variable: `imaging_ref_date`

Type: Free Text Entry (text)

Description: Reference Imaging Date (Previous Imaging of This Cancer)

Reference: N/A

Data Source: Radiology reports, imaging system (PACS)

Curation Directives:

- **Definition:** Captures the date of the most recent prior imaging study related to the same cancer type or site before the current imaging exam. This is used to assess changes in disease over time by comparison to a baseline or previous scan.
- **Permitted Values / Codes:**
Free text date field (validated in MM/DD/YYYY format)
- **Source Documents:**
 - Radiology reports referencing prior comparative scans
 - PACS/imaging system history
 - Clinical notes stating “compared to prior imaging on [date]”

- Imaging requisitions that mention recent comparative exams
- **Curation Instructions:**
 1. Record the actual imaging date of the most recent prior scan for the same cancer type.
 2. Abstract only if clearly linked to the same disease process (e.g., same organ, metastasis follow-up).
 3. Do not use the current imaging date or any future imaging; this must precede the current exam.
 4. If multiple prior imaging exams are listed, select the most recent one prior to the current scan.
 5. Leave blank if no prior imaging is referenced or available.
 6. If unclear whether prior imaging refers to the same primary cancer, err on the side of exclusion.
- **Curator Quality Review:**
 - Confirm that the date refers to a scan prior to the current imaging.
 - Ensure that the prior scan is related to the same cancer (e.g., same site or metastatic process).
 - Flag any discrepancies in scan chronology or comparisons to scans that cannot be confirmed.
- **Notes:**
 - Often documented in phrasing like “compared to prior scan dated MM/DD/YYYY.”
 - Helps assess radiographic progression or treatment response.
 - Do not confuse this with “initial” or “baseline” scan if it is not the most recent prior study.

Variable: `imaging_ref_date_int`

Type: Integer (Calculated)

Description: Reference Imaging Date in Days from DOB

Reference: N/A

Data Source: Derived from `birth_date` and `imaging_ref_date`

Curation Directives:

- **Definition:** Automatically calculates the number of days between the patient’s date of birth and the date of the most recent prior imaging, to support longitudinal analysis without exposing PHI.
- **Permitted Values / Codes:**
Integer (positive number of days)
- **Source Documents:**
 - Automatically derived from curated `birth_date` and `imaging_ref_date`
- **Curation Instructions:**
 1. No manual abstraction is required for this field.
 2. System calculates this field using: `datediff([birth_date],[imaging_ref_date],"d","mdy",true)`
 3. Confirm that input fields (`birth_date`, `imaging_ref_date`) are present and accurate.
 4. Do not override or manually edit this calculated value.
- **Curator Quality Review:**
 - Validate that `imaging_ref_date` is correct and earlier than the current imaging date.
 - Review for negative or illogical values which may suggest incorrect source dates.
 - Ensure consistency with other imaging intervals.

Variable: `imaging_modality`

Type: Categorical (Dropdown)

Description: Imaging Modality

Reference: N/A

Data Source: Radiology report header, procedure description, or imaging metadata

Curation Directives:

- **Definition:** Identifies the imaging modality used to obtain the radiologic examination. This includes both functional and anatomic imaging techniques and is essential for interpreting the findings in the context of the modality's capabilities.
- **Permitted Values / Codes:**
 - 1 = Positron Emission Tomography (PET)
 - 2 = Computed Tomography (CT)
 - 3 = PET/CT (combined PET and CT)
 - 4 = Magnetic Resonance Imaging (MRI)
 - 5 = PET/MRI (combined PET and MRI)
 - 6 = FDG-PET
 - 7 = Bone scan
 - 8 = Mammogram
 - 9 = Ultrasound
 - 10 = X-Ray
 - 11 = Other Nuclear Medicine scan
- **Source Documents:**
 - Radiology report header (e.g., “MRI of the abdomen”)
 - PACS modality type
 - Clinical notes or requisitions that describe ordered modality
- **Curation Instructions:**
 1. Review the radiology report or imaging order to identify the modality used.
 2. Choose the single code that best reflects the imaging performed.
 - Use code 3 for PET/CT and code 5 for PET/MRI when combined modalities are explicitly stated.
 - Use code 6 only if FDG-PET is specifically mentioned, regardless of whether CT was performed for attenuation correction.
 3. For other nuclear scans (e.g., MIBG, gallium), use code 11.
 4. If multiple imaging modalities were performed on the same date for the same site, abstract the most clinically relevant or comprehensive modality.
- **Curator Quality Review:**
 - Confirm that the selected code aligns with the full imaging description.
 - Check for hybrid modalities (e.g., PET/CT) and avoid double-coding.
 - Flag cases where modality is unclear or inconsistently documented across systems.
- **Notes:**
 - Use imaging requisition or metadata from PACS as supplementary sources if the report is ambiguous.
 - In most oncology cases, PET/CT or MRI are primary modalities for staging or treatment response evaluation.

Variable: imaging_body_location

Type: Categorical (Check-Box)

Description: Body Region(s) Imaged

Reference: N/A

Data Source: Radiology report headers, body region protocols, or DICOM metadata

Curation Directives:

- **Definition:** Captures the anatomic regions of the body that were imaged during the radiology procedure. This may involve one or more sites and helps contextualize findings, especially for tracking metastasis or disease spread.
- **Permitted Values / Codes:**
 - 1 = Brain
 - 2 = Head and Neck
 - 3 = Spine
 - 4 = Chest/Thorax
 - 5 = Abdomen
 - 6 = Pelvis
 - 7 = Extremities (upper/arms)
 - 8 = Extremities (lower/legs)
 - 9 = Whole Body
- **Source Documents:**
 - Radiology report title or header (e.g., “CT Chest/Abdomen/Pelvis”)
 - Imaging protocol or acquisition summary
 - Clinical note or requisition specifying scan area
 - PACS images (if needed for clarification)
- **Curation Instructions:**
 1. Select all regions imaged during the procedure.
 2. Use code 9 (Whole Body) only if the imaging spans skull to mid-thigh or more (e.g., standard oncologic PET/CT).
 3. For regional imaging, select all appropriate codes (e.g., CT Chest/Abdomen/Pelvis = 4, 5, 6).
 4. Do not select additional regions unless explicitly included in the scan field.
 5. If scan involves unclear boundaries (e.g., thoracoabdominal junction), use best clinical judgment.
- **Curator Quality Review:**
 - Ensure that whole-body scans are not redundantly coded with regional components unless instructed.
 - Cross-check modality and report to ensure consistent anatomical scope.

Variable: imaging_eod

Type: Categorical (Dropdown)

Description: Radiologic Impression – Evidence of Disease

Reference: N/A

Data Source: Final radiology report (impression section)

Curation Directives:

- **Definition:** Captures whether the radiologist’s impression section indicates evidence of disease based on the imaging findings. This high-level summary reflects whether measurable or visualizable disease was observed.
- **Permitted Values / Codes:**
 - 1 = The radiology report states that there is evidence of disease
 - 2 = The radiology report states that there is no evidence of disease
 - 3 = The radiology report does not mention evidence of disease or is indeterminate regarding evidence of disease
- **Source Documents:**
 - Radiology report – Impression section (or “Conclusion” if labeled as such)
 - Radiologist’s addenda or follow-up interpretation (if applicable)

- Oncology clinic notes summarizing imaging findings (secondary source)

- **Curation Instructions:**

1. Review the Impression section of the radiology report for an explicit statement regarding disease presence.
2. Assign:
 - **Code 1** if the report confirms evidence of disease (e.g., “There is interval progression,” “Metastatic disease is seen,” or “Recurrent mass identified”).
 - **Code 2** if the report states there is **no** radiographic evidence of disease (e.g., “No evidence of recurrent or metastatic disease,” “Negative for malignancy”).
 - **Code 3** if the report is ambiguous, does not reference disease status directly, or is equivocal (e.g., “Findings are indeterminate,” “Cannot exclude recurrence,” or “No comparison available”).
3. Abstract based only on what is reported; do not infer based on prior imaging or treatment history.
4. Do not assign code based on lesion size, lymph node appearance, or enhancement patterns unless the impression interprets these as representing disease.
5. If the impression discusses multiple findings (e.g., post-treatment changes **and** disease), code based on the radiologist’s summary assessment.

- **Curator Quality Review:**

- Confirm that the code selected reflects the final impression, not just the body of the report.
- Check for language like “suspicious,” “likely disease,” or “consistent with malignancy” as indicators of code 1.
- If multiple imaging reports exist for the same time point, use the most definitive report.

- **Notes:**

- This variable is a general indicator of disease status used in outcome monitoring, disease tracking, and longitudinal analysis.

Primary Tumor Assessment

Variable: imaging_ca_yn

Type: Binary (Yes/No)

Description: Primary Tumor Assessment – Presence of Primary Cancer Site

Reference: N/A

Data Source: Final radiology report (impression and findings sections)

Curation Directives:

- **Definition:** This variable captures whether the imaging report describes a primary cancer site, based on radiologic findings. This may reflect an initial tumor, residual disease, or recurrence at the primary site.
- **Permitted Values / Codes:**
 - 0 = No, the report does not describe any primary cancer site(s)
 - 1 = Yes, the report describes one or more primary cancer site(s)
- **Source Documents:**
 - Radiology report – Impression and Findings sections
 - Radiologist’s comparison statements with prior imaging

- Addenda or amended reports (if available)
- **Curation Instructions:**
 1. Review the full radiology report with attention to the Impression and Findings sections.
 2. Assign:
 - **Code 1** if the report describes a mass, lesion, thickening, or area interpreted as a primary tumor in the anatomical site where the malignancy originated.
 - **Code 0** if no primary site is mentioned, or if the report only describes metastatic disease, treatment effects, or unrelated findings.
 3. Do not confuse metastatic disease or lymph node involvement with primary site unless explicitly stated.
- **Curator Quality Review:**
 - Confirm that the primary site aligns with the patient’s documented diagnosis and anatomic location.
 - Verify that “primary” is based on tumor origin, not dominance of lesion or site of largest tumor burden.
- **Notes:**
 - In complex cases (e.g., multiple primaries), document presence of **any** primary site described in the report, even if multiple tumors exist.

Variable: `imaging_ca_count`

Type: Categorical (Dropdown)

Description: Number of Primary Cancer Sites Documented in Imaging Report

Reference: N/A

Data Source: Final radiology report (findings and impression sections)

Curation Directives:

- **Definition:** This variable captures the total number of distinct primary cancer sites described in the imaging report. This count includes only primary tumor locations and does not include metastatic disease or lymph node involvement.
- **Permitted Values / Codes:**
 - 1 = One primary site
 - 2 = Two primary sites
 - 3 = Three primary sites
 - 4 = Four primary sites
 - 5 = Five primary sites
- **Source Documents:**
 - Radiology report – Findings and Impression sections
 - Historical reports (if referenced explicitly for comparative purposes)
- **Curation Instructions:**
 1. This field should only be completed **if** `imaging_ca_yn = 1`, indicating that at least one primary site is mentioned in the imaging report.
 2. Review the imaging report and determine how many anatomically and biologically distinct primary tumor sites are described.
 - Example: A single breast mass in the left breast and a separate lung primary are counted as two primary sites.

- Bilateral tumors of the same cancer type (e.g., bilateral breast primaries) count as two if clearly described as distinct.
- 3. Do not count metastatic sites, lymph node involvement, or post-surgical changes unless described as primary lesions.
- 4. If the radiologist notes uncertainty between metastasis vs. second primary, and both possibilities are presented, defer to the radiologist’s dominant impression.
- 5. Count only primary sites described in this imaging report. Do not infer from previous clinical history unless referenced explicitly.
- 6. In cases where more than five primaries are clearly documented, assign **5** and flag for manual review if necessary.
- **Curator Quality Review:**
 - Ensure consistency with `imaging_ca_yn`. If `imaging_ca_yn` = 0, this field must be blank or omitted.
 - Cross-check with prior or linked reports for consistency in identifying and counting distinct primary lesions.
- **Notes:**
 - This variable helps distinguish between multifocal, bilateral, or multiple synchronous primary cancers for longitudinal tracking and clinical interpretation.
 - Useful for harmonizing imaging interpretation with pathology and genomic records when evaluating tumor heterogeneity or multiple cancer diagnoses.

Variable: `imaging_primary_site1-5`

Type: Categorical (Dropdown)

Description: Primary Cancer Site (ICD-O-3 Topography Code)

Reference: NAACCR #400

Data Source: Imaging report (Impression, Findings, or Technique sections)

Curation Directives:

- **Definition:** The primary anatomical site where cancer is identified based on the imaging findings, recorded using the ICD-O-3 topography code set. This field captures the specific location(s) of primary cancer described radiologically.
- **Permitted Values / Codes:** C00.0–C80.9 and site-specific ICD-O-3 codes as defined in the dropdown list. Include special fluid designations (e.g., F10–F50) as applicable.
- **Source Documents:**
 - Final imaging report
 - Radiologist’s impression and findings
 - Imaging technique or protocol if anatomical coverage is described
- **Curation Instructions:**
 1. Review the radiologist’s report for any mention of a primary tumor site. Focus on the Impression and Findings sections.
 2. If a primary site is described, assign the corresponding ICD-O-3 code. Use the most specific anatomical site available.
 3. If laterality is described and relevant, this will be captured in a separate field.
 4. If multiple primary sites are documented in a single report, enter each site using the appropriate repeated field (e.g., `imaging_primary_site1`, `imaging_primary_site2`, etc.), based on the value recorded in `imaging_ca_count`.
 5. If the imaging report notes “unknown primary” or does not identify a specific anatomic origin, code as C80.9 Unknown Primary Site.

6. Do not code sites of metastasis here—only primary sites should be recorded.

- **Curator Quality Review:**

- Confirm that the selected code matches the text in the report.
- Ensure correct mapping to the most specific available ICD-O-3 code.
- Verify the correct number of sites is entered in alignment with `imaging_ca_count`.

- **Notes:**

- This field is tied to the interpretation of evidence of disease at a primary site and may be referenced during later abstraction or correlation with diagnosis data.
- Always validate the anatomical location using radiologist language and avoid assuming based on cancer type or history alone.

Variable: `imaging_laterality1-5`

Type: Categorical (Dropdown)

Description: Laterality of Primary Cancer Site

Reference: NAACCR #410

Data Source: Imaging report (Impression, Findings, Technique)

Curation Directives:

- **Definition:** Laterality refers to the side of the body on which the primary cancer site is located, as identified on imaging. This field is applicable only to paired organs or structures and captures whether the cancer is right-sided, left-sided, bilateral, midline, or not otherwise specified.
- **Permitted Values / Codes:**
 - 0 = Not a paired site
 - 1 = Right: origin of primary
 - 2 = Left: origin of primary
 - 3 = Only one side involved, right or left origin unspecified
 - 4 = Bilateral involvement at time of diagnosis, lateral origin unknown for a single primary; or both ovaries involved simultaneously, single histology; bilateral retinoblastomas; bilateral Wilms tumors
 - 5 = Paired site: midline tumor (effective with 01/01/2010 dx)
 - 9 = Paired site, but no information concerning laterality
- **Source Documents:**
 - Final imaging report
 - Radiologist's impression and findings
 - Body of report including anatomic detail
- **Curation Instructions:**
 1. Determine if the primary site listed in `imaging_primary_site1-5` is a paired site (e.g., lung, breast, kidney).
 2. If not a paired site (e.g., esophagus, stomach), assign code 0 = Not a paired site.
 3. If paired, review the imaging report for language indicating side-specific involvement.
 - Use **1** or **2** if right or left is explicitly mentioned as the origin.
 - Use **3** if only one side is involved but the specific side is not stated.
 - Use **4** if both sides are involved and origin is unclear or if conditions listed in the definition apply.
 - Use **5** if the tumor is described as midline (e.g., midline prostate, nasal septum).
 - Use **9** if laterality is not mentioned and cannot be inferred.

4. Code should reflect laterality as of the imaging report date, even if clinical history or future reports provide different details.
- **Curator Quality Review:**
 - Confirm that the selected laterality aligns with the anatomical detail and terminology used in the report.
 - Cross-check the assigned laterality with the primary site to ensure it is applicable (paired vs. unpaired).
 - Validate against any inconsistencies (e.g., “right breast mass” should not be coded as bilateral).
 - **Notes:**
 - Laterality is required only for paired sites. Do not assign laterality for unpaired structures (e.g., pancreas, bladder).
 - If laterality is unclear but likely inferred (e.g., single lung mass described without side), consult supervisor before assigning code 3 or 9.
 - This field should be repeated in alignment with `imaging_ca_count` and corresponds directly to the associated `imaging_primary_site` entry.

Variable: `imaging_ca_status1-5`

Type: Categorical (Dropdown)

Description: Radiologist-Impression of Cancer Status

Reference: N/A

Data Source: Imaging report (Impression, Findings)

Curation Directives:

- **Definition:** The radiologist’s overall summary of the primary cancer’s current status as interpreted from an imaging study (e.g., CT, MRI, PET, bone scan). This reflects the radiologist’s expert interpretation of whether the disease is stable, progressing, responding, not visible, or unclear based on imaging characteristics and comparison to prior studies, when available.
- **Permitted Values / Codes:**
 - 1 = Stable Disease: No significant change in tumor size
 - 2 = Progression: Indicates worsening of the disease, with increase in tumor size.
 - 3 = Partial Response: Decrease in tumor size, but not complete disappearance.
 - 4 = No Evidence of Disease: Used when previous scans showed cancer, but current scan shows no detectable disease.
 - 5 = Indeterminate: When the scan results are unclear or cannot be compared to previous scans.
 - 6 = Not Mentioned: Radiologist does not describe the cancer’s status.
- **Source Documents:**
 - Final imaging report
 - Radiologist’s impression section (typically located at the end of the report)
 - Comparison statements to prior studies (if applicable)
- **Curation Instructions:**
 1. Review the **Impression** section of the radiology report for the radiologist’s overall assessment of disease status:
 2. Assign:
 - **Code 1 - Stable Disease** when the radiologist explicitly states that the primary tumor is “stable,” “unchanged,” “no significant interval change,” or “no progression.” Acceptable language may also include “findings are consistent with stable disease.” If prior imaging is

referenced and the comparison indicates no meaningful change in the size, shape, or enhancement of the primary lesion, select this code. **Do not assign** if stability is only noted for metastatic disease or lymph nodes.

- **Code 2 - Progression** when the radiologist impression indicates worsening of the primary tumor, using terms like “progression,” “interval increase,” “increased size,” “advancing disease,” or “worsening mass.” May also include appearance of new disease in the region of the primary tumor (e.g., new invasion or recurrence at the primary site). **Do not assign** this code if increased findings are attributed to inflammation, infection, or treatment-related change unless explicitly described as disease progression.
- **Code 3 - Partial Response** when the primary tumor has decreased in measurable size or enhancement but is still visible. Acceptable phrases include “partial response,” “interval decrease in size,” “reduction in tumor burden,” or “tumor regression noted.” **Do not assign** this code if complete resolution is documented; instead, assign Code 4.
- **Code 4 - No Evidence of Disease** when the radiologist impression states that the previously visualized primary tumor is no longer detectable. Look for phrases like “no evidence of disease,” “no residual tumor,” “no definite malignancy seen,” or “complete resolution of previously seen mass.” Ensure this refers specifically to the primary tumor site, and not to metastases or nodal disease.
- **Code 5 - Indeterminate** when the radiologist is unable to determine the status of the primary tumor, due to limited visualization, technical factors (e.g., motion artifact), or non-specific findings. Use when language includes “cannot exclude,” “difficult to assess,” “nondiagnostic,” “limited study,” or “findings are equivocal.” May also apply if a comparison study is not available and the radiologist expresses uncertainty in characterizing change. **Do not assign** this code if the impression makes no mention of the primary tumor at all, instead use Code 6 in those cases.
- **Code 6 - Not Mentioned** when the radiologist does not reference the primary tumor in the impression section of the imaging report. Use when the impression focuses solely on other findings and omits any summary statement or description related to the primary cancer site.

- **Curator Quality Review:**

- Ensure the code reflects the primary tumor only, not nodal or metastatic sites. Ignore overall disease trends unless they specifically reference the primary lesion.
- Assign codes only when the radiologist explicitly states the tumor status in the Impression. Do not interpret from findings or assume based on prior reports.
- When multiple regions are discussed, base the code solely on the primary tumor’s status even if metastases are progressing or regressing.

- **Notes:**

- This variable should be completed for each primary cancer site identified in the report, as repeated per `imaging_ca_count`.

Metastatic Site (s) Assessment

Variable: `imaging_mets_yn`

Type: Binary (Yes/No)

Description: Presence of Metastatic Site(s)

Reference: N/A

Data Source: Imaging report (Impression, Findings)

Curation Directives:

- **Definition:** Indicates whether the radiology report describes any sites of metastatic disease.
- **Source Documents:**
 - Imaging report (particularly Findings and Impression sections)
- **Curation Instructions:**
 1. Review the entire imaging report for any mention of metastatic disease, distant spread, or specific references to cancer presence outside the primary tumor site(s).
 2. Assign **Yes (1)** if the report explicitly describes metastases (e.g., “metastatic lesions to liver,” “bone metastases,” or “distant disease”).
 3. Assign **No (0)** if the report does not mention metastatic disease, or explicitly states “no evidence of metastasis.”
 4. Do not infer metastatic disease from enlarged lymph nodes or ambiguous findings unless clearly described as metastatic.
- **Curator Quality Review:**
 - Ensure terminology aligns with confirmed metastatic disease and not suspected or uncertain lesions.
 - Confirm the variable is answered consistently with related fields (e.g., mets count, sites, and status)

Variable: `imaging_mets_count`

Type: Categorical (Dropdown)

Description: Number of Distinct Metastatic Sites

Reference: N/A

Data Source: Imaging report (Findings, Impression)

Branching Logic: Show only if `imaging_mets_yn` = Yes

Curation Directives:

- **Definition:** The number of anatomically distinct metastatic sites described in the imaging report.
- **Permitted Values / Codes:**

1–15 (as listed in the dropdown)
- **Source Documents:**
 - Radiology report with site-specific metastatic findings
- **Curation Instructions:**
 1. Count the number of distinct anatomical sites (e.g., liver, bone, lung) where metastases are explicitly described.
 2. Do not count multiple lesions within the same site as separate entries.
 - Example: “Multiple hepatic lesions suggestive of metastasis” = 1 site.
 3. If the report mentions multiple metastatic regions within one broader organ system (e.g., “multiple osseous sites”), still count this as 1 site (bone).
 4. If the number of metastatic sites is unclear or vaguely described, conservatively record the number explicitly stated or implied.
- **Curator Quality Review:**
 - Verify consistency with `imaging_mets_yn` = **Yes**; this field should not be filled if `imaging_mets_yn` = No.
- **Notes:**

- This field determines how many times site-specific metastatic variables (e.g., `imaging_site_caX`, `imaging_cancer_statusX`) will be repeated.
- Do not include primary sites.

Variable: `imaging_met_site1-15`

Type: Categorical (Dropdown)

Description: Site of Metastasis (ICD-O-3 Topography Code)

Reference: NAACCR Item #400

Data Source: Imaging report (Findings, Impression)

Curation Directives:

- **Definition:** Records the anatomical site of metastasis using the corresponding ICD-O-3 Topography Code, based on radiologist interpretation of metastatic disease.
- **Permitted Values / Codes:**
 - Full list of ICD-O-3 topography codes as provided in the data dictionary (e.g., C22.0 = Liver, C34.9 = Lung NOS, C41.2 = Vertebral column, C77.2 = Intra-abdominal lymph nodes, etc.)
- **Source Documents:**
 - Radiology report, including the Findings section and the Impression section
- **Curation Instructions:**
 1. For each metastatic site identified in the imaging report, select the ICD-O-3 topography code that best corresponds to the described anatomic location.
 2. Use only explicit descriptions of metastatic involvement (e.g., “metastatic lesions to the liver,” “osseous metastases”) as the basis for site assignment.
 3. Refer to the full list of topography codes to determine the most specific code available. For example:
 - “Metastatic disease in lung” → C34.9 (Lung NOS)
 - “Metastatic spread to vertebrae” → C41.2 (Vertebral column)
 - “Retroperitoneal adenopathy consistent with metastatic disease” → C77.2 (Intra-abdominal lymph nodes)
 4. For codes beginning with C77.x (lymph nodes), use the most anatomically specific C77.x code available to enable differentiation between regional and distant lymph node involvement relative to the primary tumor site.
 5. Do **not** use codes for sites that are:
 - Mentioned without clear evidence of metastasis
 - Only suspected or equivocal (e.g., “possible,” “cannot exclude”)
 - Related to primary tumor location (this variable captures metastases only)
 6. Repeat this field for each site up to the value indicated in `imaging_mets_count`.
- **Curator Quality Review:**
 - Confirm that selected codes reflect metastatic and not primary involvement.
 - Check alignment between ICD-O-3 code and radiology description for accuracy and specificity.
- **Notes:**
 - If multiple metastatic sites are present within a general area (e.g., “multiple liver lesions”), code the site once using the appropriate topography code.
 - Always refer to the ICD-O-3 manual when uncertain about anatomical mapping.

Variable: imaging_met_ca_status1-15

Type: Categorical (Dropdown)

Description: Radiologist Impression of Cancer Status at Each Site of Metastasis

Reference: n/a

Data Source: Imaging report (Impression, Findings)

Curation Directives:

- **Definition:** Captures the radiologist's site-specific impression of cancer status for each distinct site of metastatic disease identified in the imaging report (corresponding to `imaging_site_ca1-15`). This reflects whether the metastasis is progressing, stable, regressing, absent, or not assessed.
- **Permitted Values / Codes:**
 - 1 = Stable Disease: No significant change in size of metastatic tumor/lesion
 - 2 = Progression: Indicates worsening of the metastatic tumor/lesion
 - 3 = Partial Response: Decrease/improving in metastatic tumor/lesion size, but not complete disappearance.
 - 4 = No Evidence of Disease: Used when previous scans showed metastatic tumor/lesion, but current scan shows no detectable disease at this site.
 - 5 = Indeterminate: When the scan results are unclear or cannot be compared to previous scans.
 - 6 = Not Mentioned: Radiologist does not describe the cancer's status.
- **Source Documents:**
 - Radiology report, including the Findings section and the Impression section
- **Curation Instructions:**
 1. For each site of metastasis recorded in `imaging_site_ca1-15`, review the imaging report for the radiologist's interpretation of that site's cancer status.
 2. Assign:
 - **Code 1 – Stable Disease** when the radiologist indicates no significant change in size, number, or characteristics of metastatic lesions at the specified site. Acceptable language: "stable metastases," "no interval change," "unchanged," "no progression," or "findings consistent with stable disease." **Do not assign** if stability is described for another site or generalized.
 - **Code 2 - Progression** Assign when the impression indicates worsening of metastatic disease at the given site. Acceptable terms: "progression," "increased size," "new lesions," "interval growth," "advancing disease." **Do not assign** if findings are attributed to benign causes (e.g., post-treatment inflammation) unless progression is explicitly stated. Progression must relate to the specific metastatic site, not overall status.
 - **Code 3 - Partial Response** when metastatic lesions at the site have decreased in measurable size or number, but are still present. Look for: "partial response," "interval decrease in size," "reduction in measurable disease," "tumor regression," "improvement with residual disease." **Do not assign** if the lesion is completely resolved (see Code 4). Ensure that change is site-specific and not referring to a different region.
 - **Code 4 - No Evidence of Disease** when the radiologist impression states that the previously visualized metastatic site is no longer detectable. Look for phrases like "no evidence of disease," "no residual lesion," "complete resolution," "resolved metastatic disease." Ensure this refers specifically to the primary tumor site, and not to metastases or nodal disease.
 - **Code 5 - Indeterminate** when the radiologist is unable to determine the status of the primary tumor, due to limited visualization, technical factors (e.g., motion artifact), or non-specific findings. Use when language includes "difficult to assess," "indeterminate," "nondiagnostic," "limited evaluation," or "cannot exclude." May also apply if a comparison study is not available and the radiologist expresses uncertainty in characterizing change. **Do not assign** this code if the impression makes no mention of the primary tumor at all, instead use Code 6 in those cases.

- **Code 6 - Not Mentioned** when the radiologist does not reference the cancer status of the metastatic site or lesion without any interpretive summary.
- **Curator Quality Review:**
 - Ensure each status assessment aligns with the correct metastatic site in `imaging_met_siteX`.
 - Do not infer cancer status if the report is silent, select code 6 only after confirming omission.
 - In cases where multiple sites are grouped or language is non-specific, curate conservatively (e.g., select the worst status if multiple are described together ambiguously).
- **Notes:**
 - This variable is repeated for each site up to the maximum described by `imaging_mets_count`.
 - Abstract only the current radiologic status. Do not incorporate historical context unless explicitly referenced as part of the current interpretation.

Overall Cancer Status Assessment

Variable: `imaging_ca_status_overall`

Type: Categorical (Dropdown)

Description: Overall Radiologic Impression of Cancer Status

Reference: n/a

Data Source: Imaging report (Impression section; Findings if needed)

Curation Directives:

- **Definition:** Captures the radiologist's overall assessment of cancer status across the primary site and all metastatic sites. This reflects the summary interpretation of disease activity or treatment response in the impression section. Use this variable to document the global status as described by the radiologist, rather than site-by-site abstraction.
- **Permitted Values / Codes:**
 - 1 = Stable Disease: No significant change in tumor size or number of lesions.
 - 2 = Progression (local): Indicates worsening of primary/local lymph site(s), but stable or no new distant metastatic site(s) / lesion(s).
 - 3 = Progression (distant): Indicates stable primary/local lymph, but worsening or new distant metastatic site(s) / lesion(s).
 - 4 = Overall Progression (local and distant): Indicates worsening of primary/local lymph site(s), and worsening or new distant metastatic site(s) / lesion(s).
 - 5 = Response (local): Indicates improvement of primary/local lymph site(s), but stable or worsening metastatic site(s) / lesion(s).
 - 6 = Response (distant): Indicates stable or worsening primary/local lymph, but improvement of distant metastatic site(s) / lesion(s).
 - 7 = Overall Response (local and distant): Indicates improvement of primary/local lymph site(s), and improvement of distant metastatic site(s) / lesion(s).
 - 8 = No Evidence of Disease: Used when previous scans showed cancer, but current scan shows no detectable disease.
 - 9 = Indeterminate: When the scan results are unclear or cannot be compared to previous scans.
 - 10 = Not Mentioned: Radiologist does not describe the cancer's status.
- **Source Documents:**
 - Radiology report, including the Findings section and the Impression section

- **Curation Instructions:**

1. Complete this field based on clear, overarching language from the Impression section that describes the patient's disease trajectory
2. Assign:
 - **Code 1 - Stable Disease** when the radiologist states that overall disease burden is unchanged, with no significant interval change in size, number, or character of any known lesions. Acceptable language: "overall stable disease," "no interval change," "findings remain stable," "no progression."
 - **Code 2 - Progression (local)** when the radiologist indicates worsening at the primary tumor and/or regional lymph nodes, but no new or worsening distant metastatic lesions. Example: "Enlargement of primary mass, no new distant disease." **Do not assign** if distant mets are also worsening, instead use Code 4.
 - **Code 3 - Progression (distant)** when the primary tumor and local lymph nodes are stable, but there is progression at distant metastatic sites. Example: "Stable primary tumor, new liver metastases identified." **Do not assign** if the primary tumor is also worsening, instead use Code 4.
 - **Code 4 - Overall Progression (local and distant)** when the radiologist states that there is worsening disease at both local (primary/lymph) and distant metastatic sites. Acceptable phrases: "overall disease progression," "interval increase in size of all lesions," "worsening both locally and distantly."
 - **Code 5 - Response (local)** when there is improvement in primary tumor and/or local lymph nodes, but stable or worsening disease at metastatic sites. Example: "Interval decrease in primary tumor, stable lung metastases." **Do not assign** if distant disease is also improving instead use Code 7.
 - **Code 6 - Response (distant)** when there is improvement in distant metastatic lesions, but the primary tumor and local lymph nodes are stable or worsening. Example: "Decreased size of liver metastases, stable primary tumor." **Do not assign** if primary or locoregional disease is also improving instead use Code 7.
 - **Code 7 - Overall Response (local and distant)** when the radiologist describes improvement at both the primary/local site(s) and distant metastases. Examples: "interval decrease in primary mass and hepatic lesions," "overall response across disease sites."
 - **Code 8 - No Evidence of Disease** when the radiologist states that previously seen cancer (local or metastatic) is no longer detectable. Examples: "no evidence of disease," "no residual tumor," "complete resolution of prior findings." Ensure this applies to all known sites.
 - **Code 9 - Indeterminate** when the overall cancer status cannot be determined due to technical limitations, equivocal findings, or lack of comparative study. Examples: "limited evaluation," "indeterminate study," "cannot exclude progression." **Do not assign** if the radiologist simply omits overall assessment, instead use Code 10.
 - **Code 10 - Not Mentioned** when the radiologist does not include a general or overall impression of disease status anywhere in the report.

- **Curator Quality Review:**

- Do not infer an overall status if only individual sites are described—only code based on explicit radiologist summary.

- **Notes:**

- This is a global summary variable; do not conflate with site-level variables (`imaging_ca_status_X`).
- If the report mentions multiple regions but lacks a unifying summary, and no synthesis is offered by the radiologist, use Code 10.

Clinical Visits

Preliminary Cancer Status Assessment

Curators are required to abstract clinical visit data from provider-authored notes available in the patient's medical record, including consultations, progress notes, and follow-up documentation from both internal and external institutions following patient's first cancer diagnosis.

If cancer status has been described as having changed, abstraction will then include key indicators of disease status (e.g., evidence of disease, response, progression), provider-described assessments of primary tumors, lymph nodes, and distant metastases. For all notes, performance status should be captured if documented. All information must be abstracted as explicitly stated in the clinical note; curators should not infer findings from prior or future documentation. Accurate and structured capture of provider assessments enables longitudinal evaluation of cancer trajectory, treatment response, and functional status across the care continuum.

Variable: visit_yn

Type: Binary (Yes/No)

Description: Indicates whether the patient has any recorded clinical visit notes in the EHR or abstracted documentation.

Reference: n/a

Data Source: Clinical visit documentation

Curation Directives:

- **Definition:** Record whether any clinical visit notes are available for the patient. A “Yes” response should be selected only if at least one visit note (e.g., progress note, consult, or oncology follow-up) is accessible and associated with the patient's cancer care timeline.
 - **Permitted Values:**
 - 1 = Yes
 - 0 = No
 - **Source Documents:**
 - Oncology progress notes
 - Consultation notes
 - Follow-up visit summaries
 - Multidisciplinary clinic notes
 - **Curation Instructions:**
 1. Search the EHR or source record for documented clinical visit notes.
 2. Include only visits related to the cancer diagnosis, treatment, or follow-up.
 3. If no clinical notes are available or they are unrelated to cancer care, select “No.”
 - **Curator Quality Review:**
 - Confirm that at least one clinical document is present and relevant to oncology.
-

Variable: visit_date

Type: Date (MM/DD/YYYY)

Description: Date of Clinic Visit

Reference: n/a

Data Source: Clinical visit documentation

- **Definition:** Captures the date of the patient's clinical visit.
- **Permitted Values / Codes:**
Free text date field (validated in MM/DD/YYYY format)
- **Source Documents:**
 - Oncology progress notes
 - Consultation notes
 - Follow-up visit summaries
 - Multidisciplinary clinic notes
- **Curation Instructions:**
 1. Record the date the of the clinical visit.
 2. Use the exact date found in the clinical visit note documentation, typically listed near the top.
 3. If the date is missing or ambiguous, leave the field blank and flag for review.
- **Curator Quality Review:**
 - Confirm that the recorded date matches the clinical visit documentation exactly.
 - Cross-check that the visit date aligns with the patient's cancer care timeline (e.g., diagnosis, treatment, follow-up).
 - If multiple dates are present, verify that the selected date corresponds to the actual clinical encounter, not the document creation or signature date.
- **Notes:**
 - If the visit note includes multiple dates (e.g., referral date, dictation date), prioritize the encounter date listed in the header or body of the note.
 - This variable is critical for sequencing patient care events and aligning clinical documentation with genomic data.

Variable: visit_date_int

Type: Integer (Calculated)

Description: Calculates the number of days between the patient's date of birth and the clinical visit date, providing a continuous time variable that enables age-based or temporal analysis across visits.

Reference: n/a

Data Source: Derived from birth_date and visit_date fields

Curation Directives:

- **Definition:** This field auto-calculates the interval (in days) between the patient's date of birth and the date of clinical visit, using REDCap's `datediff()` function:
`datediff([birth_date],[visit_date],"d","mdy",true)`
- **Permitted Values:**
 - Integer 0
 - Null if either birth_date or visit_date is missing or improperly formatted
- **Source Documents:**
 - Patient demographics (birth_date)

- Clinical visit documentation with clearly recorded `visit_date`

- **Curation Instructions:**

1. Ensure both `birth_date` and `visit_date` fields are entered and valid before relying on the output of this field.
2. Do not manually override this field—this is a calculated value.
3. Check for anomalous values (e.g., negative intervals or implausibly high day counts).

- **Curator Quality Review:**

- Validate that both contributing fields (`birth_date`, `visit_date`) are populated and correctly formatted (MM/DD/YYYY).
- Confirm that calculated intervals align with expected patient age and clinical context.

Variable: `visit_eod`

Type: Categorical (Dropdown)

Description: Does the clinical visit note indicate evidence of disease?

Reference: n/a

Data Source: Show only if `visit_yn = 1`

Curation Directives:

- **Definition:** Document whether the clinical visit note explicitly mentions evidence of disease, based on the provider’s report. This variable captures the provider’s assessment of cancer presence, including affirmations, denials, omissions, or uncertainty about disease status.
- **Permitted Values:**
 - **1** = The report states that there **is evidence of disease**
 - **2** = The report states that there **is no evidence of disease**
 - **3** = The report **does not mention evidence of disease**. Documentation does not address cancer status in the visit note.
 - **4** = The report is **indeterminate** regarding evidence of disease. Documentation does not clearly confirm or rule out active cancer, or clinical findings are inconclusive.
- **Source Documents:**
 - Clinical visit note (e.g., consult, progress note, or H&P)
 - Assessment & Plan or Impression section of the note
- **Curation Instructions:**
 1. Review the provider’s documentation for explicit statements regarding cancer presence or absence.
 2. **Assign:**
 - **Code 1** if the provider clearly documents evidence of active disease, such as: “Patient has active disease,” “Metastatic disease is present,” “Cancer is progressing,” “Recurrent disease,” or “Disease burden remains.” Acceptable terms include “known malignancy,” “residual cancer,” or “persistent disease.”
 - **Code 2** if the provider clearly states no evidence of disease, such as: “No evidence of disease,” “NED,” “In remission,” or “Patient is disease-free.” This includes post-treatment surveillance visits with no signs of recurrence.
 - **Code 3** if the note does not mention cancer status at all. This applies when no summary or language about malignancy status is provided.
 - **Code 4** if the documentation is indeterminate, such as: “Cannot rule out recurrence,” “Indeterminate findings,” “Awaiting scans,” “Surveillance visit,” or ambiguous impressions without firm conclusion.

- Do not infer disease status from treatment history, imaging, or labs unless the provider explicitly interprets them.
- If the note includes multiple findings, abstract based on the provider’s overall summary regarding disease status.

- **Curator Quality Review:**

- Confirm the selected code reflects the provider’s clinical judgment, not imaging or pathology alone.
- Cross-check language in the Impression, Assessment & Plan, and related sections.
- Escalate ambiguous or conflicting notes for adjudication.

Variable: visit_stable_yn

Type: Binary (Yes/No)

Description: Was the patient’s cancer status described as stable or unchanged from previous clinical visit?

Reference: n/a

Data Source: Clinical provider note (e.g., consultation, history & physical, or progress note)

Branching Logic: Show only if visit_eod = 1 (i.e., there is evidence of disease)

Curation Directives:

- **Definition:** Indicates whether the provider described the patient’s disease burden or cancer status as stable, unchanged, or without progression or response, relative to the previous visit or clinical assessment.
- **Permitted Values:**
 - 1 = **Yes**, The provider explicitly states that disease is stable, unchanged, or not worsening
 - 0 = **No**, The provider does not describe disease as stable, or describes it as responding, progressing, or changing
- **Source Documents:**
 - Clinical visit note: consult, progress note, or H&P
 - Assessment & Plan, Impression, or provider summary section
- **Curation Instructions:**
 - Review the provider’s language for explicit statements of cancer disease status.
 - * Assign **Yes (1)** if the provider clearly describes the disease as stable or unchanged since the last assessment or indeterminate.
 - Stable (e.g., “Stable disease,” “No interval change,” “Findings are unchanged,” “No new lesions,” or “Disease remains stable.”)
 - Indeterminate (e.g., “can’t tell,” “awaiting confirmation,” “uncertain significance”)
 - * Assign **No (0)** if the provider clearly describes the disease as progressing, responding or otherwise changing.
 - Progressing (e.g., “Disease progression,” “Increasing disease burden,” “New metastatic lesions,” “Worsening disease”)
 - Responding (e.g., “Partial response,” “Marked tumor shrinkage,” “Improved imaging findings,” “Tumor is decreasing”)
 - Changing (e.g., “Interval change noted,” “Mixed response,” “Evolving lesions,” “Indeterminate but possibly changing”)
 - This variable should reflect the overall provider impression, not radiology reports alone—only use radiology if directly referenced in the provider’s summary.
 - Do not infer stability based on unchanged treatment or labs unless the provider comments on disease status explicitly.

- **Curator Quality Review:**

- Confirm the assignment aligns with documented provider statements—not imaging or labs in isolation.

Variable: `ir_mentioned_yn`

Type: Boolean (Yes/No)

Description: Interventional radiology (IR) procedure mentioned in note

Reference: n/a

Data Source: Clinical Visit Note

Curation Directives:

- **Definition:** Indicates whether the clinical note mentions a referral to or completion of an IR procedure, including tumor-directed ablation or embolization therapies.
- **Permitted Values:**
 - 1 = Yes
 - 0 = No
- **Source Documents:**
 - Oncology clinic note
- **Curation Instructions:**
 1. Mark Yes if the note includes referral/recommendation, scheduling, completion, or post-procedure follow-up for an IR procedure such as:
 1. **Ablation:** cryoablation, radiofrequency ablation (RFA), microwave ablation (MWA), thermal/percutaneous/image-guided tumor ablation.
 2. **Embolization:** trans arterial embolization, bland/particle embolization, chemoembolization (TACE), radioembolization (Y-90).
 2. Mark No if the note does not mention IR procedures, or only mentions diagnostic imaging (CT/MRI/PET) without an intervention.
 3. Count both “planned/referred” and “completed” mentions; status will be captured in `ir_status`.
 4. Do not infer IR involvement from nonspecific language (for example: “local therapy,” “liver-directed therapy,” “procedure”) unless an IR-type intervention is explicitly described.
- **Curator Quality Review:**
 - Confirm the mention is about an interventional procedure (not diagnostic radiology).
 - If Yes, ensure `ir_status` is completed and consistent with note wording.
- **Notes:**
 - This variable functions as a gatekeeper for downstream IR detail fields

Interventional Radiology Information

Variable: ir_status

Type: Categorical (Radio)

Description: IR status mentioned in the note

Reference: n/a

Data Source: Clinical Visit Note

Curation Directives:

- **Definition:** Describes whether the note indicates that an IR procedure is planned/recommended versus completed.
- **Permitted Values:**
 - 1 = Referred for Procedure (planned/recommended)
 - 2 = Procedure Completed
 - 3 = Unknown
- **Source Documents:**
 - Oncology clinic note
- **Curation Instructions:**
 1. Complete this field only when ir_mentioned_yn = 1 (Yes).
 2. Select Referred for Procedure when the note indicates a recommendation, referral, order placed, consult requested, or scheduling for an IR intervention, without stating it has occurred.
 3. Select Procedure Completed when the note explicitly states the procedure occurred (for example: “underwent Y-90 last week,” “cryoablation performed on 1/15/2026,” “s/p TACE”).
 4. Select Unknown when IR is mentioned but the status is unclear (for example: “IR involved,” “consider ablation,” without a clear plan vs completion).
- **Curator Quality Review:**
 - Verify the status is supported by explicit note language (for example: “plan,” “will,” “referred” vs “s/p,” “underwent,” “completed”).
 - If status = Completed, ensure downstream fields (procedure type, date, intent) are evaluated for completion where documented.
- **Notes:**
 - Use the note’s strongest evidence: if both referral and completion are described, select Procedure Completed.

Variable: ir_proc_type

Type: Categorical (Checkbox; multi-select)

Description: IR procedure type(s) completed

Reference: n/a

Data Source: Clinical Visit Note

Curation Directives:

- **Definition:** Captures the type(s) of IR procedure reported as completed.
- **Permitted Values:**
 - 1 = Cryoablation
 - 2 = Radiofrequency ablation (RFA)
 - 3 = Microwave ablation (MWA)
 - 4 = Embolization

- 5 = Other
 - **Source Documents:**
 - Oncology clinic note
 - **Curation Instructions:**
 1. Complete this field only when `ir_status = 2` (Procedure Completed).
 2. Select all procedure types explicitly documented as completed in the note.
 3. Choose **Embolization** for TACE, Y-90 radioembolization, bland/particle embolization, transarterial embolization.
 4. Choose **Other** for procedures not represented (for example: irreversible electroporation, laser ablation, portal vein embolization) or when the procedure is clearly IR but does not fit listed categories.
 5. Do not infer procedure type from anatomic site alone (for example: “liver-directed therapy”) without explicit naming.
 - **Curator Quality Review:**
 - Ensure each selected type is explicitly supported by documentation.
 - If Other is selected, confirm the record clearly indicates an IR intervention (not surgery or radiation).
-

Variable: `ir_date`

Type: Date (MM/DD/YYYY)

Description: Date of Interventional Radiology Procedure

Reference: n/a

Data Source: Clinical visit documentation

- **Definition:** The date the IR procedure was performed, as documented in the clinical record.
- **Permitted Values / Codes:**

Free text date field (validated in MM/DD/YYYY format)
- **Source Documents:**
 - Oncology follow-up notes that document the procedure date.
- **Curation Instructions:**
 1. Complete this field only when `ir_status = 2` (Procedure Completed).
 2. Abstract the procedure performance date (not referral date, consult date, scheduling date, or imaging date).
 3. If multiple IR procedures are documented in the note and distinct dates are provided, record the date corresponding to the procedure most clearly tied to the “completed” event being abstracted in that note.
 4. If only a partial/approximate date is available (for example: “January 2026,” “last week”), apply the standardized Data Estimation Rules.
 5. Enter the date in MM-DD-YYYY format.
- **Curator Quality Review:**
 - Confirm the date aligns with a completed procedure (not a planned procedure).
 - Cross-check consistency with `ir_status` and `ir_proc_type`.
 - Verify estimated dates follow standardized rules.
- **Notes:**
 - This date supports temporal analyses (for example: time from diagnosis or treatment to local control interventions).

Variable: `ir_date_int`

Type: Integer (Calculated)

Description: Date of Interventional Procedure in Days from DOB (Interval)

Reference: n/a

Data Source: Derived from `birth_date` and `ir_date` fields

Curation Directives:

- **Definition:** This field auto-calculates the interval (in days) between the patient's date of birth and the date of ir procedure, using REDCap's `datediff()` function:
`datediff([birth_date],[ir_date],"d","mdy",true)`
- **Permitted Values:**
 - Integer 0
 - Null if either `birth_date` or `ir_date` is missing or improperly formatted
- **Source Documents:**
 - Clinical visit documentation with clearly recorded `ir_date`
- **Curation Instructions:**
 1. Ensure both `birth_date` and `ir_date` fields are entered and valid before relying on the output of this field.
 2. Do not manually override this field—this is a calculated value.
 3. Check for anomalous values (e.g., negative intervals or implausibly high day counts).
- **Curator Quality Review:**
 - Validate that both contributing fields (`birth_date`, `ir_date`) are populated and correctly formatted (MM/DD/YYYY).
 - Confirm that calculated intervals align with expected patient age and clinical context.

Variable: `ir_intent`

Type: Categorical (Radio)

Description: Intent of the IR procedure

Reference: n/a

Data Source: Clinical visit documentation

Curation Directives:

- **Definition:** Captures the clinician-stated intent of the completed IR procedure as curative/definitive versus palliative/symptom control.
- **Permitted Values:**
 - 1 = Curative / definitive
 - 2 = Palliative / symptom control
 - 3 = Unknown
- **Source Documents:**
 - Oncology clinic notes
- **Curation Instructions:**
 1. Complete this field only when `ir_status` = 2 (Procedure Completed).

2. Select Curative / definitive when documentation indicates local control/eradication of limited disease or definitive treatment intent (for example: “definitive ablation,” “curative intent,” “treat oligometastatic disease for local control”).
 3. Select Palliative / symptom control when documentation indicates symptom relief, debulking for palliation, bleeding control, pain relief, or treatment in the setting of diffuse/progressive disease without curative language.
 4. Select Unknown when intent is not stated or cannot be determined from the note.
Do not infer intent solely from procedure type (for example: embolization can be either).
- **Curator Quality Review:**
 - Confirm the selected intent is supported by explicit documentation (goal/indication statements).
 - If intent is ambiguous, use Unknown rather than inference.

Clinical Visit Information

Variable: visit_institution

Type: Categorical (dropdown)

Description: Records whether the documented clinical visit occurred at the participating GENIE institution or at an external (non-participating) institution.

Reference: n/a

Data Source: Clinical visit documentation

Curation Directives:

- **Definition:** Indicate the location of the clinical visit as either internal (within the GENIE-affiliated site) or external (at a non-participating site), based on the header, letterhead, or metadata associated with the visit note.⁰
- **Permitted Values:**
 - 1 = Internal Institution
 - 2 = External Institution
- **Source Documents:**
 - Clinical visit notes (progress notes, consultation notes, follow-up summaries)
 - Document metadata (e.g., author, institution)
 - Referral notes or scanned documents referencing other sites
- **Curation Instructions:**
 1. Review the visit note’s header, author credentials, and institutional affiliation.
 2. If the document was authored at the GENIE site or explicitly references an internal provider, select Internal Institution (1).
 3. If the visit occurred elsewhere (e.g., at a community oncology clinic or outside hospital), select External Institution (2).
- **Curator Quality Review:**
 - Verify that facility designation aligns with the author and document origin.
 - Cross-check with patient timeline and referral history to ensure consistency.
- **Notes:**

- This variable helps distinguish care received within the GENIE network versus external settings.
-

Variable: visit_provider

Type: Categorical (Dropdown)

Description: Records the type of provider seen by the patient during the clinical visit. Provider type reflects the clinical role responsible for the visit encounter.

Reference: n/a

Data Source: Clinical visit documentation, provider notes, appointment summaries

Curation Directives:

- **Definition:** Select the clinical specialty or role of the provider overseeing the patient visit. This classification helps contextualize the visit type and links the visit to the patient’s broader treatment plan.
 - **Permitted Values:**
 - 1: Surgical Oncologist
 - 2: Medical Oncologist
 - 3: Radiation Oncologist
 - **Source Documents:**
 - Clinical visit notes
 - Provider signatures or headers
 - Appointment summaries or EHR visit metadata
 - **Curation Instructions:**
 1. Review provider credentials and specialty listed in the clinical visit documentation.
 2. If the provider is clearly an oncologist with a defined specialty (e.g., medical, surgical, radiation), select the appropriate option.
 3. If multiple providers are involved, select the lead provider responsible for the visit documentation.
 - **Curator Quality Review:**
 - Confirm provider role matches supporting documentation.
 - **Notes:**
 - This variable supports stratified analyses by provider type.
-

Variable: visit_type

Type: Categorical (Dropdown)

Description: Records the type of clinical documentation or visit associated with the patient’s care, aligned to LOINC-coded categories to support standardized interpretation and data harmonization.

Reference: LOINC

Data Source: Clinical documentation headers, note titles, or structured metadata

Curation Directives:

- **Definition:** Identify the type of clinical note or visit associated with the entry, based on standard LOINC terminology. This classification supports interoperability and structured data analysis.
- **Permitted Values for Manual Curation:**
 - 1: Consultation Note (11488-4)

- 3: History & Physical Note (34117-2)
- 5: Progress Note (11506-3)
- 9: Other
- **[Do Not Use for Manual Curation] Additional Values to support programmatic collection:**
 - 2: Discharge Summary (18842-5)
 - 4: Procedures Note (28570-0)
 - 6: Imaging Narrative (18748-4)
 - 7: Laboratory Report Narrative (11502-2)
 - 8: Pathology Report Narrative (11526-1)
- **Source Documents:**
 - Note headers in EHR (e.g., “Progress Note”, “H&P”)
 - Document type metadata in clinical systems
- **Curation Instructions:**
 1. Identify the document type from the note header or file metadata.
 2. Select the LOINC-mapped category that best represents the document or visit type.
 3. If the document type does not align with the options provided, select “9: Other” and complete `visit_type_other`.
- **Curator Quality Review:**
 - Confirm that the selected category matches the documentation title or content.
 - Ensure “Other” is only selected if the document cannot reasonably be classified under any predefined LOINC categories.
- **Notes:**
 - LOINC codes included for potential OMOP or mCODE interoperability and future programmatic collection efforts.

Variable: `visit_type_other`

Type: Free Text Entry (text)

Description: Captures a free-text description of the clinical visit type or documentation when “Other” is selected in `visit_type`.

Reference: n/a

Data Source: Clinical documentation headers, provider notes, or uploaded documents

Curation Directives:

- **Definition:** Used when the clinical visit or documentation type does not align with LOINC-coded categories. Provides detail to support future categorization or ontology mapping.
- **Permitted Values:**
 - Free text
- **Source Documents:**
 - Note headers
 - Visit metadata
 - Clinic scheduling labels
- **Curation Instructions:**
 1. Only complete this field if `visit_type = 9: Other`.
 2. Transcribe or summarize the visit type or document label as it appears in the source.

3. Avoid generic terms; be specific and include role or visit intent if clear.

- **Curator Quality Review:**

- Ensure field is blank if `visit_type` is not “Other”.
- Check for spelling, clarity, and meaningful classification of the visit type.

- **Notes:**

- Use supports eventual harmonization of uncategorized documentation into new or custom domains.
- Can be used to flag novel or rare documentation types for system enhancement.

Primary Tumor Assessment

Variable: `visit_ca_yn`

Type: Binary (Yes/No)

Description: Medical Oncologist Assessment – Presence of Primary Cancer Site

Reference: n/a

Data Source: Clinical provider note (e.g., progress note, consultation, H&P)

Curation Directives:

- **Definition:** Captures whether the medical oncologist’s clinical note describes a primary cancer site as present or active at the time of assessment. This may reflect new diagnosis, residual disease, or recurrence of the tumor at its site of origin.

- **Permitted Values / Codes:**

- 0 = No, the clinical visit note does not describe any primary cancer site(s)
- 1 = Yes, the clinical visit note describes one or more primary cancer site(s)

- **Source Documents:**

- Medical oncologist’s progress note, consultation, or H&P
- Assessment & Plan, Impression, or Oncology Summary sections
- Problem list or disease status summary within the note

- **Curation Instructions:**

1. Review the oncologist’s note for explicit mention of a primary cancer site being present, persistent, or recurrent.
2. Assign:
 - **Code 1, “Yes”** if the provider documents active disease, residual tumor, recurrence, or ongoing treatment involving the primary tumor site. Phrases may include: “primary tumor remains,” “recurrent disease at the [site],” “residual disease in breast,” “ongoing disease at primary site,” or “active disease in prostate.”
 - **Code 0, “No”** if the provider does not mention the primary site; or the provider states the primary site has been resected or treated with no evidence of current disease; or only metastatic disease or nodal involvement is discussed with no indication of primary site disease.
3. Do not confuse metastatic sites or lymph nodes with primary tumors.
4. If multiple primaries exist, assign Code 1 if any primary site is described as present.

- **Curator Quality Review:**

- Confirm that the primary site aligns with the patient’s documented diagnosis and anatomic location.
- Verify that “primary” is based on tumor origin, not dominance of lesion or site of largest tumor burden.

Variable: visit_ca_count

Type: Categorical (Dropdown)

Description: Number of Primary Cancer Diagnoses Documented in Clinical Visit

Reference: n/a

Data Source: Clinical provider note (e.g., consultation, H&P, progress note)

Curation Directives:

- **Definition:** This variable captures the total number of distinct primary cancer diagnoses noted during the clinical visit. This count includes only *primary malignancies* — i.e., independent primary tumor sites and does not include metastatic lesions, lymph node involvement, or recurrences.
- **Permitted Values / Codes:**
 - 1 = One primary diagnosis
 - 2 = Two primary diagnoses
 - 3 = Three primary diagnoses
 - 4 = Four primary diagnoses
 - 5 = Five primary diagnoses
- **Source Documents:**
 - Clinical visit note (consultation, progress note, or history & physical)
 - Problem list or diagnosis summary within the visit documentation
- **Curation Instructions:**
 - Review the clinical note and count the number of anatomically and biologically distinct primary cancer diagnoses discussed during the visit.
 - Bilateral cancers (e.g., bilateral breast) count as two primaries only if clearly described as separate disease processes.
 - Do not count metastatic sites or nodal disease.
 - If the provider expresses uncertainty between metastasis and new primary, defer to their stated diagnostic impression.
 - If more than five distinct primary cancers are explicitly documented, select 5 and flag for manual review.
- **Curator Quality Review:**
 - Ensure consistency with clinic_yn; if clinic_yn = 0, this field must be blank or omitted.
- **Notes:**
 - This variable supports downstream mapping of cancer-specific fields (e.g., visit_primary_site1-5).

Variable: visit_primary_site1-5

Type: Categorical (Dropdown)

Description: Primary Cancer Site (ICD-O-3 Topography Code)

Reference: NAACCR #400

Data Source: Clinical provider note (e.g., Consultation, H&P, or Progress Note)

Curation Directives:

- **Definition:** The primary anatomical site of malignancy identified during the clinical visit, recorded using the ICD-O-3 topography code set. This field captures the clinician-documented location(s) of one or more primary cancers relevant to the visit.
- **Permitted Values / Codes:**
 - C00.0–C80.9 and site-specific ICD-O-3 topography codes as defined in the dropdown list. Include special fluid-based designations (F10–F50) when applicable.
- **Source Documents:**
 - Clinical visit note: Consultation, History & Physical, Progress Note
 - Diagnosis or Assessment section
- **Curation Instructions:**
 1. Review the provider’s note for documentation of active or historical primary cancers. Focus on the Assessment, History of Present Illness, and Plan sections.
 2. Assign the most specific ICD-O-3 topography code corresponding to the primary cancer site.
 3. Use one field per distinct primary site, following the value recorded in `visit_ca_count`.
 - Example: If `visit_ca_count` = 2, complete `clinic_primary_site1` and `clinic_primary_site2`.
 4. Code only primary sites do not include sites of recurrence, metastasis, or lymph node involvement.
 5. If the provider states the primary is unknown or uncertain, assign C80.9 (Unknown Primary Site).
 6. In cases where the provider references multiple primaries but does not clearly differentiate anatomic site, select the best available option based on context, and flag for review if necessary.
- **Curator Quality Review:**
 - Confirm that each selected ICD-O-3 code is explicitly supported by provider documentation.
 - Verify that the number of completed primary site fields aligns with `visit_ca_count`.
 - Ensure entries are distinct. Do not repeat the same site unless explicitly noted as bilateral, synchronous primaries.
 - Cross-check with prior records if referenced, but do not infer primary site unless documented during the current visit.
 - Flag ambiguous or incomplete documentation for clinical clarification or internal review.
- **Notes:**
 - This variable captures the clinician’s understanding and documentation of primary malignancy, which may differ from radiologic or pathologic interpretation.
 - This field informs longitudinal tracking of primary cancers and supports downstream abstraction for cancer site-specific variables.

Variable: `visit_laterality1-5`

Type: Categorical (Dropdown)

Description: Laterality of Primary Cancer Site

Reference: NAACCR #410

Data Source: Clinical provider note (Consultation, H&P, Progress Note)

Curation Directives:

- **Definition:** Laterality refers to the side of the body on which the primary cancer site is located, as documented in the clinical visit note. This variable is applicable only to paired organs or structures and captures whether the cancer is right-sided, left-sided, bilateral, midline, or not otherwise specified.
- **Permitted Values / Codes:**
 - 0 = Not a paired site

- 1 = Right: origin of primary
 - 2 = Left: origin of primary
 - 3 = Only one side involved, right or left origin unspecified
 - 4 = Bilateral involvement at time of diagnosis, lateral origin unknown for a single primary; or both ovaries involved simultaneously, single histology; bilateral retinoblastomas; bilateral Wilms tumors
 - 5 = Paired site: midline tumor (effective with 01/01/2010 dx)
 - 9 = Paired site, but no information concerning laterality
- **Source Documents:**
 - Clinical visit note (Consult, H&P, Progress Note)
 - Assessment or Impression section
 - Provider’s physical exam, imaging summary, or diagnostic plan
 - **Curation Instructions:**
 1. Determine whether the corresponding `clinic_primary_site` is a paired structure (e.g., breast, kidney, lung).
 2. If the site is not paired, assign code 0 = **Not a paired site**.
 3. If paired, review the note for statements indicating side-specific involvement.
 - Use code 1 or 2 when the provider specifies right or left.
 - Use code 3 when only one side is mentioned but the provider does not specify which.
 - Use code 4 if both sides are involved and side of origin is unclear, or if special cases apply (e.g., bilateral Wilms).
 - Use code 5 if tumor is described as midline (e.g., midline nasal septum, midline prostate).
 - Use code 9 if laterality is not mentioned and cannot be inferred.
 4. Base the code only on documentation from the current clinical visit. Do not infer from prior or future notes unless explicitly referenced.
 5. Complete one laterality field per primary site, in alignment with the number specified in `visit_ca_count`.
 - **Curator Quality Review:**
 - Ensure the laterality code is appropriate for a paired site; do not assign for unpaired structures.
 - Confirm that language in the note supports the selected laterality code.
 - **Notes:**
 - This variable complements `clinic_primary_site1-5` and enables detailed site-specific characterization of laterality.

Overall Cancer Status Assessment

Variable: `visit_ca_status_overall`

Type: Categorical (Radio)

Description: Physician’s overall assessment of cancer status, synthesizing findings from the primary tumor, regional lymph nodes, and distant metastatic sites.

Reference: n/a

Data Source: Clinical provider note (Consultation, H&P, Progress Note)

Curation Directives:

- **Definition:** Captures the provider’s overall clinical impression of the patient’s cancer status at the time of the visit, integrating information across all known disease sites (primary, regional, and distant) as described in the clinical note. This reflects the provider’s synthesis of symptoms, exam findings, treatment response, and any referenced imaging, pathology, or laboratory results.
- **Permitted Values / Codes:**
 - 1 = Progressing / Worsening (all sites of cancer)
 - 2 = Responding / Improving (all sites of cancer)
 - 3 = Mixed (some sites improving or stable, some sites worsening)
- **Source Documents:**
 - Oncology progress notes
 - History & physicals
 - Consult notes
 - Follow-up visit documentation
- **Curation Instructions:**
 1. Identify the provider’s global cancer status language. Prioritize statements such as:
 1. “progression,” “worsening,” “growing disease,” “increased burden,” “recurrent disease,” “new metastatic lesions”
 2. “response,” “improving,” “regressing,” “decreasing lesions,” “responding to therapy”
 3. “mixed response,” “some improved but others progressed,” “mixed picture”
 2. Assign codes as follows:
 1. **Code 1 (Progressing / Worsening):** Use when the provider’s overall assessment indicates the cancer is worsening without documentation of improvement in any sites. Examples: “overall progression,” “disease is progressing,” “worsening metastatic burden,” “new lesions consistent with progression.”
 1. If the note mentions “progression” at any site and does not also describe improvement elsewhere, code as 1.
 2. **Code 2 (Responding / Improving):** Use when the provider’s overall assessment indicates improvement/response and does not describe worsening at any site. Examples: “responding to treatment,” “overall improvement,” “tumor burden decreasing,” “partial response” described as the overall clinical impression.
 3. **Code 3 (Mixed):** Use when the provider documents a combination of directions across sites, such as one site improving/stable while another worsens, or explicitly uses “mixed response.” Examples: “mixed response,” “lung lesions improved but liver lesions progressed,” “some lesions smaller, others larger,” “oligoprogression with response elsewhere.”
- **Curator Quality Review:**
 - Confirm the selected category reflects the provider’s overall impression, not an isolated finding (for example: one lab change or symptom change without a disease assessment).
 - Ensure “Mixed” is used when there is both improvement/stability and worsening across sites, even if the provider does not explicitly say “mixed,” as long as both directions are documented.
 - Do not infer progression/response solely from a treatment change (for example: stopping therapy, switching regimens) unless the provider explicitly links the change to disease status.
 - Cross-check internal consistency with any site-specific narrative in the same note (for example: if a note states “new liver metastases” and “lung lesions shrinking,” “Mixed” should be selected).

Clinical Visit - Patient Status

Variable: visit_perf_status_type

Type: Categorical (Dropdown)

Description: Performance Status Scale Used in Clinical Note

Reference: n/a

Data Source: Clinical provider note (Consultation, H&P, Progress Note)

Curation Directives:

- **Definition:** Identifies the type of performance status scale (ECOG or KPS) used by the provider to assess the patient's functional ability during the clinical visit.
- **Permitted Values / Codes:**
 - 1 = ECOG
 - 2 = KPS
 - 3 = Not mentioned
- **Source Documents:**
 - Clinical provider note (Consult, Progress Note, H&P)
 - Assessment, Plan, or Physical Exam section
 - Performance status may be listed in a separate vitals table or narrative assessment
- **Curation Instructions:**
 - Review the full note to determine if the provider uses ECOG or KPS to document the patient's performance status.
 - If either ECOG or KPS is explicitly mentioned, assign the corresponding code.
 - If no performance status is documented, assign 3 = Not mentioned.
 - Do not infer scale type from description alone unless the format clearly matches one of the two scales.
 - This field governs display logic for visit_ecog and visit_kps.
- **Curator Quality Review:**
 - Confirm that the assigned code matches the scale name explicitly stated in the note.
 - Cross-check against performance status values elsewhere in the form to ensure consistency.
 - Avoid assigning a scale if the provider describes general function without using a formal scale.
- **Notes:**
 - This field is used to trigger downstream fields for ECOG or KPS value abstraction.
 - Required for clinical interpretation and harmonization of functional status across visits.

Variable: visit_ecog

Type: Categorical (Dropdown)

Description: ECOG Performance Status (Eastern Cooperative Oncology Group)

Reference: ECOG

Data Source: Clinical provider note (Consultation, H&P, Progress Note)

Curation Directives:

- **Definition:** Captures the ECOG performance status recorded during the clinical visit. ECOG is a 5-point scale that reflects the patient's level of functioning and ability to perform daily activities.
- **Permitted Values / Codes:**
 - 0 = Fully active, no restrictions
 - 1 = Restricted in strenuous activity, but ambulatory

- 2 = Capable of selfcare, but unable to work; up >50% of day
- 3 = Limited selfcare; confined to bed/chair >50% of day
- 4 = Completely disabled; bed/chair-bound
- **Source Documents:**
 - Clinical note sections: Assessment, Plan, or functional status
 - Often located in a performance status table or documented with ECOG label
- **Curation Instructions:**
 - Complete only if `visit_perf_status_type = 1` (ECOG)
 - Abstract the exact ECOG score reported by the provider.
 - Do not infer ECOG score from clinical status unless explicitly stated.
 - Assign value as coded in note; if multiple values are listed, use the most recent in the note.
 - Leave blank if performance status is not clearly documented or if another scale is used.
- **Curator Quality Review:**
 - Ensure ECOG value matches permitted code definitions.
 - Cross-check for consistency with functional descriptors in the same note.
 - Flag any discrepancies or ambiguous reporting in curator notes.
- **Notes:**
 - Widely used in oncology clinical trials and treatment planning.
 - Should not be completed if provider uses KPS or no scale is noted.

Variable: `visit_kps`

Type: Categorical (Dropdown)

Description: Karnofsky Performance Status

Reference: KPS Scale

Data Source: Clinical provider note (Consultation, H&P, Progress Note)

Curation Directives:

- **Definition:** Records the Karnofsky Performance Status (KPS) score used by the provider to describe the patient's ability to perform ordinary tasks and self-care. The KPS scale ranges from 100 (fully functional) to 10 (moribund).
- **Permitted Values / Codes:**
 - 100 = Normal, no complaints
 - 90 = Minor signs/symptoms, normal activity
 - 80 = Normal activity with effort
 - 70 = Cares for self, but no normal activity or work
 - 60 = Occasional assistance required
 - 50 = Considerable assistance, frequent care
 - 40 = Disabled, needs special care
 - 30 = Severely disabled, hospitalization indicated
 - 20 = Very sick, supportive treatment needed
 - 10 = Moribund, rapidly progressing fatal disease
- **Source Documents:**
 - Clinical visit note with performance scale table or score
 - Assessment or Plan section when KPS is referenced
- **Curation Instructions:**

- Complete only if `visit_perf_status_type` = 2 (KPS)
- Abstract the score exactly as documented.
- Do not infer or convert to ECOG or from clinical language unless clearly stated.
- Use most recent score in the note if multiple are provided.
- Leave blank if another performance scale is used or score is not specified.
- **Curator Quality Review:**
 - Confirm score aligns with standard KPS scale points.
 - Validate that note references KPS explicitly.
 - If KPS is not documented or appears nonstandard, do not assign a score.
- **Notes:**
 - KPS is typically used in specific cancer settings or legacy documentation.
 - Do not substitute ECOG or descriptive language for a KPS value.
 - Used for evaluating patient eligibility for certain treatments and trials.

Surgical Procedures

Surgical Procedures Information

Complete one instance of the Surgical Procedures form per discrete operative event (biopsy, excision/resection, re-excision, node dissection, etc.). Abstract primarily from the final surgical pathology report (including addenda when present). For each event, capture the procedure and pathology dates, performing institution, whether neoadjuvant therapy preceded the operation, the procedure category, margin status, and specimen-level details (topography, laterality, histology) and tumor/site-specific path elements. Match each pathology report to the correct procedure date, maintain plausible chronology with diagnosis and treatments, and do not infer unreported findings. If multiple procedures occur on different dates, create separate form instances.

Variable: `proc_yn`

Type: Binary (Yes/No)

Description: Surgical Procedures – Ever Performed

Reference: n/a

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:**

Indicates whether the patient has undergone any surgical procedure related to their cancer diagnosis at any time during their treatment course. Includes both curative and palliative surgeries, biopsies with therapeutic intent, and any procedures involving resection or excision of primary or metastatic disease.
- **Permitted Values / Codes:**
 - 0 = No
 - 1 = Yes
- **Source Documents:**

- Surgical operative reports
- Pathology reports indicating post-surgical specimen

- **Curation Instructions:**

1. Review all available clinical documentation for any reference to a surgical procedure performed in the context of cancer care.
2. Include procedures performed at external institutions if documentation is available in the record.
3. Include prior procedures even if they occurred before the reporting institution’s care, as long as they are documented.
4. Select “Yes” (1) if any eligible procedure is found; otherwise, select “No” (0).

- **Curator Quality Review:**

- Confirm procedure is explicitly documented in source material.
- Ensure procedures are related to cancer diagnosis, not unrelated surgeries (e.g., appendectomy, gallbladder removal unless relevant).

Variable: `proc_institution`

Type: Categorical (Dropdown)

Description: Institution at which the surgical procedure was performed

Reference: n/a

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Indicates whether the documented surgical procedure was performed at the GENIE-participating institution or an external facility. This variable helps distinguish institutional responsibility and documentation source quality.

- **Permitted Values / Codes:**

- 1 = Internal Institution (procedure performed at the GENIE site)
- 2 = External Institution (procedure performed outside the GENIE site)

- **Source Documents:**

- Surgical operative reports
- Pathology reports indicating post-surgical specimen

- **Curation Instructions:**

1. Identify the surgical procedure referenced in associated clinical documents.
2. Determine whether the procedure was performed at the reporting GENIE institution or at an outside facility.
3. Select “Internal Institution” (1) if the procedure occurred at the reporting site, typically indicated by detailed operative notes or institutional headers.
4. Select “External Institution” (2) if documentation suggests the procedure was performed elsewhere, or if only a summary of outside procedure is included.
5. If the location cannot be determined with available documentation, escalate for review.

- **Curator Quality Review:**

- Confirm institution using headers, provider signatures, or context within the document.
 - Ensure consistency with other variables such as `proc_date`, `proc_type`, and source document metadata.
-

Variable: `proc_date`

Type: Date (MM/DD/YYYY)

Description: Date on which the surgical procedure was performed

Reference: n/a

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Captures the calendar date on which the documented surgical procedure occurred, regardless of institution.
- **Permitted Values / Codes:**
 - Date entered in MM-DD-YYYY format
 - Leave blank only if the procedure date cannot be determined from any available documentation
- **Source Documents:**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions:**
 1. Review the primary source document that describes the surgical procedure (e.g., operative note).
 2. Extract the exact date the procedure was performed.
 3. If multiple dates are mentioned (e.g., consent, admission, procedure), use the actual procedure date.
 4. Enter the date using MM-DD-YYYY format.
- **Curator Quality Review:**
 - Confirm that the selected date matches the surgical description in the source document.
- **Notes:**
 - If multiple procedures occurred on different dates, each should be entered as a distinct entry (i.e., one form per procedure).
 - Accurate date capture is essential for aligning treatment timelines and assessing temporal relationships with imaging, labs, or recurrence.

Variable: `proc_date_int`

Type: Integer (Calculated)

Description: Interval in days between the patient's date of birth and the date the surgical procedure was performed

Reference: n/a

Data Source: Automatically calculated from `birth_date` and `proc_date` fields

Curation Directives:

- **Definition:** This calculated field represents the patient's age in days at the time of the surgical procedure. It is used to standardize timing across patients for downstream analytics and temporal alignment.
- **Permitted Values / Codes:**
 - Automatically populated by the system using: `datediff([birth_date], [proc_date], "d", "mdy", true)`
 - Value will be a non-negative integer if `birth_date` and `proc_date` are both valid
- **Source Documents:**

- Not directly abstracted—calculated from:
 - * **birth_date** (from demographics)
 - * **proc_date** (from surgical procedure documentation)
- **Curation Instructions:**
 1. Ensure both **birth_date** and **proc_date** are correctly entered and validated.
 2. This field should auto-populate upon saving the record.
 3. Do not manually modify this field or override its value.
- **Curator Quality Review:**
 - Confirm accuracy of source fields (**birth_date** and **proc_date**) if calculated value appears implausible (e.g., negative, extreme).
- **Notes:**
 - This variable is used to derive age-standardized metrics for analysis.

Variable: `path_report_date`

Type: Date (MM/DD/YYYY)

Description: Date on which the finalized pathology report was issued following a surgical procedure or biopsy

Reference: n/a

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** This field captures the official date of the final pathology report that documents pathological findings from a surgical specimen or biopsy. This is distinct from the date of the procedure itself or preliminary pathology notes.
- **Permitted Values / Codes:**
 - Date format: MM-DD-YYYY
- **Source Documents:**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions:**
 1. Locate the finalized pathology report associated with the procedure.
 2. Identify the report issuance or finalization date explicitly stated in the document header or footer.
 3. Do **not** use the date of the procedure or preliminary pathology notes unless explicitly marked as the final report date.
 4. Enter the date in MM-DD-YYYY format.
 5. Flag and resolve discrepancies if multiple dates are referenced—prefer the official final report signature/finalization date.
- **Curator Quality Review:**
 - Verify the selected report is final, not preliminary or addended.
 - Confirm that the date aligns with the timeline of the related surgical event.
- **Notes:**
 - This date is important for anchoring diagnosis confirmation and treatment initiation timelines.
 - If no finalized pathology report is available, leave this field blank

Variable: `path_report_date_int`

Type: Integer (Calculated)

Description: Number of days between the patient's date of birth and the date the final pathology report was issued

Reference: n/a

Data Source: Derived from `birth_date` and `path_report_date`

Curation Directives:

- **Definition:** This field represents the patient's age in days at the time the final pathology report was issued, calculated using their date of birth and the `path_report_date`. It is auto-populated by REDCap.
- **Source Documents:**
 - Not directly curated by the registrar—auto-calculated in the system
 - Based on:
 - * `birth_date` (abstracted elsewhere)
 - * `path_report_date` (manually curated by the registrar)
- **Curation Instructions:**
 1. Ensure that `birth_date` and `path_report_date` have been accurately entered.
 2. REDCap will automatically compute the day interval.
 3. Do not manually modify this field.
 4. If the value appears implausible (e.g., negative, or exceeding patient lifespan), re-check the source fields for errors.
- **Curator Quality Review:**
 - Confirm the input values (`birth_date`, `path_report_date`) are correctly formatted and appropriate.
 - Review outlier values that suggest possible data entry errors.
- **Notes:**
 - This interval aids in normalizing event timelines for comparative or temporal analyses across patients.

Variable: `proc_ca_type`

Type: Categorical (Dropdown)

Description: Captures the overall cancer type most relevant to the surgical procedure performed, corresponding to the primary cancer diagnosis.

Reference: n/a

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Indicates the type of cancer associated with the surgical procedure. The cancer type selected should reflect the disease site or diagnosis that prompted the surgery.
- **Permitted Values:**
 - 1 = Lung
 - 2 = Prostate
 - 3 = Stomach

- 4 = Colorectal
 - 5 = Liver
 - 6 = Melanoma
 - 7 = Bladder
 - 8 = Kidney
 - 9 = Breast
 - 10 = Ovarian
 - 11 = Pancreas
 - 12 = Brain / CNS
 - 13 = Chordoma
 - 88 = Other
- **Source Documents:**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions:**
 1. Review the surgical procedure documentation to determine the anatomical site and indication for surgery.
 2. Cross-reference with pathology findings and clinical history to confirm the cancer type.
 3. Select the cancer type that most closely aligns with the primary diagnosis targeted by the procedure.
 4. Use code **88 = Other** only if the cancer type is not listed; do not guess—refer to documentation.
 5. If multiple cancer types are addressed in one surgery, select the dominant or primary one as indicated by clinical context.
 - **Curator Quality Review:**
 - Confirm the selected cancer type matches the anatomical and diagnostic focus of the surgery.
 - Validate against tumor registry records or EHR diagnosis list, if available.
 - Check that code **88 = Other** is used only when truly warranted and not due to insufficient review.
 - **Notes:**
 - This variable supports disease-specific analyses of surgical intervention patterns and timing.
 - Required for linking procedures to cancer-specific timelines in downstream analytics.

Variable: `proc_ca_type_other`

Type: Free Text Entry (text)

Description: Captures a descriptive cancer type when “Other” is selected in the surgical procedure cancer type dropdown.

Reference: n/a

Data Source: Operative reports, pathology reports, oncology notes

Curation Directives:

- **Definition:**

Provides the specific cancer type when the dropdown variable `proc_ca_type` is coded as **88 = Other**. This allows curators to document procedures related to cancer types not listed among the predefined options.
- **Permitted Values:**
 - Free-text description of the cancer type (e.g., “Endometrial carcinoma,” “Thyroid cancer,” “Soft tissue sarcoma”)

- **Source Documents:**
 - Operative reports describing the anatomical site and intent of surgery
 - Final pathology report listing cancer diagnosis
 - Oncology or surgical consult notes confirming disease classification
- **Curation Instructions:**
 1. Only complete this field if `proc_ca_type` = 88 (“Other”) is selected.
 2. Review documentation for a clear statement of cancer type.
 3. Enter the most specific cancer type description available (e.g., “Thyroid papillary carcinoma” rather than just “thyroid”).
 4. Avoid abbreviations or vague entries such as “unspecified” or “unknown.”
 5. Use consistent terminology with other site-level classifications whenever possible.
- **Curator Quality Review:**
 - Ensure spelling and medical terminology are accurate and standardized.
 - Verify that the cancer type listed is not already included in the main dropdown list.
- **Notes:**
 - This variable enables flexibility for less common cancer types not captured in predefined categories.
 - Entries may be reviewed for potential updates to future controlled vocabulary lists.

Variable: `proc_neoadjuvant_yn`

Type: Binary (Yes/No)

Description: Indicates whether the patient received any form of neoadjuvant therapy (e.g., chemotherapy, hormone therapy, immunotherapy, or radiation) prior to the documented surgical procedure.

Reference: NAACCR Item #1632

Data Source: Oncology progress notes, surgical notes, multidisciplinary treatment plans, radiation oncology records

Curation Directives:

- **Definition:** Captures whether neoadjuvant (preoperative) therapy (either systemic or radiation) was administered with the intent to reduce tumor burden before the indexed surgical procedure.
- **Permitted Values:**
 - 0 = No
 - 1 = Yes
- **Source Documents:**
 - Oncology progress notes, surgical notes, multidisciplinary treatment plans, radiation oncology records
- **Curation Instructions:**
 - Code “Yes” if systemic therapy (e.g., chemotherapy, hormonal therapy, immunotherapy, targeted therapy) or radiation was given prior to surgery.
 - Treatment must occur before the documented procedure date.
 - Do not code as neoadjuvant if therapy was adjuvant, salvage, maintenance, or unrelated to the indexed cancer.
 - Accept partial or discontinued neoadjuvant treatment as “Yes” if initiated before surgery.
 - Default to “No” if documentation is unclear or absent unless supported by multiple sources.
- **Curator Quality Review:**
 - Verify therapy date precedes surgical procedure.

- Confirm preoperative intent, not postoperative or palliative care.
- Cross-check pathology reports for evidence of treatment effect (e.g., necrosis, fibrosis).

- **Notes:**

- Neoadjuvant status impacts surgical planning and interpretation of pathology findings.
- If “Yes,” ensure therapy details are documented elsewhere in systemic or radiation fields.

Variable: `proc_neoadjuvant_detail`

Type: Categorical (Dropdown)

Description: Specifies the category of neo-adjuvant therapy administered before the indexed surgical procedure.

Reference: Aligns with NAACCR #1632

Data Source: Oncology medication administration records, radiation-oncology summaries, multidisciplinary tumor-board notes, operative reports

Curation Directives:

- **Definition:** Capture the therapy type only when neo-adjuvant treatment was confirmed (i.e., when `proc_neoadjuvant_yn = 1`).
 - **1 — Systemic Therapy:** Chemotherapy, hormonal therapy, immunotherapy, or targeted agents.
 - **2 — Radiation:** External-beam or brachytherapy delivered pre-operatively.
 - **3 — Both:** Both systemic therapy **and** radiation given before surgery.
- **Source Documents**
 - Signed chemotherapy orders and administration logs
 - Radiation “End-of-Treatment” summary
 - Operative report pre-operative history section
 - Discharge summaries listing completed pre-surgical treatments
- **Curation Instructions**
 1. Confirm `proc_neoadjuvant_yn` is coded “Yes” (value = 1); if not, leave this field blank.
 2. Determine whether systemic, radiation, or both modalities occurred before the surgery date.
 3. Assign:
 - **Systemic Therapy** if only systemic agents were administered.
 - **Radiation** if only radiation was delivered.
 - **Both** if documentation shows use of both modalities prior to surgery.
- **Curator Quality Review**
 - Verify therapy dates precede surgery.
- **Notes**
 - If systemic therapy included concurrently administered radiation (e.g., chemoradiation), code as “Both.”
 - Exclude intra-operative radiation; it is considered part of the surgical event.

Variable: `path_type`

Type: Categorical (Check-Box)

Description: Records the pathology discipline that processed and examined the specimen associated with the indexed cancer diagnosis.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Select the single option that best represents the *primary* pathology specialty or method used for the specimen:
 1. **Cytology**
 2. **Surgical Pathology**
 3. **Hematopathology**
 4. **Flow Cytometry**
 5. **Dermatopathology**
 6. **Other** – choose only when none of the predefined categories apply.
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Open the final pathology (or cytology) report for the indexed procedure.
 2. Identify the laboratory section that issued the report.
 3. If multiple specialties issued *separate* reports, code the dominant diagnostic report (typically the one providing histologic diagnosis).
 4. If no category fits, select “Other” and populate **path_type_other**.
- **Curator Quality Review**
 - Confirm report accession numbers/specimen IDs align with the surgery date.
 - Ensure “Other” is justified and **path_type_other** is completed.
- **Notes**
 - Do not code ancillary techniques (IHC, molecular) here — capture only the overarching pathology discipline.
 - Cytology includes FNAs and exfoliative samples.
 - Flow cytometry for immunophenotyping hematologic malignancies should be coded “4.”
 - Dermatopathology applies chiefly to skin and soft-tissue specimens reviewed by a dermatopathologist.

Variable: **path_type_other**

Type: Free Text Entry (text)

Description: Describes the pathology specialty or method when “Other” is selected in **path_type**.

Reference: NAACCR Item #164

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Provide a concise, specific description (e.g., “Neuropathology,” “Molecular Pathology,” “Electron Microscopy”) when none of the predefined options adequately characterize the report type.
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Confirm **path_type** = 6 (**Other**) is selected; otherwise leave this field blank.
 2. Extract the specialty name exactly as documented (avoid uncommon abbreviations).
 3. Enter the description in sentence case without trailing punctuation.
 4. If multiple unconventional methods were used, list the primary method followed by “; see report” if clarity is needed.

5. Record the report accession number and service line in the curation log.

- **Curator Quality Review**

- Ensure free text exists only when “Other” is selected.
- Spell-check and standardize terminology across records.

- **Notes**

- Do not repeat a standard category; if one applies, correct **path_type**.
- Avoid vague phrases like “Special Pathology” — specify the discipline whenever possible.

Variable: `proc_type`

Type: Categorical (Check-Box)

Description: Identifies the surgical procedure performed to obtain a tissue sample or treat the cancer.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select the single option that best matches the documented procedure: Biopsy, Resection, Excision, Other.
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Locate the operative or procedure report for the indexed surgery.
 2. Determine the intent and extent of tissue removal:
 - *Biopsy* = diagnostic sampling (core needle, wedge, punch, incisional).
 - *Excision* = complete removal of a discrete lesion with minimal margins.
 - *Resection* = partial or total removal of an organ/segment with oncologic margins.
 3. Check the corresponding checkbox.
 4. If the documented procedure does not fit categories 1–3, select “**Other**” and complete **proc_type_other**.
 5. Record report title, date, and accession number in the curation log.
- **Curator Quality Review**
 - Confirm only **one** option is selected.
 - Cross-check procedure type with CPT/ICD-10-PCS codes when available.
 - Ensure “Other” is justified by narrative and that **proc_type_other** is populated.
- **Notes**
 - For multi-step surgeries (e.g., diagnostic biopsy followed by immediate resection), code the **definitive** procedure.
 - Exclude non-oncologic procedures unrelated to the cancer treatment plan.

Variable: `proc_type_other`

Type: Free Text Entry (text)

Description: Specifies the surgical procedure when “Other” is selected in **proc_type**.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Provide a concise, specific description of the procedure (e.g., “Debulking laparotomy,” “Endoscopic mucosal resection,” “Mohs micrographic surgery”) when none of the standard options apply.
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Confirm `proc_type = 4 (Other)` is selected; otherwise leave this field blank.
 2. Transcribe the procedure name exactly as documented (avoid unnecessary abbreviations).
 3. If multiple uncommon procedures were performed, list the primary oncologic procedure.
- **Curator Quality Review**
 - Ensure free text exists **only** when “Other” is selected.
- **Notes**
 - Avoid vague terms like “Surgery” or “Procedure”; specify the exact technique (e.g., “Laser ablation”).

Variable: `proc_type_lung`

Type: Categorical (Check-Box)

Description: Captures the specific lung-related diagnostic or therapeutic procedure(s) documented for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 - Sputum cytology
 - Bronchoscopy with biopsy $\pm$ TBNA
 - Image-guided transthoracic needle core biopsy / FNA
 - Thoracentesis
 - Mediastinoscopy
 - VATS or open surgical biopsy
 - EBUS-guided biopsy
 - EUS-guided biopsy
 - Navigational bronchoscopy
 - Robotic bronchoscopy
 - Wedge resection
 - Segmentectomy
 - VATS lobectomy
 - Lobectomy
 - Completion lobectomy
 - Sleeve lobectomy
 - Bilobectomy
 - Pneumonectomy
 - Major airway resection
 - Other – use when none of the above capture the procedure performed
 - Not specified – use only if documentation is silent or conflicting
- **Source Documents**

- Surgical operative reports
- Pathology reports indicating post-surgical specimen

- **Curation Instructions**

1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
2. If a procedure is performed that is **not** listed (e.g., *argon-plasma coagulation*), select “**Other**” and complete `proc_type_lung_other`.
3. If procedure type cannot be determined, select “**99 – Not specified.**”

- **Curator Quality Review**

- Ensure each selected option is supported by explicit documentation.
- Confirm no contradictory procedure codes are chosen.
- Validate that “Other” selections have corresponding free-text detail.

Variable: `proc_type_lung_other`

Type: Free Text Entry (text)

Description: Provides a detailed description of lung procedure(s) when “Other” is selected in `proc_type_lung`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is **not** represented by codes for `proc_type_lung`
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
- **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
- **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_lung` accordingly.

Variable: `proc_type_prostate`

Type: Categorical (Check-Box)

Description: Documents the specific prostate-related diagnostic or therapeutic procedure(s) performed for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 - **Systematic biopsy** (template/12-core, sextant, etc.)

- **Targeted biopsy** (MRI-fusion, cognitive, PET-fusion)
 - **Radical prostatectomy** (open, laparoscopic, or robotic)
 - **TURP** — Transurethral resection of the prostate
 - **Enucleation** (simple or subtotal prostatectomy)
 - **Other** – use when none of the above apply
 - **Not specified** – use only if procedure cannot be determined from documentation
 - **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_prostate_other`.
 3. If procedure type cannot be ascertained, select “99 – Not specified.”
 - **Curator Quality Review**
 - Ensure each selected option is supported by explicit documentation.
 - Confirm no contradictory procedure codes are chosen (e.g., both lobectomy and pneumonectomy for same side in one session).
 - Validate that “Other” selections have corresponding free-text detail.
-

Variable: `proc_type_prostate_other`

Type: Free Text Entry (text)

Description: Specifies prostate procedure(s) when “Other” is selected in `proc_type_prostate`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_prostate`.
 - **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 - **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
 - **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_prostate` accordingly.
-

Variable: `proc_type_stomach`

Type: Categorical (Check-Box)

Description: Records the specific stomach-related diagnostic or therapeutic procedure(s) documented for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 1. **Excisional biopsy** – includes EMR, ESD, polypectomy specimens
 2. **Endoscopic resection** – full-thickness or submucosal tunnel resections not captured in code 1
 3. **Laparoscopy with cytology**
 4. **Partial gastrectomy, proximal**
 5. **Partial gastrectomy, distal**
 6. **Partial gastrectomy, other** (e.g., sleeve, wedge)
 7. **Total gastrectomy**
 8. **Other** – use when none of the above apply
 9. **Not specified** – use only if procedure cannot be determined from documentation
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_stomach_other`
 3. If procedure type cannot be ascertained, select “99 – Not specified.”
- **Curator Quality Review**
 - Ensure each selected option is supported by explicit documentation.
 - Confirm no contradictory procedure codes are chosen.
 - Validate that “Other” selections have corresponding free-text detail.

Variable: `proc_type_stomach_other`

Type: Free Text Entry (text)

Description: Specifies stomach procedure(s) when “Other” is selected in `proc_type_stomach`.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_stomach`
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.
- **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
- **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_stomach` accordingly.

Variable: `proc_type_colorectal`

Type: Categorical (Check-Box)

Description: Captures the specific colorectal diagnostic or therapeutic procedure(s) performed for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select all documented procedures:
 1. **Excisional biopsy (polypectomy)**
 2. **Right hemicolectomy**
 3. **Transverse colectomy**
 4. **Lymphadenectomy** (isolated or en-bloc)
 5. **Proctoscopy**
 6. **Colonoscopy**
 7. **Sigmoidoscopy**
 8. **Transabdominal resection**
 9. **Transanal endoscopic microsurgery (TEM)**
 10. **Transanal minimally invasive surgery (TAMIS)**
 11. **Left hemicolectomy**
 12. **Sigmoidectomy**
 13. **Low anterior resection**
 14. **Total abdominal colectomy**
 15. **Abdominoperineal resection**
 16. **Transanal disk excision (local excision)**
 17. **Endoscopic mucosal resection (carcinoma-specific)**
 18. **Polypectomy (NET-specific)**
 19. **Other** – use when none of the above apply
 20. **Not specified** – use only if procedure cannot be determined
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_colorectal_other`
 3. If procedure type cannot be ascertained, select “**99 – Not specified.**”
- **Curator Quality Review**
 - Ensure each selected option is supported by explicit documentation.
 - Confirm no contradictory procedure codes are chosen.
 - Validate that “Other” selections have corresponding free-text detail.

Variable: `proc_type_colorectal_other`

Type: Free Text Entry (text)

Description: Describes colorectal procedure(s) when “Other” is selected in `proc_type_colorectal`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_colorectal`.
 - **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.
 - **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
 - **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_colorectal` accordingly.
-

Variable: `proc_type_liver`

Type: Categorical (Check-Box)

Description: Documents the specific liver-related diagnostic or therapeutic procedure(s) performed for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 1. **Percutaneous liver biopsy**
 2. **Laparoscopic liver biopsy**
 3. **Transjugular liver biopsy**
 4. **Wedge resection**
 5. **Partial hepatectomy, major** (≥ 3 segments)
 6. **Partial hepatectomy, minor** (< 3 segments)
 7. **Partial hepatectomy, NOS** (segments not specified)
 8. **Total hepatectomy**
 9. **Other** – use when none of the above apply
 10. **Not specified** – use only if procedure cannot be determined
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_liver_other`
 3. If procedure type cannot be ascertained, select “99 – Not specified.”
- **Curator Quality Review**
 - Ensure each selected option is supported by explicit documentation.
 - Confirm no contradictory procedure codes are chosen.
 - Validate that “Other” selections have corresponding free-text detail.

Variable: `proc_type_liver_other`

Type: Free Text Entry (text)

Description: Provides a detailed description of liver procedure(s) when “Other” is selected in `proc_type_liver`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_liver`.
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.
- **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
- **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_liver` accordingly.

Variable: `proc_type_melanoma`

Type: Categorical (Check-Box)

Description: Documents the specific melanoma-related diagnostic or therapeutic procedure(s) performed for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 1. **Biopsy – shave**
 2. **Biopsy – punch**
 3. **Biopsy – incisional**
 4. **Excision** (primary removal with intended margins)
 5. **Re-excision** (wider margins after prior biopsy/excision)
 6. **Sentinel node(s) biopsy**
 7. **Lymphadenectomy, regional nodes**
 8. **Other** – use when none of the above apply
 9. **Not specified** – use only if procedure cannot be determined
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen

- **Curation Instructions**

1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
2. If an unlisted procedure is performed, select “Other” and populate `proc_type_melanoma_other`
3. If procedure type cannot be ascertained, select “99 – Not specified.”

- **Curator Quality Review**

- Ensure each selected option is supported by explicit documentation.
- Confirm no contradictory procedure codes are chosen.
- Validate that “Other” selections have corresponding free-text detail.

Variable: `proc_type_melanoma_other`

Type: Free Text Entry (text)

Description: Specifies melanoma procedure(s) when “Other” is selected in `proc_type_melanoma`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_melanoma`
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.
- **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
- **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_melanoma` accordingly.

Variable: `proc_type_bladder`

Type: Categorical (Check-Box)

Description: Documents the specific melanoma-related diagnostic or therapeutic procedure(s) performed for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 1. **Biopsy**
 2. **TURBT** – Transurethral resection of bladder tumor
 3. **TURP** – Transurethral resection of the prostate
 4. **TUR of urethral tumor**

5. **Partial cystectomy**
 6. **Radical cystectomy** (total cystectomy)
 7. **Radical cystoprostatectomy**
 8. **Radical nephroureterectomy with cuff of bladder**
 9. **Anterior exenteration**
 10. **Other** – use when none of the above apply
 11. **Not specified** – use only if procedure cannot be determined
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_bladder_other`
 3. If procedure type cannot be ascertained, select “99 – Not specified.”
 - **Curator Quality Review**
 - Ensure each selected option is supported by explicit documentation.
 - Confirm no contradictory procedure codes are chosen.
 - Validate that “Other” selections have corresponding free-text detail.

Variable: `proc_type_bladder_other`

Type: Free Text Entry (text)

Description: Provides a detailed description of bladder procedure(s) when “Other” is selected in `proc_type_bladder`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_bladder`.
 - **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.
 - **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
 - **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_bladder` accordingly.
-

Variable: `proc_type_kidney`

Type: Categorical (Check-Box)

Description: Documents the specific melanoma-related diagnostic or therapeutic procedure(s) performed for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 1. **Needle biopsy**
 2. **Incisional biopsy, wedge**
 3. **Partial Nephrectomy**
 4. **Total Nephrectomy**
 5. **Radical Nephrectomy**
 6. **Adrenalectomy**
 7. **Other** – use when none of the above apply
 8. **Not specified** – use only if procedure cannot be determined
 - **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_kidney_other`
 3. If procedure type cannot be ascertained, select “**99 – Not specified.**”
 - **Curator Quality Review**
 - Ensure each selected option is supported by explicit documentation.
 - Confirm no contradictory procedure codes are chosen.
 - Validate that “Other” selections have corresponding free-text detail.
-

Variable: `proc_type_kidney_other`

Type: Free Text Entry (text)

Description: Provides a detailed description of kidney procedure(s) when “Other” is selected in `proc_type_kidney`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_kidney`.
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.

- **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
 - **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_kidney` accordingly.
-

Variable: `proc_type_breast`

Type: Categorical (Check-Box)

Description: Documents the specific breast-related diagnostic or therapeutic procedure(s) performed for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 1. **Needle biopsy** (core needle, vacuum-assisted, stereotactic, ultrasound/MRI-guided)
 2. **Fine-needle aspiration** (FNA)
 3. **Excision** (lumpectomy / wide local excision; anything less than total mastectomy)
 4. **Total mastectomy** (simple, skin-sparing, or nipple-sparing mastectomy)
 5. **Other** – use when none of the above apply
 6. **Not specified** – use only if procedure cannot be determined from documentation
 - **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_breast_other`
 3. If procedure type cannot be ascertained, select “**99 – Not specified.**”
 - **Curator Quality Review**
 - Ensure each selected option is supported by explicit documentation.
 - Confirm no contradictory procedure codes are chosen.
 - Validate that “Other” selections have corresponding free-text detail.
-

Variable: `proc_type_breast_other`

Type: Free Text Entry (text)

Description: Specifies breast procedure(s) when “Other” is selected in `proc_type_breast`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_breast`.
- **Source Documents**
 - Surgical operative reports

- Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.
 - **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
 - **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_breast` accordingly.
-

Variable: `proc_type_ovarian`

Type: Categorical (Check-Box)

Description: Captures the specific ovarian / gynecologic-oncology diagnostic or therapeutic procedure(s) documented for the indexed encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 1. Needle biopsy
 2. Fine-needle aspiration (FNA)
 3. Total hysterectomy + bilateral salpingo-oophorectomy (TH-BSO)
 4. Radical hysterectomy
 5. Simple hysterectomy
 6. Supracervical (subtotal) hysterectomy
 7. Bilateral salpingo-oophorectomy (BSO)
 8. Right salpingo-oophorectomy
 9. Left salpingo-oophorectomy
 10. Salpingo-oophorectomy, side NOS
 11. Right oophorectomy
 12. Left oophorectomy
 13. Oophorectomy, side NOS
 14. Bilateral salpingectomy
 15. Right salpingectomy
 16. Left salpingectomy
 17. Salpingectomy, side NOS
 18. Omentectomy
 19. Peritoneal biopsies
 20. Peritoneal tumor debulking / cytoreduction
 21. Peritoneal washing / lavage
 22. Pleurocentesis (pleural fluid)
 23. Other – use when none of the above apply
 24. Not specified – use only if procedure cannot be determined
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**

1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
2. If an unlisted procedure is performed, select “Other” and populate `proc_type_ovarian_other`
3. If procedure type cannot be ascertained, select “99 – Not specified.”

- **Curator Quality Review**

- Ensure each selected option is supported by explicit documentation.
- Confirm no contradictory procedure codes are chosen.
- Validate that “Other” selections have corresponding free-text detail.

Variable: `proc_type_ovarian_other`

Type: Free Text Entry (text)

Description: Specifies ovarian procedure(s) when “Other” is selected in `proc_type_ovarian`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_ovarian`.
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.
- **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
- **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_ovarian` accordingly.

Variable: `proc_type_pancreas`

Type: Categorical (Check-Box)

Description: Records the specific pancreas-related diagnostic or therapeutic procedure(s) performed for the indexed encounter.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select all documented procedures:
 1. **Excisional biopsy / Enucleation**
 2. **Staging laparoscopy**
 3. **Pancreaticoduodenectomy (Whipple, partial pancreatectomy of head)**
 4. **Total pancreatectomy**
 5. **Partial pancreatectomy – body** (e.g., central pancreatectomy)
 6. **Partial pancreatectomy – tail**

7. **Distal pancreatectomy** (body $\pm$ tail)
 8. **Segmental pancreatectomy – body**
 9. **Other** – use when none of the above apply
 10. **Not specified** – use only if procedure cannot be determined
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_pancreas_other`
 3. If procedure type cannot be ascertained, select “99 – Not specified.”
 - **Curator Quality Review**
 - Verify each selected code is backed by explicit source evidence.
 - Ensure mutually exclusive codes (e.g., total and distal pancreatectomy in one stage) aren’t both selected unless staged procedures occurred.
 - Confirm “Other” selections have corresponding free-text detail.
-

Variable: `proc_type_pancreas_other`

Type: Free Text Entry (text)

Description: Provides a detailed description of pancreas procedure(s) when “Other” is selected in `proc_type_pancreas`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_pancreas`
 - **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.
 - **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
 - **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_pancreas` accordingly.
-

Variable: `proc_type_cns`

Type: Categorical (Check-Box)

Description: Captures the specific central-nervous-system (CNS) diagnostic or therapeutic procedure(s) performed for the indexed encounter.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** documented procedures:
 1. **Open biopsy** (craniotomy/craniectomy with limited tissue sampling)
 2. **Stereotactic biopsy** (frame-based or frameless needle biopsy)
 3. **Resection** (gross-total or subtotal removal of the lesion)
 4. **Other** – use when none of the above apply
 5. **Not specified** – use only if procedure cannot be determined
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_cns_other`.
 3. If procedure type cannot be ascertained, select “99 – Not specified.”
- **Curator Quality Review**
 - Ensure each selected option is supported by explicit documentation.
 - Confirm no contradictory procedure codes are chosen.
 - Validate that “Other” selections have corresponding free-text detail.

Variable: `proc_type_cns_other`

Type: Free Text Entry (text)

Description: Specifies CNS procedure(s) when “Other” is selected in `proc_type_cns`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_cns`.
 - **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
 - **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
 3. Check spelling and consistency with institutional terminology.
 - **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
 - **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_cns` accordingly.
-

Variable: `proc_type_chordoma`

Type: Categorical (Check-Box)

Description: Documents the specific chordoma-related diagnostic or therapeutic procedure(s) performed for the indexed surgical encounter.

Reference: Aligns with NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – select **all** procedures explicitly documented:
 - **1 = Open biopsy**
 - **2 = Stereotactic / image-guided biopsy**
 - **3 = Excisional biopsy**
 - **4 = Curettage (specimen obtained)**
 - **5 = Intralesional resection / Debulking**
 - **6 = Marginal resection**
 - **7 = Segmental / Wide resection (en bloc)**
 - **8 = Radical resection**
 - **9 = Skull base resection (EEA / TSS / Craniotomy)**
 - **10 = Sacrectomy (partial/high/total)**
 - **11 = Sacrococcygeal amputation / Coccygectomy**
 - **12 = Vertebrectomy / Spondylectomy**
 - **13 = Laminectomy / Laminotomy (if tumor specimen sent)**
 - **14 = Other**
 - **99 = Not specified**
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Check every box that corresponds to a documented procedure; multiple selections are permissible.
 2. If an unlisted procedure is performed, select “Other” and populate `proc_type_chordoma_other`.
 3. If procedure type cannot be ascertained, select “**99 – Not specified.**”
- **Curator Quality Review**
 - Ensure each selected option is supported by explicit documentation.
 - Confirm no contradictory procedure codes are chosen.
 - Validate that “Other” selections have corresponding free-text detail.

Variable: `proc_type_chordoma_other`

Type: Free Text Entry (text)

Description: Specifies Chordoma procedure(s) when “Other” is selected in `proc_type_chordoma`.

Reference: NAACCR #1640

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Enter a concise, specific procedure name if patient underwent a procedure that is not represented by codes for `proc_type_chordoma`
- **Source Documents**
 - Surgical operative reports

- Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Transcribe the procedure name exactly as documented; avoid uncommon abbreviations.
 2. If multiple uncommon procedures were performed, list each separated by semicolons.
- **Curator Quality Review**
 - Check spelling and consistency with institutional terminology.
- **Notes**
 - Do not restate a standard option; if a listed code applies, adjust `proc_type_chordoma` accordingly.

Variable: `proc_margins`

Type: Categorical (Dropdown)

Description: Final surgical-margin status after resection of the primary tumor; used for staging, quality metrics, and recurrence risk assessment.

Reference: NAACCR Item #1320

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition** – choose the single code that best reflects the margin status documented in the *final* pathology report:
 - **0** — **No residual tumor** (R0; all margins negative).
 - **1** — **Residual tumor, NOS** (positive but not specified as microscopic vs macroscopic).
 - **2** — **Microscopic residual tumor** (R1; tumor cells present on histologic evaluation).
 - **3** — **Macroscopic residual tumor** (R2; gross residual disease noted by surgeon or pathologist).
 - **4** — **Margins not evaluable** (specimen inked/fragmented or margin status cannot be assessed).
 - **5** — **No primary site surgery** (diagnostic biopsy only or non-surgical management).
 - **6** — **Unknown or not applicable** (margin status absent from documentation).
- **Source Documents**
 - Final pathology report indicating post-surgical specimen
- **Curation Instructions**
 1. Locate the **final pathology report** for the indexed surgical specimen.
 2. Identify margin wording (e.g., “all margins negative,” “tumor present at inferior margin,” “cannot be assessed”).
 3. Map the wording to the corresponding code (0–4).
 4. If **no resection** of the primary tumor was performed, select **code 5**.
 5. If documentation is silent or conflicting, select **code 6** and note the discrepancy in the curation log.
- **Curator Quality Review**
 - Confirm code selection matches explicit language in the pathology report.
 - Cross-check with operative note to ensure resection actually occurred when codes 0–4 are used.
 - Ensure “Not evaluable” (4) is reserved for clearly stated technical limitations, not missing data.
- **Notes**
 - Do **not** infer margin status from clinical impression; rely on pathology report.
 - For **piecemeal excisions** or endoscopic resections without orientable margins, code **4**.
 - If multiple specimens, base coding on the **definitive** resection of the primary tumor.

Cancer Status - Pathology

Variable: `proc_specimen_num`

Type: Dropdown (single select; integers 1 – 30)

Description: Total count of distinct surgical specimens submitted to pathology during the indexed procedure, as documented in the final pathology report.

Reference: n/a

Data Source: Final pathology report (gross description or “Specimens Received” section), operative note “Specimens sent,” pathology accession list

Curation Directives:

- **Definition**

Select the number (1-30) that equals the **total specimens** submitted for histologic evaluation from the procedure. A specimen is any tissue piece containerized and accessioned separately (e.g., “Specimen A: Right colon,” “Specimen B: Omentum”).

- **Source Documents**

- Pathology report header or gross description enumerating specimens
- Operative report detailing specimens sent to pathology

- **Curation Instructions**

1. Open the **final pathology report** for the procedure.
2. Count each uniquely labeled specimen/container. Ignore subsamples (e.g., “A1, A2”) within a single specimen.
3. Select the matching integer (1-30) from the dropdown.
4. If the specimen count cannot be determined from pathology or operative documentation, escalate to the QA manager and leave field blank pending clarification.

- **Curator Quality Review**

- Verify the selected number matches the specimen enumeration in the pathology report.
- Ensure intraoperative frozen sections or cytology specimens are included **only** if they were accessioned as separate specimens.
- Cross-check with operative note to detect any missed specimens.

- **Notes**

- Do **not** count non-tissue samples like blood cultures or lab tests.
- If bilateral or multi-site specimens are submitted in one container and accessioned as one specimen, count it as **one**.
- For staged or multiple procedures on the same day, record specimens **per procedure record**, not cumulatively across the day.

Variable: `proc_spec1-30_topography`

Type: Dropdown (single select; ICD-O-3 site codes)

Description: ICD-O-3 topography code designating the anatomic site of **Specimens 1-30** as documented in the pathology report.

Reference: NAACCR Item #400

Data Source: Final pathology report “Specimens Received”/“Gross Description,” surgeon’s operative note

Curation Directives:

- **Definition**
Select the ICD-O-3 code that best matches the primary anatomic location of the specimen. Use the dropdown’s pre-mapped list (e.g., *C18.7 Sigmoid colon*, *C50.4 Upper-outer quadrant of breast*, etc.).
- **Source Documents**
 - Pathology report header or gross description enumerating specimens
 - Operative report detailing specimens sent to pathology
- **Curation Instructions**
 1. Identify the sequential specimen letter/number for **Specimen 1-30**.
 2. Locate the documented anatomic descriptor for that specimen.
 3. Match the descriptor to the corresponding ICD-O-3 code in the dropdown and select it.
 4. If no code reasonably matches, escalate to the QA manager; **do not** choose an approximate code.
- **Curator Quality Review**
 - Ensure selected code aligns with verbatim wording in the report.
 - Cross-check that laterality (if paired site) is compatible with the chosen site.
- **Notes**
 - For ambiguous site language (e.g., “cecal/ascending colon”), prefer the *Overlapping lesion* code when provided.

Variable: `proc_spec1-30_laterality`

Type: Categorical (Dropdown)

Description: Laterality value for Specimen 1-30 indicating right, left, midline, bilateral, etc., of a paired site.

Reference: NAACCR Item #410

Data Source: Final pathology report “Specimens Received”/“Gross Description,” surgeon’s operative note.

Curation Directives:

- **Definition**
 - Choose one of the standard laterality codes:
 - * 0 - Not a paired site
 - * 1 - Right
 - * 2 - Left
 - * 3 - Only one side involved—side unknown
 - * 4 - Bilateral
 - * 5 - Midline
 - * 9 - Paired site, laterality unknown.
- **Source Documents**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions**
 1. Determine if the coded topography is a paired site.
 2. Extract laterality language for Specimen 1-30.
 3. Select the appropriate code; if the site is not paired, choose 0.
- **Curator Quality Review**
 - Midline only applies to paired organs where midline tissue exists (e.g., prostate = 5).

Variable: proc_spec1-30_histology

Type: Categorical (Dropdown)

Description: ICD-O-3 morphology (histology / behavior) code assigned to Specimen 1-30 per pathology diagnosis.

Reference: NAACCR Items #522 (Histologic type) & #523 (Behavior)

Data Source: Final pathology report “Specimens Received”/“Gross Description,” surgeon’s operative note.

Curation Directives:

- **Definition:** Capture the most specific histologic term for the specimen’s primary diagnosis (e.g., *8140/3 Adenocarcinoma, NOS*).
- **Source Documents**
 - Pathology report header or gross description enumerating specimens
 - Operative report detailing specimens sent to pathology
- **Curation Instructions**
 1. Locate the Definitive Diagnosis section for Specimen 1-30.
 2. Note the ICD-O-3 morphology code or the verbal description; map to the dropdown choice.
 3. If the exact code is absent from the list, escalate for code addition. Do **not** substitute.
- **Curator Quality Review**
 - Cross-verify morphology is appropriate for the recorded topography (e.g., urothelial carcinoma should pair with urinary tract sites).
- **Notes**
 - Use the invasive component’s code when both in situ and invasive are present.
 - For specimens showing only benign tissue (e.g., “benign gastric mucosa”), choose the benign or “no tumor” code provided.

Site Specific Data Items

Variable: lung_invasion

Type: Categorical (Dropdown)

Description: Degree of tumor invasion into visceral or parietal pleura for primary lung cancers

Reference: NAACCR Item #3937

Data Source: Final lung resection pathology report (synoptic & narrative), surgeon’s operative note

Curation Directives:

- **Definition:** Capture the greatest pleural depth reached by the tumor, as determined histologically on the surgical specimen.
- **Permitted Values / Codes**
 - **0 – PL0** No visceral pleural invasion (tumor does not traverse elastic layer)
 - **4 – PL1 or PL2** Visceral pleural invasion present, NOS
 - * *PL1*: penetrates elastic layer
 - * *PL2*: reaches pleural surface
 - **5 – PL3** Tumor invades parietal pleura **or** chest wall

- **97** “Extends to pleura” stated, visceral vs parietal not specified
- **98** Not applicable — SSDI not collected for this case
- **99** Unknown/not assessed (no resection or margin evaluation)
- **Source Documents**
 - Pathology synoptic “Pleural Invasion (PL)” line
 - Microscopic description referencing elastic layer or pleural surface
 - Surgeon’s operative note (only if pathology absent/indeterminate)
- **Curation Instructions**
 1. Open the **final pathology report** for the lung resection.
 2. Locate any explicit PL code or descriptive statement of pleural invasion.
 3. Select the corresponding dropdown value from the list above.
 4. If statement is vague (“tumor reaches pleura”) and side not defined, code **97**.
 5. If no resection or no assessment, code **99**.
- **Curator Quality Review**
 - Verify chosen code matches exact wording (PL0-PL3) or equivalent description.
- **Notes**
 - Do **not** infer pleural invasion from radiology or clinical staging.
 - Tumors originating in pleura or metastatic deposits are exempt, code as 98.

Variable: prostate_gleason_patterns

Type: Categorical (Dropdown)

Description: Pathologic Gleason primary and secondary patterns from radical prostatectomy or autopsy (not from biopsy).

Reference: NAACCR Item #3839

Data Source: Final prostatectomy/autopsy pathology report (synoptic or narrative), pathology addenda

Curation Directives:

- **Definition:** Record the primary (most prevalent) and secondary (second most prevalent) Gleason architectural patterns for the entire prostatectomy/autopsy specimen as stated by pathology. Do **not** use biopsy information.
- **Permitted Values / Codes**
 - **11** Primary 1, Secondary 1
 - **12** Primary 1, Secondary 2
 - **13** Primary 1, Secondary 3
 - **14** Primary 1, Secondary 4
 - **15** Primary 1, Secondary 5
 - **19** Primary 1, Secondary **unknown**
 - **21** Primary 2, Secondary 1
 - **22** Primary 2, Secondary 2
 - **23** Primary 2, Secondary 3
 - **24** Primary 2, Secondary 4
 - **25** Primary 2, Secondary 5
 - **29** Primary 2, Secondary **unknown**
 - **31** Primary 3, Secondary 1
 - **32** Primary 3, Secondary 2
 - **33** Primary 3, Secondary 3
 - **34** Primary 3, Secondary 4

- **35** Primary 3, Secondary 5
 - **39** Primary 3, Secondary **unknown**
 - **41** Primary 4, Secondary 1
 - **42** Primary 4, Secondary 2
 - **43** Primary 4, Secondary 3
 - **44** Primary 4, Secondary 4
 - **45** Primary 4, Secondary 5
 - **49** Primary 4, Secondary **unknown**
 - **51** Primary 5, Secondary 1
 - **52** Primary 5, Secondary 2
 - **53** Primary 5, Secondary 3
 - **54** Primary 5, Secondary 4
 - **55** Primary 5, Secondary 5
 - **59** Primary 5, Secondary **unknown**
 - **96** Prostatectomy done; **primary unknown, secondary unknown**
 - **97** No radical prostatectomy/autopsy performed
 - **98** Not applicable — information not collected for this case
 - **99** Not documented / not assessed **or** unknown if radical prostatectomy done
- **Source Documents**
 - Pathology synoptic under “Gleason Score/Patterns (Prostatectomy)”
 - Narrative diagnosis or “Comment” noting primary/secondary patterns
 - **Curation Instructions**
 1. Open the final prostatectomy pathology report.
 2. Abstract the stated primary and secondary Gleason patterns for the whole specimen (dominant tumor if specified).
 3. Select the exact code from the permitted list above.
 4. If prostatectomy performed but patterns are **both** unstated, code **96**.
 5. If no prostatectomy/autopsy, code **97**. If documentation is absent/unclear, code **99**.
 - **Curator Quality Review**
 - Confirm the selected combination matches the pathology statement (e.g., “Gleason 3+4”).
 - If a tertiary pattern is reported, do **not** change this field, instead capture it in the tertiary-pattern variable.

Variable: prostate_gleason_score

Type: Categorical (Dropdown)

Description: Pathologic Gleason score (sum of primary + secondary patterns) from radical prostatectomy/autopsy.

Reference: NAACCR Item #3841

Data Source: Final prostatectomy pathology report (synoptic or diagnosis line)

Curation Directives:

- **Definition:** Record the sum of the pathologic primary and secondary Gleason patterns (e.g., 3+4=7). Do not use biopsy scores; do not include any tertiary pattern in the sum.
- **Permitted Values / Codes**
 - **02** Score 2
 - **03** Score 3
 - **04** Score 4
 - **05** Score 5

- **06** Score 6
- **07** Score 7
- **08** Score 8
- **09** Score 9
- **10** Score 10
- **97** No radical prostatectomy/autopsy performed
- **98** Not applicable — information not collected for this case
- **99** Not documented / not assessed **or** unknown if radical prostatectomy done
- **Source Documents**
 - Pathology synoptic “Gleason Score (Prostatectomy)”
 - Final diagnosis text (e.g., “Gleason score 4+3=7”)
- **Curation Instructions**
 1. From the prostatectomy/autopsy report, locate the Gleason score statement.
 2. Select the matching two-digit code above.
 3. If only patterns are given (e.g., 3+4), compute the sum (7) and choose the corresponding code.
 4. If no prostatectomy/autopsy, code **97**; if not documented/assessed, code **99**.
- **Curator Quality Review**
 - Verify the score corresponds to the recorded patterns (this record or the prior variable).
 - Confirm tertiary pattern, if present, was not included in the score.
- **Notes**
 - Grade Group (1–5) is not interchangeable; it may be stored elsewhere.
 - If multiple tumor foci exist, use the final score the pathologist assigns for the specimen.

Variable: prostate_gleason_tertiary_pattern

Type: Categorical (Dropdown)

Description: Pathologic tertiary Gleason pattern (highest minor component) reported on radical prostatectomy/autopsy.

Reference: NAACCR Item #3842

Data Source: Final prostatectomy pathology report (synoptic “Tertiary Pattern,” diagnosis/comment)

Curation Directives:

- **Definition:** Capture the tertiary (less prevalent) Gleason pattern when explicitly reported on the prostatectomy/autopsy specimen. Do not add this to the Gleason score.
- **Permitted Values / Codes**
 - **10** Tertiary pattern 1
 - **20** Tertiary pattern 2
 - **30** Tertiary pattern 3
 - **40** Tertiary pattern 4
 - **50** Tertiary pattern 5
 - **97** No radical prostatectomy/autopsy performed
 - **98** Not applicable — information not collected for this case
 - **99** Not documented / not assessed
- **Source Documents**
 - Pathology synoptic “Tertiary pattern (if present)”
 - Narrative diagnosis or comment noting a tertiary component (e.g., “minor tertiary pattern 5”)
- **Curation Instructions**

1. Review the prostatectomy report for an explicit **tertiary pattern** statement.
 2. If present, select the corresponding code (10–50).
 3. If tertiary pattern not mentioned, select **99** (unless no prostatectomy, then: **97**).
 4. Do **not** alter the recorded patterns/score based on tertiary pattern.
- **Curator Quality Review**
 - Ensure tertiary pattern is explicitly documented (do not infer from scattered foci).
 - Check that the score and primary/secondary patterns remain unchanged by tertiary coding.
 - **Notes**
 - Many labs only report tertiary when it is higher grade (e.g., tertiary 5 in 3+4=7); follow the report as written.
 - If pathology states “no tertiary pattern,” treat as **99** (not present/not assessed).

Variable: `crc_tumor_deposits`

Type: Categorical (Dropdown)

Description: Presence/quantity of tumor deposits in pericolic/perirectal fat or adjacent mesentery for colorectal primaries.

Reference: NAACCR Item #3934

Data Source: Final colorectal resection pathology report (synoptic & narrative), gross/microscopic descriptions, addenda, tumor board summary

Curation Directives:

- **Definition:** A tumor deposit is a discrete nodule of carcinoma in the pericolic/perirectal fat or adjacent mesentery (within lymphatic drainage of the primary), without identifiable lymph node tissue or identifiable vascular structure.
- **Permitted Values / Codes**
 - **0** — No tumor deposits.
 - **1** — *01–99 tumor deposits* (exact number required in companion field).
 - **2** — *100 tumor deposits*.
 - **3** — Tumor deposits present, **number unknown**.
 - **4** — Not applicable: information not collected for this case.
 - **5** — Any of the following: not documented; not assessed/unknown; no surgical resection performed; pathologist unable to determine.
- **Source Documents**
 - Pathology synoptic “Tumor Deposits” row (or “Tumor Deposits (TD)”)
 - Microscopic description and/or gross description enumerating deposits
- **Curation Instructions**
 1. Confirm the primary site is colorectum (`proc_ca_type = 4`).
 2. In the final pathology report, locate the tumor-deposit statement.
 3. Map the statement to one code above.
 - If 1 (*01–99*), you **must** enter the exact count in `crc_tumor_deposits_details`.
 - If statement reads “too numerous to count,” select **2**.
 4. If deposits are noted but count **not given**, select **3**.
 5. If no resection or no assessment, select **5**.
- **Curator Quality Review**
 - Verify deposits are not positive lymph nodes and not intravascular tumor.
 - Ensure consistency between the coded value and the companion details field (see below).

- **Notes**

- Do **not** infer count from phrases like “multiple deposits” unless a number is provided (then code 3).
- If both benign mesenteric nodules and tumor deposits are described, count only tumor deposits.
- When both colon and rectal segments are resected, abstract deposits for the indexed primary only.

Variable: `crc_tumor_deposits_details`

Type: Numeric (1–99; conditional)

Description: Exact number of tumor deposits when code **1 (01–99)** is selected in `crc_tumor_deposits`.

Reference: NAACCR Item #3934

Data Source: Same pathology sources as above (synoptic, gross/microscopic descriptions, addenda)

Curation Directives:

- **Definition**

Capture the exact count of tumor deposits (from **1 to 99**) when the main field indicates *1–99 tumor deposits*.

- **Permitted Values / Codes**

- Integers 1–99 only.
- Leave blank for any main-field code other than 1 (e.g., 0, 2, 3, 4, 5).

- **Source Documents**

- Pathology synoptic line giving a number (e.g., “Tumor deposits: **4**”).
- Narrative/microscopic text enumerating deposits.

- **Curation Instructions**

1. Proceed **only if** `crc_tumor_deposits` = 1.
2. Abstract the exact number stated (omit leading zero; enter **4**, not “04”).
3. If the report lists a range, select the lower bound (e.g., “4–6 deposits”: enter 4).
4. If report says “numerous/too numerous to count,” do not use this field; change main field to 2.
5. If no explicit number is provided, change main field to 3 and leave this field blank.

- **Notes**

- Do not sum deposits from separate, staged resections into a single event unless the report explicitly issues a combined count for the indexed procedure.
- If deposits are identified only on a metastasectomy or non-primary specimen, do not record here unless the pathology specifically attributes them to the primary colorectal resection.

Variable: `crc_crm`

Type: Categorical (Dropdown)

Description: Circumferential/Radial Resection Margin (CRM) status and distance for colorectal resections.

Reference: NAACCR Item #3823

Data Source: Final colorectal resection pathology report (synoptic & narrative), gross/microscopic descriptions, addenda, surgeon’s operative note

Curation Directives:

- **Definition:** Record the shortest distance (mm) from invasive carcinoma to the circumferential/radial (non-peritonealized) resection margin, or record that the margin is involved.

- **Permitted Values / Codes**

- **1** — CRM **positive**; margin **involved** (e.g., “tumor at ink,” or described as “< 1 mm”).
- **2** — **Exact distance 0.1–99.9 mm** (enter the value to the nearest **0.1 mm** in the companion details field).
- **3** — **100 mm or greater**.
- **4** — Described as “**at least 1 mm.**”
- **5** — Described as “**at least 2 mm.**”
- **6** — Described as “**at least 3 mm.**”
- **7** — Described as “**greater than 3 mm.**”
- **8** — Margins **clear**, distance **not stated**; CRM negative; no residual tumor identified.
- **9** — Margins **cannot be assessed**.
- **10** — **No resection** of primary site / specimen insufficient to measure CRM.
- **11** — **Not applicable**: information not collected for this case.
- **12** — **Not documented** in medical record; CRM not assessed or unknown if assessed.

- **Source Documents**

- Pathology synoptic: “Circumferential/Radial Resection Margin (CRM)”
- Microscopic description noting closest distance/involvement

- **Curation Instructions**

1. Confirm primary site is colorectum (`proc_ca_type` = 4), and that a resection specimen is being abstracted.
2. In the final pathology report, identify the closest CRM distance or an explicit statement of involvement.
3. Map the report to a single code above.
 - If an **exact number 0.1–99.9 mm** is given, choose **code 2** and enter the value in `crc_crm_details`.
 - If the report states “**100 mm**”, choose **code 3** and leave details **blank**.
 - If qualitative phrases are used (“at least...”, “>3 mm”), choose **codes 4–7** accordingly.
4. If CRM is negative but no distance is reported, choose **code 8**.
5. If the pathologist states cannot assess, choose **code 9**.
6. If no primary resection, choose **code 10**.

- **Curator Quality Review**

- The smallest stated distance governs coding when multiple distances are provided.
- Ensure code 1 is used when tumor is at ink or explicitly <1 mm, even if a small numeric distance is also provided.
- Verify consistency between this field and `crc_crm_details` (only populated when **code 2**).

- **Notes**

- Use the distance to invasive carcinoma (ignore dysplasia/in situ changes).
- Do not estimate or calculate distances not stated by pathology.

Variable: `crc_crm_details`

Type: Numeric (decimal; one decimal place; range 0.1–99.9)

Description: Exact CRM distance in millimeters when an exact value (0.1–99.9 mm) is reported.

Reference: NAACCR Item #3823

Data Source: Same pathology sources as above (synoptic “Margins,” microscopic description/addendum)

Curation Directives:

- **Definition**

Enter the exact CRM distance (in mm) to the nearest 0.1 mm when the main CRM field is coded **2**.

- **Permitted Values / Codes**
 - Numeric entry: **0.1–99.9 mm** (one decimal).
 - Leave **blank** for any main-field code **other than 2** (e.g., 1, 3–12).
- **Source Documents**
 - Pathology synoptic stating an exact distance (e.g., “CRM: **2.3 mm**”).
 - Microscopic description/addendum with precise measurement.
- **Curation Instructions**
 1. Proceed only if `crc_crm` = 2.
 2. Transcribe the distance to one decimal (round standardly if needed; e.g., 2.34 = 2.3; 2.35 = 2.4).
 3. If the report provides a range, record the lower bound (e.g., “2–3 mm” = 2.0).
 4. If the report states “100 mm”, change the main field to code 3 and leave details blank.
 5. If only qualitative language is present (“at least...,” “>3 mm”), code the main field 4–7 and leave details blank.
 6. Document the verbatim phrase and location in the curation log.
- **Curator Quality Review**
 - Confirm unit is mm and the value lies within **0.1–99.9**.
- **Notes**
 - Use the closest distance when multiple are listed.
 - If the pathologist later issues an addendum with a revised exact distance, update both fields accordingly.
 - Do not convert from centimeters unless the report actually gives cm.

Variable: `crc_perineural_invasion`

Type: Categorical (Dropdown)

Description: Presence of perineural invasion (PNI) in colorectal resection pathology specimens.

Reference: NAACCR Item #3909

Data Source: Final pathology report (synoptic “Perineural Invasion” row, microscopic description, addenda)

Definition: Perineural invasion is tumor involving, surrounding, or tracking along a nerve within or adjacent to the primary colorectal tumor.

- **Permitted values / codes**
 - **0 – PNI not present** (“Perineural invasion: **negative** / **not identified**”)
 - **1 – PNI present** (“Perineural invasion: **positive** / **identified**”)
 - **8 – Not applicable** (variable not collected for this case)
 - **9 – Unknown / not assessed** (no statement, indeterminate wording, or specimen not evaluable)
- **Source Documents**
 - Pathology synoptic line “Perineural Invasion”
 - Microscopic description noting tumor-nerve involvement
 - Pathology addendum clarifications
- **Curation Instructions**
 1. Confirm primary site is **colon or rectum** (`proc_ca_type` = 4).
 2. Locate the PNI statement in the final pathology report.
 3. Select the matching code from the table above.

- If PNI not addressed, select **9**.
- 4. Record the verbatim phrase and report location in the curation log.
- **Curator Quality Review**
 - If both “focal” and “extensive” PNI appear, code **1** (present).
- **Notes**
 - Do **not** confuse PNI with lymphovascular invasion (LVI); code separately.
 - If specimen is a **polypectomy** without nerves to evaluate, PNI may legitimately be **9**.

Variable: `crc_msi_result`

Type: Categorical (Dropdown)

Description: Microsatellite instability (MSI) or mismatch-repair (MMR) status for colorectal cancers.

Reference: NAACCR Item #3890

Data Source: Molecular pathology report (PCR-based MSI, IHC for MMR proteins), NGS panels, synoptic “MSI/MMR Status”

Definition

MSI testing identifies length variability of DNA microsatellites, or loss of MMR-protein nuclear expression, classifying tumors as stable, low-instability, or high-instability/MMR-deficient.

Permitted values / codes

- **0 – MSI-Stable (MSS) and/or MMR-proficient (pMMR)** (“MSI: **stable**”; “MMR intact, no loss of expression”)
- **1 – MSI-Low (MSI-L)** (“**Low-level** MSI detected”)
- **2 – MSI-High (MSI-H) and/or MMR-deficient (dMMR)** (“MSI-H”; “Loss of MLH1/PMS2 expression”)
 - • Sub-bullet example: Loss of any MMR protein on IHC → code 2.
- **8 – Not applicable** (variable not collected for this case)
- **9 – Unknown / not assessed / indeterminate** (test not mentioned, “MSI-indeterminate”)

Source documents

- Molecular/IHC appendices attached to the pathology report
- Synoptic “MMR/MSI Status” section

Curation instructions

1. Abstract the most definitive pre-treatment MSI/MMR result (prefer surgical specimen over biopsy).
2. Map the result to a single code above, prioritizing any unstable/deficient finding.
3. If test not done or not mentioned, code **9**.

Curator quality review

- Verify concordance between MSI PCR results and MMR IHC (e.g., dMMR usually correlates with MSI-H).
- Escalate conflicting or equivocal language for clarification.

Notes

- MLH1 promoter hypermethylation with BRAF mutation still codes **2** (MSI-H/dMMR).
- If multiple tests disagree, record the highest level of instability/deficiency.

Variable: `liver_fibrosis_score`

Type: Categorical (Dropdown)

Description: Degree of hepatic fibrosis/cirrhosis as determined around diagnosis from pathology (and, when no histology, clinical statement).

Reference: NAACCR Item #3835

Data Source: Liver resection/biopsy pathology report (synoptic or narrative), hepatology pathology addenda; if no histology: hepatology/oncology notes, imaging reports explicitly stating cirrhosis

Curation Directives:

- **Definition:** Code the highest fibrosis stage documented for the liver at or near diagnosis, giving precedence to histology (resection or biopsy).
- **Permitted values / codes**
 - **0 – No to moderate fibrosis** — e.g., *Ishak 0–4*; *METAVIR F0–F3*; *Batts–Ludwig 0–3*.
 - **1 – Advanced/severe fibrosis or cirrhosis (histologically confirmed)** — e.g., *Ishak 5–6*; *METAVIR F4*; *Batts–Ludwig 4*.
 - **7 – Clinical statement of advanced/severe fibrosis or cirrhosis, not histologically confirmed or unknown if confirmed** — e.g., “cirrhosis on imaging/clinical exam,” no liver histology available.
 - **8 – Not applicable** — information not collected for this case.
 - **9 – Unknown / not assessed** — not documented; scoring system not recorded; cannot map stated score to codes 0 or 1; **or** record states the patient **does not** have advanced fibrosis/cirrhosis but histologic confirmation is not available/unknown.
- **Source documents**
 - Pathology synoptic or narrative for liver fibrosis stage/score
 - Hepatology pathology addendum (if separate from oncology report)
- **Curation instructions**
 1. **First, search pathology** (resection or biopsy) for a fibrosis staging system and record the stage/score.
 - Assign:
 - * **Code 0**, if Ishak 0–4 / METAVIR F0–F3 / Batts–Ludwig 0–3
 - * **Code 1**, if Ishak 5–6 / METAVIR F4 / Batts–Ludwig 4
 - * **Code 7**, if no histology or histology is silent, and there is a clear clinical/imaging statement of advanced fibrosis/cirrhosis.
 - * **Code 9**, if documentation is insufficient to classify (or only states “no cirrhosis” without histology)
- **Curator quality review**
 - Verify the scoring system (Ishak, METAVIR, Batts–Ludwig) and ensure correct mapping to codes.
 - Ensure the fibrosis assessment is temporally appropriate (closest to diagnosis and before definitive liver treatment when possible).
- **Notes**
 - Do not confuse fibrosis stage with necroinflammatory grade or steatosis. Those do not map here.
 - Non-histologic tools (FibroScan/elastography, CT/MRI) may support **code 7** only when histology is absent.

Variable: melanoma_breslow_tt

Type: Categorical (Dropdown)

Description: Pathologist-reported Breslow tumor thickness (depth) for cutaneous melanoma, measured in millimeters.

Reference: NAACCR Item #3817

Data Source: Final dermatopathology report (synoptic & narrative), addenda, amended reports

Curation Directives:

- **Definition:** Record the maximum Breslow thickness measured from the top of the tumor (or from the base of any ulceration) to the deepest invasive melanoma cells, as stated by pathology on the definitive specimen.
- **Permitted values / codes**
 - **0 – 0.0:** No tumor identified (e.g., in situ only or no residual melanoma).
 - **1 – 0.1:** Greater than 0.0 and **0.1 mm** (microinvasion up to one tenth mm).
 - **2 – 0.2–99.9:** Exact thickness **0.2–99.9 mm** (enter the exact value in the companion *details* field).
 - **3 – XX.1: 100.0 mm.**
 - **4 – A0.1–A9.9:** Stated as “**at least**” some value **0.1–9.9 mm** (no exact number given).
 - **5 – AX.0:** Stated as “**greater than 9.9 mm**” (no exact number given).
 - **6 – XX.8:** Not applicable — information not collected for this schema/case.
 - **7 – XX.9:** Not documented/unknown/not assessed; microinvasion stated without a measurable thickness; cannot be determined by pathologist.
- **Source documents**
 - Dermatopathology synoptic (e.g., “Breslow thickness: 1.8 mm”).
 - Narrative diagnosis/comments describing ulceration baseline and depth.
 - Amended or addendum reports revising the thickness.
- **Curation instructions**
 1. Locate the Breslow thickness statement in the final path report.
 2. Assign:
 - **Code 2** if an exact value **0.2–99.9 mm** is provided, then complete `melanoma_breslow_tt_details`.
 - **Code 3** if stated as **100 mm**
 - **Code 4** if phrased as “**at least X mm (0.1–9.9)**”
 - **Code 5** if phrased as “**>9.9 mm**” without a number.
 - **Code 7** if only “microinvasion” is reported without a number.
 3. If multiple measurements appear (e.g., different blocks), abstract the greatest Breslow thickness.
 4. Record the verbatim statement, block/slide ID (if given), and report section/page in the curation log.
- **Curator quality review**
 - Verify units are millimeters
 - If ulceration is present, confirm that the measurement reference (base of ulcer) is reflected in the reported value.
- **Notes**
 - Do **not** substitute Clark level for Breslow thickness.
 - Ignore in-situ components when an invasive thickness is reported, abstract the invasive depth.
 - If the case is excision after a prior biopsy, use the final consolidated measurement from the definitive pathology.

Variable: melanoma_breslow_tt_details

Type: Numeric (decimal; mm)

Description: Exact Breslow thickness in millimeters when melanoma_breslow_tt is coded **2** (**0.2–99.9 mm**).

Reference: NAACCR Item #3817

Data Source: Same dermatopathology sources as above

Curation Directives:

- **Definition:** Enter the exact numeric thickness (mm) only when the main field is **code 2**.
- **Permitted values / codes**
 - **Numeric entry: 0.2–99.9 mm** (record to one decimal place if the report provides tenths; otherwise record as given).
 - Leave **blank** for main-field codes **0, 1, 3–7**.
- **Source documents**
 - Synoptic line with exact depth (e.g., “1.7 mm”).
 - Narrative text/addendum specifying a precise value.
- **Curation instructions**
 1. Transcribe the stated number in **mm**; convert from cm when necessary.
 2. If a range is reported (e.g., 1.6–1.8 mm), record the highest value.
 3. If the report revises the thickness in an addendum, update the value.
- **Curator quality review**
 - Confirm the value is within **0.2–99.9 mm** and in mm units.

Variable: melanoma_ulceration

Type: Categorical (Dropdown)

Description: Indicates whether ulceration is present in the primary melanoma, based on histopathological examination. Ulceration is an adverse prognostic factor and is part of melanoma staging.

Reference: NAACCR Data Item #3936

Data Source: Final pathology report of the primary melanoma specimen

Curation Directives:

- **Definition:** Ulceration is defined as the complete loss of the epidermis overlying the melanoma, observed under histopathologic examination. It reflects more aggressive tumor behavior and worsens prognosis. The determination must be made by the pathologist and should not be inferred from clinical descriptions of crusting or bleeding.
- **Permitted Values:**
 - 0 – Ulceration not identified / not present
 - 1 – Ulceration present
 - 8 – Not applicable: Information not collected for this case
 - 9 – Not documented in medical record; cannot be determined by the pathologist; not assessed or unknown if assessed
- **Source Documents:**
 - Final pathology report (gross and microscopic description)

- Dermatopathology report (if separate from general pathology)
- **Curation Instructions:**
 - Code 0 if the pathology report explicitly states that ulceration is absent, not identified, or not present.
 - Code 1 if ulceration is present, even if the extent is not quantified.
 - Code 8 if the record states that ulceration is not applicable for this schema or tumor type.
 - Code 9 if:
 - * The pathology report does not mention ulceration at all
 - * The pathologist explicitly states that assessment could not be made
 - * There was no surgical resection of the primary site
 - * It is unclear whether the tumor was evaluated for ulceration
 - Do not infer presence or absence based on clinical appearance, bleeding, or crusting. Only code based on pathologist’s statement.
- **Curator Quality Review:**
 - Ensure the term “ulceration” refers to histologic ulceration, not superficial erosion.
 - Validate that any statement of presence/absence is based on microscopic findings, not gross only.
- **Notes:**
 - Ulceration status is used in AJCC staging and prognosis; accurate abstraction is essential.
 - If both gross and microscopic descriptions are available and conflict, defer to the microscopic interpretation

Variable: melanoma_margin

Type: Categorical (Dropdown)

Description: Captures the preoperative clinical margin width, in centimeters, used for wide local excision of the primary melanoma lesion. This measurement reflects the planned peripheral margin recorded by the surgeon.

Reference: NAACCR Data Item #3961

Data Source: Surgical operative report, dermatology notes, or pathology report if it includes specimen handling description

Curation Directives:

- **Definition:** Clinical margin width refers to the documented pre-surgical measurement from the visible or previously biopsied lesion (or scar) to the edge of the area designated for excision. It is measured in centimeters and reflects the planned surgical margin for melanoma excision, typically determined by tumor depth and standard guidelines. It is not based on the pathological margin (i.e., not the post-operative histologic clearance).
- **Permitted Values:**
 - 1 – Documented as 0.1 cm or less (1 mm or less)
 - 2 – 0.2 to 9.9 cm
 - 3 – 10 cm
 - 4 – Not applicable: Information not collected for this schema
 - 5 – Not documented in medical record; no wide excision performed; Mohs or similar procedure; unknown if procedure performed
 - 6 – Not applicable due to diagnosis prior to 2023
- **Source Documents:**
 - Dermatology consultation notes or pre-surgical planning documentation

- Pathology report specimen description (only if explicitly documenting margin measurements from gross examination)
- **Curation Instructions:**
 - Code 1 if the margin is explicitly stated as 1 mm or less (e.g., “0.5 cm,” “1 mm,” “1 mm”).
 - Code 2 if a margin between 0.2 and 9.9 cm is documented; use surgeon’s planned value (not pathological clearance).
 - Code 3 for stated margin of 10 cm.
 - Code 4 if the margin width is not collected for this schema (rare).
 - Code 5 if:
 - * The record states “no wide excision performed.”
 - * A Mohs procedure or similar tissue-sparing technique is performed.
 - * Margin not documented and unclear if excision was performed.
 - Code 6 if melanoma diagnosis occurred **before 2023**, when this variable was not routinely collected.
 - Do not infer margin width from tumor size or pathological margins.
 - Round to the nearest tenth of a centimeter if needed, but do not estimate if not explicitly documented.
- **Curator Quality Review:**
 - Confirm that margin documentation refers to planned surgical margin and not pathological margin.
 - Double-check that measurement is in centimeters (cm) and not millimeters (mm); convert as appropriate.
 - Document ambiguities or missing values in internal comments for audit.
- **Notes:**
 - This is a clinician-measured value taken preoperatively.
 - Often found in dermatologic surgery records rather than pathology reports.

Variable: melanoma_mitotic_rate

Type: Categorical (Dropdown)

Description: Captures the mitotic activity of melanoma tumors, defined as the number of mitoses per square millimeter (mm²), based on pathological evaluation.

Reference: NAACCR Data Item #3893

Data Source: Primary pathology report (microscopic description or synoptic summary)

Curation Directives:

- **Definition:** Mitotic Rate (Melanoma) represents the number of mitotic figures per square millimeter of tumor tissue observed under microscopic examination. It is an important prognostic factor for melanoma of the skin. Reports may describe mitotic activity as an exact count, a range, or with qualitative terms such as “mitogenic” or “nonmitogenic.”
- **Permitted Values:**
 - 0 – 0 mitoses/mm² (mitoses absent; no mitoses present)
 - 1 – 1 to 99 mitoses/mm²
 - 2 – 100 mitoses/mm²
 - 3 – Stated as “less than 1 mitosis/mm²” or “nonmitogenic”
 - 4 – Stated as “at least 1 mitosis/mm²” or “mitogenic”
 - 5 – Rate described with a denominator other than mm² (e.g., per HPF)
 - 6 – Test ordered, results not in chart

- 7 – Not applicable: Information not collected for this case
- 8 – Not documented in the medical record; unknown; not assessed
- **Source Documents:**
 - Pathology report (microscopic findings or synoptic summary)
 - Dermatopathology addenda or second opinion pathology (if available)
- **Curation Instructions:**
 - Only abstract this field if `proc_ca_type` = 6 (melanoma).
 - Assign codes using exact text or numeric values from the pathology report:
 - * Code 0: Explicit statement of “0 mitoses,” “no mitoses present,” or “mitoses absent.”
 - * Code 1: Exact mitotic count between 1–99/mm²; also enter the value in `melanoma_mitotic_rate_details`.
 - * Code 2: Explicit mention of 100 mitoses/mm².
 - * Code 3: Described as “<1 mitosis/mm²” or “nonmitogenic.”
 - * Code 4: Described as “1 mitosis/mm²” or “mitogenic.”
 - * Code 5: Mitotic rate given with units other than mm² (e.g., per high-power field).
 - * Code 6: Test ordered but results not yet available.
 - * Code 7: Field not applicable based on schema.
 - * Code 8: Mitotic rate not reported or cannot be determined.
 - Do not assume values if wording is ambiguous, defer to code 8 when uncertain.
 - Clarify with clinical team or pathologist only if critical for staging or downstream interpretation.
- **Curator Quality Review:**
 - Ensure unit of measure is stated or implied to be mm² for numeric codes.
 - Confirm rate pertains to invasive component of the melanoma.
 - Review both synoptic and narrative portions of pathology report for consistency.

Variable: `melanoma_mitotic_rate_details`

Type: Numeric Entry (number)

Description: Melanoma – Mitotic Rate (Details)

Reference: NAACCR #3893

Data Source: Primary pathology report (microscopic description or synoptic summary)

Curation Directives:

- **Definition:** Captures the exact numeric mitotic rate (in mitoses per square millimeter) when the pathologist reports a value within the 1–99 mitoses/mm² range. This value is a direct quantitative measurement and enhances the categorical mitotic rate variable by providing granular prognostic detail.
- **Permitted Values:**
 - Any whole number between 1 and 99
- **Source Documents:**
 - Pathology report (microscopic findings or synoptic summary)
 - Dermatopathology addenda or second opinion pathology (if available)
- **Curation Instructions:**
 - If the mitotic rate was coded as 1 (1–99 mitoses/mm²) in the categorical field `melanoma__mitotic__rate`, extract and enter the exact number of mitoses per square millimeter here.
 - Transcribe the value exactly as stated in the pathology report (e.g., “4 mitoses/mm²” → enter 4).
 - Do not round, estimate, or calculate if a specific number is not provided. Leave blank unless clearly documented.

- Do not enter ranges (e.g., “3–5 mitoses/mm²”)—use the categorical variable only in such cases.
- Enter numeric values only (e.g., 7), with no units, punctuation, or text.

- **Curator Quality Review:**

- Confirm that the value is consistent with the pathology narrative and falls within the valid range (1–99).
- Check that no text, decimal points, or non-numeric characters are included.

Variable: kidney_invasion

Type: Categorical (Dropdown)

Description: Captures the extent of kidney tumor invasion beyond the renal capsule, as confirmed by pathological examination.

Reference: NAACCR Data Item #3864

Data Source: Final pathology report of the primary renal tumor

Curation Directives:

- **Definition:** This variable documents whether the tumor has invaded structures beyond the renal capsule, based solely on pathologic examination. The fibrous capsule surrounds the kidney and demarcates the limit of the parenchyma. Invasion beyond the capsule, especially into adjacent structures, is a critical prognostic and staging feature in renal cell carcinoma. Only pathological evidence should be used; do not use clinical imaging or operative impressions.

- **Permitted Values:**

- 0 – Invasion beyond capsule not identified
- 1 – Perinephric (beyond renal capsule) fat or tissue
- 2 – Renal sinus
- 3 – Gerota’s fascia
- 4 – Any combination of codes 1–3
- 5 – Invasion beyond capsule, NOS
- 8 – Not applicable: Information not collected for this case
- 9 – Not documented in medical record; invasion not assessed or unknown if assessed; no surgical resection

- **Source Documents:**

- Final pathology report (gross and microscopic sections)
- Urologic pathology synoptic report (if available)

- **Curation Instructions:**

- Only abstract this field if `proc_ca_type` = 8 (kidney cancer).
- Use code 0 if the pathology report explicitly states no invasion beyond the capsule.
- Use code 1, 2, or 3 when the report specifies one of those anatomical areas is involved.
- Use code 4 if two or more specific structures (from codes 1–3) are listed as involved.
- Use code 5 only when the pathology confirms “invasion beyond capsule” but does not specify site(s).
- Code 8 if this variable is not applicable to the case type or schema.
- Code 9 if:
 - * The pathology report makes no mention of extracapsular invasion
 - * A resection was not performed, or
 - * It is unclear whether assessment was done

- **Curator Quality Review:**

- Confirm that the information is derived strictly from **pathologic examination**, not imaging or surgeon impression.
 - **Notes:**
 - This variable informs staging for renal cell carcinoma (e.g., T3a for perinephric or sinus invasion, T4 for Gerota’s fascia).
-

Variable: kidney_major_vein_involve

Type: Categorical (Dropdown)

Description: Captures pathologically confirmed tumor involvement of major renal venous structures, including renal vein and/or inferior vena cava (IVC).

Reference: NAACCR Data Item #3886

Data Source: Final pathology report (gross and microscopic sections), synoptic pathology report (if available)

Curation Directives:

- **Definition:** Documents whether the renal tumor has invaded or extended into major venous structures (renal vein, IVC, or both) as confirmed on pathological review. Only include findings based on the surgical pathology report. Do **not** code based on clinical findings, radiology, or operative notes. Do not include small intrarenal vessel involvement (captured in LVI), or direct wall invasion of IVC.
- **Permitted Values:**
 - 0 – Major vein involvement not present/not identified
 - 1 – Renal vein or its segmental branches
 - 2 – Inferior vena cava (IVC)
 - 3 – Major vein invasion, NOS
 - 4 – Any combination of codes 1–3
 - 8 – Not applicable: Information not collected for this case
 - 9 – Not documented in medical record; vein involvement not assessed or unknown if assessed; no surgical resection
- **Source Documents:**
 - Final pathology report (gross description and microscopic findings)
 - Urologic oncology synoptic report (if available)
- **Curation Instructions:**
 - Abstract only when `proc_ca_type` = 8 (Kidney cancer).
 - Code 1 if the pathology report describes tumor involvement in the renal vein or its segmental branches.
 - Code 2 if the tumor is described as extending into the inferior vena cava (IVC).
 - Code 3 if major vein involvement is mentioned but the specific vein(s) are not identified.
 - Code 4 if there is involvement of both renal vein and IVC, or any combination of codes 1–3.
 - Code 8 if this variable is not applicable for the case (e.g., schema exclusion).
 - Code 9 if:
 - * There is no resection of the primary tumor
 - * The pathology report does not mention major vein involvement
 - * Vein involvement is not assessed or cannot be determined
- **Curator Quality Review:**
 - Confirm that vein involvement is described based on pathology, not imaging.
 - Cross-check gross and microscopic sections for any mention of tumor thrombus or extension into veins.

- Do not confuse with lymphovascular invasion (separate variable).

- **Notes:**

- Major vein involvement plays a key role in TNM staging for renal cell carcinoma.
- Tumor thrombus may be described as floating or extending intraluminally without adhering to the vessel wall.
- Terminology such as “tumor thrombus,” “intraluminal extension,” or “tumor within vein” typically indicate positive involvement.

Variable: kidney_gland_involve

Type: Categorical (Dropdown)

Description: Captures pathologically confirmed involvement of the ipsilateral adrenal gland in cases of kidney cancer, based on direct tumor extension or separate tumor nodules.

Reference: NAACCR Data Item #3861

Data Source: Final pathology report (gross and microscopic sections), urologic oncology synoptic report (if available)

Curation Directives:

- **Definition:** This field documents whether the kidney tumor involves the ipsilateral adrenal gland, either through direct contiguous invasion or via a separate (noncontiguous) metastatic nodule. Findings must be pathologically confirmed. Do not code based on clinical suspicion, imaging, or operative impressions.
- **Permitted Values:**
 - 0 – Ipsilateral adrenal gland involvement not present/not identified
 - 1 – Adrenal gland involvement by direct involvement (contiguous involvement)
 - 2 – Adrenal gland involvement by separate nodule (noncontiguous involvement)
 - 3 – Combination of code 1–2
 - 4 – Ipsilateral adrenal gland involvement, unknown if direct involvement or separate nodule
 - 8 – Not applicable: Information not collected for this case
 - 9 – Not documented in medical record; ipsilateral adrenal gland not resected or involvement not assessed; no surgical resection performed
- **Source Documents:**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions:**
 1. Complete this field only if `proc_ca_type = 8` (Kidney cancer).
 2. Code 1 if the pathology report states that the tumor directly extends into the ipsilateral adrenal gland (contiguous involvement).
 3. Code 2 if adrenal gland involvement is described as a separate nodule (noncontiguous from the primary renal tumor).
 4. Code 3 if both direct invasion and separate nodule are present.
 5. Code 4 if adrenal involvement is confirmed but the mode of involvement (direct vs. separate) is unclear or unspecified.
 6. Code 8 if this data element is not applicable for the schema or case.
 7. Code 9 if:
 - The adrenal gland was not resected
 - There is no pathology assessment of adrenal involvement
 - The pathology report does not mention the adrenal gland at all

- The primary tumor was not surgically resected
 - **Curator Quality Review:**
 - Ensure the finding is pathologically confirmed and not based on imaging or intraoperative notes.
 - Distinguish between direct extension and separate metastatic nodules when possible.
 - Confirm that the gland referenced is ipsilateral (on the same side as the primary renal tumor).
 - **Notes:**
 - Involvement of the ipsilateral adrenal gland is a significant staging factor in renal cell carcinoma.
 - “Contiguous” or “direct extension” refers to uninterrupted tumor spread, while “noncontiguous” indicates separate foci of disease.
 - If the adrenal gland is not mentioned in the pathology report, and resection status is unknown, code 9.
-

Variable: kidney_sarcomatoid

Type: Categorical (Dropdown)

Description: Captures the presence and extent (percentage) of sarcomatoid features in kidney tumors, as assessed on pathological examination.

Reference: NAACCR Data Item #3925

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Sarcomatoid features refer to the presence of malignant spindle cells in a kidney tumor, which may occur across all renal cell carcinoma subtypes. This variable captures whether sarcomatoid morphology is present and, when present, the estimated percentage of tumor area demonstrating these features.
- **Permitted Values:**
 - 0 – 000: Sarcomatoid features not present/not identified
 - 1 – 001-100: Sarcomatoid features 1–100%
 - 2 – R01: Stated as less than 10%
 - 3 – R02: Stated as 10–30%
 - 4 – R03: Stated as 31–50%
 - 5 – R04: Stated as 51–80%
 - 6 – R05: Stated as greater than 80%
 - 7 – XX5: Present in metastatic site only; not present or unknown in primary site
 - 8 – XX6: Present, percentage unknown
 - 9 – XX7: Not applicable – not a renal cell carcinoma morphology
 - 10 – XX8: Not applicable – information not collected
 - 11 – XX9: Not documented; unknown if assessed; no surgical resection
- **Source Documents:**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions:**
 1. Complete this field only if `proc_ca_type` = 8 (Kidney).
 2. Use code **0** if the pathology explicitly states sarcomatoid features are not present.
 3. Use code **1** if a specific percentage from 1% to 100% is provided. Record the exact percentage in the companion field `kidney_sarcomatoid_details`.
 4. Use codes **2–6** based on reported ranges in the pathology report:

- **2:** Stated as <10%
 - **3:** Stated as 10–30%
 - **4:** Stated as 31–50%
 - **5:** Stated as 51–80%
 - **6:** Stated as >80%
5. Use **7** if sarcomatoid features are documented in the metastatic site only and are absent or unknown in the primary tumor.
 6. Use **8** if sarcomatoid features are mentioned as present but no percentage or range is given.
 7. Use **9** if the case is not renal cell carcinoma.
 8. Use **10** if this variable is not applicable to the case.
 9. Use **11** if not documented or not assessed, or if no resection was performed.
- **Curator Quality Review:**
 - Confirm the mention of sarcomatoid features comes from pathological assessment of the primary tumor unless coding for metastatic site (code 7).
 - For code 1, ensure the exact % value is documented in the linked text field.
 - If only “sarcomatoid features present” is mentioned with no further quantification, use code 8.
 - Do not infer values from ambiguous language—select code 8 or 11 in such cases.
 - **Notes:**
 - Sarcomatoid features are associated with poorer prognosis in renal cell carcinoma and are significant in staging and treatment planning.
 - This variable captures both presence and extent; both are relevant for risk stratification.

Variable: kidney_sarcomatoid_details

Type: Numeric Entry (number)

Description: Kidney – Sarcomatoid Features (Details)

Reference: NAACCR Data Item #3925

Data Source: Surgical / pathology reports

Curation Directives:

- **Definition:** Captures the exact percentage of sarcomatoid features present in a surgically resected kidney tumor specimen, when the categorical variable indicates a value between 1–100%. Sarcomatoid transformation is a known adverse prognostic factor across all renal cell carcinoma histologies, and percentage-based reporting has clinical and staging relevance.
- **Permitted Values:**
 - Whole number between **1** and **100**
 - Do not enter values outside this range; if percentage is only described in ranges or non-numeric terms, rely on the categorical variable only.
- **Source Documents:**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions:**
 1. Enter the exact percentage of sarcomatoid features if reported explicitly (e.g., “Sarcomatoid features present in 35% of tumor” → enter 35).
 2. Do not estimate, calculate, or average from ranges. If a range is reported (e.g., “10–30%”), do not complete this field—use the categorical variable only.
 3. Enter numeric values only. Do not include the percent symbol (%), units, decimals, or text.

4. Only complete this field if `kidney_sarcomatoid` = 1 was selected (i.e., features reported as 1–100%).

- **Curator Quality Review:**

- Double-check that the value is consistent with the narrative or synoptic summary of the pathology report.
- If the primary site was not resected or sarcomatoid component is only reported from metastatic site, this field should remain blank.

- **Notes:**

- Sarcomatoid features are associated with poorer prognosis in renal cell carcinoma and are significant in staging and treatment planning.

Variable: `breast_ln_pos_axil`

Type: Categorical (Dropdown)

Description: Records the number of pathologically positive ipsilateral Level I–II axillary and intramammary lymph nodes in breast cancer cases.

Reference: NAACCR Data Item #3882

Data Source: Final pathology report (including sentinel node biopsy and axillary dissection results)

Curation Directives:

- **Definition:** Captures the presence and number of positive lymph nodes at axillary level I or II, based solely on pathological examination (not clinical or imaging findings). This includes nodes sampled from sentinel node biopsy, core needle biopsy, or axillary lymph node dissection. Intramammary nodes may be included in this count per AJCC guidance.

- **Permitted Values:**

- 0 – 00: All ipsilateral axillary nodes examined negative
- 1 – 00-99: 1–99 nodes positive
- 2 – X1: 100 or more nodes positive
- 3 – X5: Positive nodes, number unspecified
- 4 – X6: Positive aspiration or needle core biopsy of lymph node(s)
- 5 – X8: Not applicable: Information not collected for this case
- 6 – X9: Not documented in medical record; Level I-II axillary nodes not assessed or unknown if assessed

- **Source Documents:**

- Surgical operative reports
- Pathology reports indicating post-surgical specimen

- **Curation Instructions:**

1. Use **code 0** if all nodes examined were negative for metastasis.
2. Use **code 1** if between 1 and 99 positive nodes are reported; specify the exact number in the companion field `breast_ln_pos_axil_details`.
3. Use **code 2** if 100 or more nodes are stated to be positive.
4. Use **code 3** if the report states nodes are positive but does not specify how many.
5. Use **code 4** if only an aspirate or needle biopsy was performed and found positive.
6. Use **code 5** if the data is not collected for this case schema.
7. Use **code 6** if there is no documentation in the record, or if axillary node evaluation was not performed or the number could not be determined.
8. If both core biopsy and axillary dissection findings are present, prioritize the **axillary dissection** data when available.

9. Do not include contralateral lymph nodes or those from other nodal levels (e.g., supraclavicular, internal mammary) in this count.

- **Curator Quality Review:**

- Confirm that the node count is derived from pathological evaluation, not imaging or clinical exam.
- Ensure axillary level (I–II) is explicitly stated or implied based on node location.
- When pathology provides a sentinel node count, confirm whether additional dissection occurred to include in total count.

- **Notes:**

- This field is a key prognostic factor and may influence staging.
- If the exact number of positive nodes is given (e.g., “3 of 10 nodes positive”), code as **1** and specify “3” in `breast_ln_pos_axil_details`.

Variable: `breast_ln_pos_axil_details`

Type: Categorical (Dropdown)

Description: Captures the exact number of positive Level I–II axillary or intramammary lymph nodes when a range of 1–99 is selected in the primary lymph node variable.

Reference: NAACCR Data Item #3882

Data Source: Final pathology report (including sentinel node biopsy and axillary dissection results)

Curation Directives:

- **Definition:** Provides numeric detail for the count of positive ipsilateral axillary Level I–II lymph nodes when the permitted value **1 - 00-99 nodes positive** is selected in the parent field `breast_ln_pos_axil`.
- **Permitted Values:**
 - Integer values between **1** and **99**
- **Source Documents:**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions:**
 1. Enter the exact number of positive nodes as stated in the pathology report.
 2. Do not include nodes from non-axillary levels (e.g., supraclavicular or internal mammary).
 3. Do not include contralateral or distant nodal sites.
 4. If both sentinel node biopsy and axillary dissection are performed, total the number of positive nodes across all Level I–II specimens.
 5. If multiple specimens are reported separately (e.g., “2 positive sentinel nodes” and “1 positive axillary node”), sum and enter “3”.
 6. If the number is unclear or ranges are given (e.g., “approximately 3–4”), enter the lower value.
- **Curator Quality Review:**
 - Ensure value matches that reported in the pathology document and is consistent with `breast_ln_pos_axil = 1`.
- **Notes:**
 - Leave blank if the report lacks a definitive numeric value, even if nodes are stated as positive. Use permitted value 3 in `breast_ln_pos_axil` in such cases.

Variable: breast_er_range

Type: Categorical (Dropdown)

Description: Records the percent or estimated range of tumor cells expressing Estrogen Receptor (ER) positivity by IHC.

Reference: NAACCR Item #3826

Data Source: Pathology report (synoptic report, biomarker section)

Curation Directives:

- **Definition:** Captures the percentage or descriptive range of ER-positive cells as measured by immunohistochemistry (IHC), used to determine hormone receptor status for breast tumors.
- **Permitted Values:**
 - 0 – 000: ER negative, or stated as less than 1%
 - 1 – 001-100: 1–100 percent
 - 2 – R10: Stated as 1–10%
 - 3 – R20: Stated as 11–20%
 - 4 – R30: Stated as 21–30%
 - 5 – R40: Stated as 31–40%
 - 6 – R50: Stated as 41–50%
 - 7 – R60: Stated as 51–60%
 - 8 – R70: Stated as 61–70%
 - 9 – R80: Stated as 71–80%
 - 10 – R90: Stated as 81–90%
 - 11 – R99: Stated as 91–100%
 - 12 – XX7: Test done, results not in chart
 - 13 – XX8: Not applicable: Information not collected for this case
 - 14 – XX9: Not documented in medical record
- **Source Documents:**
 - Pathology report (ER immunohistochemistry section or biomarker summary)
 - Synoptic templates
 - Oncology consults summarizing pathology
- **Curation Instructions:**
 1. Identify the ER percent positivity from the pathology report.
 2. If an exact percentage is reported, choose code 1 (001–100 percent) and enter the exact value in `breast_er_range_details`.
 3. If only a descriptive range is provided (e.g., “11–20%”), code the corresponding range (e.g., R20).
 4. Code 0 if the report states ER is negative or <1%.
 5. If the ER test is noted but results are not available, use XX7.
 6. If ER testing was not applicable or not collected, use XX8.
 7. Use XX9 if there is no information regarding ER testing.
- **Curator Quality Review:**
 - Confirm that the ER result reflects tumor cell staining, not surrounding tissue or nonspecific staining.
 - Verify correct use of codes when only qualitative descriptions are available.
- **Notes:**
 - Do not infer percent from summary interpretation (e.g., “ER positive”)—only record if a numeric or descriptive range is provided.

Variable: breast_er_range_details

Type: Numeric

Description: Captures the exact ER percent positivity if breast_er_range = 1.

Reference: NAACCR Item #3826

Data Source: Pathology report (immunohistochemistry, ER biomarker section)

Curation Directives:

- **Definition:** Collects the specific percentage of ER-positive cells by IHC if the permitted value 1 - 001-100 percent is selected in breast_er_range.
- **Permitted Values:**
 - Integer values between 1 and 100
- **Source Documents:**
 - Pathology report (ER IHC details or biomarker results section)
 - Synoptic pathology templates
- **Curation Instructions:**
 1. Use only when breast_er_range = 1.
 2. Enter the exact percentage of ER positivity as documented.
 3. Do not estimate or convert descriptive ranges (e.g., “11–20%”) into an exact percent—leave blank in such cases.
 4. Record the value as a whole number (e.g., “5”, not “5%”).
- **Curator Quality Review:**
 - Ensure the entered value falls between 1–100 and corresponds to the reported figure.
 - Check for consistency with breast_er_range.
- **Notes:**
 - Leave this field blank if an exact percent is not documented.
 - Do not populate this field if a range or qualitative statement (e.g., “ER positive”) is provided without an exact percent.

Variable: breast_er_summary

Type: Categorical (Dropdown)

Description: Summary interpretation of estrogen receptor (ER) assay, used in AJCC prognostic staging.

Reference: NAACCR Item #3827

Data Source: Pathology report (biomarker summary or synoptic), oncology notes

Curation Directives:

- **Definition:** Captures the interpreted result of ER testing (positive or negative), regardless of whether percent positivity is provided.
- **Permitted Values:**
 - 0 – ER negative (0.0% or less than 1%)
 - 1 – ER positive
 - 7 – Test ordered, results not in chart
 - 9 – Not documented; Cannot be determined; ER status not assessed or unknown if assessed
- **Source Documents:**
 - Pathology report (summary, synoptic section)

- **Curation Instructions:**

1. Review the ER result interpretation from the pathology report.
2. Code 1 if ER is interpreted as positive (1%).
3. Code 0 if ER is interpreted as negative (<1%).
4. Use 7 if the test is known to have been ordered but no result is available.
5. Use 9 if no documentation is found or ER testing was not done.

- **Curator Quality Review:**

- Confirm interpretation aligns with numeric values when available (e.g., percent positivity).
- Ensure this variable does not contradict **breast_er_range**.

- **Notes:**

- Do not infer ER positivity from treatment decisions (e.g., hormone therapy prescribed).

Variable: breast_pr_range

Type: Categorical (Dropdown)

Description: Records the percent or estimated range of tumor cells staining positive for progesterone receptor (PR) by IHC.

Reference: NAACCR Item #3914

Data Source: Pathology report (biomarker section, synoptic report, immunohistochemistry results)

Curation Directives:

- **Definition:** Captures the PR immunohistochemistry result as either a specific percent positive value or a descriptive range indicating the proportion of stained tumor cells.

- **Permitted Values:**

- 0 – 000: PR negative, or stated as less than 1%
- 1 – 001-100: 1–100 percent
- 2 – R10: Stated as 1–10%
- 3 – R20: Stated as 11–20%
- 4 – R30: Stated as 21–30%
- 5 – R40: Stated as 31–40%
- 6 – R50: Stated as 41–50%
- 7 – R60: Stated as 51–60%
- 8 – R70: Stated as 61–70%
- 9 – R80: Stated as 71–80%
- 10 – R90: Stated as 81–90%
- 11 – R99: Stated as 91–100%
- 12 – XX7: Test done, results not in chart
- 13 – XX8: Not applicable: Information not collected for this case
- 14 – XX9: Not documented in medical record

- **Source Documents:**

- Surgical operative reports
- Pathology reports indicating post-surgical specimen

- **Curation Instructions:**

- Review the pathology report for the documented PR staining result.
- If an exact percentage is provided, code 1 and enter the value in **breast_pr_range_details**.
- If only a range is provided (e.g., “11–20%”), select the corresponding range code (e.g., R20).
- Code 0 if the report states “negative” or “<1%”.

- Use **XX7** if the test is confirmed to be performed but results are unavailable.
- Use **XX8** if the case is not applicable for PR assessment.
- Use **XX9** if there is no documentation of PR testing.

- **Curator Quality Review:**

- Ensure concordance between percent values and selected range codes.

- **Notes:**

- Percent staining may guide eligibility for hormone therapy.
- Range codes should only be used when percent is not explicitly stated.

Variable: `breast_pr_range_details`

Type: Numeric

Description: Captures the exact PR percent positivity if `breast_pr_range` = 1.

Reference: NAACCR Item #3914

Data Source: Pathology report (PR immunohistochemistry or biomarker result section)

Curation Directives:

- **Definition:** Records the precise percent of tumor cells staining positive for PR, only when the exact figure is provided and `breast_pr_range` is coded as 1.
- **Permitted Values:**
 - Integer values between **1** and **100**
- **Source Documents:**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions:**
 - Enter the documented value exactly as stated.
 - Leave blank if a range (e.g., “11–20%”) is provided instead of an exact percentage.
- **Curator Quality Review:**
 - Confirm the value falls within 1–100 and matches the pathology report.
 - Ensure consistency with `breast_pr_range` coding.
- **Notes:**
 - This variable provides more granular information than the range-based dropdown.

Variable: `breast_pr_summary`

Type: Categorical (Dropdown)

Description: Captures a summary interpretation of progesterone receptor (PR) expression status for staging and treatment purposes.

Reference: NAACCR Item #3915

Data Source: Pathology report (synoptic section or biomarker summary), oncology consult notes

Curation Directives:

- **Definition:** Documents whether the PR assay result is interpreted as positive or negative based on IHC staining, or if the result is unavailable or unassessable.

- **Permitted Values:**
 - 0 – PR negative (0.0% or less than 1%)
 - 1 – PR positive
 - 7 – Test ordered, results not in chart
 - 9 – Not documented; Cannot be determined; PR status not assessed or unknown if assessed
- **Source Documents:**
 - Surgical operative reports
 - Pathology reports indicating post-surgical specimen
- **Curation Instructions:**
 - Select 1 if PR is interpreted as positive (1%).
 - Select 0 if PR is interpreted as negative (<1%).
 - Use 7 if PR testing is known to have occurred but results are not available.
 - Use 9 if PR testing is not mentioned or cannot be confirmed.
- **Curator Quality Review:**
 - Ensure logical alignment with `breast_pr_range` and `breast_pr_range_details`.
- **Notes:**
 - When in doubt, prioritize clear documentation from the pathology report.

Variable: `breast_her2_summary`

Type: Categorical (Dropdown)

Description: Captures the overall interpretation of HER2 status based on immunohistochemistry (IHC) and/or in situ hybridization (ISH) testing.

Reference: NAACCR Item #3855

Data Source: Pathology report (HER2 IHC/ISH results), synoptic templates, oncology or multidisciplinary notes

Curation Directives:

- **Definition:** Records the HER2 overall summary result, representing the final interpreted status used for clinical staging and therapeutic decision-making.
- **Permitted Values:**
 - 0 – HER2 negative; equivocal
 - 1 – HER2 positive
 - 2 – Borderline
 - 7 – Test ordered, results not in chart
 - 9 – Not documented; Cannot be Determined; HER2 Overall Summary status not assessed or unknown if assessed
- **Source Documents:**
 - Final pathology report (including synoptic report)
 - HER2 IHC and/or ISH report(s)
 - Molecular diagnostics summary
- **Curation Instructions:**
 - Select 1 (HER2 positive) if explicitly stated as positive or if IHC is 3+ or ISH is amplified.
 - Select 0 (HER2 negative; equivocal) if IHC is 0 or 1+, or ISH is non-amplified or reported as negative.

- Use 2 (Borderline) if the pathology explicitly describes HER2 as “borderline” or discordant without final determination.
 - Use 7 if HER2 testing is known to have been ordered or attempted, but results are not present in the chart.
 - Use 9 if there is no documentation of HER2 testing or if assessment status is unknown.
 - Do not infer positivity or negativity from treatment decisions alone (e.g., use of trastuzumab).
 - **Curator Quality Review:**
 - Verify consistency with HER2 IHC and ISH testing results when available.
 - Ensure no contradiction between `breast_her2_summary` and other related HER2 biomarker fields.
 - **Notes:**
 - If equivocal results are ultimately interpreted as “HER2 negative,” code as 0.
-

Variable: `breast_ki67`

Type: Categorical (Dropdown)

Description: Records the Ki-67 (MIB-1) proliferative index, which quantifies the percentage of tumor cells expressing the Ki-67 protein, a marker of cellular proliferation.

Reference: NAACCR Item #3863

Data Source: Pathology report (IHC for Ki-67), synoptic reports, oncology notes

Curation Directives:

- **Definition:** Ki-67 is a nuclear antigen expressed in proliferating cells. This variable captures the reported percent positivity of tumor cells for Ki-67 via immunohistochemistry (IHC), as an indicator of proliferation. It is often used to help classify breast cancer molecular subtypes and guide treatment decisions.
- **Permitted Values:**
 - 1 – 0.0–100.0: 0.0 to 100.0 percent positive
 - 2 – XXX.7: Test done; actual percentage not stated
 - 3 – XXX.8: Not applicable: Information not collected for this case
 - 4 – XXX.9: Not documented in medical record; Ki-67 (MIB-1) not assessed or unknown if assessed
- **Source Documents:**
 - Final pathology report (Ki-67 IHC result)
 - Breast cancer synoptic report
 - Molecular profiling summaries
- **Curation Instructions:**
 - If the Ki-67 percentage is explicitly stated, select 1 and record the exact value in `breast_ki67_details`.
 - If the test was performed but the report does not state a specific percentage, select 2.
 - Select 3 if the case is out of scope for Ki-67 collection (e.g., not collected by protocol or cohort).
 - Select 4 if there is no documentation of Ki-67 testing or if it’s unclear whether it was performed.
 - Do not infer Ki-67 values from descriptive terms (e.g., “high” or “low”) unless clearly defined in the pathology report.
- **Curator Quality Review:**
 - Confirm that the Ki-67 result is from a validated IHC method and not inferred from other markers.
 - Check for discrepancies between summary staging documents and original pathology.
 - Ensure consistency between `breast_ki67` and `breast_ki67_details` when applicable.
- **Notes:**

- NCCN guidelines recommend that Ki-67 be recorded for certain breast cancer subtypes.
- This variable is often used in conjunction with ER, PR, and HER2 status to assign molecular subtype.

Variable: `breast_ki67_details`

Type: Numeric

Description: Captures the exact Ki-67 proliferation index percentage when a specific range of 1–100% is reported.

Reference: NAACCR Item #3863

Data Source: Pathology report (IHC for Ki-67), synoptic reports

Curation Directives:

- **Definition:**
Records the precise percent positivity for Ki-67 when reported as a specific value between 1% and 100%. This field is only applicable if the general Ki-67 dropdown (`breast_ki67`) is coded as 1.
- **Permitted Values:**
 - Integers between 1 and 100 (inclusive)
- **Source Documents:**
 - Final pathology report
 - IHC lab result or synoptic summary
- **Curation Instructions:**
 1. Complete this field **only** if `breast_ki67` = 1.
 2. Extract the numeric percentage exactly as reported (e.g., “Ki-67 proliferation index: 35%” = record 35).
 3. Round only if instructed in the report (e.g., “approximately 20%” → record 20).
 4. Do not average ranges (e.g., if the report states “20–30%”, leave this field blank and ensure the dropdown is coded appropriately).
 5. Do not enter decimal values, instead round to the nearest whole number if needed.
- **Curator Quality Review:**
 - Ensure consistency between `breast_ki67_details` and the main `breast_ki67` field.
 - Confirm that values fall within 1–100 and are not extrapolated from descriptions like “high” or “elevated”.
- **Notes:**
 - This field allows more granular analysis of Ki-67 values when available.
 - Use only explicitly stated percentages. If percentage is not specified, leave blank and code appropriately in the dropdown.

Variable: `cns_chrom1p`

Type: Categorical (Dropdown)

Description: Captures results of chromosome 1p deletion or loss of heterozygosity (LOH) testing in CNS tumors.

Reference: NAACCR Item #3801

Data Source: Molecular pathology report, integrated surgical pathology reports, tumor board summaries

Curation Directives:

- **Definition:** Indicates whether loss of heterozygosity (LOH) or deletion involving chromosome 1p was identified in CNS tumor testing. This test is commonly performed alongside 19q testing as part of 1p/19q codeletion status.
- **Permitted Values:**
 - 0 – Chromosome 1p deletion/LOH not identified/not present
 - 1 – Chromosome 1p deletion/LOH identified/present
 - 6 – Benign or borderline tumor
 - 7 – Test ordered, results not in chart
 - 8 – Not applicable: Information not collected for this case
 - 9 – Not documented in medical record; Cannot be determined by the pathologist; Not assessed or unknown if assessed
- **Source Documents:**
 - Final surgical pathology report with molecular diagnostics
 - Standalone molecular testing report (e.g., FISH, PCR, NGS)
- **Curation Instructions:**
 - Use value **1** if the report explicitly states chromosome 1p deletion or LOH present.
 - Use value **0** if the report states chromosome 1p LOH not identified or deletion not present.
 - Use value **6** if the tumor is benign or borderline and molecular testing is not clinically relevant.
 - Use **7** only if testing was ordered but no results are available in the chart.
 - Use **8** if the data item is not collected per institutional practice or protocol.
 - Use **9** if no information is found, or testing was not performed or cannot be determined.
- **Curator Quality Review:**
 - Verify that the selected code matches the result of molecular testing, not just a mention of the test being ordered.
 - Check for combined 1p/19q result statements and extract individual 1p status if available.
- **Notes:**
 - Chromosome 1p LOH is most relevant for gliomas and oligodendroglioma classification.
 - Often reported together with 19q LOH as part of “codeletion” status

Variable: cns_chrom19q

Type: Categorical (Dropdown)

Description: Captures results of chromosome 19q deletion or loss of heterozygosity (LOH) testing in CNS tumors.

Reference: NAACCR Item #3802

Data Source: Molecular pathology report, integrated surgical pathology reports, tumor board summaries

Curation Directives:

- **Definition:** Indicates whether loss of heterozygosity (LOH) or deletion involving chromosome 19q was identified in CNS tumor testing. Commonly assessed with chromosome 1p as part of combined 1p/19q codeletion testing.
- **Permitted Values:**
 - 0 – Chromosome 19q deletion/LOH not identified/not present
 - 1 – Chromosome 19q deletion/LOH present
 - 6 – Benign or borderline tumor
 - 7 – Test ordered, results not in chart
 - 8 – Not applicable: Information not collected for this case

- 9 – Not documented in medical record; Cannot be determined by the pathologist; Not assessed or unknown if assessed
- **Source Documents:**
 - Final surgical pathology report with molecular diagnostics
 - Standalone molecular testing report (e.g., FISH, PCR, NGS)
- **Curation Instructions:**
 - Use value **1** if 19q deletion or LOH is reported as present.
 - Use value **0** if deletion is not present or LOH is not identified.
 - Use **6** for benign or borderline tumors where testing is not clinically applicable.
 - Use **7** when test was ordered but the result is missing.
 - Use **8** if data collection for 19q is not part of the case protocol.
 - Use **9** if documentation is absent or insufficient to determine status.
- **Curator Quality Review:**
 - Ensure the LOH or deletion applies specifically to chromosome 19q, not just part of a combined statement.
 - Confirm that coding aligns with documented test results, not with assumptions or test orders alone.
- **Notes:**
 - 1p/19q codeletion is a diagnostic hallmark of oligodendrogliomas.
 - Results may be embedded in molecular pathology addenda or synoptic reports.

Biomarkers

Variable: ini1_testing

Type: Binary (Yes/No)

Description: INI1/SMARCB1 Testing Reported on Pathology Report

Reference: n/a

Data Source: Pathology report (including immunohistochemistry section and addenda)

Curation Directives

- **Definition:** Indicates whether INI1/SMARCB1 immunohistochemistry (IHC) or equivalent testing is reported on the pathology report for the indexed procedure/specimen.
- **Permitted Values:**
 - 1 = Yes
 - 0 = No
- **Source Documents:**
 - Pathology report for the indexed surgical procedure/specimen
 - Immunohistochemistry / ancillary studies section
 - Addenda / supplemental pathology reports attached to the same case
- **Curation Instructions:**
 - Review the pathology report and any addenda for explicit mention of INI1 or SMARCB1 testing.

- Select “Yes” if any of the following are documented: “INI1,” “SMARCB1,” “BAF47,” “INI-1,” or equivalent wording indicating INI1/SMARCB1 IHC/testing was performed.
 - Select “No” if INI1/SMARCB1 is not mentioned anywhere in the report or addenda.
 - Do not infer testing from chordoma subtype alone; testing must be explicitly documented.
- **Curator Quality Review:**
 - Confirm the selection is based on explicit documentation in the pathology report (not external notes).
 - If “Yes,” verify that the result (retained vs loss) is captured in the appropriate downstream variable if one exists.
 - **Notes:**
 - INI1 loss is associated with poorly differentiated chordoma; this field captures testing presence only, not the result.

Variable: `pdl1_testing`

Type: Categorical (Radio)

Description: Number of PD-L1 Tests Reported on Pathology Report

Reference: n/a

Data Source: Pathology report (including immunohistochemistry / biomarker section and addenda)

Curation Directives

- **Definition:** Captures the number of distinct PD-L1 test results documented on the pathology report for the indexed specimen, including any report addenda.
- **Permitted Values:**
 - 1 = 0
 - 2 = Yes
 - 3 = No
- **Source Documents:**
 - Pathology report for the indexed surgical procedure/specimen
 - Biomarker / immunohistochemistry section
 - Addenda / supplemental pathology reports attached to the same case
- **Curation Instructions:**
 - Review the pathology report and associated addenda for PD-L1 testing documentation. Count the number of distinct PD-L1 results reported on the case (for example: multiple assays, multiple scoring methods, or multiple specimens reported separately under the same case).
 - Assign:
 - * Code 1 (0) if PD-L1 is not mentioned anywhere in the report or addenda.
 - * Code 2 (1) if exactly one PD-L1 result is documented.
 - * Code 3 (2) if two distinct PD-L1 results are documented.
 - If the report references PD-L1 testing being ordered or pending without a reported result, do not count it as a test result; code based on results actually reported.
 - Do not count duplicate restatements of the same PD-L1 result across the main report and addenda as multiple tests.
- **Curator Quality Review:**
 - Confirm that counted tests correspond to distinct reported PD-L1 results (not repeated text).

- If PD-L1 results are present, verify downstream PD-L1 detail fields (assay, score, interpretation) are completed consistently with the count.

Variable: ini1_date

Type: Date (MM/DD/YYYY)

Description: INI1/SMARCB1 Test – Addendum Date of Pathology Report

Reference: n/a

Data Source: Pathology report (addendum date associated with INI1/SMARCB1 testing)

- **Definition:** The addendum date on the pathology report that contains the definitive INI1/SMARCB1 test result being abstracted.
- **Permitted Values / Codes:**
Free text date field (validated in MM/DD/YYYY format)
- **Source Documents:**
 - Pathology report addendum(s) for the indexed specimen/procedure
 - Ancillary studies / immunohistochemistry section and dated addenda
- **Curation Instructions:**
 1. Review the pathology report and all addenda for INI1/SMARCB1 results.
 2. If multiple INI1/SMARCB1 tests/results are reported:
 1. Select the test with the definitive result (clearly “retained/intact” vs “loss”).
 2. If more than one test is definitive, select the test documented in the most recent addendum.
 3. Enter the addendum date (not specimen collection date, procedure date, or sign-out date) in MM-DD-YYYY format.
 3. If INI1/SMARCB1 is reported only in the main report without an addendum date, enter the report date used for that result (if explicitly dated). If no date is provided anywhere, leave blank.
- **Curator Quality Review:**
 - Confirm the date corresponds to the addendum/report section where the abstracted INI1 result is documented.
- **Notes:**
 - This field timestamps the reported INI1/SMARCB1 result for longitudinal interpretation (for example, later testing on recurrence/metastasis).

Variable: ini1_date_int

Type: Integer (Calculated)

Description: Date of Addendum for INI1/SMARCB1 Test in Days from DOB (Interval)

Reference: n/a

Data Source: Derived from birth_date and ini1_date fields

Curation Directives:

- **Definition:** Automatically calculates the number of days between the patient’s date of birth and the INI1/SMARCB1 addendum date using REDCap’s datediff() function:
datediff([birth_date],[ini1_date],"d","mdy",true)
- **Permitted Values:**
 - Integer 0

- Null if either `birth_date` or `ini1_date` is missing or improperly formatted
 - **Source Documents:**
 - Patient demographics (`birth_date`)
 - Date of Addendum for INI1/SMARCB1 Test recorded as `ini1_date`
 - **Curation Instructions:**
 1. Ensure both `birth_date` and `ini1_date` fields are entered and valid before relying on the output of this field.
 2. Do not manually override this field—this is a calculated value.
 3. Check for anomalous values (e.g., negative intervals or implausibly high day counts).
 - **Curator Quality Review:**
 - Validate that both contributing fields (`birth_date`, `ini1_date`) are populated and correctly formatted (MM/DD/YYYY).
 - Confirm that calculated intervals align with expected patient age and clinical context.
-

Variable: `ini1_result`

Type: Categorical (Radio)

Description: INI1/SMARCB1 Test Result

Reference: n/a

Data Source: Pathology report (including immunohistochemistry / biomarker section and addenda)

Curation Directives

- **Definition:** The reported INI1/SMARCB1 status for the selected test result in the pathology report.
- **Permitted Values:**
 - 1 = Intact/Retained
 - 2 = Loss
 - 3 = Equivocal
 - 99 = Unknown
- **Source Documents:**
 - Pathology report for the indexed surgical procedure/specimen
 - Biomarker / immunohistochemistry section
 - Addenda / supplemental pathology reports attached to the same case
- **Curation Instructions:**
 - Abstract the INI1/SMARCB1 result exactly as interpreted in the report.
 - Assign:
 - * 1 = Intact/Retained when the report states retained/intact expression or positive nuclear staining consistent with retention.
 - * 2 = Loss when the report states loss/absent nuclear staining (loss of expression).
 - * 3 = Equivocal when the report explicitly describes the result as equivocal/indeterminate or technically limited with an inconclusive interpretation.
 - * 99 = Unknown when INI1/SMARCB1 is referenced but the final interpreted result cannot be determined from the report text available.
 - Do not infer result based on diagnosis label alone (for example, “poorly differentiated chordoma”) unless the report explicitly states INI1/SMARCB1 result.
- **Curator Quality Review:**

- Confirm the selected result is supported by explicit interpretation language (not only raw stain descriptors without interpretation).
- Ensure consistency with ini1_date (same addendum/test instance) and ini1_test_method.

Variable: ini1_specimen_type

Type: Categorical (Radio)

Description: INI1/SMARCB1 Specimen Type Tested

Reference: n/a

Data Source: The disease context of the specimen on which INI1/SMARCB1 testing was performed (primary, recurrence, or metastasis).

Curation Directives

- **Definition:** The reported INI1/SMARCB1 status for the selected test result in the pathology report.
- **Permitted Values:**
 - 1 = Primary tumor
 - 2 = Recurrence
 - 3 = Metastasis
 - 99 = Unknown
- **Source Documents:**
 - Pathology report for the indexed surgical procedure/specimen
 - Biomarker / immunohistochemistry section
 - Addenda / supplemental pathology reports attached to the same case
- **Curation Instructions:**
 - Select 1 = Primary tumor when the specimen represents the original tumor site at initial diagnosis.
 - Select 2 = Recurrence when the report describes locally recurrent disease, recurrent tumor, or re-resection at the primary site/adjacent region.
 - Select 3 = Metastasis when the specimen is from a distant metastatic site (for example, lung, liver, bone outside primary region, distant lymph node) and is described as metastatic.
 - Select 99 = Unknown when the pathology documentation does not clearly indicate whether the tested specimen is primary, recurrent, or metastatic.
 - Do not infer specimen type solely from timing; rely on pathology wording (site and clinical history).
- **Curator Quality Review:**
 - Confirm the selected specimen type aligns with the specimen site and clinical history described in the pathology report.
 - If multiple specimens are in the case, ensure the specimen type corresponds to the block/specimen used for INI1 testing when specified.

Variable: ini1_test_method

Type: Categorical (Radio)

Description: INI1/SMARCB1 Testing Method

Reference: n/a

Data Source: Pathology report (ancillary testing methodology)

Curation Directives

- **Definition:** The method used to assess INI1/SMARCB1 status for the reported result.

- **Permitted Values:**
 - 1 = IHC
 - 2 = Other (specify)
 - 99 = Unknown
- **Source Documents:**
 - Pathology report for the indexed surgical procedure/specimen
 - Biomarker / immunohistochemistry section
 - Addenda / supplemental pathology reports attached to the same case
- **Curation Instructions:**
 - Select 1= IHC when the report documents INI1/SMARCB1 immunohistochemistry / immunostain / immunoperoxidase staining.
 - Select 2 = Other when the report documents a non-IHC approach used to assess SMARCB1/INI1 status (for example, sequencing-based assessment, copy number, targeted molecular test). Populates ini1_test_method_oth.
 - Select 99 = Unknown when INI1/SMARCB1 is reported but the method is not stated.
- **Curator Quality Review:**
 - Confirm the chosen method matches the report language (for example, “immunostain” should be coded as IHC).
 - If Other is selected, ensure ini1_test_method_oth is completed.

Variable: ini1_test_method_other

Type: Free Text Entry (text)

Description: Specify Other INI1/SMARCB1 Testing Method

Reference: n/a

Data Source: Pathology report (ancillary testing methodology)

Curation Directives

- **Definition:** A concise description of the testing method used when INI1/SMARCB1 was assessed by a method other than IHC.
- **Permitted Values:**
 - 1 = IHC
 - 2 = Other (specify)
 - 99 = Unknown
- **Source Documents:**
 - Pathology report for the indexed surgical procedure/specimen
 - Biomarker / immunohistochemistry section
 - Addenda / supplemental pathology reports attached to the same case
- **Curation Instructions:**
 - Complete this field only when ini1_test_method = 2 (Other).
 - Transcribe the method name as documented (for example: “NGS,” “targeted sequencing,” “copy number analysis,” “molecular testing for SMARCB1”). Include relevant qualifiers if stated (for example, panel name or platform) but keep the entry concise.
 - If multiple non-IHC methods are listed for INI1/SMARCB1 assessment, separate entries with semicolons.
- **Curator Quality Review:**

- Confirm the text matches the pathology report language and is specific enough to distinguish the method from IHC.
- If PD-L1 results are present, verify downstream PD-L1 detail fields (assay, score, interpretation) are completed consistently with the count.

Variable: `pd11_date(1-2)`

Type: Date (MM/DD/YYYY)

Description: PD-L1 Test – Addendum Date of Pathology Report

Reference: n/a

Data Source: Pathology report (addendum date associated with PD-L1 testing)

- **Definition:** The addendum date on the pathology report that contains the definitive PD-L1 test result being abstracted.
- **Permitted Values / Codes:**
Free text date field (validated in MM/DD/YYYY format)
- **Source Documents:**
 - Pathology report addendum(s) for the indexed specimen/procedure
 - Ancillary studies / immunohistochemistry section and dated addenda
- **Curation Instructions:**
 1. Review the pathology report and any addenda for PD-L1 results.
 2. Complete this field when `pd11_testing` indicates at least one PD-L1 result is present (`pd11_testing` = 1 test or 2 tests, per instrument coding).
 3. Enter the addendum date associated with that PD-L1 result in MM-DD-YYYY format.
 4. Do not enter specimen collection date, procedure date, or report sign-out date unless that is explicitly the dated addendum/report section containing the PD-L1 result.
 5. If the PD-L1 result is reported without a dated addendum and no date is provided for the biomarker section, leave blank.
- **Curator Quality Review:**
 - Confirm the date corresponds to the addendum/report section where the abstracted PD-L1 result is documented.
- **Notes:**
 - This field timestamps the reported PD-L1 result for longitudinal interpretation (for example, later testing on recurrence/metastasis).

Variable: `pd11_date_int(1-2)`

Type: Integer (Calculated)

Description: Date of Addendum for PD-L1 Test in Days from DOB (Interval)

Reference: n/a

Data Source: Derived from `birth_date` and `pd11-2_date` fields

Curation Directives:

- **Definition:** Automatically calculates the number of days between the patient's date of birth and the PD-L1 addendum date using REDCap's `datediff()` function:
`datediff([birth_date],[pd11_date(1-2)],"d","mdy",true)`
- **Permitted Values:**

- Integer 0
 - Null if either `birth_date` or `pd11_date(1-2)` is missing or improperly formatted
 - **Source Documents:**
 - Patient demographics (`birth_date`)
 - Date of Addendum for PD-L1 Test recorded as `pd11_date(1-2)`
 - **Curation Instructions:**
 1. Ensure both `birth_date` and `pd11_date(1-2)` fields are entered and valid before relying on the output of this field.
 2. Do not manually override this field—this is a calculated value.
 3. Check for anomalous values (e.g., negative intervals or implausibly high day counts).
 - **Curator Quality Review:**
 - Validate that both contributing fields (`birth_date`, `pd11_date(1-2)`) are populated and correctly formatted (MM/DD/YYYY).
 - Confirm that calculated intervals align with expected patient age and clinical context.
-

Variable: `pd11_assay(1-2)`

Type: Categorical (Dropdown)

Description: PD-L1 Assay Type (Antibody Clone)

Reference: n/a

Data Source: Pathology report (including immunohistochemistry / biomarker section and addenda)

Curation Directives:

- **Definition:** AThe PD-L1 antibody clone (assay) used for the PD-L1 test reported on the pathology case being abstracted.
- **Permitted Values:**
 - 1 = 22C3
 - 2 = 28-8
 - 3 = SP142
 - 4 = SP263
 - 5 = E1L3N
 - 88 = Other
 - 99 = Unknown
- **Source Documents:**
 - Pathology report addendum(s) for the indexed specimen/procedure
 - Ancillary studies / immunohistochemistry section and dated addenda
- **Curation Instructions:**
 1. Review the PD-L1 result section and any methods/antibody listing for explicit documentation of the clone (commonly formatted as “PD-L1 (22C3)” or “PD-L1, clone 28-8”).
 2. Select the corresponding clone when explicitly stated.
 3. Select “88 – Other” when a PD-L1 clone is documented but is not one of the listed options.
 4. Select “99 – Unknown” when PD-L1 testing is reported but the antibody clone is not specified anywhere in the report or addenda.
 5. Do not infer the clone based on the score type, disease site, or interpretation language alone.
 6. If two PD-L1 tests are present on the same case (`pd11_testing = 2`), abstract the clone corresponding to the PD-L1 test details being captured in the associated “first test” fields (date/value/score type).

- **Curator Quality Review:**

- Confirm the selected assay is supported by explicit pathology report language.
- If the report includes multiple PD-L1 entries, verify the abstracted clone aligns with the correct PD-L1 result instance (avoid selecting the clone from a different specimen/addendum on the same case).

Variable: pdl1_assay(1-2)

Type: Categorical (Dropdown)

Description: PD-L1 Assay Type (Antibody Clone)

Reference: n/a

Data Source: Pathology report (including immunohistochemistry / biomarker section and addenda)

Curation Directives:

- **Definition:** AThe PD-L1 antibody clone (assay) used for the PD-L1 test reported on the pathology case being abstracted.
- **Permitted Values:**
 - 1 = 22C3
 - 2 = 28-8
 - 3 = SP142
 - 4 = SP263
 - 5 = E1L3N
 - 88 = Other
 - 99 = Unknown
- **Source Documents:**
 - Pathology report addendum(s) for the indexed specimen/procedure
 - Ancillary studies / immunohistochemistry section and dated addenda
- **Curation Instructions:**
 1. Review the PD-L1 result section and any methods/antibody listing for explicit documentation of the clone (commonly formatted as “PD-L1 (22C3)” or “PD-L1, clone 28-8”).
 2. Select the corresponding clone when explicitly stated.
 3. Select “88 – Other” when a PD-L1 clone is documented but is not one of the listed options.
 4. Select “99 – Unknown” when PD-L1 testing is reported but the antibody clone is not specified anywhere in the report or addenda.
 5. Do not infer the clone based on the score type, disease site, or interpretation language alone.
 6. If two PD-L1 tests are present on the same case (pdl1_testing = 2), abstract the clone corresponding to the PD-L1 test details being captured in the associated “first test” fields (date/value/score type).
- **Curator Quality Review:**
 - Confirm the selected assay is supported by explicit pathology report language.
 - If the report includes multiple PD-L1 entries, verify the abstracted clone aligns with the correct PD-L1 result instance (avoid selecting the clone from a different specimen/addendum on the same case).

Variable: pdl1_score_type(1-2)

Type: Categorical (Dropdown)

Description: PD-L1 Score Type Used (Reporting Format)

Reference: n/a

Data Source: Pathology report (including immunohistochemistry / biomarker section and addenda)

Curation Directives:

- **Definition:** The scoring system used to report the PD-L1 result documented on the pathology case being abstracted (e.g., TPS, CPS, or IC), as explicitly stated in the report.
- **Permitted Values:**
 - 1 = TPS (Tumor Proportion Score)
 - 2 = CPS (Combined Positive Score)
 - 3 = IC (Immune Cell Score)
 - 88 = Other
 - 99 = Unknown
- **Source Documents:**
 - Pathology report addendum(s) for the indexed specimen/procedure
 - Ancillary studies / immunohistochemistry section and dated addenda
- **Curation Instructions:**
 1. Review the PD-L1 result section for explicit labeling of the score type (e.g., “TPS,” “CPS,” “IC,” “Tumor Proportion Score,” “Combined Positive Score,” “Immune Cell score”).
 2. Assign:
 1. Code 1 (TPS) when the report provides TPS or describes “% tumor cells positive.”
 2. Code 2 (CPS) when the report provides CPS or describes a combined score of tumor and immune cells (often numeric and may exceed 100).
 3. Code 3 (IC) when the report provides IC or describes “% tumor area with PD-L1+ immune cells” (commonly SP142, but do not infer).
 4. Code 88 (Other) when a non-TPS/CPS/IC scoring approach is explicitly documented (e.g., H-score, “staining intensity score,” or other specified methodology).
 5. Code 99 (Unknown) when PD-L1 is reported only as a qualitative statement (e.g., “PD-L1 positive/negative”) and no scoring method is stated.
 3. If multiple PD-L1 score types are documented on the same pathology case, select the score type corresponding to the PD-L1 result instance being abstracted in the associated PD-L1 fields (date/value/interpretation).
 4. Do not infer score type from assay clone, specimen site, or treatment context.
- **Curator Quality Review:**
 - Confirm the selected score type matches the report’s explicit terminology and the numeric formatting (e.g., TPS/IC often expressed as %; CPS often presented as a score that may exceed 100).

Variable: pdl1_value(1-2)

Type: Free Text Entry (text)

Description: PD-L1 Numeric Result

Reference: n/a

Data Source: Pathology report (including immunohistochemistry / biomarker section and addenda)

Curation Directives:

- **Definition:** The numeric PD-L1 result reported for the score type captured in `pd11_score_type`, as explicitly documented in the pathology report for the PD-L1 test being abstracted.
- **Source Documents:**
 - Pathology report addendum(s) for the indexed specimen/procedure
 - Ancillary studies / immunohistochemistry section and dated addenda
- **Curation Instructions:**
 - Complete this field only when `pd11_score_type` is not Unknown (99) and a numeric PD-L1 result is reported.
 - Enter the numeric value exactly as stated, using numbers only (do not include “%”, “<”, “>”, “ ”, “ ”, or text).
 - For TPS and IC: enter the numeric percent value (typically 0–100).
 - For CPS:
 - * enter the numeric value as reported (may exceed 100).
 - * If a range or inequality is reported (e.g., “1–49%”, “<1%”, “>50%”), then enter the lower numeric value in `pd11_value`.
 - If the report provides multiple numeric values for PD-L1 within the same test instance (e.g., multiple specimens/blocks), abstract the value that corresponds to the PD-L1 test instance being captured in the associated fields (date/assay/interpretation).
 - Do not average or combine values. Do not infer a numeric value from a qualitative interpretation (e.g., “positive”) if the number is not provided.
- **Curator Quality Review:**
 - Confirm the numeric value matches the documented score type (TPS/CPS/IC) and is taken directly from the report.
 - If `pd11_range_yn` = Yes, verify `pd11_value` contains the lower value and `pd11_range_upper` contains the corresponding upper/boundary value.
 - Check for transcription errors (e.g., misplaced decimals, swapped range values).

Variable: `pd11_range_yn`(1-2)

Type: Binary (Yes/No)

Description: PD-L1 Score Type Used (Reporting Format)

Reference: n/a

Data Source: Pathology report (including immunohistochemistry / biomarker section and addenda)

Curation Directives:

- **Definition:** Indicates whether the PD-L1 numeric result is reported as a range or inequality (e.g., “1–49%”, “<1%”, “>50”) rather than a single point value.
- **Permitted Values:**
 - 1 = Yes
 - 0 = No
- **Source Documents:**
 - Pathology report addendum(s) for the indexed specimen/procedure
 - Ancillary studies / immunohistochemistry section and dated addenda
- **Curation Instructions:**
 1. Complete this field only when `pd11_score_type` is not Unknown (99) and a numeric PD-L1 result is present.

2. Select Yes when the report expresses the result using any of the following:
 1. A range with two bounds (e.g., “1–49%”, “10–20”).
 2. An inequality (e.g., “<1%”, “1%”, “>50%”, “10”).
 3. A categorical numeric band (e.g., “0–1%”, “1–5%”) rather than a single value.
 3. Select No when the report provides a single numeric value (e.g., “TPS 5%”, “CPS 20”, “IC 1%”).
 4. If Yes, ensure pdl1_value contains the lower/boundary value per its instructions and complete pdl1_range_upper when an upper/boundary value is specified or implied by the notation.
- **Curator Quality Review:**
 - Confirm the range flag matches the exact way the result is presented in the pathology report. If pdl1_range_yn = Yes, verify that downstream range handling is consistent (pdl1_value and pdl1_range_upper populated appropriately).

Variable: pdl1_range_upper(1-2)

Type: Free Text Entry (text)

Description: PD-L1 Score Type Used (Reporting Format)

Reference: n/a

Data Source: Pathology report (including immunohistochemistry / biomarker section and addenda)

Curation Directives:

- **Definition:** Indicates whether the PD-L1 numeric result is reported as a range or inequality (e.g., “1–49%”, “<1%”, “>50”) rather than a single point value.
- **Permitted Values:**
 - 1 = Yes
 - 0 = No
- **Source Documents:**
 - Pathology report addendum(s) for the indexed specimen/procedure
 - Ancillary studies / immunohistochemistry section and dated addenda
- **Curation Instructions:**
 1. Complete this field only when pdl1_range_yn = 1 (Yes).
 2. Enter the upper value as a number only (do not include “%”, inequality symbols, or text).
 3. For two-sided ranges (e.g., “1–49%”): enter the upper value (49).
 4. For inequality expressions:
 1. “<1%” or “1%”: enter 1.
 2. “>50%” or “50%”: enter 50 (the reported boundary value).
 3. Ensure pdl1_value contains the lower/boundary value corresponding to the same expression.
- **Curator Quality Review:**
 - Confirm the upper value matches the reported range/inequality and is captured as a numeric value only. Verify consistency between pdl1_value (lower/boundary) and pdl1_range_upper (upper/boundary).

Variable: pdl1_interpretation(1-2)

Type: Categorical (Dropdown)

Description: PD-L1 Summary Interpretation

Reference: n/a

Data Source: Pathology report (including immunohistochemistry / biomarker section and addenda)

Curation Directives:

- **Definition:** The qualitative PD-L1 interpretation explicitly stated in the pathology report for the PD-L1 test being abstracted (e.g., “positive,” “negative,” “high expression,” “low expression,” “indeterminate”).
- **Permitted Values:**
 - 1 = Positive
 - 2 = Negative
 - 3 = High
 - 4 = Low
 - 5 = Indeterminate
 - 99 = Not stated
- **Source Documents:**
 - Pathology report addendum(s) for the indexed specimen/procedure
 - Ancillary studies / immunohistochemistry section and dated addenda
- **Curation Instructions:**
 1. Review the PD-L1 result section and any interpretive comments for an explicit qualitative call.
 2. Assign:
 1. Positive when the report states “PD-L1 positive,” “positive expression,” or equivalent.
 2. Negative when the report states “PD-L1 negative,” “no expression,” or equivalent.
 3. High when the report explicitly states “high,” “high expression,” or similar categorical high designation.
 4. Low when the report explicitly states “low,” “low expression,” or similar categorical low designation.
 5. Indeterminate when the report explicitly states “indeterminate,” “equivocal,” “cannot be determined,” “insufficient,” or indicates a technically limited assay with no definitive call.
 6. Not stated when no qualitative interpretation is provided and only numeric values (e.g., TPS/CPS/IC) are reported.
 3. Do not infer Positive/Negative/High/Low from numeric thresholds unless the report itself provides the qualitative interpretation.
 4. If multiple interpretations are present on the same case (e.g., different specimens/addenda), abstract the interpretation that corresponds to the PD-L1 test instance captured in the associated PD-L1 fields (date/assay/score type/value).
- **Curator Quality Review:**
 - Confirm the selected interpretation is explicitly documented (not inferred).
 - Check consistency between the interpretation and the PD-L1 result instance being abstracted (same specimen/addendum as pdl1_date_1 and pdl1_value).
 - If interpretation conflicts with the numeric value, escalate to QA Manager.

Drug Exposure

Drug Exposure Information

Complete one instance of the Drug Exposure form for each distinct drug administered to the patient, regardless of whether the drug was part of a combination regimen, single-agent therapy, or multiple concurrent

treatments. For each drug, abstract only what is explicitly documented in authoritative sources (e.g., medication administration records, infusion center notes, verified pharmacy orders, or oncology provider notes).

For each drug instance, record the ordering and administration institutions, treating provider, and whether the drug was administered as part of a clinical trial. Capture the start and end dates for administration (or project using approved conventions when dates are incomplete), the drug name (standardized per project conventions), route, dose and units, frequency, and any dose modifications or discontinuations, including documented reasons.

Ensure each drug’s timeline aligns with the patient’s diagnosis, procedures, and other treatments to maintain internal chronology. Do not infer administration from lab trends, imaging, or treatment response notes unless explicitly stated. If the same drug is restarted after a gap or is given in a different phase of therapy, create a separate instance of the form for that episode.

Variable: `trt_yn`

Type: Binary (Yes/No)

Description: Indicates whether the patient received any systemic anti-cancer therapy (SACT) at any point during their course of treatment.

Reference: n/a

Data Source: Oncology clinic notes, chemotherapy administration records, infusion center records, medication reconciliation, EMR treatment summaries

Curation Directives

- **Definition:** This variable captures whether the patient received any systemic anti-cancer therapy (SACT), regardless of timing or treatment setting. SACT includes but is not limited to chemotherapy, immunotherapy, targeted therapy, endocrine therapy, radiopharmaceuticals, and cellular therapy (e.g., CAR-T). Do not include surgical or radiation-only treatments here.
- **Permitted Values:**
 - 0 – No
 - 1 – Yes
- **Source Documents:**
 - Oncology progress notes
 - Treatment plans and protocols
 - Chemotherapy or immunotherapy administration records
 - Medication lists and prescription records
 - Discharge summaries and EMR treatment history
- **Curation Instructions:**
 - Review all available documentation to determine whether the patient received any form of systemic anti-cancer therapy.
 - Code **1 (Yes)** if there is any documented administration or receipt of systemic therapy, even if it was a single dose or part of a clinical trial.
 - Code **0 (No)** only if the record clearly indicates that no systemic therapy was administered at any time or if the patient received only local therapy (e.g., surgery or radiation).
 - If documentation is ambiguous or missing, escalate for clinical review and leave the field blank until confirmation.
- **Curator Quality Review:**
 - Ensure consistency with downstream drug-level entries (e.g., `drug_regimen`, `drug_class`, or `drug_start_dt`).

- Confirm that coding of “Yes” aligns with drug exposure records in the EMR or chart.
- Document rationale in internal notes if clinical ambiguity is present.

- **Notes:**

- This field acts as a gatekeeper for populating downstream treatment fields.
- If coded as “No,” other systemic therapy fields (e.g., drug regimen details) may remain blank.
- Inclusion is irrespective of treatment intent (curative, adjuvant, neoadjuvant, palliative).

Variable: `trt_order_institution`

Type: Categorical (Dropdown)

Description: Indicates whether systemic anti-cancer therapy was ordered at the GENIE-participating institution or at an external facility.

Reference: n/a

Data Source: Oncology consult notes, EMR medication orders, external referral documents, care coordination communications

Curation Directives

- **Definition:** Captures the originating facility where systemic anti-cancer therapy was prescribed. This is distinct from the site of administration and helps assess treatment attribution and documentation quality.
- **Permitted Values:**
 - 1 – Internal Institution
 - 2 – External Institution
- **Source Documents:**
 - Initial oncology consultation or treatment plan
 - EMR medication orders (e.g., Epic Beacon, CPOE)
 - External referral records or scanned treatment orders
 - Documentation from tumor board or multidisciplinary treatment planning notes
 - Correspondence from external oncology providers
- **Curation Instructions:**
 - Code **1 (Internal Institution)** if treatment was initiated based on orders by a clinician affiliated with the GENIE-participating institution.
 - Code **2 (External Institution)** if treatment was initiated based on orders from an outside provider or facility.
 - If treatment was continued at the internal site but originally ordered externally, select **2**.
 - If unclear, review timing of first documentation of treatment plan and consult ordering provider’s affiliation.
 - Leave blank and flag for clinical review if both internal and external roles are unclear or if no documentation exists.
- **Curator Quality Review:**
 - Confirm consistency with `trt_admin_institution`, but treat each independently unless documented otherwise.
 - Ensure that the provider’s institutional affiliation is verifiable via signature blocks, referral notes, or EMR metadata.
 - If external treatment was later formally adopted under the internal institution’s plan, code as 1 (Internal),
- **Notes:**

- Discrepancies between ordering and administering institutions are expected in shared-care or transferred patients.
- This variable supports accurate assignment of treatment origin and completeness tracking for downstream data use.

Variable: `trt_admin_institution`

Type: Categorical (Dropdown)

Description: Captures whether systemic anti-cancer therapy was administered at the GENIE-participating institution or at an outside facility.

Reference: n/a

Data Source: Oncology notes, treatment summaries, external consult notes, EMR medication administration records

Curation Directives

- **Definition:** Indicates the institutional setting where systemic anti-cancer therapy was administered—either internally at the GENIE-participating site or externally at another healthcare facility. This is relevant for determining data completeness and source reliability.
 - **Permitted Values:**
 - 1 – Internal Institution
 - 2 – External Institution
 - **Source Documents:**
 - On-treatment summaries or chemotherapy administration records
 - Oncology progress notes indicating treatment location
 - Outside facility records scanned into the EMR
 - Referral notes or external consultation summaries
 - Medication reconciliation lists specifying treatment origin
 - **Curation Instructions:**
 - Code **1 (Internal Institution)** if systemic treatment was administered at the GENIE-participating site.
 - Code **2 (External Institution)** if documentation indicates the patient received therapy at a different hospital, clinic, or cancer center.
 - In cases of both internal and external administration, prioritize the location where the majority of systemic therapy occurred, or where the index regimen was delivered.
 - If the location is unclear or not documented, flag for review and leave the field blank until clarified.
 - Use other variables (e.g., `trt_regimen_name`, `trt_start_dt`) to assist in deducing the treatment site from context.
 - **Curator Quality Review:**
 - Escalate ambiguous or mixed-location cases to clinical leads for resolution.
 - **Notes:**
 - This variable supports downstream decisions about expected drug-level completeness and curation fidelity.
 - Consider implications for missing dosing or cycle details if treatment was external. r
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Variable: trt_trial_yn

Type: Binary (Yes/No)

Description: Indicates whether the documented systemic anti-cancer treatment was administered as part of a clinical trial.

Reference: n/a

Data Source: Oncology provider notes, clinical trial enrollment records, treatment plans, pharmacy records, research coordinator documentation

Curation Directives

- **Definition:** Denotes whether the treatment was part of an interventional clinical research study involving investigational or approved agents under a protocol.
 - **Permitted Values:**
 - Yes (1) – Treatment was administered under a clinical trial protocol
 - No (0) – Treatment was not associated with a clinical trial
 - **Source Documents:**
 - Oncologist notes indicating trial enrollment
 - Clinical trial consent forms or documentation in EMR
 - Research pharmacy records or investigational drug dispensation logs
 - Treatment plans labeled with protocol numbers
 - Notes from research coordinators or trial nurses
 - **Curation Instructions:**
 - Code Yes (1) if there is clear documentation that the therapy was administered under a clinical trial protocol (e.g., “patient enrolled on NCTxxxxxx,” “per protocol,” “under study drug”), even if the drug is standard-of-care.
 - Code No (0) if there is no indication the treatment was part of a clinical trial.
 - **Curator Quality Review:**
 - Confirm the presence of trial language and protocol identifiers when coding “Yes.”
 - If documentation is ambiguous, flag for adjudication or seek verification from research staff.
 - Be cautious of templated language that may suggest trial consideration without enrollment.
 - **Notes:**
 - This field supports analyses of investigational treatment patterns and outcomes.
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Variable: trt_ongoing_yn

Type: Binary (Yes/No)

Description: Treatment Ongoing at Time of Abstraction

Reference: n/a

Data Source: Oncology provider notes, clinical trial enrollment records, treatment plans, pharmacy records, research coordinator documentation

Curation Directives

- **Definition:** Indicates whether the patient is still actively receiving the documented treatment at the time the curator is abstracting the record. For clinical trial drug exposure, this determination is made at the regimen level (not per-agent), consistent with limited trial drug capture (regimen start/stop only).
- **Permitted Values:**

- 1 = Yes
- 0 = No
- **Source Documents:**
 - Oncologist notes indicating trial enrollment
 - Clinical trial consent forms or documentation in EMR
 - Research pharmacy records or investigational drug dispensation logs
 - Treatment plans labeled with protocol numbers
 - Notes from research coordinators or trial nurses
- **Curation Instructions:**
 - Determine status using documentation available up to the abstraction date.
 - * Select Yes if the record indicates the regimen is ongoing/active/continuing and there is no documentation of discontinuation. Acceptable evidence includes language such as “continue,” “ongoing,” “currently on,” “still receiving,” “on treatment,” “maintaining,” “cycle X ongoing,” or other clear indication the regimen remains active.
 - * Select No if the record indicates the regimen has ended, been stopped, discontinued, completed, permanently held, or switched to a different therapy. This includes discontinuation for any reason (progression, toxicity, patient preference, completion of planned course, hospice, death).
 - * Do not infer ongoing status solely from the absence of a stop date; use the most recent documentation to make the determination.
 - * If treatment is documented as temporarily held with intent to resume and no documentation of permanent discontinuation, code Yes (still considered active/ongoing).
 - * If documentation indicates the hold is indefinite or effectively stopped, code No.
 - Clinical trial drug exposure (regimen-level rule):
 - * Assess ongoing status at the regimen level. Do not require agent-level administration evidence.
 - * Select Yes if documentation indicates the patient is on-study/on-treatment or continuing protocol therapy (e.g., “receiving study drug,” “Cycle X Day Y,” “continue per protocol”) without documentation of end-of-treatment/off-study.
 - * Select No if documentation indicates end of treatment, off study, withdrawn, trial drug stopped, protocol therapy discontinued, or equivalent, regardless of reason.
- **Curator Quality Review:**
 - Confirm that treatment switches are not misclassified as ongoing (if a new regimen starts and the prior regimen is stopped, the prior should be “No”).

Clinical Trial Drug Exposure

Note: For drugs provided as part of a clinical trial, only limited data (regimen-level start and stop dates) is collected. Please record the start date as the first dose of the first drug administered and, if applicable, the end date as the last dose of the last drug in the regimen.

Variable: trt_trial_start_date

Type: Date (MM/DD/YYYY)

Description: Clinical Trial Regimen Start Date

Reference: n/a

Data Source: Oncology provider notes, clinical trial enrollment records, treatment plans, pharmacy records, research coordinator documentation

Curation Directives:

- **Definition:** Captures the calendar date when the patient received the first dose of any drug included in the clinical trial regimen. For clinical trial drug exposure, dates are collected at the regimen level only.
- **Permitted Values / Codes:**
 - Date entered in MM-DD-YYYY format
 - Leave blank only if the regimen start date cannot be determined from any available documentation
- **Source Documents:**
 - Oncologist notes indicating trial enrollment
 - Clinical trial consent forms or documentation in EMR
 - Research pharmacy records or investigational drug dispensation logs
 - Treatment plans labeled with protocol numbers
 - Notes from research coordinators or trial nurses
- **Curation Instructions:**
 1. Identify the clinical trial regimen being abstracted and review documentation for the earliest date the patient actually received protocol therapy.
 2. Record the start date as the first dose date of the first drug administered within the regimen (not consent date, enrollment date, randomization date, or planned start date unless it is explicitly the first administered dose date).
 3. If multiple trial drugs are given on different days at initiation, use the earliest administration date among them.
 4. If documentation includes multiple relevant dates (e.g., consent, screening, Cycle 1 Day 1 planned, Cycle 1 Day 1 administered), select the date indicating treatment administration (e.g., “C1D1 given,” “first infusion,” “started study drug”).
 5. Enter the date using MM-DD-YYYY format. Apply Data Estimation Rules when a partial date is provided.
- **Curator Quality Review:**
 - Confirm the selected date reflects first administered dose of protocol therapy (not referral/consent/screening).
 - Cross-check against visit timelines and any available infusion/dispense records for plausibility and internal consistency.
 - If multiple source documents disagree, prioritize documentation that directly indicates administration (e.g., infusion record, “C1D1 administered” note) and flag discrepancies per study workflow if applicable.
- **Notes:**
 - This field is regimen-level and may not align to individual agent start dates. It supports time-to-treatment and longitudinal trial exposure analyses.

Variable: trt_trial_start_date_int

Type: Integer (Calculated)

Description: Date of Clinical Trial Regimen Start in Days from Date of Birth (Interval)

Reference: n/a

Data Source: Automatically calculated from birth_date and trt_trial_start_date fields

Curation Directives:

- **Definition:** Automatically calculates the number of days between the patient's date of birth and the clinical trial regimen start date (`trt_trial_start_date`). This standardizes timing across patients for downstream analytics.
- **Permitted Values / Codes:**
 - Automatically populated by the system using: `datediff([birth_date], [trt_trial_start_date], "d", "mdy", true)`
 - Value will be a non-negative integer if `birth_date` and `trt_trial_start_date` are both valid
- **Source Documents:**
 - Not directly abstracted—calculated from:
 - * `birth_date`
 - * `trt_trial_start_date`
- **Curation Instructions:**
 1. Ensure both `birth_date` and `trt_trial_start_date` are correctly entered and validated.
 2. This field should auto-populate upon saving the record.
 3. Do not manually modify this field or override its value.
- **Curator Quality Review:**
 - Confirm accuracy of source fields (`birth_date` and `trt_trial_start_date`) if calculated value appears implausible (e.g., negative, extreme).
- **Notes:**
 - This variable is used to derive age-standardized metrics for analysis.

Variable: `trt_trial_end_date`

Type: Date (MM/DD/YYYY)

Description: Clinical Trial Regimen End Date

Reference: n/a

Data Source: Oncology provider notes, clinical trial enrollment records, treatment plans, pharmacy records, research coordinator documentation

Curation Directives:

- **Definition:** Captures the calendar date when the patient received the last dose of any drug included in the clinical trial regimen. For clinical trial drug exposure, dates are collected at the regimen level only (last dose of last drug in the regimen).
- **Permitted Values / Codes:**
 - Date entered in MM-DD-YYYY format
 - Leave blank only if the regimen end date cannot be determined from any available documentation
- **Source Documents:**
 - Oncologist notes indicating trial enrollment
 - Clinical trial consent forms or documentation in EMR
 - Research pharmacy records or investigational drug dispensation logs
 - Treatment plans labeled with protocol numbers
 - Notes from research coordinators or trial nurses
- **Curation Instructions:**
 1. Identify the clinical trial regimen being abstracted and review documentation for the date the patient last actually received protocol therapy.

2. Record the end date as the last dose date of the last drug administered within the regimen (not the date of consent withdrawal, off-study date, or visit date unless it explicitly corresponds to last administration).
3. If multiple trial drugs are administered on different schedules near the end of the regimen, use the latest administration date among the regimen drugs.
4. If documentation provides an end-of-treatment/off-treatment note without a clear last dose date, search for supporting administration language (e.g., “last dose given on...,” “C#D# administered,” infusion record date). If the last dose date cannot be determined, leave blank
5. Enter the date using MM-DD-YYYY format. Apply Data Estimation Rules when a partial date is provided.

- **Curator Quality Review:**

- Confirm the selected date reflects an administered last dose of protocol therapy (not administrative end/withdrawal documentation unless it specifies the last dose date).
- Cross-check consistency with `trt_ongoing_yn = 0` and with any available administration/dispense records.
- If the documented end date precedes the start date, re-check both fields and correct discrepancies.

- **Notes:**

- This field is regimen-level and may not align to individual agent stop dates. It supports duration-of-exposure and time-to-event analyses for clinical trial treatment.

Variable: `trt_trial_end_date_int`

Type: Integer (Calculated)

Description: Date of Clinical Trial Regimen End in Days from Date of Birth (Interval)

Reference: n/a

Data Source: Automatically calculated from `birth_date` and `trt_trial_end_date` fields

Curation Directives:

- **Definition:** Automatically calculates the number of days between the patient’s date of birth and the clinical trial regimen end date (`trt_trial_end_date`). This standardizes timing across patients for downstream analytics.
- **Permitted Values / Codes:**
 - Automatically populated by the system using: `datediff([birth_date], [trt_trial_end_date], "d", "mdy", true)`
 - Value will be a non-negative integer if `birth_date` and `trt_trial_end_date` are both valid
- **Source Documents:**
 - Not directly abstracted—calculated from:
 - * `birth_date`
 - * `trt_trial_end_date`
- **Curation Instructions:**
 1. Ensure both `birth_date` and `trt_trial_end_date` are correctly entered and validated.
 2. This field should auto-populate upon saving the record.
 3. Do not manually modify this field or override its value.
- **Curator Quality Review:**
 - Confirm accuracy of source fields (`birth_date` and `trt_trial_end_date`) if calculated value appears implausible (e.g., negative, extreme).

- **Notes:**
 - This variable is used to derive age-standardized metrics for analysis.
-

Drug Exposure

Variable: trt_start_date

Type: Date (MM/DD/YYYY)

Description: Calendar date when systemic anti-cancer treatment was first administered to the patient.

Reference: n/a

Data Source: Medication administration records, infusion logs, pharmacy dispense records, oncology notes, external medical records

Curation Directives

- **Definition:** Captures the initiation date of a systemic anti-cancer regimen, including chemotherapy, immunotherapy, targeted therapy, or hormonal therapy. This reflects the *first dose administered* to the patient and is distinct from the date the treatment was ordered or prescribed.
 - **Permitted Values:**
 - Valid calendar date in MM/DD/YYYY format
 - **Source Documents:**
 - Medication Administration Record (MAR)
 - Infusion center logs
 - Oncology provider or nursing progress notes
 - Pharmacy records indicating drug administration
 - External institution records if internal data are unavailable
 - **Curation Instructions:**
 - Record the exact treatment initiation date based on documentation of administration, not prescription.
 - For oral agents, use the date therapy was started per documentation (e.g., “patient started capecitabine on 4/15/2023”), not the dispense date.
 - For regimens involving multiple drugs, use the earliest administration date.
 - Do not use dates from planned future treatments or tentative schedules.
 - **Curator Quality Review:**
 - Verify date consistency across multiple documents when available.
 - Ensure that the selected date aligns with the start of the regimen being abstracted.
 - **Notes:**
 - This field supports calculation of treatment timing intervals and time-to-treatment metrics.
 - Accurate capture is critical for treatment sequence reconstruction.
-

Variable: trt_start_date_int

Type: Integer (Calculated)

Description: Number of days from birth to treatment start date, representing the patient’s age in days at

initiation of systemic anti-cancer therapy.

Reference: n/a

Data Source: Automatically calculated from `birth_date` and `trt_start_date` fields

Curation Directives

- **Definition:** Represents the time interval in days between the patient's date of birth and the documented treatment start date (`trt_start_date`). Used to support calculations involving age at treatment and treatment timelines.
- **Permitted Values:**
 - Integer 0 (representing days from birth to treatment initiation)
- **Source Documents:**
 - Not manually curated; auto-derived by the system using:
`datediff([birth_date],[treatment_start_date],"d","mdy",true)`
- **Curation Instructions:**
 - Ensure both `birth_date` and `trt_start_date` are correctly and completely curated, as this field is dependent on their accuracy.
 - Do not edit this field manually.
 - If either source field is blank or contains invalid dates, this field will return a null or error—flag for correction.
 - For data quality checks, verify that the resulting age in days aligns logically with patient records (e.g., treatment not recorded prior to birth).
- **Curator Quality Review:**
 - Confirm source dates are valid and non-null.
 - Cross-check for negative values or implausible ages (e.g., >40,000 days [~110 years]) that may indicate date entry errors.

Variable: `trt_end_date`

Type: Date (MM/DD/YYYY)

Description: Date when the cancer treatment was completed or discontinued.

Reference: n/a

Data Source: Oncology treatment summaries, medical oncology notes, infusion records, medication stop dates in EHR

Curation Directives

- **Definition:** The final calendar date on which the patient received systemic anti-cancer therapy, whether due to completion of planned treatment, discontinuation for toxicity or progression, or patient decision.
- **Permitted Values:**
 - Valid calendar date (MM/DD/YYYY)
 - Null if treatment is ongoing or if end date is not documented
- **Source Documents:**
 - Oncology progress notes detailing treatment course and end date
 - Medication administration records
 - Infusion center documentation
 - Discharge summaries or EHR medication stop dates

- **Curation Instructions:**
 - Record the documented treatment end date from the source with the highest clinical detail (e.g., note from oncologist > EHR auto-entry).
 - If the patient is still receiving treatment, leave this field blank.
 - Do not infer the end date based solely on absence of future treatment entries.
 - If the end date is noted in free text (e.g., “last dose given on March 15”), abstract as MM/DD/YYYY.
 - Ensure `trt_end_date` is chronologically after `trt_start_date` if both are present.
- **Curator Quality Review:**
 - Confirm consistency between `trt_start_date` and `trt_end_date`.
 - Verify against medication logs or oncology notes for discrepancies.
- **Notes:**
 - This variable supports time-on-treatment calculations and survival analyses.

Variable: `trt_end_date_int`

Type: Integer (Calculated)

Description: Number of days from date of birth to treatment end date.

Reference: n/a

Data Source: Derived from `birth_date` and `trt_end_date`

Curation Directives

- **Definition:** This calculated field represents the patient’s age in days at the time systemic cancer treatment ended. It is automatically computed using the REDCap function:
`datediff([birth_date], [treatment_end_date], "d", "mdy", true)`
 - **Permitted Values:**
 - Integer value representing number of days
 - Null if `trt_end_date` is missing
 - **Source Documents:**
 - Not manually curated; system-derived from existing variables
 - **Curation Instructions:**
 - No manual entry is required. This field is automatically calculated by REDCap based on values in `birth_date` and `trt_end_date`.
 - Ensure both `birth_date` and `trt_end_date` are correctly formatted and present for the calculation to proceed.
 - If either component is missing, this field will remain blank.
 - **Curator Quality Review:**
 - Confirm that the `trt_end_date` and `birth_date` fields are correctly populated and valid.
 - Investigate if the resulting value is negative, zero, or unreasonably low/high (e.g., <100 or >40,000 days), as this may indicate a data entry or calculation error.
-

Variable: trt_drug

Type: Categorical (Dropdown)

Description: Specifies the name of the systemic anti-cancer drug administered to the patient.

Reference: NCI Thesaurus / Drug_or_Substance EVS FTP (*Version Accessed: Antineoplastic_Agent.xlsx, version: 2025-03-12 17:47 (accessed 10/2/2025)*)

Data Source: Medication administration records, oncology treatment plans, pharmacy logs, eMAR

Curation Directives:

- **Definition:** Captures the standardized name of the anti-cancer drug administered to the patient as part of systemic therapy, based on the curated NCI Thesaurus list. Used to populate structured drug exposure fields in alignment with external vocabularies.
- **Permitted Values:**
 - Preloaded dropdown of systemic antineoplastic agents from NCI Thesaurus (e.g., C77333, Abiraterone)
 - Values include code (e.g., C77333) and corresponding drug name
- **Source Documents:**
 - Chemotherapy/infusion administration records
 - Oncology provider treatment plans or clinic notes
 - Pharmacy dispensing records
 - Electronic Medication Administration Records (eMAR)
 - Research drug orders (if applicable)
- **Curation Instructions:**
 - Select the exact drug name administered from the provided dropdown list based on source documentation.
 - Match both the name and NCI code (if visible) to ensure correct agent.
 - Record each distinct agent as a separate drug form in multi-drug regimens.
 - For novel agents or those not in the list complete using trt_drug_other
- **Curator Quality Review:**
 - Ensure consistency between drug name selected and source documentation.
 - Verify completeness of drug entries across the treatment course.
 - Cross-check with treatment start and end dates for consistency.
 - Confirm correct handling of combination therapies.
- **Notes:**
 - Drug names correspond to curated entities in the NCI EVS Thesaurus to facilitate downstream linkage with ontologies such as RxNorm, WHO-ATC, and MedDRA.
 - The variable is foundational for treatment classification, exposure-response studies, and harmonization in external registries (e.g., Project GENIE, OMOP Drug Exposure).

Variable: trt_drug_other

Type: Free Text Entry (text)

Description: Captures the name of the systemic anti-cancer drug if “Other” is selected in the treatment drug dropdown.

Reference: n/a

Data Source: Oncology treatment plan, infusion records, pharmacy notes, clinical provider documentation

Curation Directives:

- **Definition:** This variable provides a descriptive drug name only when the predefined dropdown list of drugs does not include the administered agent. It serves as a supplementary field to ensure full capture of all systemic anti-cancer treatments.
- **Permitted Values:**
 - Free-text entry of the treatment drug name
- **Source Documents:**
 - Chemotherapy orders or administration records
 - Pharmacy dispensing logs
 - Treatment plans or provider progress notes
 - Electronic medical record (EMR) drug administration modules
- **Curation Instructions:**
 - Only complete this field if the primary variable `trt_drug` is coded as “Other.”
 - Enter the full name of the treatment drug as stated in the medical record.
 - Avoid abbreviations unless the abbreviation is the only label available.
 - If the drug name is uncertain or illegible, flag for review or query.
- **Curator Quality Review:**
 - Confirm that “Other” is selected in the dropdown before free text is entered.
 - Validate that the entered drug name matches documentation without spelling errors.
 - Check for duplicate or conflicting entries across structured and free-text fields.
- **Notes:**
 - This field is critical for ensuring capture of newly approved, off-label, or site-specific systemic treatments not yet reflected in the standardized dropdown.

Variable: `trt_drug_route`

Type: Categorical (Dropdown)

Description: Captures the route by which the systemic anti-cancer drug was administered to the patient.

Reference: n/a

Data Source: Chemotherapy administration records, pharmacy records, provider documentation, EMR treatment modules

Curation Directives:

- **Definition:** This variable specifies the method used to deliver systemic anti-cancer therapy. It is used to assess drug bioavailability, site-specific treatment effects, and overall treatment context.
- **Permitted Values:**
 - 1 – Intravenous
 - 2 – Oral
 - 3 – Intramuscular
 - 4 – Subcutaneous
 - 5 – Topical
 - 6 – Intrathecal
 - 7 – Intrapleural
 - 8 – Intra-arterial
 - 9 – Intraperitoneal
 - 10 – Intravesical
 - 11 – Intralesional
 - 12 – Inhalation

- 13 – Other
- **Source Documents:**
 - Chemotherapy/infusion center administration records
 - EMR medication administration route fields
 - Pharmacy dispensing data
 - Oncology notes specifying treatment delivery
 - Clinical trial protocols (if applicable)
- **Curation Instructions:**
 - Select the route of administration as documented for each drug entry.
 - If multiple routes are documented for a single drug, select the most frequent or intended route.
 - If “Other” is selected, ensure supporting documentation is available and consider querying for standardization.
- **Curator Quality Review:**
 - Ensure consistency with route mentioned in the EMR or pharmacy system.
 - Confirm that free-text documentation does not contradict dropdown selection.
 - Validate route against known administration methods for the selected drug (e.g., bevacizumab is IV, not oral).
 - Flag rare or ambiguous administration methods for review.
- **Notes:**
 - This field supports analysis of treatment modality and safety monitoring. Certain administration routes may influence toxicity profiles or treatment adherence and should be accurately captured.

Variable: `trt_drug_route_other`

Type: Free Text Entry (text)

Description: Captures the specific route of drug administration when “Other” is selected from the predefined route dropdown.

Reference: n/a

Data Source: Provider notes, chemotherapy administration records, EMR medication logs

Curation Directives:

- **Definition:** This field is used to document the precise route of systemic anti-cancer drug administration if it is not covered by existing dropdown options.
- **Permitted Values:**
Free-text route of administration. This field is only enabled when the variable `trt_drug_route` is coded as 13 = Other.
- **Source Documents:**
 - Treatment administration notes or logs
 - Oncology or infusion center documentation
 - Medication administration records in the EMR
 - Clinical trial protocol (if applicable)
- **Curation Instructions:**
 - Enter the full descriptive term for the administration route as documented. Avoid abbreviations unless they are standard.
- **Curator Quality Review:**
 - Verify that the text entry is consistent with the source documentation.

- Confirm proper spelling and use of medical terminology.
- **Notes:**
 - This field supports future expansion of permissible route values and enables tracking of uncommon or emerging administration methods in oncology treatment delivery.

Variable: trt_dose

Type: Numeric

Description: Records the dose amount administered per single administration of the treatment drug.

Reference: n/a

Data Source: Chemotherapy administration records, medication orders, provider notes

Curation Directives:

- **Definition:** This field captures the numerical quantity of the drug administered in a single dose. It is not cumulative and should reflect only the amount delivered in each individual treatment session or administration event.
- **Permitted Values:**
Numeric values (e.g., 50, 200.5, 1000)
- **Source Documents:**
 - Chemotherapy or infusion administration records
 - Medication administration logs in the EMR
 - Oncology treatment plans
 - Progress notes with detailed treatment documentation
- **Curation Instructions:**
 - Abstract only the numeric dose given per administration, not cumulative dose or total across multiple administrations.
 - Do not include units (e.g., mg, IU) in this field. Units will be captured in a separate field (**trt_dose_unit**).
 - If the dose is documented as a range or varies between administrations, record the most typical value. If not possible to determine, leave blank and flag for clinical review.
 - If the dose is not specified or illegible, document as missing and flag accordingly for the QA Manager.
- **Curator Quality Review:**
 - Confirm accuracy of numeric entry against source documentation.
- **Notes:**
 - This field supports standardized analysis of drug dosing patterns and must be reviewed alongside route, frequency, and duration variables to evaluate treatment intensity and regimen adherence.

Variable: trt_dose_unit

Type: Categorical (Dropdown)

Description: Specifies the unit of measurement for the dose of drug administered per single treatment.

Reference: n/a

Data Source: Medication administration records, provider documentation, pharmacy records

Curation Directives:

- **Definition:** This variable captures the unit associated with the numeric dose entered in the `trt_dose` field. It provides context to interpret the magnitude and method of the drug administered.
- **Permitted Values:**
 - 1 = Milligrams (mg)
 - 2 = Milliliters (mL)
 - 3 = Grams (g)
 - 4 = Units (U)
 - 5 = Micrograms (µg)
 - 6 = Milliequivalents (mEq)
 - 7 = Kilograms (kg)
 - 8 = Milligrams per square meter (mg/m²)
 - 9 = Milligrams per kilogram (mg/kg)
 - 10 = Other
- **Source Documents:**
 - Medication orders and infusion records
 - Pharmacy documentation
 - Chemotherapy or treatment plans
 - Clinical notes indicating prescribed dose and unit
- **Curation Instructions:**
 - Select the unit that corresponds exactly to the numeric value recorded in `trt_dose`.
 - If the unit is not listed, select “10 = Other” and specify the unit in the associated `trt_dose_unit_other` field.
 - Avoid assumptions or conversions. Abstract as documented.
- **Curator Quality Review:**
 - Verify unit consistency with documented treatment protocol.
 - If “Other” is selected, ensure that `trt_dose_unit_other` is populated and valid.
- **Notes:**
 - This field is essential for interpreting the clinical relevance of drug dosing. Standardized capture of units enables downstream dose normalization, toxicity assessment, and pharmacologic profiling.

Variable: `trt_drug_dose_other`

Type: Free Text Entry (text)

Description: Specifies the drug dose unit when “Other” is selected from the dropdown list of predefined units.

Reference: n/a

Data Source: Medication administration records, pharmacy notes, provider documentation

Curation Directives:

- **Definition:** This field captures the textual description of the unit of measurement used for the drug dose when none of the predefined options in `trt_dose_unit` adequately describe the documented unit.
- **Permitted Values:**

Free-text entry (e.g., “cc,” “IU/kg,” “tablet count”)
- **Source Documents:**
 - Pharmacy administration records
 - Treatment protocols and infusion documentation
 - Clinical notes or orders specifying exact dose and unit

- **Curation Instructions:**
 - Only complete this field if `trt_dose_unit` is coded as “10 = Other”.
 - Enter the unit of dose exactly as documented in the source records, including compound or less common units.
 - Do not abbreviate or reformat the value unless explicitly written that way in the documentation.
 - Leave blank if `trt_dose_unit` is not “Other” or if no unit is provided in the source.
- **Curator Quality Review:**
 - Confirm that `trt_dose_unit` is correctly coded as “Other” when this field is populated.
 - Review free-text entry for consistency with clinical context and documented drug administration.
- **Notes:**
 - Use this field only to document non-standard units not covered by predefined dropdown options. When possible, escalate unclear units to clinical or pharmacology reviewers for resolution.

Variable: `trt_freq`

Type: Categorical (Dropdown)

Description: Documents the frequency at which the cancer treatment was administered to the patient.

Reference: n/a

Data Source: Chemotherapy orders, infusion records, clinical notes

Curation Directives:

- **Definition:** This field captures the interval at which a systemic cancer treatment was administered, reflecting the schedule defined in the treatment regimen (e.g., weekly, biweekly, monthly). It is based on how often the patient received the drug, not how long the treatment continued.
- **Permitted Values:**
 - 1 = Daily
 - 2 = Weekly
 - 3 = Every 2 weeks (Biweekly)
 - 4 = Every 3 weeks (Triweekly)
 - 5 = Every 4 weeks (Monthly)
 - 6 = Every 6 weeks
 - 7 = Every 8 weeks (Every 2 months)
 - 8 = Every 12 weeks (Every 3 months)
 - 9 = Single Treatment (One-time administration)
 - 10 = Other
- **Source Documents:**
 - Chemotherapy/medication administration records
 - Treatment plan and protocol documentation
 - Oncology clinic notes specifying dosing schedule
- **Curation Instructions:**
 - Select the dropdown option that best matches the documented treatment schedule.
 - If multiple drugs with different frequencies are documented under the same treatment start date, select the most frequent applicable value or enter separately by drug.
 - If a one-time treatment was administered, code as “9 = Single Treatment.”
 - Use “10 = Other” only if the schedule does not match any listed options; in that case, provide details in the corresponding “Other” field, if available.
 - Do not infer frequency if it is not explicitly stated, seek clarification from source documents.
- **Curator Quality Review:**

- Confirm the selected frequency aligns with the treatment protocol or administration records.
- Check for consistency with `trt_start_date`, `trt_end_date`, and dosing entries.
- Ensure “Other” is not used when a matching frequency is available in the dropdown.

- **Notes:**

- This field helps standardize treatment regimen data for downstream analysis and should reflect true administration intervals, not prescribing intentions or idealized regimens.

Variable: `trt_freq_other`

Type: Free Text Entry (text)

Description: Captures a descriptive treatment frequency when “Other” is selected in the treatment frequency dropdown.

Reference: n/a

Data Source: Chemotherapy orders, infusion records, clinical notes

Curation Directives:

- **Definition:** This field is used to specify the treatment frequency when the dropdown variable `trt_freq` is coded as 10 = Other.
- **Permitted Values:**
Free-text entry (e.g., “Every 5 days,” “Every 10 weeks,” “Every Monday and Thursday”)
This field is conditional on selection of “10 = Other” in the `trt_freq` field.
- **Source Documents:**
 - Chemotherapy/treatment protocols
 - Oncology clinic notes or infusion center records
 - Medication administration logs indicating exact frequency
- **Curation Instructions:**
 - Only complete this field if `trt_freq` = 10 (“Other”) is selected.
 - Transcribe the exact frequency documented in the source record without interpreting or abbreviating.
- **Curator Quality Review:**
 - Ensure the free-text entry provides a clear and specific frequency.
 - Confirm consistency with treatment documentation and dosing intervals if available.
- **Notes:**
 - This variable supports flexibility in capturing atypical or customized treatment schedules not covered in the standard dropdown. Use with discretion to maintain data harmonization.

Variable: `trt_modification`

Type: Categorical (Radio)

Description: Indicates whether a modification or premature alteration was made to the cancer-directed drug regimen, and if so, the type of modification.

Reference: n/a

Data Source: Oncology notes, adverse event reports, patient communications, pharmacy documentation

Curation Directives:

- **Definition:** Captures the specific nature of any modification to a cancer-directed systemic therapy that deviates from the originally intended regimen.
- **Permitted Values:**
 - 1 = No modification
 - 2 = Dose reduction
 - 3 = Schedule change
 - 4 = Drug substitution
 - 5 = Other modification
- **Source Documents:**
 - Clinical oncology progress notes
 - Treatment plans or flow sheets
 - Pharmacy documentation
 - Adverse event logs
 - Patient-provider communication notes
- **Curation Instructions:**
 - Code “1 = No modification” if the drug was administered exactly as originally planned without deviation.
 - Select “2 = Dose reduction” if the dose of the drug was lowered due to toxicity, side effects, or provider discretion.
 - Use “3 = Schedule change” for changes in frequency or timing (e.g., every 3 weeks → every 4 weeks) without changing the drug or dose.
 - Use “4 = Drug substitution” if one drug was replaced with a different agent, even if within the same class.
 - Select “5 = Other modification” for unlisted modifications (e.g., route of administration change, premature discontinuation not otherwise classified). Additional context should be documented elsewhere in the form or notes.
- **Curator Quality Review:**
 - Verify that any non-“No modification” selection has corresponding documentation to support the change.
 - Confirm alignment between the type of modification selected and the stated rationale (cross-check with `trt_modification_reason`).
 - Flag for clinical adjudication if intent or type of change is unclear.
- **Notes:**
 - This variable supports evaluation of treatment tolerability, adherence challenges, and real-world barriers to cancer care. Completeness and accuracy improve the utility of modification data in outcome analyses

Variable: `trt_modification_reason`

Type: Categorical (Dropdown)

Description: Specifies the primary reason for modifying or prematurely altering the cancer-directed drug treatment.

Reference: n/a

Data Source: Oncology notes, adverse event reports, patient communications, pharmacy documentation

Curation Directives:

- **Definition:** Captures the rationale for a documented treatment modification, including medical, logistical, or patient-driven factors.

- **Permitted Values:**

- 1 = Adverse events
- 2 = Disease progression
- 3 = Patient request
- 4 = Comorbidity
- 5 = Drug availability issues
- 6 = Financial reasons
- 7 = Other

- **Source Documents:**

- Clinical or oncology progress notes
- Adverse event and toxicity documentation
- Pharmacy communications
- Patient-reported reasons in chart
- Discharge summaries or treatment discontinuation notes

- **Curation Instructions:**

- Only complete this field if `trt_modification` is coded as:
 - * 2 = Dose reduction
 - * 3 = Schedule change
 - * 4 = Drug substitution
 - * 5 = Other modification
- Review documentation to determine the primary driver for the change.
- Select “1 = Adverse events” if the change was prompted by toxicity or side effects.
- Use “2 = Disease progression” when the modification was due to cancer progression despite treatment.
- Select “3 = Patient request” if the patient initiated or insisted on the change.
- Use “4 = Comorbidity” for changes prompted by non-cancer medical conditions interfering with treatment.
- Use “5 = Drug availability issues” if access to the drug was delayed or disrupted.
- Use “6 = Financial reasons” if cost or insurance coverage influenced the treatment change.
- Use “7 = Other” only if the reason does not fit the above categories. Provide supporting context in free text if applicable.

- **Curator Quality Review:**

- Confirm the rationale is clearly stated or strongly implied in clinical documentation.
- Ensure the selected reason aligns with the type of modification coded in `trt_modification`.
- If documentation is unclear, flag for clinical adjudication.

- **Notes:**

- This variable supports evaluation of treatment tolerability, adherence challenges, and real-world barriers to cancer care. Completeness and accuracy improve the utility of modification data in outcome analyses.

Variable: `trt_discontinuation`

Type: Categorical (Dropdown)

Description: Indicates whether the cancer-directed drug was discontinued prior to the intended end of treatment.

Reference: n/a

Data Source: Oncology progress notes, medication administration records, treatment summaries

Curation Directives:

- **Definition:** Captures whether the treatment drug was stopped earlier than planned, and if so, whether the discontinuation was temporary or permanent.
- **Permitted Values:**
 - 1 = No discontinuation
 - 2 = Temporary discontinuation
 - 3 = Permanent discontinuation
- **Source Documents:**
 - Oncology notes documenting treatment course
 - Nursing administration records or pharmacy logs
 - Discharge summaries or treatment plans
 - Adverse event reports or clinical trial deviation logs
- **Curation Instructions:**
 - Review documentation for explicit or implied statements about early cessation or interruption of treatment.
 - Select:
 - * “1 = No discontinuation” if the drug was administered as planned through completion or until planned transition.
 - * “2 = Temporary discontinuation” if the drug was paused and later resumed.
 - * “3 = Permanent discontinuation” if the drug was stopped and not resumed, even if the full treatment course was not completed.
 - If multiple drugs were administered, base coding on the specific drug associated with the field instance.
 - Cross-check with `trt_modification_reason` or `trt_end_date` if applicable.
- **Curator Quality Review:**
 - Ensure consistency with other treatment timeline variables and adverse event documentation.
 - If unclear whether discontinuation was temporary or permanent, escalate to QA Manager.
 - Avoid assuming discontinuation based solely on lack of further documentation unless clearly indicated.
- **Notes:**
 - This field helps identify treatment interruptions that may influence therapeutic effectiveness or indicate tolerability issues. Consistent definitions are essential for downstream analyses of real-world treatment patterns.

Variable: `trt_discontinuation_reason`

Type: Categorical (Dropdown)

Description: Indicates the primary reason why the cancer-directed drug was discontinued prematurely, either temporarily or permanently.

Reference: n/a

Data Source: Oncology progress notes, adverse event reports, clinical trial deviation forms, death records, follow-up documentation

Curation Directives:

- **Definition:** Captures the clinical or logistical rationale for ending drug administration earlier than originally intended. This field is conditional and should only be completed when discontinuation has occurred.
- **Permitted Values:**

- 1 = Disease progression
- 2 = Adverse events
- 3 = Patient withdrawal
- 4 = Death
- 5 = Loss to follow-up
- 6 = Protocol violation
- 7 = Treatment completion as planned
- 8 = Comorbidity
- 9 = Other (specify)
- **Source Documents:**
 - Oncologist or provider progress notes
 - Clinical trial forms or deviation reports
 - Death certificates or electronic medical record mortality indicators
 - Patient correspondence and consent withdrawal documentation
 - Follow-up tracking notes or registry forms
- **Curation Instructions:**
 - Complete this field only if `trt_discontinuation` is coded as 2 (Temporary discontinuation) or 3 (Permanent discontinuation).
 - Select:
 - * “1 = Disease progression” when the cancer worsened or recurred, prompting treatment termination.
 - * “2 = Adverse events” when side effects or toxicity led to discontinuation.
 - * “3 = Patient withdrawal” when the patient elected to stop therapy.
 - * “4 = Death” if the patient died during the course of treatment.
 - * “5 = Loss to follow-up” if the patient was no longer reachable or did not return for further treatment.
 - * “6 = Protocol violation” when treatment was stopped due to a deviation from a clinical protocol.
 - * “7 = Treatment completion as planned” if the discontinuation was originally scheduled as part of a planned regimen.
 - * “8 = Comorbidity” if unrelated health conditions precluded continued therapy.
 - * “9 = Other (specify)” if no other category fits and further details are captured in the corresponding free-text field.
 - Ensure alignment with supporting documentation and escalate ambiguous cases to clinical adjudication when necessary.
- **Curator Quality Review:**
 - Cross-reference with `trt_end_date`, `trt_modification`, and adverse event fields.
 - Verify consistency with treatment timeline and clinical outcome documentation.
 - Ensure “Other” is not selected without accompanying specification.
- **Notes:**
 - This field is essential for analyzing reasons behind early treatment discontinuation and understanding tolerability, safety, and real-world barriers to treatment completion.

Clinical Trial History

Clinical Trial History

Complete one instance of the Clinical Trial History form per distinct clinical trial in which the patient enrolled (one protocol/trial episode = one form). For each instance, abstract only what is explicitly documented in authoritative sources: confirm participation, record the first day on protocol as the initiation date, and capture the enrollment location (country and, if U.S., state) using project date/coding conventions. Do not infer enrollment from treatment patterns or future notes. Ensure internal chronology with the patient's diagnosis/treatment timeline, apply approved options for missing or indeterminate information, and create additional form instances for additional trial enrollments.

Variable: `clinical_trial_yn`

Type: Binary (Yes/No)

Description: Clinical Trial Participation History

Reference: n/a

Data Source: Clinical provider notes (Consultation, Progress Notes), Clinical trial enrollment documents, Oncology treatment history

Curation Directives:

- **Definition:** This variable captures whether the patient has ever been enrolled in or participated in a clinical trial for any reason (treatment, diagnostic, screening, supportive care, or prevention). It reflects a lifetime history of clinical trial involvement based on documentation available in the medical record.
- **Permitted Values / Codes:**
 - 0 = No
 - 1 = Yes
- **Source Documents:**
 - Clinical provider notes (e.g., Consultation, Progress Notes, H&P)
 - Clinical trial enrollment or consent documentation
 - Treatment history or summary notes (including “Patient was enrolled on trial XYZ”)
 - Research coordinator notes, if available
 - Prior oncology institutional records (especially from referring or outside centers)
- **Curation Instructions:**
 1. Search the available clinical notes and supporting documents for any mention of clinical trial participation.
 2. Include any enrollment regardless of the type or phase of the trial (e.g., Phase I, II, III, observational, interventional).
 3. If the note refers to an outside institution's trial participation, still code as 1 = Yes.
 4. Do not code as “Yes” if the only documentation refers to trial eligibility or consideration, but not actual participation.
 5. If it is clearly stated that the patient declined trial participation or was not enrolled, code as 0 = No.
 6. If there is no mention of clinical trials at all, code as 0 = No.
- **Curator Quality Review:**
 - Confirm consistency with any trial-specific documentation or treatment plans.
 - Ensure the note reflects actual participation or enrollment, not just discussion.
 - If the variable is coded as 1 = Yes, there should be at least one note referencing trial involvement.

Variable: `clinical_trial_date`

Type: Date (MM/DD/YYYY)

Description: Date of Clinical Trial Initiation

Reference: n/a

Data Source: Clinical provider notes, clinical trial enrollment documentation, oncology treatment timeline

Curation Directives:

- **Definition:** The calendar date the patient began participation in a clinical trial, defined as the first day the patient received protocol-specified intervention, procedure, or assessment as part of the study.
- **Permitted Values / Codes:**
Valid dates between 01-01-2000 and 01-01-2100, formatted as MM-DD-YYYY
- **Source Documents:**
 - Clinical trial enrollment or consent documentation
 - Oncology clinic notes referencing trial start date
 - Clinical trial treatment summaries or protocols
 - Institutional research records, if accessible
- **Curation Instructions:**
 1. Record the documented start date of clinical trial participation as MM-DD-YYYY.
 2. Use the date when the patient began receiving protocol-defined study intervention.
 3. If the date of randomization and date of treatment differ, use the date of first treatment or intervention.
 4. Do not record the consent date unless it is explicitly stated to coincide with treatment initiation.
 5. If trial start date is mentioned only as a month and year (e.g., “enrolled in 03/2022”), use date estimation rules.
 6. Leave blank if trial participation date is not recorded or cannot be reliably estimated.
- **Curator Quality Review:**
 - Ensure date is logically consistent with other medical record dates (e.g., diagnosis, treatment).
- **Notes:**
 - This date anchors longitudinal interpretation of trial exposure and may influence disease trajectory analyses.

Variable: `clinical_trial_date_int`

Type: Integer (Calculated)

Description: Trial Initiation Date – Interval in Days from Date of Birth

Reference: n/a

Data Source: Derived from `pt_trial_date` and patient Date of Birth

Curation Directives:

- **Definition:**
Automatically calculated field representing the number of days between the patient’s date of birth and the date they began clinical trial participation.
- **Permitted Values / Codes:**
Integer value representing day difference (e.g., 18,000 = patient was 49 years old at trial start)
- **Source Documents:**
 - Derived automatically by the data platform based on entered `pt_trial_date`
- **Curation Instructions:**

1. No manual curation required; this is a system-calculated field.
2. Ensure that `pt_trial_date` is entered correctly for accurate interval calculation.

- **Curator Quality Review:**

- Verify that this field populates as expected after entry of `pt_trial_date`.
- Review for consistency with known patient age and timeline of medical events.

- **Notes:**

- This interval facilitates time-to-event and age-based stratification analyses

Variable: `clinical_trial_country`

Type: Categorical (Dropdown)

Description: Country in Which Clinical Trial Was Conducted

Reference: n/a

Data Source: Clinical provider notes, clinical trial enrollment documentation

Curation Directives:

- **Definition:**

Captures the country in which the patient's clinical trial participation took place, based on documentation within the medical record.

- **Permitted Values / Codes:**

1–194, representing individual countries (e.g., 186 = United States, 64 = Germany, 76 = India, etc.).

- **Source Documents:**

- Clinical trial enrollment or consent forms
- Clinical provider notes referencing trial site location
- Clinical research or oncology clinic documentation
- Institutional trial summaries, if available

- **Curation Instructions:**

1. Select the country where the clinical trial occurred, as explicitly stated in the documentation.
2. If more than one country is mentioned, choose the location where the patient physically received treatment.
3. If the trial was conducted in the U.S., select code 186 = United States and proceed to `pt_trial_state`.
4. If country is not mentioned, attempt to infer based on institution name or location, but only if clearly documented.
5. Leave blank if no trial location is identifiable.

- **Curator Quality Review:**

- Verify country selection is consistent with documentation.
 - Ensure U.S.-based trials are followed by valid state selection.
-

Variable: `clinical_trial_state`

Type: Categorical (Dropdown)

Description: U.S. State in Which Clinical Trial Was Conducted

Reference: n/a

Data Source: Clinical provider notes, clinical trial enrollment documentation

Curation Directives:

- **Definition:**
Captures the U.S. state in which the patient’s clinical trial participation took place.
- **Permitted Values / Codes:**
1–50, corresponding to U.S. states (e.g., 6 = Colorado, 32 = New York, 43 = Texas).
- **Source Documents:**
 - Clinical trial enrollment or site documentation
 - Oncology notes referencing study site or referral center
 - Institutional correspondence or trial summaries
- **Curation Instructions:**
 1. Only complete this field if `clinical_trial_country` = 186 (United States).
 2. Record the state where the patient received the clinical trial intervention.
 3. Leave blank if state is not mentioned or cannot be reliably inferred.

Radiation Treatment

Radiation Treatment Information

Complete one instance of the Radiation Treatment form for each distinct course of radiation therapy administered to the patient. A separate form should be completed for each course if there were breaks, changes in modality, or distinct intent (e.g., adjuvant vs. palliative). Abstract only what is explicitly documented in authoritative sources such as the radiation oncology summary, treatment planning system (TPS) records, treatment completion summaries, or detailed radiation oncology notes.

For each course, record the primary indication/site for treatment (ICD-O-3 topography), laterality (if applicable), treating institution, start and end dates (or project dates using approved conventions if incomplete), modality, planning technique, primary treatment volume, and whether draining lymph node regions were targeted. Capture dose per fraction, total number of fractions, and total dose only when clearly stated.

Document whether treatment was completed as prescribed or discontinued early, including the reason for discontinuation if available. Align all radiation details with the patient’s broader diagnostic and treatment timeline for internal chronology. Do not infer treatment from imaging findings, follow-up notes, or outcomes unless the administration is explicitly confirmed in radiation oncology documentation. When additional courses of radiation are delivered, create a new form instance for each course.

Variable: `rad_yn`

Type: Binary (Yes/No)

Description: Indicates whether the patient has received any form of radiation therapy as part of their cancer treatment.

Reference: n/a

Data Source: Radiation oncology notes, treatment summaries, clinical encounter documentation

Curation Directives:

- **Definition:** Captures whether the patient underwent any radiation therapy, regardless of modality, intent (curative vs. palliative), or timing relative to other treatments.

- **Permitted Values:**
 - 0 = No
 - 1 = Yes
- **Source Documents:**
 - Radiation therapy consultation notes
 - Radiation oncology treatment summaries
 - Oncologist or multidisciplinary team notes
 - Discharge summaries or treatment timelines
- **Curation Instructions:**
 - Select “1 = Yes” if there is documentation confirming the patient received radiation at any point in their cancer care.
 - Select “0 = No” only if the record explicitly confirms that no radiation therapy was administered or recommended.
 - If documentation is unclear or missing, escalate for clinical adjudication rather than assuming “No.”
 - Include both external beam and internal radiation (e.g., brachytherapy) as qualifying treatments.
- **Curator Quality Review:**
 - Check for consistency with downstream radiation-specific fields (e.g., radiation start/end dates, modality, anatomical site).
 - Validate against treatment plans or summary documents for completeness.
- **Notes:**
 - This field enables capture of multimodal treatment history and is critical for understanding sequencing, outcomes, and potential toxicities related to radiation exposure.

Variable: rad_institution

Type: Categorical (Dropdown)

Description: Indicates whether radiation therapy was administered at the GENIE-participating (internal) institution, an external institution, or both.

Reference: n/a

Data Source: Oncology treatment summaries, referral records, external consult notes, institutional EMR records.

Curation Directives:

- **Definition:** Identifies the facility or facilities where radiation treatment was delivered. This helps differentiate treatments documented directly in the institution’s records from those that may be partially or wholly externally managed.
- **Permitted Values:**
 - 1 = Internal Institution
 - 2 = External Institution
 - 3 = Both Internal Institution and External Institution
- **Source Documents:**
 - Radiation therapy treatment summaries
 - External consultation notes
 - EMR treatment timeline or encounter history
 - Referral documentation or care coordination records

- **Curation Instructions:**
 - Code as “1” if all radiation therapy was administered at the GENIE-participating institution.
 - Code as “2” if all radiation therapy was delivered by an outside (non-participating) institution.
 - Code as “3” if the patient received portions of radiation therapy at both internal and external institutions (e.g., initial therapy internally, boost externally).
 - If unclear from available documentation, consult external notes or care team correspondence to determine treatment setting.
- **Curator Quality Review:**
 - If coded as “3”, verify that at least one session occurred at the GENIE site and another at an outside institution.
- **Notes:**
 - This variable enables stratification of treatment records by institutional origin, supporting completeness assessments and identification of potential gaps in treatment detail or toxicity follow-up.

Radiation Treatment Exposure

Variable: rad_start_date

Type: Date (MM/DD/YYYY)

Description: Captures the start date of radiation therapy as part of the patient’s cancer-directed treatment plan.

Reference: NAACCR Data Item #1210

Data Source: Radiation oncology records, treatment summaries, oncology progress notes

Curation Directives:

- **Definition:** The calendar date on which the patient’s first session of radiation therapy was administered, regardless of modality or intent (e.g., curative, palliative).
- **Permitted Values:**

Date in MM/DD/YYYY format

 - Must represent the first date radiation was delivered
 - Leave blank if no radiation therapy was given (i.e., if rad_yn is coded as “No”)
- **Source Documents:**
 - Radiation treatment summaries or start-of-care documents
 - Radiation therapy encounter or scheduling records
 - Oncology treatment plans or progress notes
 - EMR procedural history or structured treatment logs
- **Curation Instructions:**
 - Enter the documented date of the first fraction of radiation therapy.
 - Do not enter planning, simulation, or consent dates.
 - If multiple courses of radiation occurred, enter the date for the first course only.
 - Cross-check against rad_yn to ensure consistency.
 - If no clear start date is available, consult with clinical documentation or site coordinator for clarification.
- **Curator Quality Review:**

- Ensure the date is not in the future or implausible based on diagnosis date.
- Validate that the date precedes or matches corresponding **rad_end_date** if present.
- Confirm that radiation was actually administered (e.g., avoid capturing planned but cancelled treatments).

- **Notes:**

- This field is critical for temporal alignment of radiation therapy with other modalities (e.g., surgery, systemic therapy) and may be used to determine treatment sequencing or duration.

Variable: **rad_start_date_int**

Type: Integer (Calculated)

Description: Number of days between the patient’s date of birth and the start date of radiation therapy.

Reference: n/a

Data Source: Automatically calculated from **birth_date** and **rad_start_date** fields

Curation Directives:

- **Definition:** A calculated interval capturing the total number of days from the patient’s date of birth to the date they began radiation treatment.
- **Permitted Values:**
Positive integer values (e.g., 18250)
 - System-generated using built-in date difference calculation
 - Represents age in days at radiation start
- **Source Documents:**
 - Not directly curated from source documents; value is system-generated
 - Inputs must be sourced accurately:
 - * **birth_date** from demographic or registration records
 - * **rad_start_date** from radiation therapy documentation
- **Curation Instructions:**
 - Ensure **birth_date** and **rad_start_date** are both present and valid to generate this field.
 - Do not manually modify this field—it should auto-calculate once source dates are entered.
 - If the value appears implausible (e.g., negative, zero, or excessively high), re-check source date entries for accuracy and formatting errors.
- **Curator Quality Review:**
 - Review outliers to confirm that dates are not misentered (e.g., incorrect year or reversed MM/DD formats).
 - Cross-validate with patient’s age at treatment start for consistency.
- **Notes:**
 - This derived variable may support age-related analyses of treatment timing and sequencing, particularly in pediatric or AYA oncology populations.

Variable: **rad_end_date**

Type: Date (MM/DD/YYYY)

Description: Captures the final date of the patient’s radiation treatment course.

Reference: NAACCR #1220

Data Source: Radiation oncology records, treatment summaries, external provider documentation

Curation Directives:

- **Definition:** The date on which radiation therapy was completed. This field documents the conclusion of the administered radiation course.
- **Permitted Values:**
Date in MM/DD/YYYY format
- **Source Documents:**
 - Radiation treatment summary or end-of-treatment documentation
 - External radiation records if performed at a non-participating site
 - Oncology clinic notes confirming end of radiation
- **Curation Instructions:**
 - Enter the documented end date for the radiation treatment course, not the last scheduled session if missed.
 - If multiple radiation courses were given, record the end date of the course corresponding to the primary cancer episode under review.
 - If treatment is ongoing or date is unknown, leave blank and flag for review.
- **Curator Quality Review:**
 - Ensure date is chronologically after `rad_start_date`.
 - Review for typographical errors (e.g., year mismatches, future dates).
 - Cross-check with summary notes or follow-up visit dates for plausibility.
- **Notes:**
 - If only a partial treatment course was administered, still record the actual final date of administration (if known), and document reasons for early discontinuation elsewhere.

Variable: `rad_end_date_int`

Type: Integer (Calculated)

Description: Number of days from the patient's date of birth to the radiation treatment end date.

Reference: n/a

Data Source: Automatically calculated from `birth_date` and `rad_end_date`

Curation Directives:

- **Definition:** This field reflects the age of the patient (in days) on the date radiation treatment concluded.
- **Permitted Values:**
 - System-generated upon entry of valid `birth_date` and `rad_end_date`
- **Source Documents:**
 - Not manually curated
 - Derived from date fields:
 - * `birth_date`: from patient registration
 - * `rad_end_date`: from treatment documentation
- **Curation Instructions:**
 - Ensure both input dates are complete and accurate to enable proper calculation.
 - No manual entry; field is auto-generated by system logic.
 - Investigate values that are negative, zero, or inconsistent with age-based expectations.
- **Curator Quality Review:**
 - Compare with `rad_start_date_int` to verify the radiation course duration makes sense.

- Check for transposed dates or format errors if outliers are observed.
- **Notes:**
 - Useful for interval analyses and calculating treatment duration (e.g., subtracting `rad_start_date_int` from `rad_end_date_int`).

Variable: `rad_planning`

Type: Categorical (Dropdown)

Description: Captures the specific planning technique used for external beam radiation therapy, reflecting the precision and method of radiation delivery.

Reference: NAACCR #1502, 1512, 1522

Data Source: Radiation oncology treatment plan documents, dosimetry reports, radiation summary notes, EMR encounter notes from radiation planning

Curation Directives:

- **Definition:** Specifies the planning technique used to guide the administration of external beam radiation therapy. Planning technique refers to the strategy and technology used to shape and target the radiation dose (e.g., 3D therapy, IMRT, stereotactic radiosurgery).
 - **Permitted Values:**
 - Dropdown with predefined values representing different planning techniques, including 2D, 3D conformal, IMRT, stereotactic radiosurgery variants, and adaptive therapy methods.
 - **Source Documents:**
 - Treatment planning system summary
 - Radiation oncology procedure notes
 - Physics or dosimetry reports
 - Clinical notes documenting the radiation approach or equipment used
 - **Curation Instructions:**
 - Abstract the planning technique as explicitly stated in the source documentation.
 - If the documentation references techniques such as “IMRT,” “3D-CRT,” “Gamma Knife,” or “MR-guided adaptive,” map these to the corresponding dropdown value.
 - If no radiation treatment was delivered, select “No radiation treatment.”
 - If the technique is not documented, select “Unknown.”
 - Do not infer the planning technique based solely on the treatment modality unless explicitly stated.
 - **Curator Quality Review:**
 - Confirm alignment with `rad_modality` (e.g., planning technique should not be recorded if no radiation was delivered).
 - Check consistency with dose fractionation fields and treatment intent where applicable.
 - Flag cases where multiple techniques may be suggested but only one option is allowed for selection.
 - **Notes:**

This variable helps distinguish between conventional and advanced radiation delivery techniques. It is critical for evaluating treatment precision and potential toxicity.
-

Variable: rad_modality

Type: Categorical (Dropdown)

Description: Indicates the modality of radiation therapy delivered, including external beam, brachytherapy, radioisotopes, or no treatment.

Reference: NAACCR #1506, 1516, 1526

Data Source: Radiation oncology summaries, treatment plans, radiotherapy records, EMR treatment documentation

Curation Directives:

- **Definition:** Captures the specific type of radiation modality administered to the patient. This includes the method of energy delivery (e.g., photons, protons), radioactive sources (e.g., brachytherapy), or systemic isotopes.
- **Permitted Values:**
 - Dropdown with standardized options (e.g., External beam photons, Brachytherapy HDR, Radioisotopes Radium-223, etc.)
- **Source Documents:**
 - Radiation treatment summaries
 - Physician treatment plan documentation
 - Radiotherapy dosimetry or modality selection reports
 - Radiation oncology clinic notes
- **Curation Instructions:**
 - Select the radiation modality as documented in the treatment plan.
 - Use the most specific option available (e.g., “External beam, photons” rather than “External beam, NOS”) if modality subtype is stated.
 - If multiple modalities are given in the same treatment course, each should be curated as a separate record with corresponding dates.
 - Select “No radiation treatment” only when documentation confirms that no radiation therapy was administered.
 - Use “Radiation treatment modality unknown” only if modality is not specified and cannot be inferred from source records.
- **Curator Quality Review:**
 - Verify consistency with other radiation variables such as rad_start_date, rad_institution, and rad_yn.
 - Ensure modality selected aligns with terminology used in the documentation (e.g., “proton therapy” → “External beam, protons”).
 - Flag any ambiguous or conflicting entries for clinical review.
- **Notes:**

This field aligns with NAACCR radiation modality codes and supports interoperability with standardized cancer registry data. In cases of combination therapy (e.g., external beam followed by brachytherapy), each modality should be captured on a separate row.

Variable: rad_volume

Type: Categorical (Dropdown)

Description: Identifies the primary anatomic target or treatment volume addressed during the specific course of radiation therapy.

Reference: NAACCR #1504, 1514, 1524

Data Source: Radiation treatment plan summaries, dosimetry documents, clinical oncology notes, radiotherapy start documentation

Curation Directives:

- **Definition:** Specifies the anatomical region targeted during the treatment, such as the primary tumor site, lymph nodes, or sites of distant metastasis. This field captures whether treatment was localized or involved multiple anatomical areas.
- **Permitted Values:**
 - Dropdown list includes options for single and combined treatment volumes (e.g., primary tumor only, primary + regional lymph nodes, etc.) and non-specific or other site categories.
- **Source Documents:**
 - Radiation treatment planning summary
 - Oncologist’s treatment intent notes
 - Radiation oncology consult notes
 - EMR entries indicating body site(s) targeted during therapy
- **Curation Instructions:**
 - Select the value that most accurately reflects the anatomic sites targeted by radiation in the specified treatment course.
 - If documentation states treatment targeted multiple volumes (e.g., “primary tumor and regional lymph nodes”), select the corresponding combined option.
 - Use “Other specified site(s)” if a clearly defined but unlisted anatomical region is documented.
 - Use “Site not otherwise specified (NOS)” only if the documentation is insufficient to determine a specific or combined site.
 - Do not infer treatment volume from disease location unless clearly stated in the context of radiation targeting.
- **Curator Quality Review:**
 - Cross-check against modality and planning technique to ensure internal consistency.
 - Ensure consistency with diagnosis and staging data, where applicable.
 - Flag conflicting or unclear documentation for review.
- **Notes:**

This variable provides insight into the intent and scope of radiation therapy, which can be useful for stratifying patients by disease burden, treatment aggressiveness, or location-specific outcomes.

Variable: rad_lymph

Type: Categorical (Dropdown)

Description: Identifies the specific draining lymph node regions targeted during a course of radiation therapy.

Reference: NAACCR #1505, 1515, 1525

Data Source: Radiation oncology treatment plans, dosimetry reports, radiation field summaries, physician documentation

Curation Directives:

- **Definition:** Captures whether and which draining lymph node regions were included as part of the radiation therapy field during the specified treatment course. Applies when regional lymphatics are targeted to address nodal disease or for prophylactic purposes.
- **Permitted Values:**

- 1: Neck lymph node regions
- 2: Thoracic lymph node regions
- 3: Neck and thoracic lymph node regions
- 4: Breast/Chest wall lymph node regions
- 5: Abdominal lymph nodes
- 6: Pelvic lymph nodes
- 7: Abdominal and pelvic lymph nodes
- 8: Lymph node region, NOS (not otherwise specified)
- 88: Not applicable; radiation primary treatment volume is lymph nodes
- 99: Unknown if any radiation treatment to draining lymph nodes; unknown if radiation treatment was administered

- **Source Documents:**

- Radiation field diagrams and dose-volume histograms
- Radiation therapy plans with anatomical targets
- Clinical notes from radiation oncologists specifying nodal coverage
- Simulation or planning CT summaries

- **Curation Instructions:**

- Code all lymph node regions that were explicitly targeted in the documented radiation plan.
- If more than one region was targeted (e.g., both abdominal and pelvic nodes), choose the corresponding combined value.
- Use value **88** only if the radiation volume already specifies lymph nodes as the primary target (i.e., no additional nodal fields are involved).
- Use **99** if documentation is missing or if treatment status is unknown.
- If documentation is present but the region is described vaguely (e.g., “nodal basin irradiated”), and cannot be further classified, code as **8: Lymph node region, NOS**.
- Do not infer nodal involvement from disease site alone unless clearly documented in treatment plan.

- **Curator Quality Review:**

- Confirm consistency with **rad_volume** and anatomical radiation fields.
- Ensure lymph node regions selected are anatomically plausible for disease site.
- Flag cases with unclear documentation or conflicting radiation volume descriptions.

- **Notes:**

This variable helps characterize the anatomic scope and intent of radiation therapy, especially in diseases with nodal spread. It complements primary volume coding and supports detailed stratification of treatment fields in comparative analyses.

Variable: `rad_dose_fraction_known`

Type: Categorical (Dropdown)

Description: Indicates whether the dose per fraction delivered during a course of radiation therapy is known and documented.

Reference: NAACCR #1501, 1511, 1521

Data Source: Radiation therapy summary reports, dosimetry records, treatment prescriptions

Curation Directives:

- **Definition:** Captures whether the actual dose administered per treatment fraction is documented and extractable.
- **Permitted Values:**

- 1: Yes
- 98: Not applicable, brachytherapy administered
- 99: Dose per fraction unknown; not stated in patient record
- **Source Documents:**
 - Radiation dose summary sheets
 - Radiation prescription documents
 - Treatment delivery logs
- **Curation Instructions:**
 - Select **1** if a numeric dose per fraction (in cGy) is explicitly stated or calculable in the documentation.
 - Select **98** if the modality is brachytherapy, for which dose per fraction may not be recorded in conventional units.
 - Select **99** if documentation does not specify the dose per fraction or if the source documentation is unavailable.
 - Use this field to determine whether `rad_dose_fraction` should be completed.
- **Curator Quality Review:**
 - Cross-check radiation modality (e.g., external beam vs. brachytherapy) for consistency with coding.
 - If documentation exists but the dose is reported in total only, do not assume per-fraction dose unless number of fractions is also stated and calculation is clearly valid.
 - Flag ambiguities in planning vs. delivered dose.
- **Notes:**

This variable supports interpretation of the numeric dose field and helps distinguish missing from inapplicable entries.

Variable: `rad_dose_fraction`

Type: Numeric

Description: Records the actual dose administered per fraction (session) during the radiation treatment course, in centigray (cGy).

Reference: NAACCR #1501, 1511, 1521

Data Source: Radiation prescription records, treatment delivery logs, dosimetry plans

Curation Directives:

- **Definition:** Specifies the dose delivered in each individual treatment fraction of radiation therapy for the reported course.
- **Permitted Values:**
 - Numeric values between **1** and **99997** (in cGy)
 - Only complete if `rad_dose_fraction_known` = 1
- **Source Documents:**
 - Physician-signed radiation prescription
 - Treatment delivery summary
 - Radiation therapy plan output with fractionation details
- **Curation Instructions:**
 - Enter the dose per treatment fraction in **cGy**, not Gy (e.g., 180 cGy instead of 1.8 Gy).
 - Only complete this field when `rad_dose_fraction_known` is coded as **1 (Yes)**.

- Do not enter estimated or average values unless explicitly documented as such.
 - Leave blank if the dose per fraction is not reported or if brachytherapy was used.
 - **Curator Quality Review:**
 - Confirm units are in cGy; flag values in Gy or other units for correction.
 - Check for logical consistency with total dose and number of fractions if those values are also abstracted.
 - Ensure values fall within expected therapeutic range (e.g., typically 100–300 cGy per fraction in external beam therapy).
 - **Notes:**

Enables standardized quantification of radiation fractionation schemes across treatment courses and modalities.
-

Variable: `rad_fractions_known`

Type: Categorical (Dropdown)

Description: Indicates whether the number of radiation treatment fractions (sessions) administered during a treatment course is known.

Reference: NAACCR #1503, 1513, 1523

Data Source: Radiation treatment plans, delivery records, dosimetry summaries

Curation Directives:

- **Definition:** Captures whether the total number of fractions delivered is clearly documented and available for abstraction.
- **Permitted Values:**
 - 1: Yes
 - 99: Unknown number of fractions; not stated in patient record
&#xNAN;*Note: This is a dropdown field.*
- **Source Documents:**
 - Final radiation therapy summary or plan
 - Treatment delivery logs
 - Progress notes from radiation oncology
- **Curation Instructions:**
 - Select **1 (Yes)** if the number of fractions is explicitly reported or calculable based on other documentation.
 - Select **99 (Unknown)** if the number of fractions is not specified or cannot be confidently determined from available documents.
 - Do not infer values based solely on planned treatment; only abstract what was actually administered.
- **Curator Quality Review:**
 - Ensure consistency with other radiation parameters such as dose per fraction and total dose.
 - If number of fractions is missing from standard documents, check consult or follow-up notes for summaries.
 - Confirm consistency with start and end dates, if both are known.
- **Notes:**

Use this field to determine whether `rad_fractions` should be completed. Supports interpretation of treatment intensity and schedule.

Variable: rad_fractions

Type: Numeric

Description: Records the actual number of radiation therapy fractions delivered during the treatment course.

Reference: NAACCR #1503, 1513, 1523

Data Source: Radiation oncology treatment plans, progress notes, delivery logs

Curation Directives:

- **Definition:** Represents the count of discrete radiation treatment sessions administered during the treatment course.
- **Permitted Values:**
 - Numeric values between **0** and **998**
 - Only complete if **rad_fractions_known** = **1** (Yes)
- **Source Documents:**
 - Radiation delivery summary
 - Cumulative dose documentation
 - Oncology consult or radiation summary note
- **Curation Instructions:**
 - Enter the total number of radiation fractions administered, regardless of the planned count.
 - Only abstract this field when **rad_fractions_known** is coded as **1** (Yes).
 - Do not count setup or simulation sessions unless they involved therapeutic radiation.
 - If the patient did not receive any fractions, enter **0**.
- **Curator Quality Review:**
 - Cross-check with treatment dates and number of sessions documented in logs.
 - Ensure that high or low values are plausible within the context of the disease and modality.
 - Flag inconsistencies with other related fields, such as total dose and dose per fraction.
- **Notes:**

Enables calculation of total dose delivery and supports quality control of treatment completeness.

Variable: rad_total_dose_known

Type: Categorical (Dropdown)

Description: Indicates whether the total radiation dose administered in a specific treatment course is known and documented.

Reference: NAACCR #1507, 1517, 1527

Data Source: Radiation treatment plans, dosimetry records, radiation oncology summaries

Curation Directives:

- **Definition:** Captures whether the cumulative dose of radiation delivered to the patient is available for abstraction.
- **Permitted Values:**
 - 1: Yes
 - 98: Not applicable, brachytherapy administered
 - 99: Total dose unknown; not stated in patient record

 Note: This is a dropdown field.

- **Source Documents:**
 - Final radiation therapy summary or total dose report
 - Treatment planning documentation
 - Progress or follow-up notes from radiation oncology
- **Curation Instructions:**
 - Select **1 (Yes)** if the total dose (in cGy) is clearly documented or can be derived from dose per fraction and number of fractions.
 - Select **98** for brachytherapy cases where total external beam dose is not applicable.
 - Select **99** if total dose is not stated or cannot be confidently abstracted.
- **Curator Quality Review:**
 - Confirm alignment with radiation modality (e.g., external beam vs. brachytherapy).
 - Ensure downstream variable **rad_total_dose** is only populated when value is **1 (Yes)**.
 - Check for consistency across documents; verify that derived totals match documented summaries.
- **Notes:**
Enables or suppresses entry of **rad_total_dose**. Provides critical context for understanding treatment intensity.

Variable: **rad_total_dose**

Type: Numeric

Description: Records the total cumulative radiation dose delivered to the patient in a given treatment course, measured in centigray (cGy).

Reference: NAACCR #1507, 1517, 1527

Data Source: Radiation treatment summaries, cumulative dose reports, treatment planning system exports

Curation Directives:

- **Definition:** Reflects the sum of all delivered radiation doses within a treatment course, expressed in cGy.
- **Permitted Values:**
 - Numeric value from **0** to **9,999,899,997**
 - Complete only if **rad_total_dose_known** = 1 (Yes)
- **Source Documents:**
 - Final dose summary
 - Dose-volume histogram or dosimetry report
 - Cumulative delivery log
- **Curation Instructions:**
 - Enter the total dose in **centigray (cGy)** as documented.
 - Do not round unless explicitly stated in the source.
 - Do not estimate based solely on planned fractions and doses unless confirmed in final delivery records.
- **Curator Quality Review:**
 - Check for internal consistency with **rad_dose_fraction** × **rad_fractions** if both are available.
 - Flag entries that appear unusually high or low based on standard treatment protocols.
 - Validate against total dose fields in planning system printouts or PDF summaries.

- **Notes:**
Critical for evaluating treatment intensity and comparing across modalities. Ensure values are properly contextualized with modality and site-specific intent.

Variable: rad_complete

Type: Categorical (Dropdown)

Description: Captures whether the patient completed their prescribed course of radiation treatment, discontinued it early, or if completion status is unknown.

Reference: n/a

Data Source: Radiation oncology notes, treatment summaries, physician progress notes

Curation Directives:

- **Definition:** Indicates the final status of the patient’s radiation treatment course—whether completed as intended, stopped early, or not completed due to other circumstances.
- **Permitted Values:**
 - 1: Radiation treatment completed as prescribed
 - 2: Radiation treatment discontinued early
 - 8: Radiation treatment not completed, other reason
 - 9: Unknown if radiation treatment was completed
- **Source Documents:**
 - Final radiation oncology treatment summary
 - Discharge or handoff notes from radiation oncology
 - Treatment plan status fields in EHR or radiation planning system
 - Oncology multidisciplinary case reviews
- **Curation Instructions:**
 - Select **1** if all planned fractions/doses were delivered with no early termination.
 - Select **2** if treatment was stopped prematurely due to toxicity, patient decision, clinical deterioration, or other medical concerns.
 - Select **8** if treatment was not completed for a documented reason that does not fit “discontinued early” (e.g., transfer of care, insurance denial).
 - Select **9** if documentation is insufficient to determine treatment completion.
- **Curator Quality Review:**
 - Ensure value is consistent with total number of fractions, dose delivered, and end date.
 - If “completed as prescribed” is selected, verify documentation of final dose delivery.
 - If “discontinued,” confirm cause and note if applicable for downstream QA or comment fields.
- **Notes:**
Treatment completion status is important for interpreting downstream outcomes, response assessments, and overall intent-to-treat analyses. When in doubt, opt for conservative coding (i.e., unknown) unless a clear statement is present.

Variable: rad_discontinuation

Type: Categorical (Dropdown)

Description: Captures the documented reason for early discontinuation of radiation therapy when treatment was not completed as initially prescribed.

Reference: n/a

Data Source: Radiation oncology progress notes, treatment summaries, clinician correspondence

Curation Directives:

- **Definition:** Documents the specific reason radiation therapy was discontinued earlier than intended by the treating physician. This variable is conditional and only populated when `rad_complete = 2` (Radiation treatment discontinued early).
- **Permitted Values:**
Dropdown selection with the following options:
 - 1: Radiation treatment completed as prescribed
 - 2: Discontinued early – toxicity
 - 3: Discontinued early – contraindicated due to other patient risk factors (e.g., comorbid conditions, advanced age, tumor progression)
 - 4: Discontinued early – patient decision
 - 5: Discontinued early – family decision
 - 6: Discontinued early – patient expired
 - 7: Discontinued early – reason not documented
 - 99: Unknown if radiation treatment discontinued; Unknown whether radiation therapy administered
- **Source Documents:**
 - Radiation oncology treatment summary
 - Provider notes detailing treatment status and reasons for deviation
 - Progress notes or care team communications indicating decision-making
 - Patient or family documentation (if applicable)
- **Curation Instructions:**
 - Only complete this field if `rad_complete = 2`.
 - Select the most specific applicable reason for early discontinuation.
 - If more than one reason is noted, prioritize clinical contraindications (toxicity, comorbidities) over social factors unless clearly stated.
 - If no documentation is found for early discontinuation reason, select 7.
 - Select 99 only when there is no documentation regarding whether radiation was discontinued or completed.
- **Curator Quality Review:**
 - Confirm consistency between this field and radiation dose/fraction variables (e.g., low total fractions with early discontinuation reason).
 - If completion status and discontinuation reason are conflicting, flag for secondary review.
- **Notes:**
This variable helps contextualize deviations from intended radiation therapy plans and may inform analyses of treatment tolerance, patient preferences, and outcomes. It is not intended to capture planned shorter regimens (e.g., hypofractionation).

Tumor Sample Information

Complete one instance of the Tumor Sample Information form for each unique tumor sample collected from the patient for sequencing or molecular profiling. Abstract only what is explicitly documented in authoritative sources such as pathology reports, laboratory information systems (LIS), sequencing laboratory reports, or clinical trial documentation.

For GENIE samples, piped-in Tier 1A data elements (e.g., sample ID, OncoTree code, sample type, sequencing year) will populate automatically from Sage. Curators should verify these values for accuracy but should not overwrite piped-in data.

For non-GENIE samples, assign a unique site-specific identifier, and record internal pathology accession numbers and internal sequencing IDs whenever available. Accurately document the sample collection date, sequencing date, and the anatomical sample type using the controlled vocabulary provided.

When available, link each sample to the appropriate pathology report and specimen from the Surgical Procedures form to maintain internal consistency. Align the sequencing and collection dates with the patient's treatment and diagnostic timelines, and apply approved conventions for approximated dates (e.g., mid-month, seasonal, or June 15 defaults).

Do not infer sample source, collection date, or sequencing status from provider notes, treatment patterns, or follow-up reports if not explicitly stated. Create additional form instances for each additional sample obtained or sequenced, ensuring accurate tracking of both GENIE and non-GENIE samples across the patient record.

Piped In - Tier1A Data Elements: Sample

The following variables are *piped in* from the existing Tier1A dataset and are not manually curated. These fields are auto-populated via data integration from the GENIE Tier1A repository, sourced and maintained by Sage Bionetworks. They are included for reference, audit trails, or linkage purposes, and are typically marked as @READONLY in the data collection system. Curators should not modify these fields, but may use them to validate or contextualize other data points.

Piped-in Tier1A Variables:

- `age_at_seq_report` – Age at Which Sequencing was Reported (Years)
- `oncotree_code` – Oncotree Code
- `sample_id` – GENIE Sample ID
- `sample_type` – Sample Type
- `sample_type_detailed` – Sample Type Detailed
- `seq_asaay_id` – Sequence Assay ID
- `cancer_type` – Cancer Type
- `cancer_type_detailed` – Cancer Type Detailed
- `sample_class` – Sample Class
- `seq_year` – Sequence Year
- `age_at_seq_report_days` – Age at Which Sequencing was Reported (Days)

Variable: `sample_genie_yn`

Type: Binary (Yes/No)

Description: Indicates whether the tumor sample was sequenced and successfully submitted to AACR Project GENIE.

Reference: n/a

Data Source: Molecular pathology reports, sequencing submission logs, GENIE submission metadata

Curation Directives:

- **Definition:** Identifies whether the specified tumor sample was included in the institution’s genomic data submission to GENIE.
- **Permitted Values:**
 - 1 = Yes
 - 0 = No
 - (Binary Yes/No dropdown)
- **Source Documents:**
 - Internal sequencing tracking logs or laboratory information management systems (LIMS)
 - Submission manifest files uploaded to GENIE
 - Molecular pathology reports linked to the tumor sample
 - GENIE cohort metadata files confirming submitted samples
- **Curation Instructions:**
 - Select “Yes” if the tumor sample was sequenced and the genomic data were submitted to GENIE.
 - Select “No” if the sample was not sequenced, or if sequencing occurred but the data were not included in GENIE submission.
 - Confirm sequencing status through institutional tracking systems or GENIE-specific submission files, not solely from presence of sequencing in the medical record.
 - Each sample should be reviewed independently, especially if multiple samples exist for a single patient.
- **Curator Quality Review:**
 - Cross-check against GENIE sequencing manifest or institutional submission logs.
 - Ensure accurate linkage between sample identifiers and submission status.
 - Flag cases where sequencing is performed but submission status is unclear.
- **Notes:**

This variable serves as the foundation for identifying which tumor samples are represented in GENIE genomic datasets. It should be used in conjunction with sample-level identifiers and sequencing platform metadata for downstream analyses.

Variable: `sample_non_genie_id`

Type: Free Text Entry (text)

Description: Captures a unique identifier for tumor samples that were not submitted to GENIE.

Reference: n/a

Data Source: Internal institutional sample tracking logs, pathology reports, tumor registries

Curation Directives:

- **Definition:** This field is used to assign or capture a unique identifier for tumor samples that were *not* sequenced or submitted to GENIE. It enables tracking of non-GENIE samples that are relevant for local clinical annotation or research purposes.
- **Permitted Values:**
 - Free-text unique sample identifier
 - Conditional field, completed only if `seq_yn` = 0 (No)
- **Source Documents:**
 - Local biorepository records or sample tracking databases
 - Internal pathology or specimen IDs
 - Clinical research systems containing sample-level identifiers for non-sequenced material

- **Curation Instructions:**

- Complete this field only when `seq_yn` = 0, indicating the sample was not submitted to GENIE.
- Enter the locally assigned unique identifier associated with the tumor sample.
- Ensure that the value is specific to the individual sample and not reused across multiple patients or samples.
- Avoid using PHI (e.g., patient name or MRN); use de-identified institutional sample codes or barcodes where available.
- This variable may support linkage to other local datasets or supplemental analyses outside of GENIE.

- **Curator Quality Review:**

- Verify the uniqueness of the identifier within the dataset.
- Confirm consistency with internal sample documentation.
- Flag duplicate entries or ambiguous identifiers for resolution.

- **Notes:**

This field facilitates structured tracking of tumor samples that fall outside the scope of GENIE genomic data submission but may still be clinically or analytically relevant. It supports transparency in sample selection and inclusion.

For Site Internal Use Only – Site-Specific Sample Identifiers The following variables are entered by the institution to assist with internal traceability and sample provenance documentation. These fields are not shared externally and are included solely for site-level reference, quality control, and linkage with local pathology or molecular laboratory systems. Although they may be prepopulated from institutional sources, they are considered optional. These variables are especially useful for resolving questions about tissue origin, sequencing concordance, and institutional sample tracking.

Institutional Sample Identifier Variables:

- **sample_path_acc_num**– Internal Pathology Sample Accession Number: Site-specific pathology accession number corresponding to the specimen used for sequencing. Typically assigned by the pathology department and used to locate associated slides or tissue in the LIS or EMR.
- **sample_block_acc_num**– Internal Pathology Block Accession Number: Block-level identifier linked to the exact tissue block (e.g., FFPE block) used for DNA/RNA extraction. Helps clarify sample origin and enables re-cuts or re-extraction if needed.
- **sample_requisition_id**– Internal Requisition ID: Local tracking number associated with the internal requisition for sequencing. May be generated by the genomics lab, clinical team, or requisition platform. Useful for reconciling sample shipments or orders.

Variable: `sample_coll_date`

Type: Date (MM/DD/YYYY)

Description: Captures the calendar date when the tumor sample used for NGS was collected.

Reference: n/a

Data Source: Pathology reports, laboratory requisitions, specimen tracking logs

Curation Directives:

- **Definition:** The date on which the tumor specimen was physically obtained from the patient for sequencing purposes (e.g., biopsy, surgical resection).

- **Permitted Values:**
 - Date format: MM/DD/YYYY
 - Acceptable date range: Must be on or after January 1, 2000
 - Cannot be a future date or precede the patient’s date of birth
- **Source Documents:**
 - Pathology report with specimen collection date
 - Laboratory requisition forms
 - Clinical notes describing biopsy or surgical procedure
 - Tumor sample tracking systems
- **Curation Instructions:**
 - Abstract the date the tumor sample was collected, not the date it was received, processed, or sequenced.
 - If multiple dates are documented, prioritize the date corresponding to the specific sample sequenced.
 - Ensure alignment with internal sample ID, pathology accession, and sequencing metadata where applicable.
 - If the date cannot be determined from available documentation, leave the field blank.
- **Curator Quality Review:**
 - Confirm consistency with sample identifier and collection procedure
 - Review for any potential data entry or logic errors (e.g., invalid dates, future dates)
 - Cross-reference with other timestamped events (e.g., diagnosis, surgery) for plausibility
- **Notes:**

This variable is critical for evaluating sequencing timeliness, tumor evolution, and temporal alignment with treatment history.

Variable: `sample_coll_year`

Type: Integer (Calculated)

Description: Calculated year of tumor sample collection date.

Reference: n/a

Data Source: Automatically calculated from the `sample_coll_date` field.

Curation Directives:

- **Definition:** Captures the calendar year during which the tumor sample was collected, extracted from the full `sample_coll_date`. This field supports year-based analyses of sample acquisition patterns and sequencing timelines.
- **Permitted Values:**
 - Auto-calculated using REDCap function: `[sample_coll_date]`
 - Must be a valid four-digit year (e.g., 1995–Present)
- **Source Documents:**
 - Not manually curated
 - Automatically calculated from valid entries in `birth_date` and `sample_coll_date`
- **Curator Quality Review:**
 - Cross-check this field against known sequencing years (`seq_year`) or documented diagnosis dates to ensure internal consistency.

Variable: `sample_coll_date_int`

Type: Integer (Calculated)

Description: Number of days from the patient's date of birth to the tumor sample collection date.

Reference: n/a

Data Source: Automatically derived from `birth_date` and `sample_coll_date` entries

Curation Directives:

- **Definition:** A calculated variable indicating the number of elapsed days between the patient's birth and the tumor sample collection date.
- **Permitted Values:**
 - Auto-calculated using REDCap function: `datediff([birth_date], [sample_coll_date], "d", "mdy", true)`
 - Result is a non-negative integer
 - Values should fall within the range of realistic patient ages at time of sample collection (e.g., >0 and <50,000)
- **Source Documents:**
 - Not manually curated
 - Automatically calculated from valid entries in `birth_date` and `sample_coll_date`
- **Curation Instructions:**
 - Ensure both input fields are present and valid for calculation
 - No direct editing required; validate inputs for logical accuracy
 - If result appears implausible, verify and correct source dates

Variable: `sample_seq_order_date`

Type: Date (MM/DD/YYYY)

Description: Captures the calendar date when the tumor sample used for NGS was collected.

Reference: n/a

Data Source: Entered based on institutional electronic health record (EHR), laboratory information system (LIS), or molecular test ordering platform

Curation Directives:

- **Definition:** Represents the date on which the NGS test was formally ordered for the sample, typically as documented in the institutional molecular ordering or laboratory workflow system. This variable enables tracking of sequencing pipeline timelines and quality metrics.
- **Permitted Values:**
 - Format: MM/DD/YYYY
 - Must reflect a valid, calendar-based date
- **Source Documents:**
 - Institutional test ordering system
 - Molecular tumor board documentation (if available)
 - Laboratory accession forms or electronic LIS records
 - Clinical note or pathology addendum indicating sequencing request
- **Curation Instructions:**

- Record the date when the NGS panel was formally ordered, not the date it was collected, sequenced, or resulted.
 - If multiple sequencing panels were ordered, enter the earliest valid sequencing order date associated with the submitted sample.
 - Leave blank if no order date can be confirmed; do not infer based on result or collection dates. Abstract the date the tumor sample was collected, not the date it was received, processed, or sequenced.
- **Curator Quality Review:**
 - Flag entries where order date appears anomalously early, late, or out of sequence relative to expected workflow.
 - If sequencing was conducted outside the institution, mark as unknown if institutional records do not contain a verifiable order date.
 - **Notes:**
 - This field supports sequencing pipeline QA/QC metrics, turnaround time audits, and time-to-sequencing analyses across patients by providing a consistent reference point for when genomic testing was formally initiated.

Variable: `sample_seq_order_date_int`

Type: Integer (Calculated)

Description: Number of days from the patient's date of birth to the institutional order date of next-generation sequencing (NGS) for the tumor sample.

Reference: n/a

Data Source: Automatically derived from `birth_date` and `sample_seq_order_date` entries

Curation Directives:

- **Definition:** A calculated interval variable representing the number of elapsed calendar days between the patient's date of birth and the date the NGS test was formally ordered.
- **Permitted Values:**
 - Auto-calculated using REDCap function:

```
datediff([birth_date], [sample_seq_order_date], "d", "mdy", true)
```
 - May be blank if either source date is missing or invalid
- **Source Documents:**
 - Not manually curated
 - Automatically calculated from valid entries in `birth_date` and `sample_seq_order_date`
- **Curation Instructions:**
 - Ensure both input fields are present and valid for calculation
 - No direct editing required; validate inputs for logical accuracy
 - If result appears implausible, verify and correct source dates

Variable: `sample_seq_completion_date`

Type: Date (MM/DD/YYYY)

Description: Date on which the next-generation sequencing (NGS) process was formally completed by the sequencing laboratory or documented in the institutional system.

Reference: n/a

Data Source: Sequencing laboratory reports, EMR sequencing records, molecular pathology documentation

Curation Directives:

- **Definition:** Captures the finalized date of sequencing completion, marking the point at which molecular profiling results became available or finalized for interpretation and downstream clinical or research use.
- **Permitted Values:**
 - Any valid calendar date in MM/DD/YYYY format
 - Must be equal to or later than the NGS order date
- **Source Documents:**
 - Pathology reports or molecular diagnostics records
 - NGS lab tracking forms
 - EMR timestamps from molecular results sections
- **Curation Instructions:**
 - Record the date listed as the finalization or completion date of sequencing, not the specimen receipt or processing date.
 - If multiple sequencing phases exist (e.g., panel upgrade or rerun), use the final report date corresponding to the version intended for clinical interpretation.
 - If unavailable, leave blank and flag for query if required for derived calculations.
- **Curator Quality Review:**
 - Confirm temporal consistency with `sample_seq_order_date` (completion must be after or on same day).
 - Validate against documented sequencing report release dates.
 - Review for default or placeholder entries (e.g., 01/01/1900 or today's date).
- **Notes:**
 - This field anchors the timing of sequencing-based interventions and supports evaluation of turnaround time metrics, sequencing timelines, and clinical utility analyses.

Variable: `sample_seq_completion_date_int`

Type: Integer (Calculated)

Description: Number of days from the patient's date of birth to the sequencing completion date.

Reference: n/a

Data Source: Automatically derived from `birth_date` and `sample_seq_completion_date` entries

Curation Directives:

- **Definition:** A calculated variable measuring the number of days between the patient's birth date and the date NGS testing was completed. Facilitates timeline normalization across patients for sequencing metrics.
- **Permitted Values:**
 - Auto-calculated using REDCap function:

```
datediff([birth_date], [sample_seq_completion_date], "d", "mdy", true)
```
- **Source Documents:**

- Not manually curated
- Automatically calculated from valid entries in `birth_date` and `sample_seq_completion_date`

- **Curation Instructions:**

- Ensure both input fields are present and valid for calculation
- No direct editing required; validate inputs for logical accuracy
- If result appears implausible, verify and correct source dates

Variable: `sample_seq_report_date`

Type: Date (MM/DD/YYYY)

Description: Date on which the final next-generation sequencing (NGS) report was formally issued by the sequencing laboratory or documented in the institutional system.

Reference: n/a

Data Source: Molecular pathology reports, sequencing laboratory report PDFs, or EMR sequencing documentation

Curation Directives:

- **Definition:** Captures the date the finalized NGS report was made available, typically indicating completion of sequencing analysis and generation of interpretive results.
- **Permitted Values:**
 - Any valid calendar date in MM/DD/YYYY format
 - Must occur after or on the same date as the sequencing completion date
- **Source Documents:**
 - Finalized sequencing report (PDF, HTML, or EMR upload)
 - Institutional molecular diagnostics documentation
 - EMR molecular pathology note sections
- **Curation Instructions:**
 - Abstract the date when the final interpretive sequencing report was issued
 - Do not use specimen receipt or intermediate QC dates
 - If multiple reports are issued, use the final version intended for clinical review
- **Curator Quality Review:**
 - Verify temporal logic with `sample_seq_order_date` and `sample_seq_completion_date`
 - Cross-check EMR vs lab report timestamps
- **Notes:**
 - Anchors molecular diagnostic timelines and is critical for calculating sequencing turnaround time, defining clinical utility windows, and aligning biomarker findings with treatment decisions.

Variable: `sample_seq_report_year`

Type: Integer (Calculated)

Description: Four-digit calendar year in which the final NGS report was issued.

Reference: n/a

Data Source: Automatically derived from `sample_seq_report_date`

Curation Directives:

- **Definition:** Derived variable that extracts the year component from the finalized sequencing report date to facilitate stratified analyses and temporal aggregation.
- **Permitted Values:**
 - Auto-calculated using REDCap function:
`year([sample_seq_report_date])`
 - Result is a 4-digit year (e.g., 2020, 2021, etc.)
- **Source Documents:**
 - Not manually curated
 - Automatically calculated from valid `sample_seq_report_date` entry
- **Curation Instructions:**
 - Ensure parent date field is complete and logically valid
 - No manual entry needed
 - Confirm calculation updates correctly when source date is modified
- **Curator Quality Review:**
 - Cross-check this field against known sequencing years (`seq_year`) or documented diagnosis dates to ensure internal consistency.

Variable: `sample_seq_report_date_int`

Type: Integer (Calculated)

Description: Number of days from the patient's date of birth to the final NGS report date.

Reference: n/a

Data Source: Automatically derived from `birth_date` and `sample_seq_report_date`

Curation Directives:

- **Definition:** A calculated variable measuring the time interval in days between a patient's birth and the issuance of their NGS report.
 - **Permitted Values:**
 - Auto-calculated using REDCap function:
`datediff([birth_date], [sample_seq_report_date], "d", "mdy", true)`
 - **Source Documents:**
 - Not manually curated
 - Requires valid `birth_date` and `sample_seq_report_date` values
 - **Curation Instructions:**
 - Ensure both input fields are present and valid for calculation
 - No direct editing required; validate inputs for logical accuracy
 - If result appears implausible, verify and correct source dates
-

Variable: sample_path_report

Type: Categorical (Dropdown)

Description: Links the sequenced tumor sample to the corresponding pathology report from the Surgical Procedures section.

Reference: n/a

Data Source: Surgical pathology reports, sequencing lab documentation, internal specimen tracking systems

Curation Directives:

- **Definition:** Identifies which surgical pathology report documents the tumor sample that was sequenced and submitted to GENIE.
- **Permitted Values:**
 - Dropdown menu with values 1–40 corresponding to discrete pathology reports (e.g., “Pathology Report 1”)
 - 99 = Cannot associate sequencing to any pathology report
- **Source Documents:**
 - Surgical pathology reports (previously curated in Surgical Procedures section)
 - Internal tumor sample tracking logs
 - Laboratory requisition forms
 - Sequencing metadata or test reports
- **Curation Instructions:**
 - Review the available pathology reports and match the sample collection date, anatomical site, and procedure to determine the appropriate report.
 - Use institutional specimen identifiers or accession numbers to confirm sample-source linkage.
 - If the sequenced sample was not linked to any documented surgical pathology report (e.g., external source, archival sample), select 99.
 - This field may be updated post hoc if future reconciliation clarifies the correct pathology report.
- **Curator Quality Review:**
 - Validate the selected pathology report by confirming the match with sequencing metadata and tumor sample date.
 - Review discrepancies or uncertain associations with the project data manager or site contact.
 - Ensure consistency with the previously abstracted pathology report accession number (sample_path_acc_num).
- **Notes:**

This variable supports traceability between sequenced samples and the clinical pathology documentation, which is essential for interpreting molecular results in the context of tumor histology and surgical origin.

Variable: sample_path_specimen

Type: Categorical (Dropdown)

Description: Identifies the specific pathology specimen associated with the sequenced tumor sample, as documented in the pathology report referenced in the Surgical Procedures form.

Reference: n/a

Data Source: Pathology reports, surgical procedure forms, specimen tracking logs, sequencing documentation

Curation Directives:

- **Definition:** Links the tumor sample submitted for sequencing to the specific specimen collected during a surgical procedure and documented in the pathology report.
- **Permitted Values:**
 - Dropdown options 1-30 = Specimen 1-30
 - 99 = Cannot associate sequencing to any specific specimen
- **Source Documents:**
 - Pathology reports listing specimen numbers and descriptions
 - Surgical Procedures form (specimen documentation)
 - Internal laboratory specimen tracking systems
 - Molecular pathology or sequencing reports referencing specimen IDs
- **Curation Instructions:**
 - Match the tumor sample to the correct specimen by reviewing the pathology report selected in the field `sample_path_report`.
 - Use details such as anatomic site, histologic diagnosis, collection date, and internal specimen or accession numbers to confirm the correct match.
 - Select the corresponding specimen number (e.g., 2 = Specimen 2) that was the source of the sequenced tissue.
 - If the tumor sample cannot be confidently linked to a specific specimen, select 99.
 - Do **not** select a value based on assumption; rely on documented evidence for all associations.
- **Curator Quality Review:**
 - Ensure that the selected specimen is documented in the pathology report linked via `sample_path_report`.
 - Cross-reference internal accession numbers and sample collection dates where applicable.
 - Flag cases with ambiguous or missing documentation for review by data management staff or site coordinator.
- **Notes:**

This field supports traceability between sequencing data and the histopathologic source of the tumor sample, which is critical for data integrity, reproducibility, and clinical interpretation.

Variable: `sample_class_curated`

Type: Categorical (Dropdown)

Description: Curator-assigned classification of the sample type as either a tumor-derived specimen or cell-free DNA (cfDNA) specimen based on pathology report and/or sequencing metadata.

Reference: n/a

Data Source: Molecular pathology reports, specimen accession metadata, sequencing report headers, and laboratory documentation

Curation Directives:

- **Definition:** Discrete classification of the biological source of the sample submitted for sequencing, indicating whether the material originated from solid or cytologic tumor tissue or was derived from a plasma cfDNA (liquid biopsy) source.
- **Permitted Values:**
 - 1, Tumor – Sample derived from solid tumor, cytology, core needle, surgical resection, etc.
 - 2, cfDNA – Sample derived from circulating tumor DNA (ctDNA) isolated from plasma (liquid biopsy)
- **Source Documents:**

- Molecular sequencing report (sample description, specimen type field)
- Pathology accession report or cover sheet
- Sample metadata from institutional sequencing system (e.g., EPIC Beaker, Tempus portal)
- **Curation Instructions:**
 - Review the sample label, type, and source across sequencing reports and pathology documentation
 - Assign **Tumor** if the sample is from tissue or cytologic fluid (e.g., FNA, biopsy, resection, effusion)
 - Assign **cfDNA** only when clearly stated that plasma was used and ctDNA was sequenced
 - If ambiguous or unclassified, escalate to sequencing coordinator or site PI for clarification
 - Do not assume based solely on collection date or test type
- **Curator Quality Review:**
 - Confirm alignment with sequencing panel expectations (e.g., cfDNA-specific panels vs tissue-based panels)
 - Verify consistency with other fields like **specimen_type**, **sample_seq_report_date**, and NGS platform notes
 - Flag conflicting classifications across sources for resolution
- **Notes:**
 - This curated field is critical for downstream analysis, particularly when distinguishing tumor mutation burden (TMB), variant allele frequency (VAF) distributions, and the interpretive context of sequencing results derived from tissue vs plasma.

Variable: `sample_type_detailed_curated`

Type: Categorical (Dropdown)

Description: Categorizes the anatomical origin of the sequenced tumor sample, indicating whether the specimen was collected from a primary tumor site, a metastatic site, or a recurrence.

Reference: n/a

Data Source: Pathology reports, operative reports, radiology reports, oncology clinic notes

Curation Directives:

- **Definition:** Specifies the anatomical classification of the tumor sample based on its site of origin at the time of collection.
- **Permitted Values:**
 - 1 = Primary tumor
 - 2 = Lymph node metastasis
 - 3 = Distant organ metastasis
 - 4 = Metastasis site unspecified
 - 5 = Local recurrence
 - 99 = Unknown
- **Source Documents:**
 - Pathology reports with specimen site and diagnosis
 - Operative or surgical notes describing collection site
 - Radiology reports indicating location of tumor burden
 - Oncology progress notes discussing disease course and sampling rationale
- **Curation Instructions:**
 - Assign 1 (Primary tumor) if the specimen is from the original tumor site (e.g., breast, colon, skin) without evidence of spread.

- Assign 2 (Lymph node metastasis) if the sample was taken from a lymph node with confirmed tumor involvement.
- Assign 3 (Distant organ metastasis) for samples from non-nodal metastatic sites (e.g., liver, lung, bone).
- Assign 4 (Metastasis site unspecified) if the record confirms metastatic disease but does not indicate the anatomical site.
- Assign 5 (Local recurrence) if the sample was collected from a known site of recurrence near the original tumor site.
- Assign 99 (Unknown) if anatomical origin cannot be determined from available documentation.
- Use best clinical judgment supported by documentation to distinguish between local recurrence and metastasis when relevant.

- **Curator Quality Review:**

- Verify consistency with the associated pathology report and specimen description.
- Review longitudinal records to confirm metastatic status and site, if unclear from the immediate documentation.
- Flag ambiguous or conflicting records for adjudication by clinical leads.

- **Notes:**

Accurate classification of anatomical sample type is critical for stratifying genomic data by tumor site and understanding disease progression and sampling bias across the cohort.

Variable: `sample_cancer_type_detailed_curated`

Type: Categorical (Dropdown, Autocomplete)

Description: Curated assignment of the specific cancer type associated with the tumor sample, using a controlled vocabulary aligned with GENIE, OncoTree, and disease-specific conventions.

Reference: n/a

Data Source: Next-generation sequencing (NGS) report diagnosis field, pathology reports, clinical documentation, and OncoTree mapping guidelines

Curation Directives:

- **Definition:** Indicates the specific cancer diagnosis associated with the submitted tumor sample, using a detailed dropdown list curated to reflect real-world clinical usage
- **Permitted Values:**
 - Extensive dropdown list of over 800 harmonized diagnostic categories (e.g., 1 = Lung Adenocarcinoma, 10 = Bladder Urothelial Carcinoma, 30 = Renal Clear Cell Carcinoma, 113 = Other, etc.)
- **Source Documents:**
 - Final NGS sequencing report (e.g., FoundationOne, Tempus, in-house reports)
 - Associated surgical pathology or cytopathology reports
- **Curation Instructions:**
 - Review the tumor sample’s diagnosis or indication for testing in the sequencing report header or associated clinical documentation
 - Select the closest matching term from the dropdown; prefer specificity over generality (e.g., “Breast Invasive Ductal Carcinoma” over “Breast Cancer”)
 - Use “Other” (113) only if the cancer type is truly not represented
 - Do not use NOS or unspecified terms if a more specific subtype is clearly available
 - For hematologic malignancies or rare tumors, consult disease classification guides or OncoTree for proper mapping

- For mixed histologies (e.g., adenocarcinoma + squamous), use the most dominant or clinically relevant subtype
- **Curator Quality Review:**
 - Validate that the assigned cancer type is consistent with both the NGS indication and specimen histology
 - If conflicting documentation exists, flag for adjudication by site clinical lead or registry coordinator
 - Ensure that rare and ambiguous histologies are not misclassified due to abbreviation mismatches
- **Notes:**
 - This variable is critical for downstream stratification, mutation frequency analysis, and linkage to GENIE Tier1A

Hospitalizations

Hospitalizations Information

Complete one instance of the hospitalizations form for each distinct inpatient admission that occurred after the patient’s initial cancer diagnosis. Abstract only what is explicitly documented in authoritative sources such as hospitalization encounter records, inpatient progress notes, or admission/discharge summaries.

For each hospitalization, document the admission and discharge dates, the type of facility (internal institution, external institution, or both in the case of a transfer), and the primary reason for admission (illness, treatment, or other). If the reason is coded as “Other,” provide a clear, specific explanation using exact terminology from the source documentation.

If the hospitalization was related to treatment toxicity, indicate whether toxicity was present and, if reported, capture the toxicity grade based on the Common Terminology Criteria for Adverse Events (CTCAE). Maintain internal consistency with the patient’s treatment timeline and diagnosis history.

Do not infer hospitalizations from vague provider notes or medication logs without a clear, documented inpatient admission. If the patient was hospitalized multiple times, complete a separate form instance for each hospitalization.

Variable: hosp_yn

Type: Binary (Yes/No)

Description: Indicates whether the patient has been hospitalized at any point following the initial cancer diagnosis.

Reference: NAACCR Item #605

Data Source: Oncology progress notes, hospitalization records, discharge summaries, EMR encounter lists

Curation Directives:

- **Definition:** Captures any instance of inpatient hospitalization that occurred after the patient’s initial cancer diagnosis, regardless of the cause or duration of stay.
- **Permitted Values:**
 - Yes
 - No

- **Source Documents:**
 - Oncology progress notes referencing hospitalization events
 - Inpatient encounter lists in the EHR
 - Hospital admission/discharge summaries
 - Correspondence with care teams documenting external hospitalizations
- **Curation Instructions:**
 - Select **Yes** if any documentation confirms an inpatient hospital admission occurring after the date of cancer diagnosis, regardless of whether it was cancer-related.
 - Select **No** if there is clear documentation indicating no hospitalizations since diagnosis, or if no such hospitalizations are recorded in available records.
 - Do not consider emergency room visits or outpatient infusions/procedures unless they resulted in inpatient admission.
 - If documentation of hospitalization is found, complete downstream hospitalization fields if available in the form logic.
- **Curator Quality Review:**
 - Cross-check with diagnosis and treatment timelines to ensure hospitalizations occur post-diagnosis.
 - Validate against any narrative entries that may reference “recent admission,” “discharged from hospital,” or similar language.
 - If uncertain and no corroborating documents exist, escalate for review.
- **Notes:**
 - This variable establishes the presence of post-diagnosis hospitalization and should be used as a trigger for deeper abstraction of hospitalization details, such as admission reason, duration, and location. It supports outcomes research, health resource utilization, and clinical course documentation.

Variable: hosp_institution

Type: Categorical (Dropdown)

Description: Identifies the type of institution at which the patient was hospitalized following initial cancer diagnosis.

Reference: n/a

Data Source: Hospital admission records, transfer documentation, institutional encounter lists, oncology progress notes

Curation Directives:

- **Definition:** Indicates whether the patient was hospitalized at the reporting GENIE site (internal institution), at an external facility, or at both due to inter-facility transfer.
- **Permitted Values:**
 - 1 = Internal Institution
 - 2 = External Institution
 - 3 = Both Internal and External Institutions
- **Source Documents:**
 - Encounter summaries listing hospitalization sites
 - Transfer documentation
 - Progress notes specifying site of hospitalization
 - Discharge summaries and admission notes from internal and external facilities

- **Curation Instructions:**
 - Select **1 (Internal Institution)** if documentation confirms that hospitalization occurred solely at the GENIE-participating institution.
 - Select **2 (External Institution)** if hospitalization occurred exclusively at a facility outside the reporting institution.
 - Select **3 (Both Internal and External Institutions)** if there is evidence of inter-facility transfer or multiple hospitalizations spanning both internal and external sites.
 - If source documentation includes ambiguous facility names, confirm whether the facility is affiliated with the reporting institution.
- **Curator Quality Review:**
 - Review EHR documentation for facility identifiers, admission notes, and transfer summaries.
- **Notes:**
 - This variable supports institutional attribution of hospitalization events and facilitates downstream queries related to internal versus external care coordination, outcomes comparison, and continuity of care tracking.

Variable: hosp_start_date

Type: Date (MM/DD/YYYY)

Description: Captures the calendar date on which the patient was admitted to the hospital

Reference: n/a

Data Source: Admission records, discharge summaries, encounter logs, EHR hospitalization events

Curation Directives:

- **Definition:** Represents the formal admission date of a hospitalization episode for any reason occurring after the patient’s initial cancer diagnosis. This may include inpatient treatment, observation, emergency care leading to admission, or procedural hospitalization.
- **Permitted Values:**
 - Date in MM/DD/YYYY format
- **Source Documents:**
 - Admission or encounter summary in the EHR
 - Discharge summary referencing admission date
 - Hospital course narratives or treatment notes
 - Transfer documentation including initial hospitalization date
- **Curation Instructions:**
 - Record the first date of inpatient hospital admission.
 - Confirm the admission date corresponds to a true hospitalization and not an outpatient or emergency department encounter unless it led to an inpatient stay.
 - If the patient was transferred between facilities, use the admission date at the **first** facility in the hospitalization episode.
 - Leave blank if the patient was never hospitalized or if no date is available in source documents.
- **Curator Quality Review:**
 - Verify admission status by checking EHR encounter type (e.g., “inpatient,” “admitted,” “hospital stay”).
 - Ensure the start date is after the initial cancer diagnosis date.

- **Notes:**
This field enables time-to-event analyses (e.g., time from diagnosis to hospitalization) and supports linkage with outcomes and treatment milestones.
-

Variable: `hosp_start_date_int`

Type: Integer (Calculated)

Description: Represents the number of days between the patient's date of birth and the date of hospital admission.

Reference: n/a

Data Source: Automatically calculated from recorded values in `birth_date` and `hosp_start_date`

Curation Directives:

- **Definition:** This calculated variable provides the interval, in days, from the patient's date of birth to the date of their hospitalization (`hosp_start_date`). It is used to derive age-based temporal measures without directly exposing identifiable date values.
 - **Permitted Values:**
 - Positive integer (number of days between birth and hospitalization)
 - **Source Documents:**
 - Derived from the curated values of `birth_date` and `hosp_start_date`
 - **Curation Instructions:**
 - Do not manually enter or modify this field; it is calculated automatically using the REDCap `datediff()` function.
 - Ensure that both `birth_date` and `hosp_start_date` are accurately populated before this interval value can be trusted.
 - If either of the source fields is missing or contains an invalid date, the calculated value may be blank or erroneous and should be flagged for correction.
 - **Curator Quality Review:**
 - Confirm that the interval aligns with the patient's expected age at hospitalization.
 - Check for negative or implausible values (e.g., interval less than zero or over 50,000 days).
 - Investigate and resolve discrepancies by reviewing the underlying dates.
 - **Notes:** This interval is useful for de-identification and age-at-event analyses. It supports temporal ordering of clinical events and enables time-to-event calculations while preserving patient confidentiality.
-

Variable: `hosp_end_date`

Type: Date (MM/DD/YYYY)

Description: Captures the discharge date corresponding to the end of a hospitalization episode following the patient's initial cancer diagnosis.

Reference: n/a

Data Source: Discharge summary, inpatient progress notes, encounter logs, or hospitalization episode records

Curation Directives:

- **Definition:** The calendar date on which the patient was formally discharged from the hospital following an inpatient stay for any clinical reason after their cancer diagnosis. This includes discharges due to recovery, transfer, death, or other completion of inpatient care.

- **Permitted Values:**
 - Date in MM/DD/YYYY format
 - Must be on or after the corresponding hospitalization start date (`hosp_start_date`)
- **Source Documents:**
 - Discharge summaries
 - Inpatient hospitalization records or encounter documentation
 - Electronic health record (EHR) admission-discharge-transfer (ADT) logs
- **Curation Instructions:**
 - Enter the date the patient was officially discharged from the hospital.
 - Ensure the date is complete and legible in the source documentation.
 - The field should only be completed if a valid `hosp_start_date` exists.
 - If the patient was not discharged (e.g., deceased during hospitalization), still enter the documented end date of the hospitalization episode.
 - Do not use estimated or inferred dates unless clearly documented as discharge dates.
- **Curator Quality Review:**
 - Cross-check with `hosp_start_date` to confirm chronological consistency.
 - Validate that the discharge date corresponds to the same hospitalization episode.

Variable: `hosp_end_date_int`

Type: Integer (Calculated)

Description: Automatically calculated number of days from the patient's date of birth (DOB) to the end date of a hospitalization episode.

Reference: n/a

Data Source: Derived from `birth_date` and `hosp_end_date` fields

Curation Directives:

- **Definition:** This calculated field represents the number of days from the patient's date of birth to the hospitalization end date (i.e., discharge date). It provides a standardized temporal metric to support longitudinal analyses and temporal sequencing across patients.
- **Permitted Values:**
 - Auto-generated integer values
 - Must be greater than or equal to the corresponding hospitalization start interval (`hosp_start_date_int`), if available
 - Cannot be negative
- **Source Documents:**
 - Computed using the formula: `datediff([birth_date], [hosp_end_date], "d", "mdy", true)`
- **Curation Instructions:**
 - No manual data entry required. This is a system-calculated field.
 - Ensure that the underlying fields (`birth_date` and `hosp_end_date`) are correctly populated and formatted.
 - If `hosp_end_date` is missing or invalid, this interval field may be null or error-prone.
 - Do not override or manually adjust the calculated value.
- **Curator Quality Review:**

- Confirm that the hospitalization end date occurs after birth and after hospitalization start.
- Report and investigate values that appear inconsistent or implausible based on medical history.

- **Notes:**

- This field is designed to support age-based and time-sequenced analyses without exposing direct calendar dates. It is useful for de-identified datasets and temporal modeling across patients.

Variable: hosp_reason

Type: Categorical (Dropdown)

Description: Captures the primary reason for a patient’s hospitalization following their initial cancer diagnosis.

Reference: n/a

Data Source: Discharge summaries, admission notes, oncology progress notes

Curation Directives:

- **Definition:** Indicates the primary clinical reason for the patient’s hospital admission based on medical documentation.
- **Permitted Values:**

- 1 = Illness
- 2 = Cancer Treatment
- 3 = Treatment Related Event
- 4 = Monitoring
- 5 = Other

- **Source Documents:**

- Hospital admission records
- Discharge summaries
- Oncology or inpatient progress notes
- Emergency department notes (if admission originated there)

- **Curation Instructions:**

- Select the value that best reflects the documented reason for hospitalization.
 - * **“1 = Illness”:** Select when the admission is unplanned and primarily due to an acute medical issue not explicitly framed as cancer-treatment-related, such as:
 - Infection (e.g., pneumonia, sepsis), fever not labeled as treatment-related
 - Cardiopulmonary events (e.g., CHF exacerbation, PE) unless explicitly attributed to treatment
 - GI bleed, dehydration, electrolyte abnormalities, uncontrolled diabetes, etc.
 - * **“2 = Cancer Treatment”:** Select when the admission is planned/scheduled for delivery of cancer-directed therapy or a clearly planned oncology admission, such as:
 - Inpatient chemotherapy administration, stem cell transplant conditioning, planned immunotherapy admission
 - Planned cancer-directed procedure/admission explicitly for treatment delivery
 - Do not use for complications from treatment, use “3 = Treatment Related Event” instead.
 - * **“3 = Treatment Related Event”:** Select when the primary reason is a complication/toxicity or adverse outcome explicitly attributed to cancer therapy (systemic therapy, radiation, surgery, or procedural intervention), such as:
 - Neutropenic fever, chemo-induced nausea/vomiting/dehydration, mucositis
 - Immune-related adverse events (e.g., colitis, pneumonitis) documented as ICI-related

- Post-op complication or procedural complication documented as such. Explicit documentation required (e.g., “chemo-related,” “treatment toxicity,” “post-op complication,” “immune-mediated”). Do not infer causality.
 - * **“4 = Monitoring”**: Select when the admission is primarily for observation/surveillance, evaluation, or supportive monitoring without a documented acute illness and without treatment delivery as the main purpose, such as:
 - Observation admission for symptom monitoring, workup, or rule-out (e.g., “admit for observation,” “monitor labs,” “monitor neurologic status”)
 - Inpatient monitoring required by protocol or safety monitoring (when described as monitoring rather than treatment administration) If the stay includes treatment but the stated purpose is monitoring only, code as Monitoring; if treatment delivery is the purpose, code as Cancer Treatment.
 - * **“5 = Other”**: Select only when documentation clearly indicates the primary reason does not fit Illness, Cancer Treatment, Treatment Related Event, or Monitoring. When selected, populate [hosp_reason_other] with the documented reason.
- **Curator Quality Review:**
 - Ensure that the selected category aligns with the narrative in the source documents.
 - If documentation is ambiguous, flag for clinical review or mark as unknown if permitted.
 - **Notes:**
 - Accurate classification helps differentiate between emergent vs. planned admissions and supports health services analysis.

Variable: hosp_reason_other

Type: Free Text Entry (text)

Description: Captures a specific reason for hospitalization when “Other” is selected in hosp_reason.

Reference: n/a

Data Source: Same as above

Curation Directives:

- **Definition:** Provides a detailed explanation of the hospitalization reason when the predefined options do not apply.
- **Permitted Values:**
 - Free-text description of the reason for hospitalization (e.g., “Social admission due to caregiver loss,” “Psychiatric evaluation,” “COVID-19 isolation”)
 - Field is conditionally shown only if hosp_reason = 3 (“Other”)
- **Source Documents:**
 - Same as for hosp_reason: admission notes, discharge summaries, inpatient progress notes
- **Curation Instructions:**
 - Only complete this field if hosp_reason is coded as “3, Other”.
 - Enter the documented explanation exactly or paraphrased clearly.
 - Avoid vague entries (e.g., “miscellaneous”) unless documentation lacks clarity.
 - Use standard clinical terminology where possible.
- **Curator Quality Review:**
 - Ensure consistency between the text entry and the context in the medical record.
 - Confirm that this field is not completed unless triggered by “Other” in hosp_reason.

- **Notes:**

- This field supports granular review of atypical hospitalization causes and improves classification completeness.
-

Variable: hosp_tox_yn

Type: Categorical (Dropdown)

Description: Indicates whether a hospitalization following cancer diagnosis was due to a treatment-related toxicity or adverse event.

Reference: n/a

Data Source: Discharge summaries, inpatient progress notes, oncology treatment notes

Curation Directives:

- **Definition:** Captures whether the hospitalization was explicitly linked to a treatment-related toxicity, such as adverse effects from chemotherapy, immunotherapy, radiation, or other cancer-directed interventions.
 - **Permitted Values:**
 - 1 = Yes
 - 2 = No
 - 3 = Unknown
 - **Source Documents:**
 - Discharge summary
 - Admission or consult notes
 - Oncology progress notes
 - Nursing or toxicology documentation
 - **Curation Instructions:**
 - Code “**Yes**” (**1**) only if documentation clearly attributes hospitalization to a treatment-related adverse event or toxicity (e.g., febrile neutropenia, immune-related colitis, cytokine release syndrome).
 - Code “**No**” (**2**) if the hospitalization was not due to treatment toxicity (e.g., unrelated infections, pre-existing conditions).
 - Code “**Unknown**” (**3**) if no clear attribution is made in the documentation or the records are insufficient to determine the cause.
 - Do not infer causality unless explicitly stated in clinical documentation.
 - **Curator Quality Review:**
 - Verify consistency between toxicity attribution and clinical course described in the hospitalization records.
 - Ensure that treatment dates and types align with timing of hospitalization if coding “Yes.”
 - Flag conflicting or ambiguous notes for clinical review when necessary.
 - **Notes:**
 - This field supports pharmacovigilance and toxicity monitoring efforts across cancer therapies. High-quality abstraction can inform safety profiles and treatment planning.
-

Variable: hosp_tox_grade

Type: Categorical (Dropdown)

Description: Captures the documented severity grade of treatment-related toxicity that resulted in hospitalization, based on the Common Terminology Criteria for Adverse Events (CTCAE).

Reference: Toxicity Grade (CTCAE v5.0)

Data Source: Discharge summaries, inpatient progress notes, oncology toxicity assessments

Curation Directives:

- **Definition:** Indicates the highest recorded CTCAE grade of toxicity responsible for the patient's hospitalization, if explicitly documented in the medical record.
- **Permitted Values:**
Dropdown menu options include:
 - 1 = Grade 1 (Mild)
 - 2 = Grade 2 (Moderate)
 - 3 = Grade 3 (Severe)
 - 4 = Grade 4 (Life-threatening)
 - 5 = Grade 5 (Death related to toxicity)
- **Source Documents:**
 - Discharge summaries or attending physician notes referencing toxicity grade
 - Inpatient consults or oncology notes documenting CTCAE assessments
 - Toxicity monitoring templates or forms within the EHR
- **Curation Instructions:**
 - Only populate this field if the toxicity grade is explicitly mentioned in the clinical documentation.
 - Do not assign or infer a grade based on symptoms, lab values, or clinical judgment if not stated.
 - If multiple toxicity grades are reported during hospitalization, abstract the highest CTCAE grade linked to the treatment-related cause.
 - Leave blank if no toxicity grade is recorded or if the hospitalization was unrelated to treatment toxicity.
- **Notes:**
 - Accurate grading supports downstream analyses of treatment safety and adverse event burden. Do not extrapolate grades from symptoms or complications unless CTCAE grade is explicitly reported by the clinical team.

Laboratory Testing (Piped-In Only)

Piped In - Laboratory Testing

The following variables are piped in from external laboratory testing sources and are not manually curated by clinical data abstractors. These fields are auto-populated via data integration pipelines that extract structured laboratory test results (e.g., from EHR systems, LIS, or lab compendia). Fields are typically marked as @READONLY in the data collection platform to prevent manual edits. Curators may view but should not modify these fields. They are intended to support downstream clinical interpretation, quality audits, or correlation with diagnosis, treatment, and outcomes.

Piped-in Laboratory Testing Variables:

- **lab_collection_date – Specimen Collection Date**
Calendar date on which the biological specimen (e.g., blood, urine, tissue) was collected from the patient for laboratory testing.
- **lab_collection_date_int – Date of Specimen Collection in Days from DOB (Interval)**
Auto-calculated number of days between patient birth date and the specimen collection date.
- **lab_test – Laboratory Test – Name**
Full name of the lab test performed (e.g., CBC, CMP, TSH).
- **lab_loinc – Laboratory Test – LOINC Code**
Logical Observation Identifiers Names and Codes (LOINC) standard code identifying the laboratory test.
- **lab_result – Laboratory Test – Result**
Numerical or textual result value of the laboratory test performed on the specimen.
- **lab_result_unit – Laboratory Test – Unit**
Unit of measurement associated with the result value (e.g., g/dL, ng/mL, IU/L).
- **lab_lower_range – Laboratory Test – Lower Range Normal**
Lower bound of the reference range for the test result, typically provided by the performing lab.
- **lab_upper_range – Laboratory Test – Upper Range Normal**
Upper bound of the reference range for the test result.
- **lab_result_status – Laboratory Test – Result Comparison to Range**
Interpretation of the test result relative to the normal reference range, with typical values including: Low, In Range, High, or Out of Range.

Date Estimation Rules

General Principles

- Always record the **most complete date available** from the chart.
- Never guess or infer beyond what is documented.
- Apply the standardized estimation rules below when partial dates are encountered to assist in interval calculations.
- If no component of the date is documented, leave the field blank.

Estimation Rules

1. **Full date known (Year, Month, Day)**
 - Enter as documented in MM-DD-YYYY format.
 - *Example:* “05/07/2022” → 05-07-2022
2. **Year and Month known, but Day missing**
 - Enter the **15th** day of that month.
 - *Example:* “May 2023” → 05-15-2023
3. **Year known, but Month missing (Day may or may not be mentioned - always code as 15)**

- Record the date as **June 15** of the documented year.
- *Example:* “In 2021” → 06-15-2021

4. **Year unknown**

- Leave the date field **blank**. Do not impute month or day.
- *Example:* “Occurred prior to surgery” (no year documented) → leave blank.

Additional Notes

- These rules apply uniformly across **all date fields** in the dataset (e.g., diagnosis date, procedure date, hospitalization dates, treatment dates).
- Quality review should confirm that curators apply these estimation rules consistently across records.